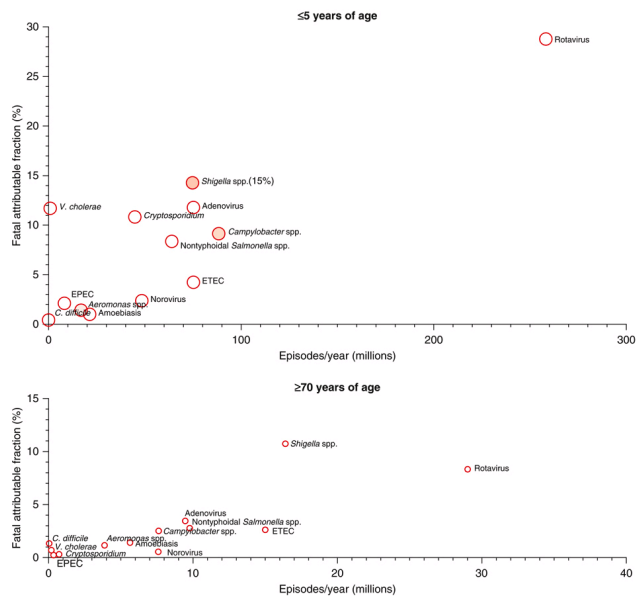


Gastroenteritis is one of the most common infectious disease syndromes. An estimated 4.5 billion episodes occur worldwide each year,¹⁻⁸ with children younger than 5 years old living in resource-limited settings accounting for most of these infections and nearly 500,000 deaths yearly.^{3,5,7} Despite decreasing global mortality rates, the collective morbidities caused by these infections in resource-limited settings remain significant.⁵⁻⁸ Also, the emergence of *Clostridioides difficile* (formerly *Clostridium difficile*) has been associated with increasing diarrhea-associated mortality in the United States and other resource-abundant countries.^{3,4} Despite considerable morbidity and mortality, most episodes of acute gastroenteritis are self-limited. In most cases there is no clinically recognizable inflammatory process (i.e., fever or dysentery),^{9,10} suggesting viral, toxigenic bacterial, or noninvasive parasitic etiologies.^{1,2,5-8} With the notable exceptions of profuse purging (cholera gravis) as a potentially fatal manifestation of *Vibrio cholerae* and diarrhea persisting for greater than 2 weeks caused by parasitic causes, clinical symptoms rarely indicate a specific etiology in an afebrile patient. Fever can accompany many etiologies of gastroenteritis; the absence of fever or blood in the stool, or both, diminishes, but does not rule-out, the likelihood of an invasive bacterial or amebic process and the urgency to evaluate the cause (see Chapter 101). Local and seasonal prevalence, outbreaks, host factors (e.g., age and immune compromise), antibiotic use, and the environment (varying across industrialized and nonindustrialized, and community and health care environments), influence the likelihood of the attributable pathogen.⁵⁻⁸ Fluid and electrolyte replacement appropriate to the severity of illness is the mainstay of treatment. Pathogen-directed diagnostic tests should guide antimicrobial therapy when indicated.

Epidemiology of Acute Noninflammatory Diarrhea

Despite declining global incidence with improving sanitation and the provision of clean water globally, a broad group of gastrointestinal (GI) pathogens remains that cause typically self-limited, watery diarrhea with or without nausea and vomiting, often accompanied by other abdominal complaints. Collectively, in 2016 these pathogens caused more than 1.6 million deaths globally, were the eighth leading cause of death for all ages, and were the fifth leading cause of death in children less than 5 years of age.³ Pathogen-attributable burden varies geographically and throughout the human life span and is largely influenced by economic indicators, social and biological influences on maternal-child dyads, and rotavirus vaccine availability.^{3,4,7,11-14} Abrupt and sweeping changes in social infrastructure and behavior as a result of sociopolitical disruptions, extreme weather events, and pandemic-related interventions can exert additional influences on exposures to these pathogens.¹⁵ The burden of GI infections is overwhelmingly skewed toward young children in resource-limited countries, particularly children with malnutrition. Therefore, diarrhea-related mortality in children less than 5 years old is almost exclusive to regions of sub-Saharan Africa, South Asia, and Haiti.³ Several areas of Latin America also report excessive diarrhea-attributable mortality in this age group, but these mortality trends have experienced greater reductions than sub-Saharan Africa. In these settings, increased mortality is seen not only with moderate-to-severe diarrhea but also with less severe diarrhea, and cases are

Description of FIG. 100.1



The graph is divided into two panels, with the top panel presenting data for children aged five or younger and the bottom panel for adults aged seventy or older. In both panels, the horizontal axis represents the number of episodes per year in millions, ranging from zero to approximately three hundred million episodes. The vertical axis represents the fatal attributable fraction, measured in percentages, ranging from zero to twenty-five percent. Each pathogen is represented by a circle, and the size of the circle indicates the number of episodes per year for that pathogen. The larger the circle, the more frequent the pathogen is in causing episodes of acute gastroenteritis. The top panel shows pathogens, namely *Shigella* species, *Cryptosporidium*, and *Vibrio cholerae*, with a specific note that *Shigella* is the most common cause of dysentery in children, with about fifteen percent of its cases presenting with dysentery. Other pathogens in this panel include EPEC, ETEC, *Campylobacter* species, and non-typhoidal *Salmonella* species. The data also includes pathogens like *Adenovirus* and *Norovirus*. The bottom panel focuses on adults aged seventy or older, with pathogens like *Shigella* spp., *Rotavirus*, and *Vibrio cholerae* shown again, but with different rates of episodes per year. This panel also includes similar pathogens, namely *Clostridioides difficile* and *Campylobacter* spp. The fatal attributable fraction for many pathogens is higher in this group, especially for *Shigella* species and *Rotavirus*.

attributable to a diverse group of pathogens, inclusive of viruses, bacteria, and protozoa.^{11,16} Further, even (identifiable) pathogen carriage-absent gastroenteritis associates with diminished childhood development in these settings.¹⁷ In contrast, pediatric gastroenteritis in the United States is seldom fatal, total diarrhea-attributable mortality across all ages in the United States increased by 36.8% from 2005 to 2015, largely due to the emergence of health care-associated and antibiotic-associated *C. difficile* diarrhea, which disproportionately affects older adults. In the United States, there are an estimated 179 million outpatient visits and nearly 500,000 hospitalizations due to foodborne or waterborne gastroenteritis per year.¹⁸

Community-Acquired Diarrhea Acute Pediatric Diarrhea

The rapid decline of infant mortality rates during the 20th century coincided with increased availability of piped water and chlorination systems. In areas of the world that continue to have unsafe water and sanitation, gastroenteritis remains a major cause of infant mortality.³ The highest attack rate of diarrhea occurs at the time of weaning (weanling diarrhea), usually between 6 and 24 months of age.^{19,20} The increased susceptibility of a recently weaned infant is related to several factors.²¹ First, human breast milk appears to convey a level of resistance to some GI pathogens and may protect an infant who is still breastfeeding.²²⁻²⁶ In contrast, in resource-limited settings, weaning and complementary foods are often prepared under conditions of poor hygiene and are frequently found to be contaminated with large numbers of potential diarrheal pathogens.^{27,28} Children may also ingest soil contaminated with diarrheagenic bacteria.²⁹ Undernutrition can influence host defenses and pathogen-attributable diarrhea risk.³⁰⁻³³ Finally, emerging studies indicate shifts in microbiota may mediate susceptibility to particular gut pathogens.³⁴

Early childhood diarrhea, regardless of etiology, often manifests as acute, watery, noninflammatory diarrhea. Cases are sporadic, but there are seasonal patterns, often peaking in the summer months or periods of high precipitation in resource-limited settings.⁷ In a well-nourished infant, the disease is usually short-lived and resolves within 2 to 3 days with adequate hydration.^{20,35} Diarrhea in a malnourished or immunocompromised child tends to persist or to recur and is often much more severe.

Early childhood diarrhea has been most commonly associated with rotaviruses³⁵⁻³⁷ and enterotoxigenic *Escherichia coli* (ETEC).^{38,39,40} Use of molecular diagnostics in longitudinal and case-control studies has refined these pathogen-attributable estimates across multiple global regions.^{5-7,41,42} Using quantitative polymerase chain reaction (PCR) methods, up to 90% of moderate-to-severe acute diarrheal episodes were pathogen-attributable in low- to middle-income settings.⁵ Rotavirus is the most common global etiology of severe acute watery diarrhea, and remains one of the most common identified agents even in the postvaccine era.^{5,7,41,42} Other viruses—predominately norovirus GII and adenovirus types 40 and 41, and to a lesser extent, sapovirus and astrovirus—are significant contributors, especially where rotavirus vaccine is established.^{5,7,42-44} Cases due to *Shigella* spp. slightly outnumber cases caused by heat-stable enterotoxin-producing ETEC (ST-ETEC).^{5,7,42} These studies have also revealed the emerging importance of previously difficult to detect or noncultivable pathogens such as *Campylobacter* spp. and the protozoan *Cryptosporidium*. In some settings, *Cryptosporidium* is second only to viral etiologies and, together with rotavirus and *Shigella* spp., may combine to account for more than 50% of diarrhea-attributable mortality in children less than 5 years of age.⁴² Although these studies have clarified the major drivers of early childhood diarrhea at the population level, the clinical application of these tools for individual cases is more challenging. First, molecular diagnostics are seldom available. Second, 40% or more of diarrheal stools may harbor two or more pathogens simultaneously.⁵

Diarrhea Caused by Rotavirus

Rotaviruses are the most common cause of severe diarrhea in infants and young children (see Chapter 157).^{42,45-48} Worldwide, rotaviruses cause more than 100 million cases of gastroenteritis and up to 150,000 deaths each year in children younger than 5 years old.^{3,46,49} Of these rotavirus-induced deaths, more than 90% occur in children residing in low- or middle-income countries.⁴⁹ Although most adults have antibody to rotaviruses that may be protective, unvaccinated children younger than 2 years are highly susceptible to rotavirus diarrhea.⁴⁶

In the United States, few deaths are attributable to rotavirus infection; however, before inclusion of rotavirus vaccine in childhood immunization programs, almost every child had been infected by the age of 5 years, and rotavirus-induced gastroenteritis caused about 400,000 physician encounters, almost 200,000 emergency department visits, and about 70,000 hospitalizations each year.⁴⁶ Because routine vaccination began in 2008, rotavirus-related diarrhea has declined by 62% to 71%, and hospitalization for all-cause gastroenteritis has declined by 31% to 33%.⁵⁰⁻⁵² Similar trends are emerging in the >100 countries that have implemented rotavirus vaccination campaigns, resulting in an estimated median reduction of 59% for rotavirus-specific hospitalizations and 36% for all-cause gastroenteritis mortality.⁵³ Nonetheless, among children in the United States admitted for acute gastroenteritis, rotavirus remains important.⁴¹ Infection occurs more frequently in the winter or in cooler, dry months^{35,36,54-59} than in the summer months.^{7,60}

There are five major rotavirus serotypes, A through E, with group A strains further subdivided based on the combination of G (the VP7 glycoprotein) and P (the VP4 protease-cleaved protein) types.^{46,47,61-64} Worldwide, five serotypes (G1–G4 and G9) predominate.^{46,47,61-63,65} A particular strain (the number in brackets indicates the genotype), P1B[4]G2, is associated with a more severe form of infection in children.^{66,67} Such epidemiologic information was important in designing rotavirus vaccines. Before rotavirus vaccination in the United States, six rotavirus strains were most commonly found: G1P[8], G2P[4], G3P[8], G4P[8], G9P[8], and G9P[6].^{63-65,68} In the postvaccine era, G3P[8] was predominant through 2011 and has since been replaced by G12P[8]. Vaccine-derived strains occur in 1.7% of cases.^{69,70} Recent outbreaks of milder illness have been associated with more unusual serotypes; for example, strain G8P[8], which affected vaccinated and nonvaccinated individuals equally.⁷¹ Introduction of rotavirus vaccines in other settings has also corresponded with emergences of G3P[8] and other less common strains.^{72,73}

Rotavirus diarrhea is usually mild, without high-grade fever, and is often associated with vomiting; however, clinical syndromes range from asymptomatic carriage to severe gastroenteritis with profound dehydration, shock, and death.^{35,36,45,46,74} Serious complications of rotavirus gastroenteritis include necrotizing enterocolitis in neonates.^{68,75,76} Severe cases are more common in resource-limited settings^{46,75} and among severely immunocompromised patients. Large symptomatic outbreaks of moderately severe rotavirus diarrhea can occur. Although outbreaks are geographically distinct, they are often caused by the same rotavirus strain.

The diagnosis of rotavirus diarrhea is established with the use of a variety of molecular assays designed to detect virus-specific antigens, antibodies, or nucleic acids in feces of infected patients.⁷⁷⁻⁸⁰ Rotaviruses can also be cultured directly from stool samples, although this method is time-consuming, expensive, and limited primarily to research-based laboratories.

The human rotaviruses are nonenveloped RNA viruses of approximately 70 nm belonging to the Reoviridae family.^{46,61,62} The major Page 1282 rotavirus-mediated effects on the host include disruption of intestinal epithelial cell brush-border enzymes, cytotoxicity, and alteration of the enteric nervous system.^{47,48,81-83} Normal columnar epithelium at the villus tips is replaced by irregular cuboidal, cryptlike cells, leading to multiple defects in fluid and electrolyte regulation. Many of these effects are believed to be mediated by a calcium ion-dependent enterotoxin.

The loss of absorptive villus tip cells may be responsible for the fluid imbalance and nutritional effect of rotavirus infections. The degree of microvillus damage roughly parallels the severity of diarrhea and dehydration.⁸⁴ Certain patients with rotavirus gastroenteritis also exhibit extraintestinal symptoms in association with detectable antigenemia and viremia, supporting a systemic phase to this infection.⁸⁵⁻⁸⁷

Oral or intravenous rehydration is the cornerstone of therapy for rotavirus gastroenteritis. No definitive antiviral therapy is established for rotavirus infection; however, the antiparasitic drug nitazoxanide was shown in randomized, controlled studies to shorten the duration of illness in children and adults with symptomatic rotavirus diarrhea.^{88,89} Possible mechanistic explanations for an antiviral function of nitazoxanide include interrupting of cellular processes underlying viral replication^{90,91} or antibacterial effects that result in enhanced antiviral immunity. Although multiple different probiotics are available in many parts of the world, adjuvant probiotic use may

reduce repeated diarrhea episodes and promote intestinal function after rotavirus infection,⁹² but the role of specific probiotics regarding duration of shedding and symptoms⁹³ and severity of illness across different populations requires further investigation in controlled clinical trials.⁹⁴

Because of the great global burden of childhood rotavirus gastroenteritis, significant multinational efforts have been directed toward the development of a safe and effective rotavirus vaccine. The first rotavirus vaccine approved for use in the United States was the rhesus-human reassortant rotavirus tetravalent vaccine (RotaShield; Wyeth-Lederle, Madison, NJ). It was approved and released in 1998 for use in infants. Despite high efficacy, the vaccine was removed less than 1 year later because of concerns about possible cases of intussusception in vaccinated infants.^{95,96} After the withdrawal of the rhesus-human reassortant rotavirus tetravalent vaccine, two new rotavirus vaccines became licensed and approved for use in infants in the United States^{46,62,97,98}: first, a live oral pentavalent rotavirus vaccine (RotaTeq; Merck Vaccines, Whitehouse Station, NJ) comprising bovine-human recombinant strains in 2006,^{46,97,98,99} followed by a monovalent human rotavirus vaccine (Rotarix; GlaxoSmithKline, London, United Kingdom) derived from an attenuated P1A[8]G1 rotavirus strain in 2008.^{46,97,98,99-101} Both are administered starting between 6 and 14 weeks of age in a three-dose (pentavalent rotavirus vaccine) or two-dose (human rotavirus vaccine) schedule and are highly efficacious. No significant increased risk of intestinal intussusception has been observed to date. Neither vaccine should be administered to infants with severe combined immunodeficiency syndrome¹⁰²; however, despite a theoretical risk of transmission of vaccine-derived strains within a household, living with an immunosuppressed patient is not an absolute contraindication.¹⁰³

Routine rotavirus vaccination has dramatically reduced rotavirus-diarrhea in the United States, and use of the rotavirus vaccine has been estimated to have saved \$1.2 billion in annual health care costs associated with acute gastroenteritis.⁵¹ Rotavirus vaccine is now available in more than 100 countries; however, vaccine efficacy varies geographically and is lower in resource-limited countries where the bulk of rotavirus infections still occur.³ Among the potential reasons for decreased rotavirus efficacy is as yet poorly understood differences in mucosal immune responses in children in these settings, particularly children with evidence of undernutrition, and possibly differences in intestinal microbial ecology or the presence of high maternal antirotavirus immunoglobulin A, or both.¹⁰⁴⁻¹⁰⁸ However, prenatal nutritional supplementation did not improve rotavirus vaccine immunogenicity.¹⁰⁹ A parenterally administered P2-VP8-P[8] subunit vaccine appears to be safe and immunogenic.¹¹⁰

Acute Nausea and Vomiting (Winter Vomiting Disease)

The syndrome of acute nausea and vomiting (intestinal flu, viral gastroenteritis, or what had been called winter vomiting disease), sometimes with low-grade fever, commonly occurs in winter months in temperate climates.^{111,112} Although this syndrome overlaps with rotavirus-associated infantile gastroenteritis, rotaviruses are relatively uncommon causes of winter vomiting disease in older children and adults. The Cleveland family studies of Dingle and coworkers¹¹³ showed GI illnesses were most common between the ages of 1 and 10 years, when approximately two illnesses occurred per person per year. Illnesses peaked from November through February, with June being the month of lowest frequency. Most illnesses lasted 1 to 3 days; 20% occurred with respiratory symptoms, and 20% involved only diarrhea.

Illnesses tended to occur in one of two patterns: (1) a mild afebrile illness with watery diarrhea or (2) a more severe febrile illness with vomiting, headache, and constitutional symptoms. Although etiologic agents were rarely identified, these two patterns of illness also developed among volunteers who ingested filtrates prepared from the feces of ill patients.¹¹⁴⁻¹¹⁶ Studies done in Charlottesville, Virginia, confirmed this pattern of wintertime gastroenteritis including clustering in families, highest attack rates in children, and absence of identifiable etiologic agents in most cases, despite the application of techniques for virologic and enterotoxin studies.^{117,118}

Caliciviruses are now recognized as the etiologies of most cases of gastroenteritis occurring in winter months. The Caliciviridae family comprises four genera^{119,120}: Norovirus, Sapovirus, Lagovirus, and Vesivirus (see Chapter 182). Viruses belonging to Norovirus (from Norwalk) and Sapovirus (from Sapporo) genera are currently the most common causative agents for viral gastroenteritis in humans.¹¹⁹⁻¹²³ Noroviruses and sapoviruses may predominate as causes of early childhood diarrhea in rotavirus-vaccinated populations.^{5,6,42,124} Noroviruses, in particular,

cause major outbreaks, especially in older children and adults (see Chapter 182). Immunocompromised patients can develop chronic norovirus infection lasting for months or years (see “Diarrhea in Immunocompromised Patients”).¹²⁵

Norovirus

Norovirus accounts for more than one-third of the outbreaks of nonbacterial gastroenteritis in the United States.^{126,127} Initially these viruses were referred to as Norwalk-like viruses. Their individual names were originally derived from the site of origin of each particular outbreak and included Norwalk,¹²⁸⁻¹³⁰ Hawaii,¹³¹ Snow Mountain,¹³² Taunton, and W agents.¹³³ Person-to-person spread is common, and secondary attack rates can be high. Transmission is predominately fecal-oral, but salivary transmission has been increasingly recognized.¹³⁴

Norovirus outbreaks have been observed in a wide variety of settings,¹³⁵ including hospitals,¹³⁶ extended-care facilities,¹³⁷ child care centers,¹³⁸ cruise ships,¹³⁹ refugee centers associated with natural disasters (e.g., Hurricane Katrina¹⁴⁰), and military combat areas including Afghanistan¹⁴¹ and Iraq.¹⁴² Several genotypes are known to infect humans^{121,135}; in the last decade, newly identified variant strains of norovirus genotype GII.4 have caused outbreaks worldwide¹⁴³⁻¹⁴⁵ including in the United States, United Kingdom, Europe, New Zealand, and Australia.

The pathophysiologic features of norovirus gastroenteritis parallel features of rotavirus in several respects.^{119,120,121} Each type is essentially noninflammatory and causes villus disruption and transient brush-border enzyme deficiencies in the upper small intestine, without any alteration in adenylate cyclase activity.^{63,146,147} This enteropathy can be severe in patients with underlying immunocompromise, especially patients with common variable immunodeficiency.¹⁴⁸ The roles of enzyme deficiencies, malabsorption of xylose and lactose, and an increase in intestinal bacteria during norovirus illness remain unclear.^{146,149}

Sapovirus

Global studies on etiology of pediatric diarrhea in low- to middle-income countries have unmasked a greater burden of sapovirus diarrhea than previously recognized.^{5,6,150} Sapovirus was the second most common all-cause etiology of diarrhea in the community setting,⁶ and cumulative incidence may exceed 80% by 2 years of age.¹⁵¹ Sapoviruses were initially called typical human caliciviruses or Sapporo-like viruses after the prototype strain found in an outbreak in Sapporo, Japan, in 1982. Similar to norovirus, diverse sapovirus genogroups and genotypes cause infections in humans. Repeated infections are typically due to different genotypes, suggesting immunity is genotype specific.¹⁵¹ Sapovirus symptoms overlap with norovirus symptoms, possibly with more prolonged duration of vomiting.¹⁵² Similar to norovirus, sapovirus can cause outbreaks in long-term care facilities, but fewer laboratories specifically test for sapovirus. Sapovirus should be considered if the cause for an outbreak is otherwise unexplained.¹⁵³ Sapovirus also occurs as a coinfection with other pathogens.

Diarrhea Caused by Astrovirus, Adenovirus, and Other Possible Viral Etiologies

Several other groups of viral pathogens have been identified in endemic and epidemic gastroenteritis (Table 100.1), including^{45,77,154-156} (1) other types of caliciviruses with characteristic, “chalice-like” surface; (2) astroviruses, with a five- or six-pointed starlike surface structure¹⁵⁷; (3) enteric adenoviruses (especially serotypes 40 and 41)^{5,158}; and (4) human enteric coronaviruses, including SARS-CoV-2,^{159,160,161} as well as other miscellaneous or less well-characterized viral agents. More detailed discussions of these agents may be found in Chapters 149, 162, 178, 180, 183, and 184.

Astroviruses are among the most common causes of pediatric viral gastroenteritis.¹⁶²⁻¹⁶⁵ Astrovirus occasionally occurs in association with other enteric pathogens; in these cases the illness is more severe and protracted.¹⁶⁴ After rotavirus, adenovirus 40/41 was the most common viral etiology of moderate-to-severe acute diarrhea in children in low-income settings and the third most common cause overall.⁵ Adenovirus 40/41 was also third only to rotavirus and norovirus as a cause of acute gastroenteritis among children hospitalized in a US health care system.⁴¹ Increased emphasis on adenovirus serotyping in infant-child cohort studies is revealing a likely widespread greater burden of disease caused by this virus.^{166,167} In a case-control study in

China, adenovirus serotype 3 was also associated with diarrhea, whereas asymptomatic detection of other serotypes (1, 2, 5, 6, 31, and 56) were more common in controls without diarrhea.¹⁶⁸

Vomiting and diarrhea can also manifest during infections with respiratory viruses, including in the absence of a respiratory syndrome. Enteric coronaviruses have clear pathology in animals (i.e., transmissible gastroenteritis in pigs). Although human coronaviruses, including the novel SARS-CoV-2 sarbecovirus, are predominately respiratory pathogens, these viruses can be isolated from intestinal tissues and stools.^{159,160,169,161} SARS-CoV-2 and all four of the endemic human coronaviruses (HCoV-229E, HCoV-NL63, HCoV-HKU1, and HCoV-OC43) have been detected in cases of acute gastroenteritis. Acute gastroenteritis from SARS-CoV-2 (see Chapter 135) infection¹⁶¹ can be severe enough to require hospitalization, however, the role of endemic human coronavirus as a primary etiology of diarrhea is likely minor; codetection with other common viruses such as rotavirus and norovirus is frequent.¹⁶⁹ Up to 10% to 20% of children with seasonal influenza present with diarrhea, although this is uncommon in adults. Diarrhea may be more prominent with Asian avian influenza A (H5N1) infection.¹⁷⁰

Bocaviruses,¹⁷¹ pestiviruses,¹⁷² toroviruses,^{173,174} picobirnaviruses,¹⁷⁵ and protoparvoviruses¹⁷⁶ are becoming better recognized as pathogens in children and adults, but their cooccurrence with conventional intestinal viruses and their independent roles in causing diarrheal illnesses are uncertain.^{176,177} Similarly, although enteroviruses may cause systemic illnesses, their role as diarrheal pathogens is minimal. Unbiased viral metagenomics approaches have further identified sequences of obscure viruses in diarrheal stools that require further clarification.^{178,179} In addition, a hallmark characteristic of the 2014 Ebola virus disease epidemic caused by Zaire ebolavirus in West Africa was nausea and abdominal pain, with voluminous vomiting and diarrhea that led to life-threatening cholera-like GI volume losses through unclear mechanisms.^{180,181}

Nucleic acid–based assays (e.g., reverse-transcriptase PCR) of stool specimens, either pathogen-specific or as part of multipathogen panels, are the standard method of GI virus detection. Rotavirus-specific and norovirus-specific reverse-transcriptase PCR are also available. In the typical outpatient clinical situation, norovirus-specific diagnostic tests are rarely performed because results of this assay do not alter management of the illness.

Cryptosporidiosis

Cryptosporidium is another cause of acute, noninflammatory diarrhea, second only to viruses in some countries.^{5,7,182} Sporadic cryptosporidiosis typically occurs through direct exposure to either infected people and to a lesser extent animals.^{183,184} Outbreaks occur via contaminated food, drinking water, or recreational water including swimming pools.^{185–190} Of the more than 20 species of this tiny coccidian intracellular protozoal parasite, two, *C. parvum* and *C. hominis*, cause the majority of human disease.^{189–193} Cryptosporidiosis is commonly asymptomatic or mild in normal hosts and usually results from waterborne outbreaks, often related to contaminated drinking water or public swimming pools.^{186,189–193} However, infection can be severe in infants and toddlers, accounting for an estimated 50,000 to >200,000 deaths per year.^{182,194} Malnourished children and immunocompromised hosts, especially patients with advanced human immunodeficiency virus (HIV) infection, can manifest severe diarrhea.^{185,187,192–195} Cryptosporidium, either with or without diarrhea, in some children can also contribute to early childhood growth impairment.^{182,196} Parasite shedding even after recovery from acute diarrhea can be prolonged in hosts with immunocompromise such as those with uncontrolled HIV.¹⁹⁷

Cryptosporidium causes a secretory form of diarrhea associated with dysregulated intestinal absorption that may be less responsive to oral rehydration than other diarrhea etiologies.¹⁹⁰ In addition to gastroenteritis, biliary infection with cholangitis has been reported. The pathogenesis of cryptosporidiosis is incompletely understood; the organism primarily alters villus structure and function and can promote an inflammatory enteropathy that may persist after parasite clearance.¹⁹⁰ No specific enterotoxin has been identified. Diagnosis depends on identification of oocysts in stool specimens by standard light microscopy, immunofluorescence assays, or nucleic acid–based molecular tests.^{198–200} In addition to supportive care, nitazoxanide has been shown to be moderately successful in immunocompetent patients with cryptosporidiosis. Data are

inconclusive about universally recommended duration of therapy, especially in immunocompromised and malnourished patients.²⁰¹⁻²⁰⁷ Alternative therapeutic agents for immunosuppressed patients include combinations of paromomycin and azithromycin. Ongoing searches for novel drugs have identified several with anti-Cryptosporidium activity through different mechanisms, but all remain in preclinical stages of development.²⁰⁸⁻²¹⁵ Despite resurging interest, vaccine development remains in early stages, in part because determinants of host immunity, especially in malnourished populations, are poorly understood.

Diarrhea Caused by Enteropathogenic Bacteria

Several diarrheagenic *E. coli* pathotypes (ETEC, enteroaggregative *E. coli* [EAEC], enteropathogenic *E. coli* [EPEC], enteroinvasive *E. coli* [EIEC], and Shiga toxin-producing *E. coli* [STEC], *Shigella* spp., *Salmonella* spp., *Campylobacter* spp., and both *V. cholerae* and noncholera *Vibrio* spp.) can cause noninflammatory diarrhea (Fig. 100.1). ETEC, *Campylobacter*, and *Shigella* are the most common bacterial etiologies of acute gastroenteritis worldwide.^{42,3} The presence of blood in stool (dysentery) increases the likelihood of diarrhea due to *Shigella*, *Campylobacter*, or EIEC; however, all bacterial pathogens most commonly present as a noninflammatory syndrome.^{5,7} Dysentery is present in <5% of community-associated diarrhea, and inflammatory manifestations of these bacterial gastroenteritides are discussed in greater detail in Chapter 101. Among bacterial etiologies that manifest only as noninflammatory diarrhea, ETEC and *V. cholerae* combined account for 9% to 24% of diarrhea-attributable mortality in children younger than 5 years of age.³

Viral Pathogens Causing Gastroenteritis

ESTABLISHED PATHOGENS	LIKELY AND EMERGING PATHOGENS
Adenoviruses (enteric types) Astroviruses Caliciviruses (including noroviruses and sapoviruses) Rotaviruses groups A–C Cytomegalovirus	Bocaviruses Coronaviruses (including SARS-CoV-2) Enteroviruses (various) Picobirnaviruses, picornaviruses Pestiviruses Toroviruses Filovirus (Ebola virus) Human protoparvoviruses

Data from references 45, 68, 77, 122, 123, 135, 138, 155, 157, 179, 440, 441, 176, 232, 233.

Nucleic acid–based assays (e.g., reverse-transcriptase PCR) of stool specimens, either pathogen-specific or as part of multipathogen panels, are the standard method of GI virus detection. Rotavirus-specific and norovirus-specific reverse-transcriptase PCR are also available. In the typical outpatient clinical situation, norovirus-specific diagnostic tests are rarely performed because results of this assay do not alter management of the illness.

Cryptosporidium

The greatest risk for pediatric ETEC diarrhea is between age 12 and 24 months, and severity is increased in undernourished children.^{216,42} Antibodies directed against the heat-labile enterotoxin (LT) of *E. coli* in human colostrum may account for lower burden of ETEC in infancy.^{22,23,217} Diarrhea produced by LT shares the adenylate cyclase-activating mechanism with cholera toxin.²¹⁸⁻²²² Certain strains of *Klebsiella*, *Citrobacter*, and *Aeromonas* produce an LT-like toxin and may cause a small percentage of diarrheal disease in children.²²³

Molecular methods have refined estimates of the contributions of other diarrheagenic *E. coli* on burdens of diarrhea. Typical EPEC is a contributor to diarrhea-related morbidity and mortality in the first 2 years of life.²²⁴ EAEC, although frequently subclinical, is among the most commonly identified enteropathogens. EAEC infection is nearly universal in children in resource-limited settings. Despite no clear association with diar

Community-Acquired Diarrhea

Acute Pediatric Diarrhea

The rapid decline of infant mortality rates during the 20th century coincided with increased availability of piped water and chlorination systems. In areas of the world that continue to have unsafe water and sanitation, gastroenteritis remains a major cause of infant mortality.³ The highest attack rate of diarrhea occurs at the time of weaning (weanling diarrhea), usually between 6 and 24 months of age.^{19,20} The increased susceptibility of a recently weaned infant is related to several factors.²¹ First, human breast milk appears to convey a level of resistance to some GI pathogens and may protect an infant who is still breastfeeding.²²⁻²⁶ In contrast, in resource-limited settings, weaning and complementary foods are often prepared under conditions of poor hygiene and are frequently found to be contaminated with large numbers of potential diarrheal pathogens.^{27,28} Children may also ingest soil contaminated with diarrheagenic bacteria.²⁹ Undernutrition can influence host defenses and pathogen-attributable diarrhea risk.³⁰⁻³³ Finally, emerging studies indicate shifts in microbiota may mediate susceptibility to particular gut pathogens.³⁴

Early childhood diarrhea, regardless of etiology, often manifests as acute, watery, noninflammatory diarrhea. Cases are sporadic, but there are seasonal patterns, often peaking in the summer months or periods of high precipitation in resource-limited settings.⁷ In a well-nourished infant, the disease is usually short-lived and resolves within 2 to 3 days with adequate hydration.^{20,35} Diarrhea in a malnourished or immunocompromised child tends to persist or to recur and is often much more severe.

Early childhood diarrhea has been most commonly associated with rotaviruses³⁵⁻³⁷ and enterotoxigenic *Escherichia coli* (ETEC).^{38,39,40} Use of molecular diagnostics in longitudinal and case-control studies has refined these pathogen-attributable estimates across multiple global regions.^{5-7,41,42} Using quantitative polymerase chain reaction (PCR) methods, up to 90% of moderate-to-severe acute diarrheal episodes were pathogen-attributable in low- to middle-income settings.⁵ Rotavirus is the most common global etiology of severe acute watery diarrhea, and remains one of the most common identified agents even in the postvaccine era.^{5,7,41,42} Other viruses—predominately norovirus GII and adenovirus types 40 and 41, and to a lesser extent, sapovirus and astrovirus—are significant contributors, especially where rotavirus vaccine is established.^{5,7,42-44} Cases due to *Shigella* spp. slightly outnumber cases caused by heat-stable enterotoxin-producing ETEC (ST-ETEC).^{5,7,42} These studies have also revealed the emerging importance of previously difficult to detect or noncultivable pathogens such as *Campylobacter* spp. and the protozoan *Cryptosporidium*. In some settings, *Cryptosporidium* is second only to viral etiologies and, together with rotavirus and *Shigella* spp., may combine to account for more than 50% of diarrhea-attributable mortality in children less than 5 years of age.⁴² Although these studies have clarified the major drivers of early childhood diarrhea at the population level, the clinical application of these tools for individual cases is more challenging. First, molecular diagnostics are seldom available. Second, 40% or more of diarrheal stools may harbor two or more pathogens simultaneously.⁵

Diarrhea Caused by Rotavirus

Rotaviruses are the most common cause of severe diarrhea in infants and young children (see Chapter 157).^{42,45-48} Worldwide, rotaviruses cause more than 100 million cases of gastroenteritis and up to 150,000 deaths each year in children younger than 5 years old.^{3,46,49} Of these rotavirus-induced deaths, more than 90% occur in children residing in low- or middle-income countries.⁴⁹ Although most adults have antibody to rotaviruses that may be protective, unvaccinated children younger than 2 years are highly susceptible to rotavirus diarrhea.⁴⁶

In the United States, few deaths are attributable to rotavirus infection; however, before inclusion of rotavirus vaccine in childhood immunization programs, almost every child had been infected by the age of 5 years, and rotavirus-induced gastroenteritis caused about 400,000 physician encounters, almost 200,000 emergency department visits, and about 70,000 hospitalizations each year.⁴⁶ Because routine vaccination began in 2008, rotavirus-related diarrhea has declined by 62% to 71%, and hospitalization for all-cause gastroenteritis has declined by 31% to 33%.⁵⁰⁻⁵² Similar trends are emerging in the >100 countries that have implemented rotavirus vaccination campaigns, resulting in an estimated median reduction of 59% for rotavirus-specific hospitalizations and 36% for all-cause gastroenteritis mortality.⁵³ Nonetheless, among children in the United States admitted for acute gastroenteritis, rotavirus remains important.⁴¹ Infection occurs more frequently in the winter or in cooler, dry months^{35,36,54-59} than in the summer months.^{7,60}

There are five major rotavirus serotypes, A through E, with group A strains further subdivided based on the combination of G (the VP7 glycoprotein) and P (the VP4 protease-cleaved protein) types.^{46,47,61-64} Worldwide, five serotypes (G1–G4 and G9) predominate.^{46,47,61-63,65} A particular strain (the number in brackets indicates the genotype), P1B[4]G2, is associated with a more severe form of infection in children.^{66,67} Such epidemiologic information was important in designing rotavirus vaccines. Before rotavirus vaccination in the United States, six rotavirus strains were most commonly found: G1P[8], G2P[4], G3P[8], G4P[8], G9P[8], and G9P[6].^{63-65,68} In the postvaccine era, G3P[8] was predominant through 2011 and has since been replaced by G12P[8]. Vaccine-derived strains occur in 1.7% of cases.^{69,70} Recent outbreaks of milder illness have been associated with more unusual serotypes; for example, strain G8P[8], which affected

vaccinated and nonvaccinated individuals equally.⁷¹ Introduction of rotavirus vaccines in other settings has also corresponded with emergences of G3P[8] and other less common strains.^{72,73}

Rotavirus diarrhea is usually mild, without high-grade fever, and is often associated with vomiting; however, clinical syndromes range from asymptomatic carriage to severe gastroenteritis with profound dehydration, shock, and death.^{35,36,45,46,74} Serious complications of rotavirus gastroenteritis include necrotizing enterocolitis in neonates.^{68,75,76} Severe cases are more common in resource-limited settings^{46,75} and among severely immunocompromised patients. Large symptomatic outbreaks of moderately severe rotavirus diarrhea can occur. Although outbreaks are geographically distinct, they are often caused by the same rotavirus strain.

The diagnosis of rotavirus diarrhea is established with the use of a variety of molecular assays designed to detect virus-specific antigens, antibodies, or nucleic acids in feces of infected patients.⁷⁷⁻⁸⁰ Rotaviruses can also be cultured directly from stool samples, although this method is time-consuming, expensive, and limited primarily to research-based laboratories.

The human rotaviruses are nonenveloped RNA viruses of approximately 70 nm belonging to the Reoviridae family.^{46,61,62} The major Page 1282rotavirus-mediated effects on the host include disruption of intestinal epithelial cell brush-border enzymes, cytotoxicity, and alteration of the enteric nervous system.^{47,48,81-83} Normal columnar epithelium at the villus tips is replaced by irregular cuboidal, cryptlike cells, leading to multiple defects in fluid and electrolyte regulation. Many of these effects are believed to be mediated by a calcium ion-dependent enterotoxin.

The loss of absorptive villus tip cells may be responsible for the fluid imbalance and nutritional effect of rotavirus infections. The degree of microvillus damage roughly parallels the severity of diarrhea and dehydration.⁸⁴ Certain patients with rotavirus gastroenteritis also exhibit extraintestinal symptoms in association with detectable antigenemia and viremia, supporting a systemic phase to this infection.⁸⁵⁻⁸⁷

Oral or intravenous rehydration is the cornerstone of therapy for rotavirus gastroenteritis. No definitive antiviral therapy is established for rotavirus infection; however, the antiparasitic drug nitazoxanide was shown in randomized, controlled studies to shorten the duration of illness in children and adults with symptomatic rotavirus diarrhea.^{88,89} Possible mechanistic explanations for an antiviral function of nitazoxanide include interrupting of cellular processes underlying viral replication^{90,91} or antibacterial effects that result in enhanced antiviral immunity. Although multiple different probiotics are available in many parts of the world, adjuvant probiotic use may reduce repeated diarrhea episodes and promote intestinal function after rotavirus infection,⁹² but the role of specific probiotics regarding duration of shedding and symptoms⁹³ and severity of illness across different populations requires further investigation in controlled clinical trials.⁹⁴

Because of the great global burden of childhood rotavirus gastroenteritis, significant multinational efforts have been directed toward the development of a safe and effective rotavirus vaccine. The first rotavirus vaccine approved for use in the United States was the rhesus-human reassortant rotavirus tetravalent vaccine (RotaShield; Wyeth-Lederle, Madison, NJ). It was approved and released in 1998 for use in infants. Despite high efficacy, the vaccine was removed less than 1 year later because of concerns about possible cases of intussusception in vaccinated infants.^{95,96} After the withdrawal of the rhesus-human reassortant rotavirus tetravalent vaccine, two new rotavirus vaccines became licensed and approved for use in infants in the United States^{46,62,97,98}: first, a live oral pentavalent rotavirus vaccine (RotaTeq; Merck Vaccines, Whitehouse Station, NJ) comprising bovine-human recombinant strains in 2006,^{46,97,98,99} followed by a monovalent human rotavirus vaccine (Rotarix; GlaxoSmithKline, London, United Kingdom) derived from an attenuated P1A[8]G1 rotavirus strain in 2008.^{46,97,98,99-101} Both are administered starting between 6 and 14 weeks of age in a three-dose (pentavalent rotavirus vaccine) or two-dose (human rotavirus vaccine) schedule and are highly efficacious. No significant increased risk of intestinal intussusception has been observed to date. Neither vaccine should be administered to infants with severe combined immunodeficiency syndrome¹⁰²; however, despite a theoretical risk of transmission of vaccine-derived strains within a household, living with an immunosuppressed patient is not an absolute contraindication.¹⁰³

Routine rotavirus vaccination has dramatically reduced rotavirus-diarrhea in the United States, and use of the rotavirus vaccine has been estimated to have saved \$1.2 billion in annual health care costs associated with acute gastroenteritis.⁵¹ Rotavirus vaccine is now available in more than 100 countries; however, vaccine efficacy varies geographically and is lower in resource-limited countries where the bulk of rotavirus infections still occur.³ Among the potential reasons for decreased rotavirus efficacy is as yet poorly understood differences in mucosal immune responses in children in these settings, particularly children with evidence of undernutrition, and possibly differences in intestinal microbial ecology or the presence of high maternal antirotavirus immunoglobulin A, or both.¹⁰⁴⁻¹⁰⁸ However, prenatal nutritional supplementation did not improve rotavirus vaccine immunogenicity.¹⁰⁹ A parenterally administered P2-VP8-P[8] subunit vaccine appears to be safe and immunogenic.¹¹⁰

Acute Nausea and Vomiting (Winter Vomiting Disease)

The syndrome of acute nausea and vomiting (intestinal flu, viral gastroenteritis, or what had been called winter vomiting disease), sometimes with low-grade fever, commonly occurs in winter months in temperate climates.^{111,112} Although this syndrome overlaps with rotavirus-associated infantile gastroenteritis, rotaviruses are relatively uncommon causes of winter vomiting disease in older children and adults. The Cleveland family studies of Dingle and coworkers¹¹³ showed GI illnesses were most common between the ages of 1 and 10 years, when approximately two illnesses occurred per person per year. Illnesses peaked from November through February, with June being the month of lowest frequency. Most illnesses lasted 1 to 3 days; 20% occurred with respiratory symptoms, and 20% involved only diarrhea.

Illnesses tended to occur in one of two patterns: (1) a mild afebrile illness with watery diarrhea or (2) a more severe febrile illness with vomiting, headache, and constitutional symptoms. Although etiologic agents were rarely identified, these two patterns of illness also developed among volunteers who ingested filtrates prepared from the feces of ill patients.¹¹⁴⁻¹¹⁶ Studies done in Charlottesville, Virginia, confirmed this pattern of wintertime gastroenteritis including clustering in families, highest attack rates in children, and absence of identifiable etiologic agents in most cases, despite the application of techniques for virologic and enterotoxin studies.^{117,118}

Caliciviruses are now recognized as the etiologies of most cases of gastroenteritis occurring in winter months. The Caliciviridae family comprises four genera^{119,120}: Norovirus, Sapovirus, Lagovirus, and Vesivirus (see Chapter 182). Viruses belonging to Norovirus (from Norwalk) and Sapovirus (from Sapporo) genera are currently the most common causative agents for viral gastroenteritis in humans.¹¹⁹⁻¹²³ Noroviruses and sapoviruses may predominate as causes of early childhood diarrhea in rotavirus-vaccinated populations.^{5,6,42,124} Noroviruses, in particular, cause major outbreaks, especially in older children and adults (see Chapter 182). Immunocompromised patients can develop chronic norovirus infection lasting for months or years (see “Diarrhea in Immunocompromised Patients”).¹²⁵

Norovirus

Norovirus accounts for more than one-third of the outbreaks of nonbacterial gastroenteritis in the United States.^{126,127} Initially these viruses were referred to as Norwalk-like viruses. Their individual names were originally derived from the site of origin of each particular outbreak and included Norwalk,¹²⁸⁻¹³⁰ Hawaii,¹³¹ Snow Mountain,¹³² Taunton, and W agents.¹³³ Person-to-person spread is common, and secondary attack rates can be high. Transmission is predominately fecal-oral, but salivary transmission has been increasingly recognized.¹³⁴

Norovirus outbreaks have been observed in a wide variety of settings,¹³⁵ including hospitals,¹³⁶ extended-care facilities,¹³⁷ child care centers,¹³⁸ cruise ships,¹³⁹ refugee centers associated with natural disasters (e.g., Hurricane Katrina¹⁴⁰), and military combat areas including Afghanistan¹⁴¹ and Iraq.¹⁴² Several genotypes are known to infect humans^{121,135}; in the last decade, newly identified variant strains of norovirus genotype GII.4 have caused outbreaks worldwide¹⁴³⁻¹⁴⁵ including in the United States, United Kingdom, Europe, New Zealand, and Australia.

The pathophysiologic features of norovirus gastroenteritis parallel features of rotavirus in several respects.^{119,120,121} Each type is essentially noninflammatory and causes villus disruption and transient brush-border enzyme deficiencies in the upper small intestine, without any alteration in

adenylate cyclase activity.^{63,146,147} This enteropathy can be severe in patients with underlying immunocompromise, especially patients with common variable immunodeficiency.¹⁴⁸ The roles of enzyme deficiencies, malabsorption of xylose and lactose, and an increase in intestinal bacteria during norovirus illness remain unclear.^{146,149}

Sapovirus

Global studies on etiology of pediatric diarrhea in low- to middle-income countries have unmasked a greater burden of sapovirus diarrhea than previously recognized.^{5,6,150} Sapovirus was the second most common all-cause etiology of diarrhea in the community setting,⁶ and cumulative incidence may exceed 80% by 2 years of age.¹⁵¹ Sapoviruses were initially called typical human caliciviruses or Sapporo-like viruses after the prototype strain found in an outbreak in Sapporo, Japan, in 1982. Similar to norovirus, diverse sapovirus genogroups and genotypes cause infections in humans. Repeated infections are typically due to different genotypes, suggesting immunity is genotype specific.¹⁵¹ Sapovirus symptoms overlap with norovirus symptoms, possibly with more prolonged duration of vomiting.¹⁵² Similar to norovirus, sapovirus can cause outbreaks in long-term care facilities, but fewer laboratories specifically test for sapovirus. Sapovirus should be considered if the cause for an outbreak is otherwise unexplained.¹⁵³ Sapovirus also occurs as a coinfection with other pathogens.

Diarrhea Caused by Astrovirus, Adenovirus, and Other Possible Viral Etiologies

Several other groups of viral pathogens have been identified in endemic and epidemic gastroenteritis (Table 100.1), including^{45,77,154-156} (1) other types of caliciviruses with characteristic, “chalice-like” surface; (2) astroviruses, with a five- or six-pointed starlike surface structure¹⁵⁷; (3) enteric adenoviruses (especially serotypes 40 and 41)^{5,158}; and (4) human enteric coronaviruses, including SARS-CoV-2,^{159,160,161} as well as other miscellaneous or less well-characterized viral agents. More detailed discussions of these agents may be found in Chapters 149, 162, 178, 180, 183, and 184.

Astroviruses are among the most common causes of pediatric viral gastroenteritis.¹⁶²⁻¹⁶⁵ Astrovirus occasionally occurs in association with other enteric pathogens; in these cases the illness is more severe and protracted.¹⁶⁴ After rotavirus, adenovirus 40/41 was the most common viral etiology of moderate-to-severe acute diarrhea in children in low-income settings and the third most common cause overall.⁵ Adenovirus 40/41 was also third only to rotavirus and norovirus as a cause of acute gastroenteritis among children hospitalized in a US health care system.⁴¹ Increased emphasis on adenovirus serotyping in infant-child cohort studies is revealing a likely widespread greater burden of disease caused by this virus.^{166,167} In a case-control study in China, adenovirus serotype 3 was also associated with diarrhea, whereas asymptomatic detection of other serotypes (1, 2, 5, 6, 31, and 56) were more common in controls without diarrhea.¹⁶⁸

Vomiting and diarrhea can also manifest during infections with respiratory viruses, including in the absence of a respiratory syndrome. Enteric coronaviruses have clear pathology in animals (i.e., transmissible gastroenteritis in pigs). Although human coronaviruses, including the novel SARS-CoV-2 sarbecovirus, are predominately respiratory pathogens, these viruses can be isolated from intestinal tissues and stools.^{159,160,169,161} SARS-CoV-2 and all four of the endemic human coronaviruses (HCoV-229E, HCoV-NL63, HCoV-HKU1, and HCoV-OC43) have been detected in cases of acute gastroenteritis. Acute gastroenteritis from SARS-CoV-2 (see Chapter 135) infection¹⁶¹ can be severe enough to require hospitalization, however, the role of endemic human coronavirus as a primary etiology of diarrhea is likely minor; codetection with other common viruses such as rotavirus and norovirus is frequent.¹⁶⁹ Up to 10% to 20% of children with seasonal influenza present with diarrhea, although this is uncommon in adults. Diarrhea may be more prominent with Asian avian influenza A (H5N1) infection.¹⁷⁰

Bocaviruses,¹⁷¹ pestiviruses,¹⁷² toroviruses,^{173,174} picobirnaviruses,¹⁷⁵ and protoparvoviruses¹⁷⁶ are becoming better recognized as pathogens in children and adults, but their cooccurrence with conventional intestinal viruses and their independent roles in causing diarrheal illnesses are uncertain.^{176,177} Similarly, although enteroviruses may cause systemic illnesses, their role as diarrheal pathogens is minimal. Unbiased viral metagenomics approaches have further identified sequences of obscure viruses in diarrheal stools that require further clarification.^{178,179} In addition, a hallmark characteristic of the 2014 Ebola virus disease epidemic caused by Zaire ebolavirus in West Africa was nausea and abdominal pain, with

voluminous vomiting and diarrhea that led to life-threatening cholera-like GI volume losses through unclear mechanisms.^{180,181}

TABLE 100.1

Viral Pathogens Causing Gastroenteritis

ESTABLISHED PATHOGENS LIKELY AND EMERGING PATHOGENS

Adenoviruses (enteric types)

Astroviruses

Caliciviruses (including noroviruses and sapoviruses)

Rotaviruses groups A–C

Cytomegalovirus

Bocaviruses

Coronaviruses (including SARS-CoV-2)

Enteroviruses (various)

Picobirnaviruses, picornaviruses

Pestiviruses

Toroviruses

Filovirus (Ebola virus)

Human protoparvoviruses

Data from references 45, 68, 77, 122, 123, 135, 138, 155, 157, 172, 440, 441, 176, 232, 233.

Nucleic acid–based assays (e.g., reverse-transcriptase PCR) of stool specimens, either pathogen-specific or as part of multipathogen panels, are the standard method of GI virus detection. Rotavirus-specific and norovirus-specific reverse-transcriptase PCR are also available. In the typical outpatient clinical situation, norovirus-specific diagnostic tests are rarely performed because results of this assay do not alter management of the illness.

Cryptosporidiosis

Cryptosporidium is another cause of acute, noninflammatory diarrhea, second only to viruses in some countries.^{5,7,182} Sporadic cryptosporidiosis typically occurs through direct exposure to either infected people and to a lesser extent animals.^{183,184} Outbreaks occur via contaminated food, drinking water, or recreational water including swimming pools.^{185–190} Of the more than 20 species of this tiny coccidian intracellular protozoal parasite, two, *C. parvum* and *C. hominis*, cause the majority of human disease.^{189–193} Cryptosporidiosis is commonly asymptomatic or mild in normal hosts and usually results from waterborne outbreaks, often related to contaminated drinking water or public swimming pools.^{186,189–193} However, infection can be severe in infants and toddlers, accounting for an estimated 50,000 to >200,000 deaths per year.^{182,194} Malnourished children and immunocompromised hosts, especially patients with advanced human immunodeficiency virus (HIV) infection, can manifest severe diarrhea.^{185,187,192–195} Cryptosporidium, either with or without diarrhea, in some children can also contribute to early childhood growth impairment.^{182,196} Parasite shedding even after recovery from acute diarrhea can be prolonged in hosts with immunocompromise such as those with uncontrolled HIV.¹⁹⁷

Cryptosporidium causes a secretory form of diarrhea associated with dysregulated intestinal absorption that may be less responsive to oral rehydration than other diarrhea etiologies.¹⁹⁰ In addition to gastroenteritis, biliary infection with cholangitis has been reported. The pathogenesis of cryptosporidiosis is incompletely understood; the organism primarily alters villus structure and function and can promote an inflammatory enteropathy that may persist after parasite clearance.¹⁹⁰ No specific enterotoxin has been identified. Diagnosis depends on identification of oocysts in stool specimens by standard light microscopy, immunofluorescence assays, or nucleic acid–based molecular tests.^{198–200} In addition to supportive care, nitazoxanide has been shown to be moderately successful in immunocompetent patients with cryptosporidiosis. Data are inconclusive about universally recommended duration of therapy, especially in immunocompromised and malnourished patients.^{201–207} Alternative therapeutic agents for immunosuppressed patients include combinations of paromomycin and azithromycin. Ongoing searches for novel drugs have identified several with anti-Cryptosporidium activity through different mechanisms, but all remain in preclinical stages of development.^{208–215} Despite

resurging interest, vaccine development remains in early stages, in part because determinants of host immunity, especially in malnourished populations, are poorly understood.

Diarrhea Caused by Enteropathogenic Bacteria

Several diarrheagenic *E. coli* pathotypes (ETEC, enteroaggregative *E. coli* [EAEC], enteropathogenic *E. coli* [EPEC], enteroinvasive *E. coli* [EIEC], and Shiga toxin–producing *E. coli* [STEC], *Shigella* spp., *Salmonella* spp., *Campylobacter* spp., and both *V. cholerae* and noncholera *Vibrio* spp.) can cause noninflammatory diarrhea (Fig. 100.1). ETEC, *Campylobacter*, and *Shigella* are the most common bacterial etiologies of acute gastroenteritis worldwide.^{42,3} The presence of blood in stool (dysentery) increases the likelihood of diarrhea due to *Shigella*, *Campylobacter*, or EIEC; however, all bacterial pathogens most commonly present as a noninflammatory syndrome.^{5,7} Dysentery is present in <5% of community-associated diarrhea, and inflammatory manifestations of these bacterial gastroenteritides are discussed in greater detail in Chapter 101. Among bacterial etiologies that manifest only as noninflammatory diarrhea, ETEC and *V. cholerae* combined account for 9% to 24% of diarrhea-attributable mortality in children younger than 5 years of age.³

FIG. 100.1 Global pathogen-attributable etiologies of acute gastroenteritis and estimated fatal attributable fractions. Pathogen-attributable etiologies (millions of episodes/year) of acute gastroenteritis measured in the Global Burden of Diseases (through 2016) and attributable fatal fractions for (top) children ≤5 years of age and (bottom) ≥70 years of age. Data include cases of dysentery (<5% of total cases of acute gastroenteritis) in addition to noninflammatory cases. (Top) Red shading indicates proportion of pathogen-specific cases presenting with dysentery. *Shigella* is the most common cause of dysentery and ~15% of *Shigella* cases present with dysentery. *C. difficile*, *Clostridioides difficile*; EPEC, enteropathogenic *Escherichia coli*; ETEC, enterotoxigenic *Escherichia coli*; *V. cholerae*, *Vibrio cholerae*. Adapted from Troeger C, et al. GBD 2016 Diarrhoeal Diseases Collaborators. Estimates of the global, regional, and national morbidity, mortality, and aetiologies of diarrhoea in 195 countries: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Infect Dis.* 2018;18(11):1211–1228; and Liu J, et al. Use of quantitative molecular diagnostic methods to identify causes of diarrhoea in children: a reanalysis of the GEMS case-control study. *Lancet.* 2016;288:129–301.

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The greatest risk for pediatric ETEC diarrhea is between age 12 and 24 months, and severity is increased in undernourished children.^{216,42} Antibodies directed against the heat-labile enterotoxin (LT) of *E. coli* in human colostrum may account for lower burden of ETEC in infancy.^{22,23,217} Diarrhea produced by LT shares the adenylate cyclase–activating mechanism with cholera toxin.^{218–222} Certain strains of *Klebsiella*, *Citrobacter*, and *Aeromonas* produce an LT-like toxin and may cause a small percentage of diarrheal disease in children.²²³

Molecular methods have refined estimates of the contributions of other diarrheagenic *E. coli* on burdens of diarrhea. Typical EPEC is a contributor to diarrhea-related morbidity and mortality in the first 2 years of life.²²⁴ EAEC, although frequently subclinical, is among the most commonly identified enteropathogens. EAEC infection is nearly universal in children in resource-limited settings. Despite no clear association with diarrhea, EAEC, even in the absence of diarrhea, is associated with growth impairment,²²⁵ particularly when found as a coinfection with LT-ETEC, EPEC, and *Campylobacter* spp.^{225,226} Similarly, independent of diarrhea, cumulative and diverse pathogen exposures are associated with growth impairment in the first 2 years of life; these findings highlight the substantial potential cost of early-life GI infection in low-resource settings (see Chapter 104).^{227,228,229}

Diarrhea in Adults

An emerging increasing relative burden of diarrhea and diarrhea mortality in adults has emerged in the last several decades, concomitant with an aging global population.³ In resource-abundant settings, acute gastroenteritis in adults is most often caused by noroviruses. Less common Page 1285 viruses in adults include rotaviruses,⁷⁰ adenoviruses, coxsackieviruses, and coronaviruses.^{32,233} *C. difficile* and its potential for recurrence is of increasing concern in patients with recent antibiotic use (see Chapters 101 and 249).²³⁴ Bacterial gastroenteritides such as nontyphoidal *Salmonella* spp. are important causes of foodborne diarrhea outbreaks, often with fever (see Chapter 101). In addition, several agents of food poisoning such as *Clostridium*

perfringens, *Bacillus cereus*, and *Staphylococcus aureus* commonly cause diarrheal syndromes in adults (see Chapter 103).

In adults living in resource-limited settings, several pathogens are known to cause sporadic noninflammatory diarrhea, but none has had as much historical effect or notoriety as *V. cholerae*, causing cholera.^{235,236} Cholera has long been endemic in South Asia. With the increased infection-to-case ratio of El Tor cholera, the current seventh pandemic swept most of the continents of the Eastern Hemisphere including Asia, Africa, and the Mediterranean portions of Europe in the 1960s.²³⁷ Rare isolated cases occurred in the United States^{238,239} and Europe related to contaminated mineral water²⁴⁰ and to undercooked shellfish.^{241,242} Beginning in Madras, India, in late 1992 and rapidly spreading to Calcutta and Bangladesh in 1993, a new strain of non-O1 *V. cholerae*—O139, called Bengal—caused epidemic cholera gravis.^{243–247} Most recently, natural disasters and social conflict have preceded large outbreaks including the introduction of *V. cholerae* into previously cholera-free Haiti after the devastating earthquake in 2010 that may have reemerged²⁴⁸ as endemic in 2022, and more than 1.2 million cases in conflict-stricken Yemen since 2017.

Severe, watery diarrhea in a patient residing in or having recently traveled to an endemic area should raise suspicion of cholera. The disease can be so fulminant as to cause hypovolemic shock and death from the outpouring of fluid into the upper portion of the small bowel before the first diarrheal stool occurs.²⁴⁹ As discussed in detail in Chapter 219, the entire dehydrating syndrome of cholera appears to be related to the activation of intestinal adenylate cyclase by the potent cholera enterotoxin.^{218,219,250,251} To make the diagnosis bacteriologically, stool specimens should be cultured onto thiosulfate citrate bile salts sucrose agar. Of prime importance in therapy is fluid replacement, accomplished either intravenously with isotonic fluids or orally with glucose-electrolyte solutions.

Patients from whom *V. cholerae* cannot be isolated may have a cholera-like syndrome caused by certain strains of ETEC. In 1956, De and associates²⁵² demonstrated that *E. coli* isolated from adults and children with this syndrome caused fluid accumulation similar to that seen with *V. cholerae* infection in ligated rabbit ileal loops. In the early 1960s, Trabulsi,²⁵³ working in São Paulo, Brazil, reported similar findings with toxigenic *E. coli*. Various investigators then demonstrated that acute undifferentiated diarrhea in adults was caused by ETEC strains that usually were not of the classically recognized pathogenic serotypes.^{254–257} The toxic material present in the culture filtrate of these *E. coli* strains was demonstrated to be heat labile and nondialyzable. Subsequent studies demonstrated two types of enterotoxin produced by ETEC—LT and heat-stable enterotoxin (ST).²⁵⁸ LT activates adenylate cyclase^{220–222} and is antigenically and mechanistically similar to cholera toxin in many ways. In contrast, ST activates guanylate cyclase,^{259–261} has an earlier onset of action,²⁵⁸ has greater tissue specificity,²⁶¹ and has a much lower molecular weight than LT.²⁶²

Several studies have shown that ETEC strains producing LT only, ST only, or LT plus ST are associated with episodes of diarrhea in adults. Adults in resource-limited settings who carry LT-producing *E. coli* are often asymptomatic. In contrast, ST-producing strains are significantly associated with diarrheal disease. Diarrhea due to ETEC is less common in the United States.^{54,117}

Diarrhea in Immunocompromised Patients Patients With HIV Infection

Patients with HIV infection often develop or present with diarrhea. With increased access to antiretroviral therapy (ART), the incidence of infectious diarrhea among HIV-infected patients decreased markedly.^{263,264} However, among HIV-infected patients in the United States meeting criteria for acquired immunodeficiency syndrome (AIDS), 30% to 60% present with diarrhea,^{263,265–268,269} and this figure can reach 95% in parts of Africa or Haiti. In many patients with advanced HIV infection and immunocompromise, diarrhea becomes prolonged, causes severe malnutrition, predisposes to other types of serious infections, and can be life threatening. In addition, symptomatic management of chronic diarrhea in such patients poses major difficulties. The importance of the interactions of HIV with the components of the intestinal immune system in modulating systemic HIV infection has come to light in recent years.^{270–273}

Although some investigators have reported an enteropathy without identifiable infectious pathogens^{274,275} or with primary HIV infection of enterochromaffin cells in the bowel mucosal crypts and lamina propria,²⁷⁶ others have reported one or more enteric pathogens in 55% to 85% of patients with AIDS and diarrhea.^{277,278} As shown in Table 100.2, the leading agents found in these patients are *Cryptosporidium*, cytomegalovirus (CMV), microsporidia, *Entamoeba histolytica*, *Giardia lamblia*, *Salmonella*, *Campylobacter*, *Shigella*, *C. difficile*, *Vibrio parahaemolyticus*, and *Mycobacterium* spp.^{277-281,282-287} *Cyclospora* and *Cystoisospora belli* also cause persistent diarrhea in patients with AIDS.²⁸³ *Pneumocystis jirovecii* infection can occasionally involve the intestinal tract.²⁸⁴ Men with HIV who have sex with men are at increased risk of *G. lamblia*, *E. histolytica*, *Campylobacter jejuni*, *Shigella*, *Chlamydia trachomatis*, *C. difficile* (with proctitis), *Neisseria gonorrhoeae*, herpes simplex virus, or *Treponema pallidum*.²⁸⁷

Although eradication treatment may be difficult, most patients respond to specific antimicrobial or antiparasitic therapy and ART, emphasizing the need to identify an etiologic agent whenever possible.^{207,288} The antiviral agent ganciclovir often controls intestinal CMV,^{285,286,289,290} and most bacterial and parasitic infections can be treated with the expectation of some improvement. *Cryptosporidium*, which infects 3% to 21% of patients with AIDS in the United States, can be found in 50% of patients with AIDS and diarrhea in Africa and Haiti.^{185,188,279} *Cryptosporidium*-mediated biliary infection occurs in this patient population as well. Similarly, certain microsporidia organisms, such as *Encephalitozoon intestinalis* or *Enterocytozoon bieneusi*, can cause persistent and severe diarrhea in HIV-infected individuals. *Cryptosporidium* and microsporidial infections are associated with villus atrophy, crypt hyperplasia, increased intraepithelial lymphocytes, and D-xylose malabsorption.²⁷⁹

The same acid-fast stain that detects *Cryptosporidium* or *Mycobacterium* in fecal specimens may also reveal *C. belli* and *Cyclospora* in approximately 2% of patients with AIDS with diarrhea in the United States and in 15% of patients in Africa.²⁷⁹⁻²⁸³ Nontyphoidal *Salmonella* infections occur with an estimated 20-fold increase in frequency and with increased severity in patients with AIDS.^{291-294,295} Enteric viruses also are associated with diarrhea in HIV-infected individuals. In one study, astrovirus, picobirnavirus, calicivirus, and adenovirus were found in 6% to 12% of HIV-positive patients with diarrhea.²⁹⁶ In addition, ART regimens can cause diarrhea, often complicating the workup of an HIV-infected patient with persistent diarrhea.^{264,295,297} EHEC, even in the absence of diarrhea, is associated with growth impairment.²²⁵

Several practical algorithmic approaches to the diagnosis and management of diarrhea in HIV-infected patients have been published.²⁰⁶ These strategies favor the use of early, noninvasive stool studies and practical empirical treatment trials, followed by more invasive tests (e.g., endoscopy with biopsy) for patients with refractory or more severe presentations. ART has decreased the incidence of several important opportunistic infections including certain causes of gastroenteritis. Of particular note, there has been a dramatic decline in the incidence of tissue-invasive infections caused by CMV GI disease.^{289,290} An effort to judiciously adhere to safe food and water guidelines in higher-risk patients can substantially help in preventing many of these serious enteric infections.²⁹⁸ Addressing etiologies of diarrhea among mothers with HIV may also confer benefit to their HIV-exposed, but uninfected infants.²⁹⁹

Patients With Solid-Organ Transplants and Others Receiving Immunosuppressants
Chemotherapy for malignancy, autoimmune disease, and especially antirejection medications after solid-organ transplantation (SOT) or hematopoietic cell transplantation (HCT) increase risk of serious enteric infections.^{300,301,302} Frequent encounters with the health care system, prolonged hospitalizations, and antimicrobials contribute to high rates of *C. difficile* infection (CDI) in these populations.^{303,304} Transplantation of any organ is associated with increased risk of CDI. CDI occurs in 9% to 13% of patients after HCT and up to 20% after SOT.³⁰¹ Hospital-onset CDI after SOT can prolong hospitalization and is associated with allograft dysfunction.³⁰⁵

In addition to CDI, several other bacteria, viruses, and parasites can cause diarrhea after SOT or HCT. Viral infections are common in this population and can result in long-term shedding, particularly in cases of norovirus.¹²⁵ There is less seasonal variation in norovirus infections in immunocompromised hosts, and these infections manifest with more prolonged symptoms and

viral shedding than infections in immunocompetent hosts. Up to one-third of these patients have diarrhea persisting for 2 weeks or longer, and shedding has been reported beyond a year.³⁰⁶ Symptoms may occur at lower levels of viral genome copies per gram of stool. Prolonged viral shedding in these patients is associated with an increased diversity of genomic variants in an infected individual.³⁰⁷ The pathogenicity of these divergent variants is unclear, and therefore the clinical significance of continued norovirus detection over prolonged periods of time in patients with mild symptoms is uncertain. Flattening of the villus border and loss of villin colocalizes with viral particles on duodenal biopsy specimens, but other histologic changes described in chronic norovirus infections in immunocompromised patients (edema, cellular debris, and metaplasia) are nonspecific and overlap with noninfectious etiologies of enteritis in these patients.³⁰⁸ No specific therapy is available. Combinations of oral immunoglobulins, nitazoxanide,³⁰⁹ and decreasing immunosuppressive therapy³¹⁰ may benefit some patients, but results are inconsistent.

CMV enteritis should be considered in the differential diagnosis of diarrhea after transplantation, especially in donor-recipient mismatch cases. CMV enteritis can occur in the absence of peripheral viremia, and direct intestinal sampling is recommended to rule out this condition. Diarrhea may also be a presenting symptom of disseminated adenovirus in transplant recipients. Reduction of immunosuppression is part of management for any infectious diarrhea in this population. Ganciclovir is an effective treatment for most cases of CMV enteritis; however, options for adenovirus are more limited. Brincidofovir, a prodrug of cidofovir, is conjugated to a lipid to enhance intracellular delivery with lower plasma concentrations and less renal toxicity than cidofovir. This drug has been prescribed on a compassionate use basis for disseminated adenovirus infection in this population, but efficacy data are lacking, and diarrhea itself may be a significant side effect of brincidofovir.

In addition to increased risk of CDI, infections with other bacterial etiologies are more common in this population. These infections may present with greater symptom severity, higher pathogen burdens, and increased antimicrobial resistance compared with those in immunocompetent patients.³¹¹

Possible Enteric Pathogens in Patients With Acquired Immunodeficiency Syndrome

PATHOGEN	% WITH DIARRHEA (n = 181)	% WITH NO DIARRHEA (n = 28)
Cytomegalovirus	12–45	15
<i>Cryptosporidium</i>	14–30	0
<i>Microsporidia</i>	7.5–33	0
<i>Entamoeba histolytica</i>	0–15	0
<i>Giardia lamblia</i>	2–15	5
<i>Salmonella</i> spp.	0–15	0
<i>Campylobacter</i> spp.	2–11	8
<i>Shigella</i> spp.	5–10	0
<i>Clostridioides difficile</i> (formerly <i>Clostridium difficile</i>) toxin	6–7	0
<i>Vibrio parahaemolyticus</i>	4	0
<i>Mycobacterium</i> spp.	2–25	0
<i>Cystoisospora belli</i>	2–6	0
<i>Cyclospora</i>	0–11	0
<i>Blastocystis hominis</i>	2–15	16
<i>Candida albicans</i>	6–53	24
Herpes simplex	5–18	40
<i>Chlamydia trachomatis</i>	11	13
<i>Strongyloides</i>	0–6	0
Intestinal spirochetes	11	11
One or more pathogens	55–86	39

Data from references 277–281, 283–287, 291–298, 303, 304.

Although less common, Cryptosporidiosis and other protozoa can also be severe in these patients and may mimic graft-versus-host disease on endoscopy and intestinal biopsy. The clinical presentation of these infectious etiologies overlaps with medication side effects, especially mycophenolate and chemotherapy, which cause direct epithelial cell injury.^{301,302,312}

For the individual patient, a specific pathogen etiology is seldom predictable based on symptoms and exposure history alone. In addition, noninfectious etiologies underlie the high burden of diarrhea, and may contribute up to 66%, of diarrheal etiologies in solid organ transplant recipients. Direct drug toxicities on epithelial cells, disruptions in intestinal microbial communities, and immune-mediated pathologies like graft-versus-host-disease are common in immunocompromised hosts with diarrhea.³¹³ Therefore, multipathogen testing, where available, is recommended to guide pathogen-targeted treatment in this population.³⁰¹ This strategy avoids unnecessary use of empiric anti-infectives in this population that is also at high risk of acquiring multidrug-resistant infections.

Travel-Associated Diarrhea (Turista)

Whether it “arouses one from bed with a start at 4 a.m. for a record-breaking race to the bathroom to begin a staccato ballet”³¹⁴ or produces the poetry of the psalmist (“I am poured out like water ... my heart like wax is melted in the midst of my bowels”³¹⁵), traveler’s diarrhea has a major effect each year on the 300 million to 500 million international travelers and probably on the distribution of well over \$100 billion in international tourism receipts.^{316,317}

Diarrhea is the most common and among the most disconcerting illnesses that threaten the traveler.³¹⁸⁻³²⁴ North American and northern European travelers appear to be at greatest risk when they travel to Africa, Asia, Latin America, and to a lesser extent southern Europe.³²⁵⁻³²⁸ Most symptoms of traveler’s diarrhea begin 5 to 15 days after arrival, although a range from 3 to 31 days has been noted.^{292,318,329-334} The illness is typically manifested by malaise, anorexia, and abdominal cramps followed by the sudden onset of watery diarrhea. Nausea and vomiting occur in 10% to 25% of cases. The diarrhea is usually noninflammatory, although a low-grade fever is present approximately one-third of the time. The duration is usually 1 to 5 days, but up to 50% of patients have illness that continues 5 to 10 days and sometimes beyond.

The attack rate for traveler’s diarrhea varies from 5% to 50%, depending on the destination and duration of the trip. Destinations with increased risk for acquiring traveler’s diarrhea include portions of Africa and Asia (including India), and certain areas of Central and South America.^{318,334-336} The attack rate also appears to decrease with age after 25 years, an observation that may reflect different habits and exposures rather than inherent susceptibility.^{314,327} Host genetics and intestinal microbiota factors may also predispose to greater susceptibility.³³⁷ Expatriate residents living in certain countries appear to be at some level of persistent risk for diarrhea of infectious causes; for instance, an attack rate of 49% per month is observed during the first 2 years of residence in Nepal.³³⁸

For many years, the etiology of turista was an enigma; only infrequently had parasites or bacteria, such as amoebas, *Giardia*, *Salmonella*, or *Shigella*, been identified, and viral studies had failed to elucidate significant causes. The first suggestion that an infectious process was likely came from the effective reduction in the attack rate achieved by the use of prophylactic antimicrobial agents.^{329,334,339,340} Studies by Kean³¹⁴ suggested that certain EPEC serotypes may be involved in up to one-third of the cases. The involvement of *E. coli* was further confirmed in an outbreak of traveler’s diarrhea among British Page 1287troops in Aden, where *E. coli* O148 was identified in 54% of the British troops with diarrhea.³³⁰

Later studies demonstrated ETEC in approximately 50% (range, 5%–75%) of cases of traveler’s diarrhea in Latin America, Africa, and Asia (Table 100.3).³⁴¹⁻³⁴³ The attack rate ranged from 20% to 100% (median, 52%–54%) (see Table 100.3).^{318,336,340,344} ETEC was almost never present before travel, and such organisms were acquired by only 12.6% of fellow travelers who did not become ill.^{292,332,333} ETEC producing LT, ST, or both can cause traveler’s diarrhea (Table 100.4). In some areas, EAEC and EPEC are emerging as the major cause (see Chapter 223), although EAEC is also commonly identified in asymptomatic controls.^{318,320,343,345-348}

A number of other microbial agents have also been identified in small subsets of patients with traveler’s diarrhea. *Salmonella*, *Shigella*, *Campylobacter*, *Vibrio*, and *Aeromonas* spp. are increasingly recognized in travelers with diarrhea.^{344,349-351} Pathogen-specific prevalence appears to vary by destination. *Campylobacter* was present in 25% to 35% of cases during travel to South Asia and 15% to 25% of cases related to travel in Southeast Asia, but accounted for less than 5% of cases from Africa or Latin America.³¹⁸ In US military personnel deployed to Southeast Asia, *Campylobacter* spp. were detected in 35% and nontyphoidal *Salmonella* in 23% of cases of diarrhea.³⁴⁹ In Thailand, *Campylobacter* was the most common etiology (28%) of pathogen-attributable diarrhea cases.³⁵¹ *Shigella* accounted for 5% to 15% of cases from Latin America, Africa, and South Asia and was less frequent in travelers to Southeast Asia.³¹⁸ *Salmonella* infections were more common during travel to Africa and Southeast Asia than to South Asia or Latin America.³¹⁸ *V. cholerae* is rarely a problem for most US travelers, but infection can occur among military personnel or relief workers in endemic cholera locales.^{327,351,352} Single-dose live lyophilized CVD 103-HgR (Vaxchora) oral cholera vaccine is approved for travel to areas with active cholera transmission.

Etiology of Acute Traveler's Diarrhea in Three Locales ^a

CHARACTERISTIC	LATIN AMERICA	AFRICA	ASIA
Duration of stay (days)	21 (2–42)	28 (28–35)	(28–42)
Attack rate (%)	52 (21–100)	54 (36–62)	(39–57)
Organism (%)			
No bacterial pathogen	28	—	35
Bacterial pathogen	72	—	80
Enterotoxigenic <i>Escherichia coli</i>	46 (28–72)	36 (31–75)	15 (5–25)
Enteropathogenic <i>Escherichia coli</i>	30 (25–35)	<5	17 (15–25)
<i>Shigella</i>	5 (0–30)	5 (0–15)	5 (4–15)
<i>Salmonella</i>	<5	10 (5–15)	10 (5–15)
<i>Campylobacter jejuni</i>	<5	<5	15 (2–35)
<i>Vibrio parahaemolyticus</i>	—	—	7 (1–13)
Rotavirus	23 (0–36)	10 (5–15)	5 (0–15)
Norovirus	10	—	3
Protozoa (<i>Giardia</i> , <i>Cryptosporidium</i> , <i>Entamoeba histolytica</i> , others) ^b	7	<5	11

^a Values shown are median (range) from multiple studies. ^{317–319–340–343–396–490}

^b Parasitic etiology may increase to 65% among returned travelers seeking medical care. ³⁶²

Intestinal protozoa also may be important causes of traveler's diarrhea. Agents such as *C. parvum*, *Cyclospora cayetanensis*, *G. lamblia*, and various microsporidia should be considered in the workup of subacute or chronic noninflammatory diarrhea in returned international travelers.^{359–361} In the GeoSentinel Surveillance Network global database of 2902 unwell travelers seeking medical care, parasitic etiologies (predominantly *G. lamblia* [27%], *E. histolytica* [14%]) accounted for 65% of chronic diarrhea cases, whereas bacteria (*Campylobacter* [13%], *Shigella* [6%]) were identified in 31%, and viruses (hepatitis A and hepatitis E) were identified in only 3%. *G. lamblia* was the most commonly identified pathogen regardless of travel destination.³⁶² Other protozoal parasites such as *Blastocystis hominis* can be commonly identified in the stools of persons traveling to developing countries. However, it has been difficult to ascertain whether these organisms actually cause disease or are merely commensal.^{363–365} A subset of patients have persistent diarrhea for which no infectious agent can be implicated. Chronic idiopathic diarrhea (see “Brainerd Diarrhea”) has been reported to occur in a few small travel-related outbreaks^{366–369} and may cause prolonged diarrhea in sporadic cases.

In contrast to the frequent identification of potential etiologic agents among travelers to tropical areas who develop diarrhea, travelers to temperate areas often develop a mild syndrome of diarrhea for which no infectious cause can be identified.³⁷⁰ Environmentally distributed *Cryptosporidium* and *Giardia* and oocysts are hardy in temperate regions, have zoonotic reservoirs in these regions, and are relatively tolerant to chlorination. Travelers to certain areas such as Russia and national parks in the United States who drink unfiltered or unboiled water may therefore be susceptible to development of the more insidious watery diarrhea seen with giardiasis or cryptosporidiosis.^{371–374} Strongyloidiasis acquired during travel may cause noninflammatory diarrhea, abdominal pain, and eosinophilia.³⁷⁵

Several other potentially serious infections manifested initially by diarrhea or abdominal pain may be acquired by travelers. Malaria may begin as gastroenteritis with nausea, vomiting, diarrhea, or abdominal pain in 30% to 50% of cases.³⁷⁶ The physician should also remember to consider typhoid fever and other infections that may be manifested with a typhoidal pattern including plague, melioidosis, typhus, and arboviral hemorrhagic fevers.^{376,377}

The desire to control the bothersome problem of diarrhea in travelers has led to a variety of medications used for management.³¹⁴ Commonly used remedies, such as diphenoxylate-atropine (Lomotil) and kaolin-pectin suspension, were found to have little or no efficacy in most studies.³⁷⁸ Lomotil and other antimotility agents may actually worsen the illness in inflammatory

processes such as shigellosis.³⁷⁹ Bismuth subsalicylate (Pepto-Bismol) was shown to inhibit enterotoxin activity in experimental animal models, and it has been recommended for symptomatic therapy, as well as prophylaxis.³⁷⁹⁻³⁸² As in any type of diarrheal illness, the mainstay of therapy is adequate hydration with an oral glucose-electrolyte or sucrose-electrolyte solution.

Prevention of traveler's diarrhea should be directed toward reducing the consumption of infectious agents in food and water. Foods that are handled but not cooked (e.g., salads, raw vegetables) are high-risk foods.³¹⁸⁻³²⁰ Bottled, noncarbonated water cannot be considered safe because outbreaks of cholera²⁴⁰ have been traced to bottled water, and typhoid fever^{383,384} has been traced to other bottled beverages. Care in eating and drinking may reduce one's risk, even in highly endemic areas, to less than 15%.³⁸⁵⁻³⁸⁷

The efficacy of prophylactic antimicrobial agents has been documented in several studies.^{329,334,388-390} The increased risk of acquiring a more severe infection (e.g., salmonellosis), the risk of drug side effects (e.g., photosensitivity in the tropics), and the emergence of drug-resistant organisms³⁹¹ should preclude the widespread use of antibiotic prophylaxis at this time.³⁹² Because treatment regimens that combine loperamide with an antibiotic are rapidly effective in controlling traveler's diarrhea (<10 hours), most experts consider prophylactic therapy only for travelers with special issues such as individuals with a high risk of infection and individuals for whom it is important to remain disease-free during the trip.^{393,394}

Empirical self-treatment of traveler's diarrhea with a fluoroquinolone such as ciprofloxacin or levofloxacin for 1 to 3 days or a single dose of 1000 mg of azithromycin can significantly reduce the duration and severity of the disease (see Chapters 324 and 325).^{318-320,344,395-397} Therapy with trimethoprim-sulfamethoxazole had traditionally been suggested for children, but because of resistance, most experts currently recommend the use of a macrolide such as azithromycin.^{318-320,398} Fluoroquinolone, and to a lesser extent azithromycin, resistance has emerged among all bacterial etiologies.³⁹⁹ *Campylobacter* spp. resistant to quinolones is of special concern for travelers to many areas of the world, most notably Thailand and other parts of Southeast Asia.⁴⁰⁰⁻⁴⁰³ Nearly 90% of *Campylobacter* diarrhea cases in US military personnel deployed to Southeast Asia were resistant to ciprofloxacin.³⁴⁹ Resistance to quinolones has also increased among ETEC and EAEC strains, especially strains from India and sub-Saharan Africa. Extended-spectrum β -lactamase ETEC and EAEC strains expressing blaCTX-M-15 and blaCTX-M-27 genes are emerging.^{350,404} Fluoroquinolone drugs are contraindicated in pregnant women^{318-320,394,396} and in children younger than 16 years. Also, US Food and Drug Administration black box warnings exist for arthropathy, neuropathy, tendinopathy, and central nervous system side effects associated with fluoroquinolones.⁴⁰⁵ The greater recognition of adverse reactions associated with quinolones combined with increasing resistance have made azithromycin the preferred first choice for empirical self-treatment of traveler's diarrhea in many individuals. Resistance to azithromycin among ETEC and EAEC, however, is greater than 25% in parts of Southeast Asia, India, and Africa. When necessary, rifaximin, a nonabsorbable relative of rifampin, is useful for treatment of less severe cases of traveler's diarrhea in the absence of fever. Levofloxacin, azithromycin, and rifaximin in combination with loperamide have similar efficacies of 91% and 96% cure at 48 and 72 hours, respectively.⁴⁰⁶ Because of ongoing changing resistance patterns among bacterial etiologies of traveler's diarrhea, pretravel counseling, empirical acute treatment, and posttravel diagnostic and treatment recommendations should consider up-to-date reported susceptibility patterns that are specific to geographic destinations, such as can be found at the Centers for Disease Control and Prevention (CDC) website (cdc.gov/travel). The benefits of treatment also need to be weighed against risk of acquisition of colonization with extended-spectrum β -lactamase-producing Enterobacteriales during travel and increased susceptibility to pathogens after the antibiotic course is over but before the microbiome has recovered. Nonabsorbable rifamycins may have an advantage over fluoroquinolones in having less toxicity, and limiting acquisition of multidrug-resistant bacteria.⁴⁰⁷ For all of these reasons, antibiotics are not recommended for mild cases of traveler's diarrhea, which most often are self-limiting.

Diarrhea in Institutions

Institutions provide unique settings that promote the acquisition of certain enteric pathogens. As with diarrhea in patients with AIDS and traveler's diarrhea, many cases of institution-acquired diarrhea are noninflammatory. Given the rise *C. difficile*-associated disease (CDAD), new-onset

diarrhea in patients in the hospital or long-term care facilities should prompt stool immunoassay or PCR assaying for the presence of *C. difficile* toxins.⁴⁰⁸

Epidemic Nosocomial Diarrhea in Newborns

Epidemic nosocomial neonatal diarrhea has long been recognized as a potentially serious problem with mortality rates of 24% to 50%.^{409,410} Although the risk of this entity has substantially decreased, its occurrence is an important reminder of the unusual susceptibility of newborns to pathogens. This increased susceptibility may be explained by their underdeveloped (developing) intestinal microbiota and lack of specific immunity. Prematurity or congenital cardiac or pulmonary disease complicates this situation. The consequences of diarrhea in newborns are unusually severe, partially because they have poorly developed homeostatic mechanisms with limited water and electrolyte reserves. Crowded conditions in newborn nurseries promote nosocomial transmission.⁴¹¹ In addition, delays in recognizing a nursery outbreak may occur because infants may not develop diarrhea until after they have been discharged from the hospital.

The onset of illness is often insidious, with the development of listlessness, irritability, and poor feeding over a period of 3 to 6 days.^{410,412,413} Vomiting and fever are infrequent, and the stools tend to be watery; yellow-green in color; and usually without mucus, pus, or blood.^{414,415} Early signs such as failure to gain weight or a slight weight loss and abdominal distention may be subtle. The disease may progress to more severe signs of dehydration and shock with depressed sensorium, drowsiness, coma, sunken eyes, circumoral cyanosis, and grayish discoloration of the skin. Shock without hyperpnea often occurs in this setting despite the development of severe acidosis. Poorly nourished infants may develop severe hypokalemia, hyponatremic dehydration, or paradoxical edema. Appropriate antibiotic therapy must be tailored to the specific sensitivity pattern of the organism isolated.⁴¹⁴

The typical illness usually lasts 5 to 15 days, but persistence or relapse may occur over the course of several weeks after the initial onset of symptoms. Complications may include otitis media, pneumonia, bacteremia, peritonitis, and renal vein or cerebral sinus thrombosis. Several potentially life-threatening processes may mimic this infantile diarrhea syndrome. So-called parenteral diarrhea refers to the well-recognized but poorly understood tendency for systemic infections or localized infections elsewhere (e.g., otitis, meningitis) to be manifested clinically with diarrhea. Likewise, a strangulated hernia, intussusception, or torsion of an ovary or testis may be manifested by abdominal pain or diarrhea.

Epidemic diarrhea among hospitalized newborns has been commonly associated with certain enteropathogenic serotypes of *E. coli* (EPEC). In many areas such as South Africa and southern Brazil, EPEC organisms are also among the most common causes of sporadic diarrhea in infants and young children, especially during the summer months.⁴¹⁶⁻⁴¹⁸ Up to 20% of cases of endemic childhood diarrheal illness, even in temperate, more developed areas of the world such as England and Canada, are also known to be caused by EPEC organisms.^{416,417}Page 1289

The association of a certain strain of *E. coli* with infantile diarrhea was first demonstrated by slide agglutination by Bray and Beavan⁴¹⁹ in 1945 and reported in further detail in 1948. Excluding certain invasive serotypes (see Table 101.2 in Chapter 101), there are now at least 14 classically recognized EPEC *E. coli* serotypes, including O111-K58 (α), O55-K59 (β), O127-K63, O128-K67, O26-B6, O86-K61, O119-K69, O125-K70, O126-K71, O20-B7, and O44-K74. Additional serotypes recognized as causes of epidemic infantile diarrhea include O114,^{409,420,421} O142,^{411,422} and O158.⁴²³ The mechanism by which most EPEC organisms cause disease involves a complex set of attachment and effacement traits, as detailed in Chapter 101. Although most are not invasive and do not produce conventionally recognized LT or ST enterotoxins, these organisms are capable of causing diarrheal disease in human volunteers, from whom the organism can be reisolated and in whom an antibody response can be documented.^{412,424}

In addition to EPEC organisms, certain enterohemorrhagic (EHEC) and ETEC strains of *E. coli* have caused outbreaks of diarrhea in infants. *E. coli* O157-H7 and other EHEC strains have been associated with a syndrome of hemorrhagic diarrhea with only minimal inflammation and mild severity. Most EHEC organisms secrete a Shiga-like toxin, and certain strains of EHEC have also been associated with outbreaks and sporadic cases of hemorrhagic diarrhea that are more severe and can include a form of hemolytic-uremic syndrome.⁴²⁵⁻⁴³⁰

Diarrheagenic *E. coli* serotypes are not the only bacterial causes of epidemic infantile diarrhea. For example, an outbreak of diarrheal illness was described in which multiple serotypes of different organisms (*E. coli*, *Klebsiella*, and *Citrobacter*) were found to be transiently enterotoxigenic.⁴³¹ This observation suggests the transmission of enterotoxigenic potential between susceptible strains, likely by certain plasmids⁴³² or by bacteriophages.⁴³³ Another report of sporadic diarrhea in infants and children in Africa showed that several enteric organisms other than *E. coli* may produce an enterotoxin under certain conditions.³⁸ This finding is relevant to other situations as well, as typified by an outbreak of watery diarrhea occurring on a cruise ship and found to be caused by enterotoxigenic strains of *E. coli* as well as *Klebsiella* spp. and *Citrobacter* spp.⁴³⁴ Many other well-known bacterial causes of diarrhea such as shigellosis⁴³⁵ and epidemic salmonellosis may also spread readily in the newborn nursery.^{436,437} Viral causes of epidemic infantile diarrhea include, but are not limited to, specific types of echoviruses,⁴³⁸ coxsackieviruses,⁴³⁹ adenoviruses,⁴⁴⁰ and rotaviruses.^{45,46,441-444}

Hospitals

Nosocomial diarrhea is often due to noninfectious causes, but infectious etiologies must be considered. Nosocomial outbreaks are reported to the CDC.^{445,446} Nosocomial diarrhea appears to be a significant factor predisposing to other nosocomial infections such as urinary tract infections.⁴⁴⁷ Overall rates range from 2.3 to 4.1 illnesses per 100 admissions on pediatric wards^{408,446} and from 7.7 per 100 admissions to 41% of adults hospitalized in intensive care units.^{446,448}

C. difficile remains the most common, most serious, and most costly infectious cause of nosocomial diarrhea in hospitalized patients⁴⁴⁹⁻⁴⁵¹ and ranks among the most commonly reported hospital-acquired pathogens.⁴⁵² In particular, CDAD is an important emerging nosocomial infection worldwide, especially among elderly hospitalized patients, patients with immunocompromise, and patients occupying beds in surgical wards or intensive care units.⁴⁵³⁻⁴⁵⁵ Most sporadic and outbreak cases of CDAD appear to be caused by exposure to contaminated environmental surfaces rather than direct contact with an index case.⁴⁵⁶⁻⁴⁵⁸

Other enteric pathogens are occasionally identified in outbreaks of nosocomial diarrhea. *Klebsiella oxytoca* can produce a toxin that inhibits DNA synthesis and, similar to *C. difficile*, causes diarrhea and sometimes hemorrhagic colitis associated with recent antibiotic use. Enterotoxin-producing methicillin-resistant *S. aureus* has been implicated as a rare cause of antibiotic-associated diarrhea.⁴⁴⁸ *Salmonella* is a common cause in reported outbreaks of nosocomial gastroenteritis and may rarely cause pseudomembranous colitis after antibiotic exposure. *Cryptosporidium* may be associated with cases of nosocomial diarrhea involving chronically ill, pediatric, and elderly patients, as well as HIV-infected patients and other immunocompromised patients.⁴⁵⁹ In young children and in immunocompromised hosts, viral agents (rotaviruses, adenoviruses, coxsackieviruses, and others) are often found.³⁰¹ In addition, there has been a newfound appreciation for the roles of certain viruses (e.g., norovirus, rotaviruses, adenoviruses, coxsackieviruses, and others) as causes of nosocomial diarrhea, especially in neonates, children, and patients in intensive care units.^{448,460,461}

Long-Term Care Facilities

Diarrheal illnesses constitute a significant problem in extended-care facilities for elderly patients. A conservative estimate based on passively reported illness rates is that one-third of patients in long-term care facilities experience diarrhea each year.⁴⁶²⁻⁴⁶⁴ CDAD remains a common and important cause of diarrhea in these facilities.^{408,465,466} Sporadic cases and epidemic outbreaks of CDAD have been reported in many types of long-term care facilities. In other instances, viral causes of gastroenteritis or diarrhea have been identified in certain outbreaks occurring in these settings. The frequency of potentially transmissible enteric pathogens emphasizes the importance of careful hand washing in situations in which hygiene is often difficult.

Daycare Centers

Another special institutional setting in which hygiene is difficult and enteric infections are increasingly appreciated is daycare centers. Numerous outbreaks have been reported in association with viruses, bacteria, or parasites. The most common etiologic agents in infants and

children younger than 2 years are rotaviruses, whereas older toddlers have been more likely to acquire *G. lamblia*.⁴⁶⁷ Diagnostic tests based on immunoassays and reverse-transcriptase PCR have been used to detect additional agents such as astrovirus in many diarrhea outbreaks in daycare centers.^{468,469} A clinical syndrome of prolonged noninflammatory diarrhea may be associated with *Cryptosporidium* in daycare centers.⁴⁷⁰⁻⁴⁷² Outbreaks of inflammatory diarrhea caused by *Shigella*, *Campylobacter*, and *C. difficile* have also been reported.^{341,342}

Treatment of Acute Noninflammatory Diarrhea

Treatment of diarrhea from any cause in adults and children consists primarily of rehydration. If glucose or sucrose accompanies oral isotonic fluid, the coupled absorption of sodium and water is often sufficient to replace fluid loss. Bismuth subsalicylate may reduce enterotoxin action,²⁸⁸ and if there is no significant febrile or inflammatory process, low doses of antimotility agents may offer some relief with minimal risk in adults. Some studies also suggest that novel analogues of glutamine may be beneficial in reducing

Community-Acquired Diarrhea

Acute Pediatric Diarrhea

The rapid decline of infant mortality rates during the 20th century coincided with increased availability of piped water and chlorination systems. In areas of the world that continue to have unsafe water and sanitation, gastroenteritis remains a major cause of infant mortality.³ The highest attack rate of diarrhea occurs at the time of weaning (weanling diarrhea), usually between 6 and 24 months of age.^{19,20} The increased susceptibility of a recently weaned infant is related to several factors.²¹ First, human breast milk appears to convey a level of resistance to some GI pathogens and may protect an infant who is still breastfeeding.²²⁻²⁶ In contrast, in resource-limited settings, weaning and complementary foods are often prepared under conditions of poor hygiene and are frequently found to be contaminated with large numbers of potential diarrheal pathogens.^{27,28} Children may also ingest soil contaminated with diarrheagenic bacteria.²⁹ Undernutrition can influence host defenses and pathogen-attributable diarrhea risk.³⁰⁻³³ Finally, emerging studies indicate shifts in microbiota may mediate susceptibility to particular gut pathogens.³⁴

Early childhood diarrhea, regardless of etiology, often manifests as acute, watery, noninflammatory diarrhea. Cases are sporadic, but there are seasonal patterns, often peaking in the summer months or periods of high precipitation in resource-limited settings.⁷ In a well-nourished infant, the disease is usually short-lived and resolves within 2 to 3 days with adequate hydration.^{20,35} Diarrhea in a malnourished or immunocompromised child tends to persist or to recur and is often much more severe.

Early childhood diarrhea has been most commonly associated with rotaviruses³⁵⁻³⁷ and enterotoxigenic *Escherichia coli* (ETEC).^{38,39,40} Use of molecular diagnostics in longitudinal and case-control studies has refined these pathogen-attributable estimates across multiple global regions.^{5-7,41,42} Using quantitative polymerase chain reaction (PCR) methods, up to 90% of moderate-to-severe acute diarrheal episodes were pathogen-attributable in low- to middle-income settings.⁵ Rotavirus is the most common global etiology of severe acute watery diarrhea, and remains one of the most common identified agents even in the postvaccine era.^{5,7,41,42} Other viruses—predominately norovirus GII and adenovirus types 40 and 41, and to a lesser extent, sapovirus and astrovirus—are significant contributors, especially where rotavirus vaccine is established.^{5,7,42-44} Cases due to *Shigella* spp. slightly outnumber cases caused by heat-stable enterotoxin-producing ETEC (ST-ETEC).^{5,7,42} These studies have also revealed the emerging importance of previously difficult to detect or noncultivable pathogens such as *Campylobacter* spp. and the protozoan *Cryptosporidium*. In some settings, *Cryptosporidium* is second only to viral etiologies and, together with rotavirus and *Shigella* spp., may combine to account for more than 50% of diarrhea-attributable mortality in children less than 5 years of age.⁴² Although these studies have clarified the major drivers of early childhood diarrhea at the population level, the clinical application of these tools for individual cases is more challenging. First, molecular diagnostics are seldom available. Second, 40% or more of diarrheal stools may harbor two or more pathogens simultaneously.⁵

Diarrhea Caused by Rotavirus

Rotaviruses are the most common cause of severe diarrhea in infants and young children (see Chapter 157).^{42,45-48} Worldwide, rotaviruses cause more than 100 million cases of

gastroenteritis and up to 150,000 deaths each year in children younger than 5 years old.^{3,46,49} Of these rotavirus-induced deaths, more than 90% occur in children residing in low- or middle-income countries.⁴⁹ Although most adults have antibody to rotaviruses that may be protective, unvaccinated children younger than 2 years are highly susceptible to rotavirus diarrhea.⁴⁶

In the United States, few deaths are attributable to rotavirus infection; however, before inclusion of rotavirus vaccine in childhood immunization programs, almost every child had been infected by the age of 5 years, and rotavirus-induced gastroenteritis caused about 400,000 physician encounters, almost 200,000 emergency department visits, and about 70,000 hospitalizations each year.⁴⁶ Because routine vaccination began in 2008, rotavirus-related diarrhea has declined by 62% to 71%, and hospitalization for all-cause gastroenteritis has declined by 31% to 33%.⁵⁰⁻⁵² Similar trends are emerging in the >100 countries that have implemented rotavirus vaccination campaigns, resulting in an estimated median reduction of 59% for rotavirus-specific hospitalizations and 36% for all-cause gastroenteritis mortality.⁵³ Nonetheless, among children in the United States admitted for acute gastroenteritis, rotavirus remains important.⁴¹ Infection occurs more frequently in the winter or in cooler, dry months^{35,36,54-59} than in the summer months.^{7,60}

There are five major rotavirus serotypes, A through E, with group A strains further subdivided based on the combination of G (the VP7 glycoprotein) and P (the VP4 protease-cleaved protein) types.^{46,47,61-64} Worldwide, five serotypes (G1–G4 and G9) predominate.^{46,47,61-63,65} A particular strain (the number in brackets indicates the genotype), P1B[4]G2, is associated with a more severe form of infection in children.^{66,67} Such epidemiologic information was important in designing rotavirus vaccines. Before rotavirus vaccination in the United States, six rotavirus strains were most commonly found: G1P[8], G2P[4], G3P[8], G4P[8], G9P[8], and G9P[6].^{63-65,68} In the postvaccine era, G3P[8] was predominant through 2011 and has since been replaced by G12P[8]. Vaccine-derived strains occur in 1.7% of cases.^{69,70} Recent outbreaks of milder illness have been associated with more unusual serotypes; for example, strain G8P[8], which affected vaccinated and nonvaccinated individuals equally.⁷¹ Introduction of rotavirus vaccines in other settings has also corresponded with emergences of G3P[8] and other less common strains.^{72,73}

Rotavirus diarrhea is usually mild, without high-grade fever, and is often associated with vomiting; however, clinical syndromes range from asymptomatic carriage to severe gastroenteritis with profound dehydration, shock, and death.^{35,36,45,46,74} Serious complications of rotavirus gastroenteritis include necrotizing enterocolitis in neonates.^{68,75,76} Severe cases are more common in resource-limited settings^{46,75} and among severely immunocompromised patients. Large symptomatic outbreaks of moderately severe rotavirus diarrhea can occur. Although outbreaks are geographically distinct, they are often caused by the same rotavirus strain.

The diagnosis of rotavirus diarrhea is established with the use of a variety of molecular assays designed to detect virus-specific antigens, antibodies, or nucleic acids in feces of infected patients.⁷⁷⁻⁸⁰ Rotaviruses can also be cultured directly from stool samples, although this method is time-consuming, expensive, and limited primarily to research-based laboratories.

The human rotaviruses are nonenveloped RNA viruses of approximately 70 nm belonging to the Reoviridae family.^{46,61,62} The major Page 1282rotavirus-mediated effects on the host include disruption of intestinal epithelial cell brush-border enzymes, cytotoxicity, and alteration of the enteric nervous system.^{47,48,81-83} Normal columnar epithelium at the villus tips is replaced by irregular cuboidal, cryptlike cells, leading to multiple defects in fluid and electrolyte regulation. Many of these effects are believed to be mediated by a calcium ion-dependent enterotoxin.

The loss of absorptive villus tip cells may be responsible for the fluid imbalance and nutritional effect of rotavirus infections. The degree of microvillus damage roughly parallels the severity of diarrhea and dehydration.⁸⁴ Certain patients with rotavirus gastroenteritis also exhibit extraintestinal symptoms in association with detectable antigenemia and viremia, supporting a systemic phase to this infection.⁸⁵⁻⁸⁷

Oral or intravenous rehydration is the cornerstone of therapy for rotavirus gastroenteritis. No definitive antiviral therapy is established for rotavirus infection; however, the antiparasitic drug nitazoxanide was shown in randomized, controlled studies to shorten the duration of illness in

children and adults with symptomatic rotavirus diarrhea.^{88,89} Possible mechanistic explanations for an antiviral function of nitazoxanide include interrupting of cellular processes underlying viral replication^{90,91} or antibacterial effects that result in enhanced antiviral immunity. Although multiple different probiotics are available in many parts of the world, adjuvant probiotic use may reduce repeated diarrhea episodes and promote intestinal function after rotavirus infection,⁹² but the role of specific probiotics regarding duration of shedding and symptoms⁹³ and severity of illness across different populations requires further investigation in controlled clinical trials.⁹⁴

Because of the great global burden of childhood rotavirus gastroenteritis, significant multinational efforts have been directed toward the development of a safe and effective rotavirus vaccine. The first rotavirus vaccine approved for use in the United States was the rhesus-human reassortant rotavirus tetravalent vaccine (RotaShield; Wyeth-Lederle, Madison, NJ). It was approved and released in 1998 for use in infants. Despite high efficacy, the vaccine was removed less than 1 year later because of concerns about possible cases of intussusception in vaccinated infants.^{95,96} After the withdrawal of the rhesus-human reassortant rotavirus tetravalent vaccine, two new rotavirus vaccines became licensed and approved for use in infants in the United States^{46,62,97,98}: first, a live oral pentavalent rotavirus vaccine (RotaTeq; Merck Vaccines, Whitehouse Station, NJ) comprising bovine-human recombinant strains in 2006,^{46,97,98,99} followed by a monovalent human rotavirus vaccine (Rotarix; GlaxoSmithKline, London, United Kingdom) derived from an attenuated P1A[8]G1 rotavirus strain in 2008.^{46,97,98,99-101} Both are administered starting between 6 and 14 weeks of age in a three-dose (pentavalent rotavirus vaccine) or two-dose (human rotavirus vaccine) schedule and are highly efficacious. No significant increased risk of intestinal intussusception has been observed to date. Neither vaccine should be administered to infants with severe combined immunodeficiency syndrome¹⁰²; however, despite a theoretical risk of transmission of vaccine-derived strains within a household, living with an immunosuppressed patient is not an absolute contraindication.¹⁰³

Routine rotavirus vaccination has dramatically reduced rotavirus-diarrhea in the United States, and use of the rotavirus vaccine has been estimated to have saved \$1.2 billion in annual health care costs associated with acute gastroenteritis.⁵¹ Rotavirus vaccine is now available in more than 100 countries; however, vaccine efficacy varies geographically and is lower in resource-limited countries where the bulk of rotavirus infections still occur.³ Among the potential reasons for decreased rotavirus efficacy is as yet poorly understood differences in mucosal immune responses in children in these settings, particularly children with evidence of undernutrition, and possibly differences in intestinal microbial ecology or the presence of high maternal antirotavirus immunoglobulin A, or both.¹⁰⁴⁻¹⁰⁸ However, prenatal nutritional supplementation did not improve rotavirus vaccine immunogenicity.¹⁰⁹ A parenterally administered P2-VP8-P[8] subunit vaccine appears to be safe and immunogenic.¹¹⁰

Acute Nausea and Vomiting (Winter Vomiting Disease)

The syndrome of acute nausea and vomiting (intestinal flu, viral gastroenteritis, or what had been called winter vomiting disease), sometimes with low-grade fever, commonly occurs in winter months in temperate climates.^{111,112} Although this syndrome overlaps with rotavirus-associated infantile gastroenteritis, rotaviruses are relatively uncommon causes of winter vomiting disease in older children and adults. The Cleveland family studies of Dingle and coworkers¹¹³ showed GI illnesses were most common between the ages of 1 and 10 years, when approximately two illnesses occurred per person per year. Illnesses peaked from November through February, with June being the month of lowest frequency. Most illnesses lasted 1 to 3 days; 20% occurred with respiratory symptoms, and 20% involved only diarrhea.

Illnesses tended to occur in one of two patterns: (1) a mild afebrile illness with watery diarrhea or (2) a more severe febrile illness with vomiting, headache, and constitutional symptoms. Although etiologic agents were rarely identified, these two patterns of illness also developed among volunteers who ingested filtrates prepared from the feces of ill patients.¹¹⁴⁻¹¹⁶ Studies done in Charlottesville, Virginia, confirmed this pattern of wintertime gastroenteritis including clustering in families, highest attack rates in children, and absence of identifiable etiologic agents in most cases, despite the application of techniques for virologic and enterotoxin studies.^{117,118}

Caliciviruses are now recognized as the etiologies of most cases of gastroenteritis occurring in winter months. The Caliciviridae family comprises four genera^{119,120}: Norovirus, Sapovirus,

Lagovirus, and Vesivirus (see Chapter 182). Viruses belonging to Norovirus (from Norwalk) and Sapovirus (from Sapporo) genera are currently the most common causative agents for viral gastroenteritis in humans.¹¹⁹⁻¹²³ Noroviruses and sapoviruses may predominate as causes of early childhood diarrhea in rotavirus-vaccinated populations.^{5,6,42,124} Noroviruses, in particular, cause major outbreaks, especially in older children and adults (see Chapter 182). Immunocompromised patients can develop chronic norovirus infection lasting for months or years (see “Diarrhea in Immunocompromised Patients”).¹²⁵

Norovirus

Norovirus accounts for more than one-third of the outbreaks of nonbacterial gastroenteritis in the United States.^{126,127} Initially these viruses were referred to as Norwalk-like viruses. Their individual names were originally derived from the site of origin of each particular outbreak and included Norwalk,¹²⁸⁻¹³⁰ Hawaii,¹³¹ Snow Mountain,¹³² Taunton, and W agents.¹³³ Person-to-person spread is common, and secondary attack rates can be high. Transmission is predominately fecal-oral, but salivary transmission has been increasingly recognized.¹³⁴

Norovirus outbreaks have been observed in a wide variety of settings,¹³⁵ including hospitals,¹³⁶ extended-care facilities,¹³⁷ child care centers,¹³⁸ cruise ships,¹³⁹ refugee centers associated with natural disasters (e.g., Hurricane Katrina¹⁴⁰), and military combat areas including Afghanistan¹⁴¹ and Iraq.¹⁴² Several genotypes are known to infect humans^{121,135}; in the last decade, newly identified variant strains of norovirus genotype GII.4 have caused outbreaks worldwide¹⁴³⁻¹⁴⁵ including in the United States, United Kingdom, Europe, New Zealand, and Australia.

The pathophysiologic features of norovirus gastroenteritis parallel features of rotavirus in several respects.^{119,120,121} Each type is essentially noninflammatory and causes villus disruption and transient brush-border enzyme deficiencies in the upper small intestine, without any alteration in adenylate cyclase activity.^{63,146,147} This enteropathy can be severe in patients with underlying immunocompromise, especially patients with common variable immunodeficiency.¹⁴⁸ The roles of enzyme deficiencies, malabsorption of xylose and lactose, and an increase in intestinal bacteria during norovirus illness remain unclear.^{146,149}

Sapovirus

Global studies on etiology of pediatric diarrhea in low- to middle-income countries have unmasked a greater burden of sapovirus diarrhea than previously recognized.^{5,6,150} Sapovirus was the second most common all-cause etiology of diarrhea in the community setting,⁶ and cumulative incidence may exceed 80% by 2 years of age.¹⁵¹ Sapoviruses were initially called typical human caliciviruses or Sapporo-like viruses after the prototype strain found in an outbreak in Sapporo, Japan, in 1982. Similar to norovirus, diverse sapovirus genogroups and genotypes cause infections in humans. Repeated infections are typically due to different genotypes, suggesting immunity is genotype specific.¹⁵¹ Sapovirus symptoms overlap with norovirus symptoms, possibly with more prolonged duration of vomiting.¹⁵² Similar to norovirus, sapovirus can cause outbreaks in long-term care facilities, but fewer laboratories specifically test for sapovirus. Sapovirus should be considered if the cause for an outbreak is otherwise unexplained.¹⁵³ Sapovirus also occurs as a coinfection with other pathogens.

Diarrhea Caused by Astrovirus, Adenovirus, and Other Possible Viral Etiologies

Several other groups of viral pathogens have been identified in endemic and epidemic gastroenteritis (Table 100.1), including^{45,77,154-156} (1) other types of caliciviruses with characteristic, “chalice-like” surface; (2) astroviruses, with a five- or six-pointed starlike surface structure¹⁵⁷; (3) enteric adenoviruses (especially serotypes 40 and 41)^{5,158}; and (4) human enteric coronaviruses, including SARS-CoV-2,^{159,160,161} as well as other miscellaneous or less well-characterized viral agents. More detailed discussions of these agents may be found in Chapters 149, 162, 178, 180, 183, and 184.

Astroviruses are among the most common causes of pediatric viral gastroenteritis.¹⁶²⁻¹⁶⁵ Astrovirus occasionally occurs in association with other enteric pathogens; in these cases the illness is more severe and protracted.¹⁶⁴ After rotavirus, adenovirus 40/41 was the most common viral etiology of moderate-to-severe acute diarrhea in children in low-income settings and the third most common cause overall.⁵ Adenovirus 40/41 was also third only to rotavirus and norovirus as

a cause of acute gastroenteritis among children hospitalized in a US health care system.⁴¹ Increased emphasis on adenovirus serotyping in infant-child cohort studies is revealing a likely widespread greater burden of disease caused by this virus.^{166,167} In a case-control study in China, adenovirus serotype 3 was also associated with diarrhea, whereas asymptomatic detection of other serotypes (1, 2, 5, 6, 31, and 56) were more common in controls without diarrhea.¹⁶⁸

Vomiting and diarrhea can also manifest during infections with respiratory viruses, including in the absence of a respiratory syndrome. Enteric coronaviruses have clear pathology in animals (i.e., transmissible gastroenteritis in pigs). Although human coronaviruses, including the novel SARS-CoV-2 sarbecovirus, are predominately respiratory pathogens, these viruses can be isolated from intestinal tissues and stools.^{159,160,169,161} SARS-CoV-2 and all four of the endemic human coronaviruses (HCoV-229E, HCoV-NL63, HCoV-HKU1, and HCoV-OC43) have been detected in cases of acute gastroenteritis. Acute gastroenteritis from SARS-CoV-2 (see Chapter 135) infection¹⁶¹ can be severe enough to require hospitalization, however, the role of endemic human coronavirus as a primary etiology of diarrhea is likely minor; codetection with other common viruses such as rotavirus and norovirus is frequent.¹⁶⁹ Up to 10% to 20% of children with seasonal influenza present with diarrhea, although this is uncommon in adults. Diarrhea may be more prominent with Asian avian influenza A (H5N1) infection.¹⁷⁰

Bocaviruses,¹⁷¹ pestiviruses,¹⁷² toroviruses,^{173,174} picobirnaviruses,¹⁷⁵ and protoparvoviruses¹⁷⁶ are becoming better recognized as pathogens in children and adults, but their cooccurrence with conventional intestinal viruses and their independent roles in causing diarrheal illnesses are uncertain.^{176,177} Similarly, although enteroviruses may cause systemic illnesses, their role as diarrheal pathogens is minimal. Unbiased viral metagenomics approaches have further identified sequences of obscure viruses in diarrheal stools that require further clarification.^{178,179} In addition, a hallmark characteristic of the 2014 Ebola virus disease epidemic caused by Zaire ebolavirus in West Africa was nausea and abdominal pain, with voluminous vomiting and diarrhea that led to life-threatening cholera-like GI volume losses through unclear mechanisms.^{180,181}

TABLE 100.1

Viral Pathogens Causing Gastroenteritis

ESTABLISHED PATHOGENS LIKELY AND EMERGING PATHOGENS

Adenoviruses (enteric types)

Astroviruses

Caliciviruses (including noroviruses and sapoviruses)

Rotaviruses groups A–C

Cytomegalovirus

Bocaviruses

Coronaviruses (including SARS-CoV-2)

Enteroviruses (various)

Picobirnaviruses, picornaviruses

Pestiviruses

Toroviruses

Filovirus (Ebola virus)

Human protoparvoviruses

Data from references 45, 68, 77, 122, 123, 135, 138, 155, 157, 172, 440, 441, 176, 232, 233.

Nucleic acid–based assays (e.g., reverse-transcriptase PCR) of stool specimens, either pathogen-specific or as part of multipathogen panels, are the standard method of GI virus detection. Rotavirus-specific and norovirus-specific reverse-transcriptase PCR are also available. In the typical outpatient clinical situation, norovirus-specific diagnostic tests are rarely performed because results of this assay do not alter management of the illness.

Cryptosporidiosis

Cryptosporidium is another cause of acute, noninflammatory diarrhea, second only to viruses in some countries.^{5,7,182} Sporadic cryptosporidiosis typically occurs through direct exposure to either infected people and to a lesser extent animals.^{183,184} Outbreaks occur via contaminated food, drinking water, or recreational water including swimming pools.^{185–190} Of the more than 20

species of this tiny coccidian intracellular protozoal parasite, two, *C. parvum* and *C. hominis*, cause the majority of human disease.¹⁸⁹⁻¹⁹³ Cryptosporidiosis is commonly asymptomatic or mild in normal hosts and usually results from waterborne outbreaks, often related to contaminated drinking water or public swimming pools.^{186,189-193} However, infection can be severe in infants and toddlers, accounting for an estimated 50,000 to >200,000 deaths per year.^{182,194} Malnourished children and immunocompromised hosts, especially patients with advanced human immunodeficiency virus (HIV) infection, can manifest severe diarrhea.^{185,187,192-195} Cryptosporidium, either with or without diarrhea, in some children can also contribute to early childhood growth impairment.^{182,196} Parasite shedding even after recovery from acute diarrhea can be prolonged in hosts with immunocompromise such as those with uncontrolled HIV.¹⁹⁷

Cryptosporidium causes a secretory form of diarrhea associated with dysregulated intestinal absorption that may be less responsive to oral rehydration than other diarrhea etiologies.¹⁹⁰ In addition to gastroenteritis, biliary infection with cholangitis has been reported. The pathogenesis of cryptosporidiosis is incompletely understood; the organism primarily alters villus structure and function and can promote an inflammatory enteropathy that may persist after parasite clearance.¹⁹⁰ No specific enterotoxin has been identified. Diagnosis depends on identification of oocysts in stool specimens by standard light microscopy, immunofluorescence assays, or nucleic acid-based molecular tests.¹⁹⁸⁻²⁰⁰ In addition to supportive care, nitazoxanide has been shown to be moderately successful in immunocompetent patients with cryptosporidiosis. Data are inconclusive about universally recommended duration of therapy, especially in immunocompromised and malnourished patients.²⁰¹⁻²⁰⁷ Alternative therapeutic agents for immunosuppressed patients include combinations of paromomycin and azithromycin. Ongoing searches for novel drugs have identified several with anti-Cryptosporidium activity through different mechanisms, but all remain in preclinical stages of development.²⁰⁸⁻²¹⁵ Despite resurging interest, vaccine development remains in early stages, in part because determinants of host immunity, especially in malnourished populations, are poorly understood.

Diarrhea Caused by Enteropathogenic Bacteria

Several diarrheagenic *E. coli* pathotypes (ETEC, enteroaggregative *E. coli* [EAEC], enteropathogenic *E. coli* [EPEC], enteroinvasive *E. coli* [EIEC], and Shiga toxin-producing *E. coli* [STEC], *Shigella* spp., *Salmonella* spp., *Campylobacter* spp., and both *V. cholerae* and noncholera *Vibrio* spp.) can cause noninflammatory diarrhea (Fig. 100.1). ETEC, *Campylobacter*, and *Shigella* are the most common bacterial etiologies of acute gastroenteritis worldwide.^{42,3} The presence of blood in stool (dysentery) increases the likelihood of diarrhea due to *Shigella*, *Campylobacter*, or EIEC; however, all bacterial pathogens most commonly present as a noninflammatory syndrome.^{5,7} Dysentery is present in <5% of community-associated diarrhea, and inflammatory manifestations of these bacterial gastroenteritides are discussed in greater detail in Chapter 101. Among bacterial etiologies that manifest only as noninflammatory diarrhea, ETEC and *V. cholerae* combined account for 9% to 24% of diarrhea-attributable mortality in children younger than 5 years of age.³

FIG. 100.1 Global pathogen-attributable etiologies of acute gastroenteritis and estimated fatal attributable fractions. Pathogen-attributable etiologies (millions of episodes/year) of acute gastroenteritis measured in the Global Burden of Diseases (through 2016) and attributable fatal fractions for (top) children ≤5 years of age and (bottom) ≥70 years of age. Data include cases of dysentery (<5% of total cases of acute gastroenteritis) in addition to noninflammatory cases. (Top) Red shading indicates proportion of pathogen-specific cases presenting with dysentery. *Shigella* is the most common cause of dysentery and ~15% of *Shigella* cases present with dysentery. *C. difficile*, *Clostridioides difficile*; EPEC, enteropathogenic *Escherichia coli*; ETEC, enterotoxigenic *Escherichia coli*; *V. cholerae*, *Vibrio cholerae*. Adapted from Troeger C, et al. GBD 2016 Diarrhoeal Diseases Collaborators. Estimates of the global, regional, and national morbidity, mortality, and aetiologies of diarrhoea in 195 countries: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Infect Dis.* 2018;18(11):1211–1228; and Liu J, et al. Use of quantitative molecular diagnostic methods to identify causes of diarrhoea in children: a reanalysis of the GEMS case-control study. *Lancet.* 2016;288:129–301. Follow for extended description

The greatest risk for pediatric ETEC diarrhea is between age 12 and 24 months, and severity is increased in undernourished children.^{216,42} Antibodies directed against the heat-labile enterotoxin (LT) of *E. coli* in human colostrum may account for lower burden of ETEC in infancy.^{22,23,217} Diarrhea produced by LT shares the adenylate cyclase-activating mechanism with cholera toxin.^{218–222} Certain strains of *Klebsiella*, *Citrobacter*, and *Aeromonas* produce an LT-like toxin and may cause a small percentage of diarrheal disease in children.²²³

Molecular methods have refined estimates of the contributions of other diarrheagenic *E. coli* on burdens of diarrhea. Typical EPEC is a contributor to diarrhea-related morbidity and mortality in the first 2 years of life.²²⁴ EAEC, although frequently subclinical, is among the most commonly identified enteropathogens. EAEC infection is nearly universal in children in resource-limited settings. Despite no clear association with diarrhea, EAEC, even in the absence of diarrhea, is associated with growth impairment,²²⁵ particularly when found as a coinfection with LT-ETEC, EPEC, and *Campylobacter* spp.^{225,226} Similarly, independent of diarrhea, cumulative and diverse pathogen exposures are associated with growth impairment in the first 2 years of life; these findings highlight the substantial potential cost of early-life GI infection in low-resource settings (see Chapter 104).^{227,228,229}

Diarrhea in Adults

An emerging increasing relative burden of diarrhea and diarrhea mortality in adults has emerged in the last several decades, concomitant with an aging global population.³ In resource-abundant settings, acute gastroenteritis in adults is most often caused by noroviruses.^a Less common Page 1285 viruses in adults include rotaviruses,⁷⁰ adenoviruses, coxsackieviruses, and coronaviruses.^{32,233} *C. difficile* and its potential for recurrence is of increasing concern in patients with recent antibiotic use (see Chapters 101 and 249).²³⁴ Bacterial gastroenteritides such as nontyphoidal *Salmonella* spp. are important causes of foodborne diarrhea outbreaks, often with fever (see Chapter 101). In addition, several agents of food poisoning such as *Clostridium perfringens*, *Bacillus cereus*, and *Staphylococcus aureus* commonly cause diarrheal syndromes in adults (see Chapter 103).

In adults living in resource-limited settings, several pathogens are known to cause sporadic noninflammatory diarrhea, but none has had as much historical effect or notoriety as *V. cholerae*, causing cholera.^{235,236} Cholera has long been endemic in South Asia. With the increased infection-to-case ratio of El Tor cholera, the current seventh pandemic swept most of the continents of the Eastern Hemisphere including Asia, Africa, and the Mediterranean portions of Europe in the 1960s.²³⁷ Rare isolated cases occurred in the United States^{238,239} and Europe related to contaminated mineral water²⁴⁰ and to undercooked shellfish.^{241,242} Beginning in Madras, India, in late 1992 and rapidly spreading to Calcutta and Bangladesh in 1993, a new strain of non-O1 *V. cholerae*—O139, called Bengal—caused epidemic cholera gravis.^{243–247} Most recently, natural disasters and social conflict have preceded large outbreaks including the introduction of *V. cholerae* into previously cholera-free Haiti after the devastating earthquake in 2010 that may have reemerged²⁴⁸ as endemic in 2022, and more than 1.2 million cases in conflict-stricken Yemen since 2017.

Severe, watery diarrhea in a patient residing in or having recently traveled to an endemic area should raise suspicion of cholera. The disease can be so fulminant as to cause hypovolemic shock and death from the outpouring of fluid into the upper portion of the small bowel before the first diarrheal stool occurs.²⁴⁹ As discussed in detail in Chapter 219, the entire dehydrating syndrome of cholera appears to be related to the activation of intestinal adenylate cyclase by the potent cholera enterotoxin.^{218,219,250,251} To make the diagnosis bacteriologically, stool specimens should be cultured onto thiosulfate citrate bile salts sucrose agar. Of prime importance in therapy is fluid replacement, accomplished either intravenously with isotonic fluids or orally with glucose-electrolyte solutions.

Patients from whom *V. cholerae* cannot be isolated may have a cholera-like syndrome caused by certain strains of ETEC. In 1956, De and associates²⁵² demonstrated that *E. coli* isolated from adults and children with this syndrome caused fluid accumulation similar to that seen with *V. cholerae* infection in ligated rabbit ileal loops. In the early 1960s, Trabulsi,²⁵³ working in São Paulo, Brazil, reported similar findings with toxigenic *E. coli*. Various investigators then demonstrated that acute undifferentiated diarrhea in adults was caused by ETEC strains that

usually were not of the classically recognized pathogenic serotypes.²⁵⁴⁻²⁵⁷ The toxic material present in the culture filtrate of these *E. coli* strains was demonstrated to be heat labile and nondialyzable. Subsequent studies demonstrated two types of enterotoxin produced by ETEC—LT and heat-stable enterotoxin (ST).²⁵⁸ LT activates adenylate cyclase²²⁰⁻²²² and is antigenically and mechanistically similar to cholera toxin in many ways. In contrast, ST activates guanylate cyclase,²⁵⁹⁻²⁶¹ has an earlier onset of action,²⁵⁸ has greater tissue specificity,²⁶¹ and has a much lower molecular weight than LT.²⁶²

Several studies have shown that ETEC strains producing LT only, ST only, or LT plus ST are associated with episodes of diarrhea in adults. Adults in resource-limited settings who carry LT-producing *E. coli* are often asymptomatic. In contrast, ST-producing strains are significantly associated with diarrheal disease. Diarrhea due to ETEC is less common in the United States.^{54,117}

Diarrhea in Immunocompromised Patients Patients With HIV Infection

Patients with HIV infection often develop or present with diarrhea. With increased access to antiretroviral therapy (ART), the incidence of infectious diarrhea among HIV-infected patients decreased markedly.^{263,264} However, among HIV-infected patients in the United States meeting criteria for acquired immunodeficiency syndrome (AIDS), 30% to 60% present with diarrhea,^{263,265-268,269} and this figure can reach 95% in parts of Africa or Haiti. In many patients with advanced HIV infection and immunocompromise, diarrhea becomes prolonged, causes severe malnutrition, predisposes to other types of serious infections, and can be life threatening. In addition, symptomatic management of chronic diarrhea in such patients poses major difficulties. The importance of the interactions of HIV with the components of the intestinal immune system in modulating systemic HIV infection has come to light in recent years.²⁷⁰⁻²⁷³

Although some investigators have reported an enteropathy without identifiable infectious pathogens^{274,275} or with primary HIV infection of enterochromaffin cells in the bowel mucosal crypts and lamina propria,²⁷⁶ others have reported one or more enteric pathogens in 55% to 85% of patients with AIDS and diarrhea.^{277,278} As shown in Table 100.2, the leading agents found in these patients are *Cryptosporidium*, cytomegalovirus (CMV), microsporidia, *Entamoeba histolytica*, *Giardia lamblia*, *Salmonella*, *Campylobacter*, *Shigella*, *C. difficile*, *Vibrio parahaemolyticus*, and *Mycobacterium* spp.^{277-281,282-287} *Cyclospora* and *Cystoisospora belli* also cause persistent diarrhea in patients with AIDS.²⁸³ *Pneumocystis jirovecii* infection can occasionally involve the intestinal tract.²⁸⁴ Men with HIV who have sex with men are at increased risk of *G. lamblia*, *E. histolytica*, *Campylobacter jejuni*, *Shigella*, *Chlamydia trachomatis*, *C. difficile* (with proctitis), *Neisseria gonorrhoeae*, herpes simplex virus, or *Treponema pallidum*.²⁸⁷

Although eradication treatment may be difficult, most patients respond to specific antimicrobial or antiparasitic therapy and ART, emphasizing the need to identify an etiologic agent whenever possible.^{207,288} The antiviral agent ganciclovir often controls intestinal CMV,^{285,286,289,290} and most bacterial and parasitic infections can be treated with the expectation of some improvement. *Cryptosporidium*, which infects 3% to 21% of patients with AIDS in the United States, can be found in 50% of patients with AIDS and diarrhea in Africa and Haiti.^{185,188,279} *Cryptosporidium*-mediated biliary infection occurs in this patient population as well. Similarly, certain microsporidia organisms, such as *Encephalitozoon intestinalis* or *Enterocytozoon bieneusi*, can cause persistent and severe diarrhea in HIV-infected individuals. *Cryptosporidium* and microsporidial infections are associated with villus atrophy, crypt hyperplasia, increased intraepithelial lymphocytes, and D-xylose malabsorption.²⁷⁹

The same acid-fast stain that detects *Cryptosporidium* or *Mycobacterium* in fecal specimens may also reveal *C. belli* and *Cyclospora* in approximately 2% of patients with AIDS with diarrhea in the United States and in 15% of patients in Africa.²⁷⁹⁻²⁸³ Nontyphoidal *Salmonella* infections occur with an estimated 20-fold increase in frequency and with increased severity in patients with AIDS.^{291-294,295} Enteric viruses also are associated with diarrhea in HIV-infected individuals. In one study, astrovirus, picobirnavirus, calicivirus, and adenovirus were found in 6% to 12% of HIV-positive patients with diarrhea.²⁹⁶ In addition, ART regimens can cause diarrhea, often complicating the workup of an HIV-infected patient with persistent diarrhea.^{264,295,297}

TABLE 100.2

Possible Enteric Pathogens in Patients With Acquired Immunodeficiency Syndrome

PATHOGEN	% WITH DIARRHEA (n = 181)	% WITH NO DIARRHEA (n = 28)
Cytomegalovirus	12–45	15
Cryptosporidium	14–30	0
Microsporidia	7.5–33	0
Entamoeba histolytica	0–15	0
Giardia lamblia	2–15	5
Salmonella spp.	0–15	0
Campylobacter spp.	2–11	8
Shigella spp.	5–10	0
Clostridioides difficile (formerly Clostridium difficile) toxin	6–7	0
Vibrio parahaemolyticus	4	0
Mycobacterium spp.	2–25	0
Cystoisospora belli	2–6	0
Cyclospora	0–11	0
Blastocystis hominis	2–15	16
Candida albicans	6–53	24
Herpes simplex	5–18	40
Chlamydia trachomatis	11	13
Strongyloides	0–6	0
Intestinal spirochetes	11	11
One or more pathogens	55–86	39

Data from references 277–281, 283–287, 291–298, 303, 304.

Several practical algorithmic approaches to the diagnosis and management of diarrhea in HIV-infected patients have been published.²⁰⁶ These strategies favor the use of early, noninvasive stool studies and practical empirical treatment trials, followed by more invasive tests (e.g., endoscopy with biopsy) for patients with refractory or more severe presentations. ART has decreased the incidence of several important opportunistic infections including certain causes of gastroenteritis. Of particular note, there has been a dramatic decline in the incidence of tissue-invasive infections caused by CMV GI disease.^{289,290} An effort to judiciously adhere to safe food and water guidelines in higher-risk patients can substantially help in preventing many of these serious enteric infections.²⁹⁸ Addressing etiologies of diarrhea among mothers with HIV may also confer benefit to their HIV-exposed, but uninfected infants.²⁹⁹

Patients With Solid-Organ Transplants and Others Receiving Immunosuppressants

Chemotherapy for malignancy, autoimmune disease, and especially antirejection medications after solid-organ transplantation (SOT) or hematopoietic cell transplantation (HCT) increase risk of serious enteric infections.^{300,301,302} Frequent encounters with the health care system, prolonged hospitalizations, and antimicrobials contribute to high rates of *C. difficile* infection (CDI) in these populations.^{303,304} Transplantation of any organ is associated with increased risk of CDI. CDI occurs in 9% to 13% of patients after HCT and up to 20% after SOT.³⁰¹ Hospital-onset CDI after SOT can prolong hospitalization and is associated with allograft dysfunction.³⁰⁵

In addition to CDI, several other bacteria, viruses, and parasites can cause diarrhea after SOT or HCT. Viral infections are common in this population and can result in long-term shedding, particularly in cases of norovirus.¹²⁵ There is less seasonal variation in norovirus infections in immunocompromised hosts, and these infections manifest with more prolonged symptoms and viral shedding than infections in immunocompetent hosts. Up to one-third of these patients have diarrhea persisting for 2 weeks or longer, and shedding has been reported beyond a year.³⁰⁶ Symptoms may occur at lower levels of viral genome copies per gram of stool. Prolonged viral shedding in these patients is associated with an increased diversity of genomic variants in an infected individual.³⁰⁷ The pathogenicity of these divergent variants is unclear, and therefore the clinical significance of continued norovirus detection over prolonged periods of time in patients with mild symptoms is uncertain. Flattening of the villus border and loss of villin colocalizes with viral particles on duodenal biopsy specimens, but other histologic changes described in chronic norovirus infections in immunocompromised patients (edema, cellular debris, and metaplasia) are

nonspecific and overlap with noninfectious etiologies of enteritis in these patients.³⁰⁸ No specific therapy is available. Combinations of oral immunoglobulins, nitazoxanide,³⁰⁹ and decreasing immunosuppressive therapy³¹⁰ may benefit some patients, but results are inconsistent.

CMV enteritis should be considered in the differential diagnosis of diarrhea after transplantation, especially in donor-recipient mismatch cases. CMV enteritis can occur in the absence of peripheral viremia, and direct intestinal sampling is recommended to rule out this condition. Diarrhea may also be a presenting symptom of disseminated adenovirus in transplant recipients. Reduction of immunosuppression is part of management for any infectious diarrhea in this population. Ganciclovir is an effective treatment for most cases of CMV enteritis; however, options for adenovirus are more limited. Brincidofovir, a prodrug of cidofovir, is conjugated to a lipid to enhance intracellular delivery with lower plasma concentrations and less renal toxicity than cidofovir. This drug has been prescribed on a compassionate use basis for disseminated adenovirus infection in this population, but efficacy data are lacking, and diarrhea itself may be a significant side effect of brincidofovir.

In addition to increased risk of CDI, infections with other bacterial etiologies are more common in this population. These infections may present with greater symptom severity, higher pathogen burdens, and increased antimicrobial resistance compared with those in immunocompetent patients.³¹¹

Although less common, Cryptosporidiosis and other protozoa can also be severe in these patients and may mimic graft-versus-host disease on endoscopy and intestinal biopsy. The clinical presentation of these infectious etiologies overlaps with medication side effects, especially mycophenolate and chemotherapy, which cause direct epithelial cell injury.^{301,302,312}

For the individual patient, a specific pathogen etiology is seldom predictable based on symptoms and exposure history alone. In addition, noninfectious etiologies underlie the high burden of diarrhea, and may contribute up to 66%, of diarrheal etiologies in solid organ transplant recipients. Direct drug toxicities on epithelial cells, disruptions in intestinal microbial communities, and immune-mediated pathologies like graft-versus-host-disease are common in immunocompromised hosts with diarrhea.³¹³ Therefore, multipathogen testing, where available, is recommended to guide pathogen-targeted treatment in this population.³⁰¹ This strategy avoids unnecessary use of empiric anti-infectives in this population that is also at high risk of acquiring multidrug-resistant infections.

Travel-Associated Diarrhea (Turista)

Whether it “arouses one from bed with a start at 4 a.m. for a record-breaking race to the bathroom to begin a staccato ballet”³¹⁴ or produces the poetry of the psalmist (“I am poured out like water ... my heart like wax is melted in the midst of my bowels”³¹⁵), traveler’s diarrhea has a major effect each year on the 300 million to 500 million international travelers and probably on the distribution of well over \$100 billion in international tourism receipts.^{316,317}

Diarrhea is the most common and among the most disconcerting illnesses that threaten the traveler.³¹⁸⁻³²⁴ North American and northern European travelers appear to be at greatest risk when they travel to Africa, Asia, Latin America, and to a lesser extent southern Europe.³²⁵⁻³²⁸ Most symptoms of traveler’s diarrhea begin 5 to 15 days after arrival, although a range from 3 to 31 days has been noted.^{292,318,329-334} The illness is typically manifested by malaise, anorexia, and abdominal cramps followed by the sudden onset of watery diarrhea. Nausea and vomiting occur in 10% to 25% of cases. The diarrhea is usually noninflammatory, although a low-grade fever is present approximately one-third of the time. The duration is usually 1 to 5 days, but up to 50% of patients have illness that continues 5 to 10 days and sometimes beyond.

The attack rate for traveler’s diarrhea varies from 5% to 50%, depending on the destination and duration of the trip. Destinations with increased risk for acquiring traveler’s diarrhea include portions of Africa and Asia (including India), and certain areas of Central and South America.^{318,334-336} The attack rate also appears to decrease with age after 25 years, an observation that may reflect different habits and exposures rather than inherent susceptibility.^{314,327} Host genetics and intestinal microbiota factors may also predispose to greater susceptibility.³³⁷ Expatriate residents living in certain countries appear to be at some level

of persistent risk for diarrhea of infectious causes; for instance, an attack rate of 49% per month is observed during the first 2 years of residence in Nepal.³³⁸

For many years, the etiology of turista was an enigma; only infrequently had parasites or bacteria, such as amoebas, *Giardia*, *Salmonella*, or *Shigella*, been identified, and viral studies had failed to elucidate significant causes. The first suggestion that an infectious process was likely came from the effective reduction in the attack rate achieved by the use of prophylactic antimicrobial agents.^{329,334,339,340} Studies by Kean³¹⁴ suggested that certain EPEC serotypes may be involved in up to one-third of the cases. The involvement of *E. coli* was further confirmed in an outbreak of traveler's diarrhea among British Page 1287troops in Aden, where *E. coli* O148 was identified in 54% of the British troops with diarrhea.³³⁰

Later studies demonstrated ETEC in approximately 50% (range, 5%–75%) of cases of traveler's diarrhea in Latin America, Africa, and Asia (Table 100.3).^{341–343} The attack rate ranged from 20% to 100% (median, 52%–54%) (see Table 100.3).^{318,336,340,344} ETEC was almost never present before travel, and such organisms were acquired by only 12.6% of fellow travelers who did not become ill.^{292,332,333} ETEC producing LT, ST, or both can cause traveler's diarrhea (Table 100.4). In some areas, EAEC and EPEC are emerging as the major cause (see Chapter 223), although EAEC is also commonly identified in asymptomatic controls.^{318,320,343,345–348}

A number of other microbial agents have also been identified in small subsets of patients with traveler's diarrhea. *Salmonella*, *Shigella*, *Campylobacter*, *Vibrio*, and *Aeromonas* spp. are increasingly recognized in travelers with diarrhea.^{344,349–351} Pathogen-specific prevalence appears to vary by destination. *Campylobacter* was present in 25% to 35% of cases during travel to South Asia and 15% to 25% of cases related to travel in Southeast Asia, but accounted for less than 5% of cases from Africa or Latin America.³¹⁸ In US military personnel deployed to Southeast Asia, *Campylobacter* spp. were detected in 35% and nontyphoidal *Salmonella* in 23% of cases of diarrhea.³⁴⁹ In Thailand, *Campylobacter* was the most common etiology (28%) of pathogen-attributable diarrhea cases.³⁵¹ *Shigella* accounted for 5% to 15% of cases from Latin America, Africa, and South Asia and was less frequent in travelers to Southeast Asia.³¹⁸ *Salmonella* infections were more common during travel to Africa and Southeast Asia than to South Asia or Latin America.³¹⁸ *V. cholerae* is rarely a problem for most US travelers, but infection can occur among military personnel or relief workers in endemic cholera locales.^{327,351,352} Single-dose live lyophilized CVD 103-HgR (Vaxchora) oral cholera vaccine is approved for travel to areas with active cholera transmission.

TABLE 100.3

Etiology of Acute Traveler's Diarrhea in Three Locales^a

CHARACTERISTIC	LATIN AMERICA	AFRICA	ASIA
Duration of stay (days)	21 (2–42)	28 (28–35)	(28–42)
Attack rate (%)	52 (21–100)	54 (36–62)	(39–57)
Organism (%)			
No bacterial pathogen			
28	—	35	
Bacterial pathogen			
72	—	80	
Enterotoxigenic <i>Escherichia coli</i>			
46 (28–72)	36 (31–75)	15 (5–25)	
Enterotoxigenic <i>Escherichia coli</i>			
30 (25–35)	<5	17 (15–25)	
<i>Shigella</i>			
5 (0–30)	5 (0–15)	5 (4–15)	
<i>Salmonella</i>			
<5	10 (5–15)	10 (5–15)	
<i>Campylobacter jejuni</i>			
<5	<5	15 (2–35)	
<i>Vibrio parahaemolyticus</i>			
—	—	7 (1–13)	
Rotavirus			

23 (0–36) 10 (5–15) 5 (0–15)

Norovirus

10 — 3

Protozoa (Giardia, Cryptosporidium, Entamoeba histolytica, others)^b

7 <5 11

a Values shown are median (range) from multiple studies.^{317,318,340,343,396,490}

b Parasitic etiology may increase to 65% among returned travelers seeking medical care.³⁶²

Rotavirus or calicivirus infections have been described in 0% to 36% of cases but were often found in association with either bacterial or parasitic pathogens.³⁵³ With greater use of culture-independent detection methods, noroviruses in particular are being seen as greater contributors to traveler's diarrhea than previously recognized.^{354,355} These viruses have been implicated in several large outbreaks of gastroenteritis or diarrhea on large cruise ships.^{356–358} In many respects, however, cruise ship-associated outbreaks of gastroenteritis share more epidemiologic characteristics with institution-associated gastroenteritis than with classic traveler's diarrhea.

Intestinal protozoa also may be important causes of traveler's diarrhea. Agents such as *C. parvum*, *Cyclospora cayetanensis*, *G. lamblia*, and various microsporidia should be considered in the workup of subacute or chronic noninflammatory diarrhea in returned international travelers.^{359–361} In the GeoSentinel Surveillance Network global database of 2902 unwell travelers seeking medical care, parasitic etiologies (predominantly *G. lamblia* [27%], *E. histolytica* [14%]) accounted for 65% of chronic diarrhea cases, whereas bacteria (*Campylobacter* [13%], *Shigella* [6%]) were identified in 31%, and viruses (hepatitis A and hepatitis E) were identified in only 3%. *G. lamblia* was the most commonly identified pathogen regardless of travel destination.³⁶² Other protozoal parasites such as *Blastocystis hominis* can be commonly identified in the stools of persons traveling to developing countries. However, it has been difficult to ascertain whether these organisms actually cause disease or are merely commensal.^{363–365} A subset of patients have persistent diarrhea for which no infectious agent can be implicated. Chronic idiopathic diarrhea (see “Brainerd Diarrhea”) has been reported to occur in a few small travel-related outbreaks^{366–369} and may cause prolonged diarrhea in sporadic cases.

In contrast to the frequent identification of potential etiologic agents among travelers to tropical areas who develop diarrhea, travelers to temperate areas often develop a mild syndrome of diarrhea for which no infectious cause can be identified.³⁷⁰ Environmentally distributed *Cryptosporidium* and *Giardia* and oocysts are hardy in temperate regions, have zoonotic reservoirs in these regions, and are relatively tolerant to chlorination. Travelers to certain areas such as Russia and national parks in the United States who drink unfiltered or unboiled water may therefore be susceptible to development of the more insidious watery diarrhea seen with giardiasis or cryptosporidiosis.^{371–374} Strongyloidiasis acquired during travel may cause noninflammatory diarrhea, abdominal pain, and eosinophilia.³⁷⁵

Several other potentially serious infections manifested initially by diarrhea or abdominal pain may be acquired by travelers. Malaria may Page 1288begin as gastroenteritis with nausea, vomiting, diarrhea, or abdominal pain in 30% to 50% of cases.³⁷⁶ The physician should also remember to consider typhoid fever and other infections that may be manifested with a typhoidal pattern including plague, melioidosis, typhus, and arboviral hemorrhagic fevers.^{376,377}

TABLE 100.4

Frequency of Enterotoxigenic *Escherichia coli* in Association With Traveler's Diarrhea in Latin America, Africa, and Asia

FEATURE REPORTED FREQUENCY (%)

Gastroenterologists in Mexico³³¹ Peace Corps Volunteers in Kenya³³² Yale Glee Club in Latin America³³³ Japanese Travelers Returning to Tokyo From India, Southeast Asia, Orient³⁴⁴ Total

Illness attack rate 49% in 16 days 69% in 5weeks 74% in 1month —

Type of enterotoxin

LT only

16 33 25 4.8 21

LT and ST							
16	15	12.5	11.8	38			
ST only							
9.8	2	19	13.6	41			
% of ill patients with ETEC	41 (21/51 cases)	52 (14/27 cases)	56 (9/16 cases)	32			
(270/843 cases)	33.5						
ETEC, Enterotoxigenic Escherichia coli; LT, heat-labile; ST, heat-stable.							

The desire to control the bothersome problem of diarrhea in travelers has led to a variety of medications used for management.³¹⁴ Commonly used remedies, such as diphenoxylate-atropine (Lomotil) and kaolin-pectin suspension, were found to have little or no efficacy in most studies.³⁷⁸ Lomotil and other antimotility agents may actually worsen the illness in inflammatory processes such as shigellosis.³⁷⁹ Bismuth subsalicylate (Pepto-Bismol) was shown to inhibit enterotoxin activity in experimental animal models, and it has been recommended for symptomatic therapy, as well as prophylaxis.³⁷⁹⁻³⁸² As in any type of diarrheal illness, the mainstay of therapy is adequate hydration with an oral glucose-electrolyte or sucrose-electrolyte solution.

Prevention of traveler's diarrhea should be directed toward reducing the consumption of infectious agents in food and water. Foods that are handled but not cooked (e.g., salads, raw vegetables) are high-risk foods.³¹⁸⁻³²⁰ Bottled, noncarbonated water cannot be considered safe because outbreaks of cholera²⁴⁰ have been traced to bottled water, and typhoid fever^{383,384} has been traced to other bottled beverages. Care in eating and drinking may reduce one's risk, even in highly endemic areas, to less than 15%.³⁸⁵⁻³⁸⁷

The efficacy of prophylactic antimicrobial agents has been documented in several studies.^{329,334,388-390} The increased risk of acquiring a more severe infection (e.g., salmonellosis), the risk of drug side effects (e.g., photosensitivity in the tropics), and the emergence of drug-resistant organisms³⁹¹ should preclude the widespread use of antibiotic prophylaxis at this time.³⁹² Because treatment regimens that combine loperamide with an antibiotic are rapidly effective in controlling traveler's diarrhea (<10 hours), most experts consider prophylactic therapy only for travelers with special issues such as individuals with a high risk of infection and individuals for whom it is important to remain disease-free during the trip.^{393,394}

Empirical self-treatment of traveler's diarrhea with a fluoroquinolone such as ciprofloxacin or levofloxacin for 1 to 3 days or a single dose of 1000 mg of azithromycin can significantly reduce the duration and severity of the disease (see Chapters 324 and 325).^{318-320,344,395-397} Therapy with trimethoprim-sulfamethoxazole had traditionally been suggested for children, but because of resistance, most experts currently recommend the use of a macrolide such as azithromycin.^{318-320,398} Fluoroquinolone, and to a lesser extent azithromycin, resistance has emerged among all bacterial etiologies.³⁹⁹ *Campylobacter* spp. resistant to quinolones is of special concern for travelers to many areas of the world, most notably Thailand and other parts of Southeast Asia.⁴⁰⁰⁻⁴⁰³ Nearly 90% of *Campylobacter* diarrhea cases in US military personnel deployed to Southeast Asia were resistant to ciprofloxacin.³⁴⁹ Resistance to quinolones has also increased among ETEC and EAEC strains, especially strains from India and sub-Saharan Africa. Extended-spectrum β -lactamase ETEC and EAEC strains expressing blaCTX-M-15 and blaCTX-M-27 genes are emerging.^{350,404} Fluoroquinolone drugs are contraindicated in pregnant women^{318-320,394,396} and in children younger than 16 years. Also, US Food and Drug Administration black box warnings exist for arthropathy, neuropathy, tendinopathy, and central nervous system side effects associated with fluoroquinolones.⁴⁰⁵ The greater recognition of adverse reactions associated with quinolones combined with increasing resistance have made azithromycin the preferred first choice for empirical self-treatment of traveler's diarrhea in many individuals. Resistance to azithromycin among ETEC and EAEC, however, is greater than 25% in parts of Southeast Asia, India, and Africa. When necessary, rifaximin, a nonabsorbable relative of rifampin, is useful for treatment of less severe cases of traveler's diarrhea in the absence of fever. Levofloxacin, azithromycin, and rifaximin in combination with loperamide have similar efficacies of 91% and 96% cure at 48 and 72 hours, respectively.⁴⁰⁶ Because of ongoing changing resistance patterns among bacterial etiologies of traveler's diarrhea, pretravel counseling, empirical acute treatment, and posttravel diagnostic and treatment recommendations should consider up-to-date reported susceptibility patterns that are specific to geographic destinations, such as can be found

at the Centers for Disease Control and Prevention (CDC) website (cdc.gov/travel). The benefits of treatment also need to be weighed against risk of acquisition of colonization with extended-spectrum β -lactamase-producing Enterobacteriales during travel and increased susceptibility to pathogens after the antibiotic course is over but before the microbiome has recovered. Nonabsorbable rifamycins may have an advantage over fluoroquinolones in having less toxicity, and limiting acquisition of multidrug-resistant bacteria.⁴⁰⁷ For all of these reasons, antibiotics are not recommended for mild cases of traveler's diarrhea, which most often are self-limiting.

Diarrhea in Institutions

Institutions provide unique settings that promote the acquisition of certain enteric pathogens. As with diarrhea in patients with AIDS and traveler's diarrhea, many cases of institution-acquired diarrhea are noninflammatory. Given the rise *C. difficile*-associated disease (CDAD), new-onset diarrhea in patients in the hospital or long-term care facilities should prompt stool immunoassay or PCR assaying for the presence of *C. difficile* toxins.⁴⁰⁸

Epidemic Nosocomial Diarrhea in Newborns

Epidemic nosocomial neonatal diarrhea has long been recognized as a potentially serious problem with mortality rates of 24% to 50%.^{409,410} Although the risk of this entity has substantially decreased, its occurrence is an important reminder of the unusual susceptibility of newborns to pathogens. This increased susceptibility may be explained by their underdeveloped (developing) intestinal microbiota and lack of specific immunity. Prematurity or congenital cardiac or pulmonary disease complicates this situation. The consequences of diarrhea in newborns are unusually severe, partially because they have poorly developed homeostatic mechanisms with limited water and electrolyte reserves. Crowded conditions in newborn nurseries promote nosocomial transmission.⁴¹¹ In addition, delays in recognizing a nursery outbreak may occur because infants may not develop diarrhea until after they have been discharged from the hospital.

The onset of illness is often insidious, with the development of listlessness, irritability, and poor feeding over a period of 3 to 6 days.^{410,412,413} Vomiting and fever are infrequent, and the stools tend to be watery; yellow-green in color; and usually without mucus, pus, or blood.^{414,415} Early signs such as failure to gain weight or a slight weight loss and abdominal distention may be subtle. The disease may progress to more severe signs of dehydration and shock with depressed sensorium, drowsiness, coma, sunken eyes, circumoral cyanosis, and grayish discoloration of the skin. Shock without hyperpnea often occurs in this setting despite the development of severe acidosis. Poorly nourished infants may develop severe hypokalemia, hyponatremic dehydration, or paradoxical edema. Appropriate antibiotic therapy must be tailored to the specific sensitivity pattern of the organism isolated.⁴¹⁴

The typical illness usually lasts 5 to 15 days, but persistence or relapse may occur over the course of several weeks after the initial onset of symptoms. Complications may include otitis media, pneumonia, bacteremia, peritonitis, and renal vein or cerebral sinus thrombosis. Several potentially life-threatening processes may mimic this infantile diarrhea syndrome. So-called parenteral diarrhea refers to the well-recognized but poorly understood tendency for systemic infections or localized infections elsewhere (e.g., otitis, meningitis) to be manifested clinically with diarrhea. Likewise, a strangulated hernia, intussusception, or torsion of an ovary or testis may be manifested by abdominal pain or diarrhea.

Epidemic diarrhea among hospitalized newborns has been commonly associated with certain enteropathogenic serotypes of *E. coli* (EPEC). In many areas such as South Africa and southern Brazil, EPEC organisms are also among the most common causes of sporadic diarrhea in infants and young children, especially during the summer months.⁴¹⁶⁻⁴¹⁸ Up to 20% of cases of endemic childhood diarrheal illness, even in temperate, more developed areas of the world such as England and Canada, are also known to be caused by EPEC organisms.^{416,417}Page 1289

The association of a certain strain of *E. coli* with infantile diarrhea was first demonstrated by slide agglutination by Bray and Beavan⁴¹⁹ in 1945 and reported in further detail in 1948. Excluding certain invasive serotypes (see Table 101.2 in Chapter 101), there are now at least 14 classically recognized EPEC *E. coli* serotypes, including O111-K58 (α), O55-K59 (β), O127-K63, O128-K67, O26-B6, O86-K61, O119-K69, O125-K70, O126-K71, O20-B7, and O44-K74. Additional serotypes recognized as causes of epidemic infantile diarrhea include O114,^{409,420,421}

O142,411,422 and O158.423 The mechanism by which most EPEC organisms cause disease involves a complex set of attachment and effacement traits, as detailed in Chapter 101. Although most are not invasive and do not produce conventionally recognized LT or ST enterotoxins, these organisms are capable of causing diarrheal disease in human volunteers, from whom the organism can be reisolated and in whom an antibody response can be documented.412,424

In addition to EPEC organisms, certain enterohemorrhagic (EHEC) and ETEC strains of *E. coli* have caused outbreaks of diarrhea in infants. *E. coli* O157-H7 and other EHEC strains have been associated with a syndrome of hemorrhagic diarrhea with only minimal inflammation and mild severity. Most EHEC organisms secrete a Shiga-like toxin, and certain strains of EHEC have also been associated with outbreaks and sporadic cases of hemorrhagic diarrhea that are more severe and can include a form of hemolytic-uremic syndrome.425-430

Diarrheagenic *E. coli* serotypes are not the only bacterial causes of epidemic infantile diarrhea. For example, an outbreak of diarrheal illness was described in which multiple serotypes of different organisms (*E. coli*, *Klebsiella*, and *Citrobacter*) were found to be transiently enterotoxigenic.431 This observation suggests the transmission of enterotoxigenic potential between susceptible strains, likely by certain plasmids432 or by bacteriophages.433 Another report of sporadic diarrhea in infants and children in Africa showed that several enteric organisms other than *E. coli* may produce an enterotoxin under certain conditions.38 This finding is relevant to other situations as well, as typified by an outbreak of watery diarrhea occurring on a cruise ship and found to be caused by enterotoxigenic strains of *E. coli* as well as *Klebsiella* spp. and *Citrobacter* spp.434 Many other well-known bacterial causes of diarrhea such as shigellosis435 and epidemic salmonellosis may also spread readily in the newborn nursery.436,437 Viral causes of epidemic infantile diarrhea include, but are not limited to, specific types of echoviruses,438 coxsackieviruses,439 adenoviruses,440 and rotaviruses.45,46,441-444

Hospitals

Nosocomial diarrhea is often due to noninfectious causes, but infectious etiologies must be considered. Nosocomial outbreaks are reported to the CDC.445,446 Nosocomial diarrhea appears to be a significant factor predisposing to other nosocomial infections such as urinary tract infections.447 Overall rates range from 2.3 to 4.1 illnesses per 100 admissions on pediatric wards408,446 and from 7.7 per 100 admissions to 41% of adults hospitalized in intensive care units.446,448

C. difficile remains the most common, most serious, and most costly infectious cause of nosocomial diarrhea in hospitalized patients449-451 and ranks among the most commonly reported hospital-acquired pathogens.452 In particular, CDAD is an important emerging nosocomial infection worldwide, especially among elderly hospitalized patients, patients with immunocompromise, and patients occupying beds in surgical wards or intensive care units.453-455 Most sporadic and outbreak cases of CDAD appear to be caused by exposure to contaminated environmental surfaces rather than direct contact with an index case.456-458

Other enteric pathogens are occasionally identified in outbreaks of nosocomial diarrhea. *Klebsiella oxytoca* can produce a toxin that inhibits DNA synthesis and, similar to *C. difficile*, causes diarrhea and sometimes hemorrhagic colitis associated with recent antibiotic use. Enterotoxin-producing methicillin-resistant *S. aureus* has been implicated as a rare cause of antibiotic-associated diarrhea.448 *Salmonella* is a common cause in reported outbreaks of nosocomial gastroenteritis and may rarely cause pseudomembranous colitis after antibiotic exposure. *Cryptosporidium* may be associated with cases of nosocomial diarrhea involving chronically ill, pediatric, and elderly patients, as well as HIV-infected patients and other immunocompromised patients.459 In young children and in immunocompromised hosts, viral agents (rotaviruses, adenoviruses, coxsackieviruses, and others) are often found.301 In addition, there has been a newfound appreciation for the roles of certain viruses (e.g., norovirus, rotaviruses, adenoviruses, coxsackieviruses, and others) as causes of nosocomial diarrhea, especially in neonates, children, and patients in intensive care units.448,460,461

Long-Term Care Facilities

Diarrheal illnesses constitute a significant problem in extended-care facilities for elderly patients. A conservative estimate based on passively reported illness rates is that one-third of patients in

long-term care facilities experience diarrhea each year.⁴⁶²⁻⁴⁶⁴ CDAD remains a common and important cause of diarrhea in these facilities.^{408,465,466} Sporadic cases and epidemic outbreaks of CDAD have been reported in many types of long-term care facilities. In other instances, viral causes of gastroenteritis or diarrhea have been identified in certain outbreaks occurring in these settings. The frequency of potentially transmissible enteric pathogens emphasizes the importance of careful hand washing in situations in which hygiene is often difficult.

Daycare Centers

Another special institutional setting in which hygiene is difficult and enteric infections are increasingly appreciated is daycare centers. Numerous outbreaks have been reported in association with viruses, bacteria, or parasites. The most common etiologic agents in infants and children younger than 2 years are rotaviruses, whereas older toddlers have been more likely to acquire *G. lamblia*.⁴⁶⁷ Diagnostic tests based on immunoassays and reverse-transcriptase PCR have been used to detect additional agents such as astrovirus in many diarrhea outbreaks in daycare centers.^{468,469} A clinical syndrome of prolonged noninflammatory diarrhea may be associated with *Cryptosporidium* in daycare centers.⁴⁷⁰⁻⁴⁷² Outbreaks of inflammatory diarrhea caused by *Shigella*, *Campylobacter*, and *C. difficile* have also been reported.^{341,342}

Treatment of Acute Noninflammatory Diarrhea

Treatment of diarrhea from any cause in adults and children consists primarily of rehydration. If glucose or sucrose accompanies oral isotonic fluid, the coupled absorption of sodium and water is often sufficient to replace fluid loss. Bismuth subsalicylate may reduce enterotoxin action,²⁸⁸ and if there is no significant febrile or inflammatory process, low doses of antimotility agents may offer some relief with minimal risk in adults. Some studies also suggest that novel analogues of glutamine may be beneficial in reducing the severity and extent of symptoms associated with certain forms of infectious diarrhea.⁴⁷³

The potential utility of probiotic compounds for the treatment of various forms of diarrhea has also garnered attention. The current data are not sufficient to issue a general recommendation on the use of probiotics for the management of infectious diarrhea, because of the lack of a sufficient number of randomized control trials for particular agents; probiotic compositions are quite heterogeneous, and there has been little consistent testing of individual agents in rigorous trials. Further, the development of long-term growth sequelae from early life exposures to these agents has not been well-studied (see Chapter 3).⁴⁷⁴⁻⁴⁷⁹

There is interest in the roles of various micronutrients in the management of diarrheal disease, especially in young children residing in underdeveloped areas.⁴⁸⁰⁻⁴⁸³ Several agents have been examined, but the strongest support has emerged relating to zinc supplementation. Several randomized, controlled international studies as well as a Cochrane Database Review found a beneficial effect of oral zinc supplementation in the prevention and management of a variety of infectious forms of diarrhea, especially in children.⁴⁸⁴⁻⁴⁸⁷

Chronic Noninflammatory Diarrhea

Syndromes of chronic noninflammatory diarrhea of infectious etiology include giardiasis, tropical sprue-like syndromes, syndromes of bacterial overgrowth, and *Cryptosporidium* or *C. belli* infection (especially in immunocompromised hosts).^{269,276,488,489} In a patient with weight loss, malaise, and watery or fatty stools, giardiasis or some other cause of a malabsorption syndrome should be suspected. This syndrome may also be associated with hypocalcemia; with iron or folate deficiency anemia; or with deficiency of vitamin D, vitamin K, or protein.

Giardiasis is endemic throughout most of the world, but still may often go undiagnosed for weeks of illness.^{189,191,490-493} Transmission is through water or food during outbreaks and by direct person-to-person contact. Clinical syndromes range in severity from asymptomatic infection to severe, persistent diarrhea associated with anorexia, weight loss, and malnutrition.¹⁸⁹ Current immunoassays and molecular tests have markedly improved diagnostic sensitivity.^{189,191} Recommended therapy classically included metronidazole, with a reported 70% cure rate.⁴⁹¹ More recently, tinidazole and nitazoxanide have shown improved efficacy for treatment and are associated with far fewer side effects than standard courses of metronidazole (see Chapters 27 and 45).⁴⁹⁴

Other infectious agents of chronic noninflammatory diarrhea include *Cryptosporidium*, *C. belli*, and *Cyclospora*. Each of these agents can be identified by standard stool analyses using ova and parasite testing combined with specific immunoassay or nucleic-acid detection assay as needed. Of note, *Cyclospora* was identified as the etiologic agent causing a large, multistate outbreak of gastroenteritis in 2013 in the United States.⁴⁹⁵ *Cyclospora* infection is treated with trimethoprim-sulfamethoxazole.^{285,490}

Bacterial Overgrowth Syndromes

Many syndromes have been described in which impaired absorption has been attributed to abnormal bacterial colonization in the upper small intestine.⁴⁸⁹ Whether these organisms are pathogens or part of the normal intestinal microbiota abnormally distributed is unclear at the present time.

Normally the upper portion of the small bowel is relatively sparsely populated, with less than 10⁵ organisms/mL; these are predominantly facultative gram-positive organisms (diphtheroids, streptococci, and lactobacilli).⁴⁹⁶ The organisms most often incriminated in bacterial overgrowth syndromes in the small bowel are aerobic enteric coliforms (members of the Enterobacteriales), anaerobic gram-negative fecal microbiota (*Bacteroides* and other genera), and miscellaneous other organisms such as *Plesiomonas shigelloides*.⁴⁹⁷

Bacterial colonization in the upper part of the small bowel may be associated with malabsorption or chronic diarrhea in the absence of significant histopathologic changes. Small bowel overgrowth is usually associated with a predisposing bowel abnormality, such as achlorhydria (from gastritis, pernicious anemia, or gastric surgery), blind-loop syndromes, cholangitis, impaired motility (scleroderma, diabetic neuropathy, vagotomy), surgery, strictures, diverticula, or radiation damage.^{498,499} Protein, folate, or vitamin B12 deficiency may also render the bowel more susceptible to microbial colonization and injury.^{496,500} An episode of acute infectious diarrhea may be the initiating event in the establishment of small bowel colonization and chronic diarrhea.^{496,501,502} Lindenbaum and colleagues⁵⁰³ described spruelike morphologic changes in the upper portion of the small bowel in association with increased numbers of bacteria and malabsorption among Peace Corps volunteers living in Pakistan.

Small intestinal bacterial overgrowth may cause malabsorption through bacterial binding (e.g., vitamin B12) or utilization (e.g., carbohydrates) of nutrients, deconjugation of bile salts by bacteria such as enterococci and anaerobes,⁵⁰⁴ or the toxic effects of bacterial products such as fatty acids or amines.⁴⁹⁶ *E. coli* strains that lack other recognized virulence traits but colonize the bowel were shown to cause prolonged diarrhea in a rabbit model,⁵⁰⁵ with an associated impairment in water and electrolyte absorption and disaccharidase activity.⁵⁰⁶

The approach to a patient with suspected bacterial overgrowth as a cause of malabsorption or chronic diarrhea should include quantitative aerobic and anaerobic cultures of the upper small bowel contents, obtained by intubation or string passage. Because the critical number of organisms appears to be approximately 10⁵/mL, semiquantitative estimates from a Gram stain (analogous to the urine Gram stain) may also prove valuable. In addition, the ¹⁴C-glycocholic acid breath test for bacterial deconjugation of bile salts has been shown to be helpful for diagnosis in some patients.⁵⁰⁷

Patients with symptomatic bacterial overgrowth are potential candidates for antibiotic therapy, especially if a predisposing condition such as achlorhydria, scleroderma, or diabetes is present. Depending on the results of quantitative cultures of upper small bowel aspirates, therapy may need to be directed against anaerobes, as well as aerobic coliform organisms.^{496,501}

Brainerd Diarrhea

Brainerd diarrhea is a clinical syndrome of chronic, self-limited, idiopathic diarrhea lasting for >4 weeks and up to several months.⁵⁰⁸ Symptoms often occur with an acute onset, but an infectious etiology is not found, and other noninfectious causes are also absent. Stool frequency ranges from 5 to 25 bowel movements per day and fecal electrolyte and osmolality testing are consistent with secretory diarrhea.⁵⁰⁸ Cases can be sporadic without evidence of household clustering; however, outbreaks including the first description of the entity in Brainerd, Minnesota,

in 1983 also occur.³⁶⁹ Travel and consumption of untreated water or unpasteurized milk have been associated with this entity, but infectious, chemical, or other toxins have not been identified as etiologies.^{367,509} Although symptoms often are prolonged, recovery is nearly universal. Relapse or progression to chronic sequelae suggests an alternative diagnosis such as lymphocytic colitis, collagenous colitis, malignancy, or drug reactions.

Other Noninfectious Mimics of Gastroenteritis

Acute noninflammatory diarrhea may be a consequence of several noninfectious processes. Exposures to antibiotics and other medications are common causes of noninfectious diarrhea, and although diarrhea is typically mild, it can result in additional costs related to diarrhea workups and medication nonadherence, and in some cases necessitate a change in therapy. As with agents that effect an osmotic diuresis, nonabsorbable agents such as sorbitol may cause diarrhea if consumed in excess. Heavy metal poisoning (i.e., arsenic, tin, iron, cadmium, mercury, or lead) is often associated with diarrhea, probably as a result of toxic effects on the mucosal epithelium.

Endocrine causes of diarrhea that may share the adenylate cyclase-activating mechanism with enterotoxins include non- β -islet cell tumors, medullary carcinoma of the thyroid, carcinoid tumors, and others that are associated with increased serum levels of prostaglandins or vasoactive intestinal polypeptide.⁵¹⁰ Patients with thyrotoxicosis or adrenal or parathyroid insufficiency may have diarrhea. Congenital and acquired enzyme deficiencies include lactase deficiency and pancreatic or biliary insufficiency, in which inadequately degraded or absorbed nutrients may promote an osmotic diarrhea. In a child who has diarrhea and edema, hypertension, or petechiae, hemolytic-uremic syndrome with or without enterohemorrhagic *E. coli* O157-H7 should be suspected. Patients with dermatitis herpetiformis may also have diarrhea that may respond to sulfone or sulfapyridine therapy or to a gluten-free diet.

Noninfectious causes should also be considered in the differential diagnosis of chronic noninflammatory diarrhea. Examples include congenital deficiency syndromes and food allergies, certain neoplastic and endocrine processes, and less well-understood functional disorders. Causative disorders to be considered in the first two categories are milk allergies, disaccharidase deficiencies, gluten enteropathy, acrodermatitis enteropathica, β -lipoprotein deficiency, familial hyperchloremic alkalosis (congenital chloridorrhea), Leiner disease, and Wiskott-Aldrich syndrome. Neoplastic and endocrine causes of diarrhea include carcinoid tumors, Werner syndrome (multiple endocrine adenomatosis), Zollinger-Ellison syndrome (gastrinoma), pancreatic cholera syndromes, medullary carcinoma of the thyroid, and thyrotoxicosis. Patients with partial mechanical bowel obstruction or pellagra may also have chronic diarrhea. Milder forms of inflammatory bowel disease and irritable bowel disease, including postinfectious subtypes, can also be associated with a variety of types of chronic diarrhea. Although a thorough search for an infectious cause of any form of chronic diarrhea is usually warranted, most often the specific diagnosis of one of these etiologies requires referral to a gastroenterologist.

Acknowledgment

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Definition

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Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.
- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.
- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis*, or *Treponema pallidum* as a cause of proctitis, or *Campylobacter* spp., *Shigella* spp., *C. trachomatis* (lymphogranuloma venereum serotypes), or *C. difficile* as a cause of colitis.
- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest *C. difficile* infection (CDI).

Microbiology

- Genomic studies of *Shigella* spp. have indicated that *Shigella* spp. and enteroinvasive *E. coli* are derived from multiple phyletic origins of *E. coli* despite their microbiological and clinical similarities.
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- *Campylobacter* spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.
- The primary virulence factors that are known to cause disease in CDI are the two large toxins: *C. difficile* toxin (TcdA), or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.
- *Vibrio parahaemolyticus* has been recognized since 1950 in Japan and is a cause of seafood poisoning.
- *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β –converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.
- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.
- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

- Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although *Shigella* and perhaps *Campylobacter* durations of infection and symptoms

may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic *Salmonella* infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

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- 101
- Acute Dysentery Syndromes (Diarrhea With Fever)
- Aldo A.M. Lima
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-
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- **Epidemiology**
 - Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.
 - The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus,

Campylobacter spp., astrovirus, and Cryptosporidium spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were Campylobacter spp., norovirus GII, rotavirus, astrovirus, and Shigella spp.⁶ Campylobacter spp. and EIEC were associated with dysentery. Rotavirus and Shigella spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

-
- A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, Chlamydia trachomatis (including lymphogranuloma venereum serotypes), or Treponema pallidum as a cause of proctitis, or Campylobacter, Shigella, or E. histolytica as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care facility would strongly suggest C. difficile infection (CDI), although community-acquired infection has been well recognized as well.
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- Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}
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- Potential causes of acute dysentery are listed in Table 101.1.
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- Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes
- Shigella spp. and Enteroinvasive Escherichia coli
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- Bacillary dysentery is caused by Shigella spp. and EIEC. It is estimated that Shigella spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four Shigella species are classified in subgroups A to D—Shigella dysenteriae, Shigella flexneri, Shigella boydii, and Shigella sonnei, respectively. Genomic studies of Shigella spp. have indicated that Shigella and EIEC are derived from multiple phyletic origins within E. coli.¹¹ S. dysenteriae (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ S. dysenteriae is responsible for severe and epidemic disease, and S. boydii is mostly seen in the Indian subcontinent. S. flexneri and S. sonnei are the most common species around the world.¹² Shigellae are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. Shigella spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² Shigella spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with Shigella, and other enteric pathogens such as EAEC, Campylobacter, and Giardia, were associated with negative linear growth, which was sustained up to 5 years.¹⁴
-
- Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All Shigella species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by S. dysenteriae type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ Shigella infection, especially with S. dysenteriae type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹
-
- Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with S. dysenteriae type 1,^{24,25} but may be seen with any Shigella infection. A serious arthritis similar to

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes
Bacillary dysentery (<i>Shigella dysenteriae</i> , <i>Shigella flexneri</i> , <i>Shigella sonnei</i> , <i>Shigella boydii</i> ; invasive <i>Escherichia coli</i>)
Campylobacteriosis (<i>Campylobacter jejuni</i>)
Amebic dysentery (<i>Entamoeba histolytica</i>)
Ciliary dysentery (<i>Balantidium coli</i>)
Vibriosis (<i>Vibrio parahaemolyticus</i>)
Salmonellosis (<i>Salmonella enterica</i> serovar Typhimurium)
Typhoid fever (<i>S. enterica</i> serovar Typhi)
Enteric fever (<i>S. enterica</i> serovar Choleraesuis, <i>S. enterica</i> serovar Paratyphi)
Yersiniosis (<i>Yersinia enterocolitica</i>)
Proctitis
Gonococcal (<i>Neisseria gonorrhoeae</i>)
Herpetic (herpes simplex virus)
Chlamydial (<i>Chlamydia trachomatis</i>)
Syphilitic (<i>Treponema pallidum</i>)
Other Syndromes
Necrotizing enterocolitis of the newborn
Enteritis necroticans
Pseudomembranous colitis or <i>Clostridioides difficile</i> (formerly <i>Clostridium difficile</i>) colitis without overt pseudomembranes (<i>C. difficile</i>)
Diverticulitis
Typhlitis
Chronic Inflammatory Processes
Enteropathogenic and enteroaggregative <i>E. coli</i>
Syphilis
Gastrointestinal tuberculosis
Gastrointestinal mycosis (including <i>Basidiobolus ranarum</i>)
Parasitic enteritis
Syndromes Without Known Infectious Cause
Idiopathic ulcerative colitis
Crohn disease
Radiation enteritis
Ischemic colitis
Allergic enteritis
Brainerd diarrhea

that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation Page 1294of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and Page 1292

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Acute Dysentery Syndromes (Diarrhea With Fever)

Aldo A.M. Lima

Jae Hyun Shin

Circle A. Warren

Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

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- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.
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Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis* (including lymphogranuloma venereum serotypes), or *Treponema pallidum* as a cause of proctitis, or *Campylobacter*, *Shigella*, or *E. histolytica* as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care facility would strongly suggest *C. difficile* infection (CDI), although community-acquired infection has been well recognized as well.

Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}

Potential causes of acute dysentery are listed in Table 101.1.

Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes
Shigella spp. and Enteroinvasive *Escherichia coli*

Bacillary dysentery is caused by *Shigella* spp. and EIEC. It is estimated that *Shigella* spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four *Shigella* species are classified in subgroups A to D—*Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*, respectively. Genomic studies of *Shigella* spp. have indicated that *Shigella* and EIEC are derived from multiple phyletic origins within *E. coli*.¹¹ *S. dysenteriae* (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ *S. dysenteriae* is responsible for severe and epidemic disease, and *S. boydii* is mostly seen in the Indian subcontinent. *S. flexneri* and *S. sonnei* are the most common species around the world.¹² *Shigellae* are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. *Shigella* spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² *Shigella* spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with *Shigella*, and other enteric pathogens such as EAEC, *Campylobacter*, and *Giardia*, were associated with negative linear growth, which was sustained up to 5 years.¹⁴

Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

TABLE 101.1

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes

Bacillary dysentery (*Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Shigella boydii*; invasive *Escherichia coli*)

Campylobacteriosis (*Campylobacter jejuni*)

Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis
 Gonococcal (*Neisseria gonorrhoeae*)
 Herpetic (herpes simplex virus)
 Chlamydial (*Chlamydia trachomatis*)
 Syphilitic (*Treponema pallidum*)
 Other Syndromes
 Necrotizing enterocolitis of the newborn
 Enteritis necroticans
 Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)
 Diverticulitis
 Typhlitis
 Chronic Inflammatory Processes
 Enteropathogenic and enteroaggregative *E. coli*
 Syphilis
 Gastrointestinal tuberculosis
 Gastrointestinal mycosis (including *Basidiobolus ranarum*)
 Parasitic enteritis
 Syndromes Without Known Infectious Cause
 Idiopathic ulcerative colitis
 Crohn disease
 Radiation enteritis
 Ischemic colitis
 Allergic enteritis
 Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin–producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:H11, O103:H2, O91:H2, O145:H–, O111:H58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:H75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial

Enteroinvasive *Escherichia coli* Serotypes

SEROTYPE	DIFCO SEROGROUP TEST *
O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

* Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including aggR, aggA (encoding for the biofilm-producing aggregative adherence fimbriae I [AAF/I]), sigA, sepA, pic, aatA, aaiC, and aap, but it was negative for the eae (*E. coli* attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

Enteropathogenic and Enteraggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.⁵⁸⁻⁶⁰ EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.⁶¹⁻⁶³ EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other *Campylobacter* spp. have been published.⁶⁹⁻⁷¹ Although there currently are more than 16 *Campylobacter* spp. and 6 subspecies, *C. jejuni* and *Campylobacter coli* are the most prevalent species associated with enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of *Campylobacter* bloodstream infections in humans are with *Campylobacter fetus* (former subspecies, *C. fetus* subsp. *intestinalis*),⁷² *C. jejuni* commonly

causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ *Campylobacter* enteritis results from oral transmission, and a variable but relatively small dose (5–8 × 10² cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of *Campylobacter* enteritis have been associated with ingestion of contaminated water, raw milk, and uncooked meat and poultry. A review of common source outbreaks of *Campylobacter* infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by *Campylobacter* spp., and the specific balance between these genes and host susceptibility to *Campylobacter* enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; Page 1292

101

Acute Dysentery Syndromes (Diarrhea With Fever)

Aldo A.M. Lima

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Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.

- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.

- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis*, or *Treponema pallidum* as a cause of proctitis, or

Campylobacter spp., Shigella spp., C. trachomatis (lymphogranuloma venereum serotypes), or C. difficile as a cause of colitis.

- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest C. difficile infection (CDI).

Microbiology

- Genomic studies of Shigella spp. have indicated that Shigella spp. and enteroinvasive E. coli are derived from multiple phyletic origins of E. coli despite their microbiological and clinical similarities.

- The cause of an outbreak of bloody diarrhea and severe hemolytic-uremic syndrome, unlike prior EHEC strains that had exhibited enteropathogenic E. coli traits of attachment and effacement, is a Shiga toxin-producing enteroaggregative E. coli strain.

- Campylobacter spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.

- The primary virulence factors that are known to cause disease in CDI are the two large toxins: C. difficile toxin (TcdA), or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.

- Vibrio parahaemolyticus has been recognized since 1950 in Japan and is a cause of seafood poisoning.

- Salmonella flagellin is regulated by the fliC gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.

- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.

- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

- Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although Shigella and perhaps Campylobacter durations of infection and symptoms may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic Salmonella infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

Definition

Diarrhea associated with fever typically suggests an acute inflammatory enteritis caused by one of several specific infectious agents. However, noninflammatory enteritides caused by viral or other agents may also present with diarrhea and fever. This chapter will address bacterial and amebic colitis as potential major causes of this syndrome. Viral and other parasitic etiologies will be addressed in other chapters. The acute inflammatory enteritides include several specific, distal small bowel, and colonic infections, such as those caused by Campylobacter spp., Salmonella spp., Shigella spp., and enteroinvasive Escherichia coli (EIEC), as well as the syndromes of necrotizing enteritis (NE) and antibiotic-associated pseudomembranous enterocolitis (Clostridioides difficile [formerly Clostridium difficile]). Several other infectious agents cause chronic enteric inflammatory processes that may result in syndromes of abdominal pain, weight loss, diarrhea, or malabsorption. These involve disease processes such as gastrointestinal (GI) mycoses; mycobacterioses; amoebiasis, or bacterial infections, such as those caused by enteropathogenic E. coli (EPEC), enterohemorrhagic (EHEC), and enteroaggregative E. coli (EAEC); and syphilis.

Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

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Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

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Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis

Gonococcal (*Neisseria gonorrhoeae*)

Herpetic (herpes simplex virus)

Chlamydial (*Chlamydia trachomatis*)

Syphilitic (*Treponema pallidum*)

Other Syndromes

Necrotizing enterocolitis of the newborn

Enteritis necroticans

Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)

Diverticulitis

Typhlitis

Chronic Inflammatory Processes

Enteropathogenic and enteroaggregative *E. coli*

Syphilis

Gastrointestinal tuberculosis

Gastrointestinal mycosis (including *Basidiobolus ranarum*)

Parasitic enteritis

Syndromes Without Known Infectious Cause

Idiopathic ulcerative colitis

Crohn disease

Radiation enteritis

Ischemic colitis

Allergic enteritis

Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin–producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:K60:H11, O103:H2, O91:H2, O145:H–, O111:K58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:K75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including *aggR*, *aggA* (encoding for the biofilm–producing aggregative adherence fimbriae I [AAF/I]), *sigA*,

sepA, pic, aatA, aaiC, and aap, but it was negative for the eae (*E. coli* attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

TABLE 101.2

Enteroinvasive *Escherichia coli* Serotypes
 SEROTYPE DIFCO SEROGROUP TEST^a

O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

^a Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

Enteropathogenic and Enteroaggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.⁵⁸⁻⁶⁰ EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.⁶¹⁻⁶³ EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other Campylobacter spp. have been published.⁶⁹⁻⁷¹ Although there currently are more than 16 Campylobacter spp. and 6 subspecies, *C. jejuni* and Campylobacter coli are the most prevalent species associated with enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of Campylobacter bloodstream infections in humans are with Campylobacter fetus (former subspecies, *C. fetus* subsp. *intestinalis*),⁷² *C. jejuni* commonly causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ Campylobacter enteritis results from oral transmission, and a variable but relatively small dose (5–8 × 10² cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of Campylobacter enteritis have been associated with ingestion of contaminated water, raw milk, and uncooked meat and poultry. A review of common source outbreaks of Campylobacter

infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by *Campylobacter* spp., and the specific balance between these genes and host susceptibility to *Campylobacter* enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; secretion of cytokines; chemokines and arachidonic acid metabolites; and production of substance P and reactive oxygen intermediates. Some *C. difficile* strains, including the epidemic ribotype 027, produce binary toxin that may have a role in increased virulence through host attachment and colonization, which type 2 immunity in the host may protect against.⁸³ Clinical presentation ranges from asymptomatic colonization to severe disease with pseudomembranous colitis (Fig. 101.1) and septic shock. Increased fecal lactoferrin, IL-1 β , and IL-8 were observed in patients with CDI.⁸⁴ Leukocytosis suggests severe infection, and leukemoid reactions have been described in colitis caused by *C. difficile*.⁸⁵ Further details of CDI are provided in Chapter 249.

Vibriosis

In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90} Vibriosis In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

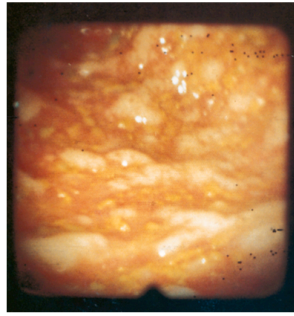


FIG. 101.1 Proctoscopic view of pseudomembranous colitis in a patient who received clindamycin. Note the 4- to 8-mm raised white plaques overlying an erythematous mucosa. From Tedesco FJ, Barton RW, Alpers DH. Clindamycin-associated colitis. *Ann Intern Med.* 1974;81:429–433.

[low for extended description](#)

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90}

Salmonellosis and Enteric Fever

Salmonella enterocolitis is characterized by fever, cramping, abdominal pain, and diarrhea that begin 8 to 48 hours after ingestion of an infective dose, usually with food, and usually lasts 3 to 5 days (see Chapter 228). The diarrheal stools of patients with salmonellosis often contain a moderate number of polymorphonuclear leukocytes, usually fewer than is typical of shigellosis.

Although *Salmonella enteritis* predominantly involves the lamina propria in the small bowel, several reports have noted *Salmonella enterica* serovar Typhimurium as a cause of colitis, with crypt abscesses and erosion and ulcerations of the colonic mucosa, which result in blood and pus in the stool.⁹¹ Certain other strains of *Salmonella*—including *S. enterica* serovar Choleraesuis and *S. enterica* serovar Paratyphi, and especially *S. enterica* serovar Typhi—elicit a mononuclear response and cause the bacteremia that is characteristic of enteric fever. Animal models for *S. Typhimurium* continue to advance our understanding of the pathogenesis of intestinal and systemic *Salmonella* infections.⁹² *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5 (TLR5), nucleotide oligomerization domain–like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein. TLR5 is located in the plasma and phagosomal membranes, and the other two are located in the cytoplasm of the host's Peyer patch cells. Flagellin activates TLR5 and leads to the production of IL-12, whereas the activation of IPAF leads to the production of IL-1 β and IL-18. Both pathways act synergistically to activate interferon- γ -producing Th1 cells. These proinflammatory cytokines play important roles in the induction of inflammation with *Salmonella* infections.

Typhoid fever may lead to an erosion of the blood vessels in Peyer patches that, without appropriate treatment, could result in gross blood in the feces in 10% to 20% of patients. Severe intestinal hemorrhage may complicate approximately 2% of cases late in the course of untreated typhoid fever. Such intestinal bleeding may precede perforation, another complication of typhoid fever.⁹³ However, typhoid fever is primarily a systemic illness, not a diarrheal disease (see Chapter 102).

Yersiniosis

Y. enterocolitica (see Chapter 235) is an enteric pathogen that may be responsible for an enteric fever–like illness, mesenteric adenitis (which may mimic acute appendicitis), or an inflammatory ileitis or ulcerative colitis syndrome with fecal neutrophils and mononuclear cells.⁹⁴ *Y. enterocolitica* and *Yersinia pseudotuberculosis* may cause similar symptoms and signs. The virulence of *Yersinia* is associated with the direct injection of proteins, such as *Yersinia* protein kinase A (YpkA) and *Yersinia* outer protein E (YopE), into eukaryotic cells.^{95,96} *Yersinia* infection may also be associated with migratory polyarthritis, reactive arthritis, or erythema nodosum. A syndrome of acute diarrhea and vomiting is especially common in young children. The organism may cause disseminated abscesses in the liver and spleen⁹⁷ or an inflammatory colitis.⁹⁸ The causative agent, a gram-negative member of the family Enterobacteriaceae, is in the same genus as the plague bacillus *Yersinia pestis* and is sometimes mistaken for *Proteus* on initial culture plates.

Amebiasis

Entamoeba histolyticum is one of the classic causes of dysenteric syndromes. Although the course and severity of the infection are variable, amebiasis is one infection that can cause sustained febrile diarrhea with stools becoming grossly bloody. For patients living in endemic areas or travelers who have been to such endemic areas and who have dysenteric symptoms, amebiasis should be considered in the differential diagnosis. For further details about amebiasis, please see Chapter 278.

Diagnosis

Diarrhea with fever, blood in the stool, severe abdominal pain, or signs of sepsis necessitates evaluation for enteric pathogens to guide disease management.⁹⁹ Clinical and epidemiologic information direct the specific tests to be requested. Examination for fecal leukocytes often reveals sheets of polymorphonuclear leukocytes in clumps of mucus, even in the absence of gross blood in the stool specimen (Fig. 101.2).^{100,101} Fewer pyknotic leukocytes are reported in amebic dysentery¹⁰²⁻¹⁰⁴; this may be attributable to the deeper undermining ulcers characteristic of intestinal amebiasis or to a contact-dependent cytolytic effect of the amoeba on leukocytes. The diagnosis of shigellosis is made using bacterial cultures or polymerase chain reaction (PCR) assay for the invasiveness plasmid in stool, rectal swab, or endoscopic biopsy specimens.¹⁰⁵ The use of fresh specimens promptly plated onto appropriate enteric culture media is very important to ensure the isolation of shigellae.¹⁰⁶

Specialized techniques are required to isolate *Vibrio* (thiosulfate citrate bile salts [TCBS] agar),¹⁰⁷ *Yersinia* (cold enrichment),¹⁰⁸ or *C. jejuni*.⁷³ The identification of toxigenic *C. difficile* is done by immunoassay for either *C. difficile* toxin B or PCR assay for *C. difficile* toxin B.¹⁰⁹

The diagnosis of infection with EHEC depends on culturing stool on MacConkey culture medium, on which the original O157:H7 EHEC strain yields sorbitol-negative colonies.¹¹⁰ The isolated *E. coli* is serotyped for somatic (O) and flagellar (H) antigens. Rapid methods for detection of Shiga toxins in the stool samples are now in use.^{111,112} The pathogenic Kanagawa-positive strains from patients produce β -hemolysis on special (Wagatsuma) medium—in contrast to environmental isolates—and are best isolated as blue-green colonies (alkaline) on TCBS agar.¹⁰⁷ A PCR method provides a species-specific probe for *V. parahaemolyticus*, *V. alginolyticus*, *Vibrio anguillarum*, and a cluster of related species.¹¹³ Cultivation may require cold enrichment techniques.¹⁰⁸ PCR methods for detection of pathogenic *Y. enterocolitica* are more sensitive than culture methods.¹¹⁴

Molecular approaches, including multiple pathogen nucleic acid amplification tests (NAAT) for simultaneous detection of multiple enteropathogens, have high sensitivity and specificity and are widely used for surveillance or clinical purposes.¹¹⁵ In outpatient settings, these have largely replaced culture methods in the United States, Canada, and western Europe. Not only do the NAAT approaches allow for higher sensitivity, but the turnaround time is also faster, facilitating hospital infection control measures and initiation of appropriate intervention. The broad, multiplex approach allows for a screening diagnostic approach for enteric pathogens, including viruses, bacteria, and parasites, without having to test for individual pathogens separately.¹¹⁶ There are several limitations inherent to this advanced molecular assay. In addition to relatively high cost, interpretation of the results of the assay is challenging. Detection of a pathogen does not necessarily mean identification of the cause of dysentery, as the test is not able to differentiate colonization from clinically relevant infection. Another reason is the high frequency of polymicrobial infections detected by this method, with up to 30% of positive samples detecting multiple pathogens, which can make determination of the true pathogen challenging.^{38,117,118} Additionally, specific pathogens such as EPEC or EAEC tend to be detected more commonly as observed in one study where both organisms were positive in 84% of the multipathogen samples.¹¹⁷ Another study showed a false positive rate around 20% with NAAT for organisms like *C. difficile*, EPEC, EAEC, and *Campylobacter*, which was largely caused by cross-reactivity with commensal organisms.³⁸ Quantitation of the burden of infection may assist in the interpretation of results from these molecular assays.¹¹⁹ Moreover, current commercially available GI enteric panels do not allow for detection of antibiotic resistance. The steady increase in the prevalence of antimicrobial resistant diarrhea-causing pathogens indicates the continuing need for traditional microbiologic methods for susceptibility testing to guide treatment.

Amebic dysentery is traditionally diagnosed by direct examination of wet mounts of fresh fecal or proctoscopic specimens, which reveal *E. histolytica* trophozoites or cysts characterized by four or fewer delicate nuclei with central karyosomes. Additional patients may be diagnosed by biopsy, which may reveal periodic acid–Schiff stain (PAS)-positive trophozoites or cysts in the undermining ulcer in the lamina propria, or by a fecal or serum *E. histolytica*-specific antigen test, which distinguishes virulent *E. histolytica* from avirulent *Entamoeba dispar* infections.¹²⁰ The use of multiplex NAAT panels including *E. histolytica* is largely replacing the historical diagnostic approaches to amoebiasis.¹²¹

Sigmoidoscopic examination, especially with biopsy, may be useful in the diagnosis of a pseudomembranous enterocolitis or in the identification of parasites such as *E. histolytica* (with a special PAS stain) or *B. coli*. Amebic colitis is associated with discrete small ulcerations with undermined edges amid relatively normal mucosa. Acute shigellosis causes more widespread, shallow, 3- to 7-mm ulcers with a more intense inflammatory exudate.

Other considerations in the differential diagnosis of inflammatory colitis include the potentially rapidly progressive necrotizing enterocolitis (NEC) syndromes (see Chapters 81 and 249) and noninfectious syndromes such as inflammatory bowel diseases, ulcerative colitis, and Crohn disease. These diagnoses are suggested by the setting, clinical course, history, and findings on radiologic and endoscopic examinations.

Finally, the presence of biomarkers of acute or prolonged intestinal inflammation, such as fecal myeloperoxidase and neopterin, may be associated with growth shortfalls in young children in developing areas.¹²²

Therapy

Therapy consists of supportive fluid and electrolyte management with antimicrobial therapy directed at a specific pathogen if suspected on Page 1297the basis of the epidemiologic setting or microbiologic test results. Fluid resuscitation is the primary treatment of all acute diarrheal disease syndromes. Oral rehydration salt solution (World Health Organization formula) is recommended for oral rehydration therapy.¹²³ Presumptive therapy for the inflammatory colitides varies greatly with the different organisms and is influenced by the increasing resistance of many enteric pathogens to antimicrobial therapy.¹²⁴ For example, an acute febrile dysenteric illness in a young child with daycare exposure or in an area in which shigellosis is common should be treated with an

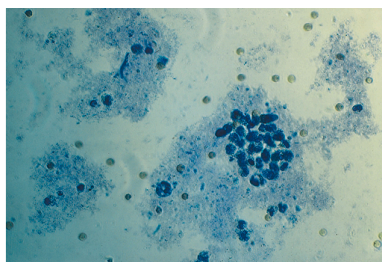


FIG. 101.2 Methylene blue stain of fecal leukocytes from a patient with colitis. This exudative response may be seen in any active colitis syndrome, including those caused by the inflammatory or invasive pathogens discussed in this chapter.

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Acute Dysentery Syndromes (Diarrhea With Fever)

Aldo A.M. Lima

Jae Hyun Shin

Circle A. Warren

Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.
- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.
- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis*, or *Treponema pallidum* as a cause of proctitis, or *Campylobacter* spp., *Shigella* spp., *C. trachomatis* (lymphogranuloma venereum serotypes), or *C. difficile* as a cause of colitis.
- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest *C. difficile* infection (CDI).

Microbiology

- Genomic studies of *Shigella* spp. have indicated that *Shigella* spp. and enteroinvasive *E. coli* are derived from multiple phyletic origins of *E. coli* despite their microbiological and clinical similarities.
- The cause of an outbreak of bloody diarrhea and severe hemolytic-uremic syndrome, unlike prior EHEC strains that had exhibited enteropathogenic *E. coli* traits of attachment and effacement, is a Shiga toxin-producing enteroaggregative *E. coli* strain.
- *Campylobacter* spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.
- The primary virulence factors that are known to cause disease in CDI are the two large toxins: *C. difficile* toxin (Tcd)A, or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.
- *Vibrio parahaemolyticus* has been recognized since 1950 in Japan and is a cause of seafood poisoning.
- *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.
- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.
- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

- Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although *Shigella* and perhaps *Campylobacter* durations of infection and symptoms may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic *Salmonella* infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

Definition

Diarrhea associated with fever typically suggests an acute inflammatory enteritis caused by one of several specific infectious agents. However, noninflammatory enteritides caused by viral or other

agents may also present with diarrhea and fever. This chapter will address bacterial and amebic colitis as potential major causes of this syndrome. Viral and other parasitic etiologies will be addressed in other chapters. The acute inflammatory enteritides include several specific, distal small bowel, and colonic infections, such as those caused by *Campylobacter* spp., *Salmonella* spp., *Shigella* spp., and enteroinvasive *Escherichia coli* (EIEC), as well as the syndromes of necrotizing enteritis (NE) and antibiotic-associated pseudomembranous enterocolitis (*Clostridioides difficile* [formerly *Clostridium difficile*]). Several other infectious agents cause chronic enteric inflammatory processes that may result in syndromes of abdominal pain, weight loss, diarrhea, or malabsorption. These involve disease processes such as gastrointestinal (GI) mycoses; mycobacterioses; amoebiasis, or bacterial infections, such as those caused by enteropathogenic *E. coli* (EPEC), enterohemorrhagic (EHEC), and enteroaggregative *E. coli* (EAEC); and syphilis.

Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis* (including lymphogranuloma venereum serotypes), or *Treponema pallidum* as a cause of proctitis, or *Campylobacter*, *Shigella*, or *E. histolytica* as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care facility would strongly suggest *C. difficile* infection (CDI), although community-acquired infection has been well recognized as well.

Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}

Potential causes of acute dysentery are listed in Table 101.1.

Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes
Shigella spp. and Enteroinvasive *Escherichia coli*

Bacillary dysentery is caused by *Shigella* spp. and EIEC. It is estimated that *Shigella* spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four *Shigella* species are classified in subgroups A to D—*Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*, respectively. Genomic studies of *Shigella* spp. have indicated that *Shigella* and EIEC are derived from multiple phyletic origins within *E. coli*.¹¹ *S. dysenteriae* (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ *S. dysenteriae* is responsible for severe and epidemic disease, and *S. boydii* is mostly seen in the Indian subcontinent. *S. flexneri* and *S. sonnei* are the most common species around the world.¹² *Shigellae* are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. *Shigella* spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² *Shigella* spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with *Shigella*, and other enteric pathogens such as EAEC, *Campylobacter*, and *Giardia*, were associated with negative linear growth, which was sustained up to 5 years.¹⁴

Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

TABLE 101.1

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes

Bacillary dysentery (*Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Shigella boydii*; invasive *Escherichia coli*)

Campylobacteriosis (*Campylobacter jejuni*)

Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis
 Gonococcal (*Neisseria gonorrhoeae*)
 Herpetic (herpes simplex virus)
 Chlamydial (*Chlamydia trachomatis*)
 Syphilitic (*Treponema pallidum*)
 Other Syndromes
 Necrotizing enterocolitis of the newborn
 Enteritis necroticans
 Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)
 Diverticulitis
 Typhlitis
 Chronic Inflammatory Processes
 Enteropathogenic and enteroaggregative *E. coli*
 Syphilis
 Gastrointestinal tuberculosis
 Gastrointestinal mycosis (including *Basidiobolus ranarum*)
 Parasitic enteritis
 Syndromes Without Known Infectious Cause
 Idiopathic ulcerative colitis
 Crohn disease
 Radiation enteritis
 Ischemic colitis
 Allergic enteritis
 Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin–producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:H11, O103:H2, O91:H2, O145:H–, O111:H58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:H75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including aggR, aggA (encoding for the biofilm-producing aggregative adherence fimbriae I [AAF/I]), sigA, sepA, pic, aatA, aaiC, and aap, but it was negative for the eae (*E. coli* attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

TABLE 101.2

Enteroinvasive *Escherichia coli* Serotypes
 SEROTYPE DIFCO SEROGROUP TEST^a

O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

^a Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

Enteropathogenic and Enteraggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.⁵⁸⁻⁶⁰ EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.⁶¹⁻⁶³ EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other *Campylobacter* spp. have been published.⁶⁹⁻⁷¹ Although there currently are more than 16 *Campylobacter* spp. and 6 subspecies, *C. jejuni* and *Campylobacter coli* are the most prevalent species associated with

enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of *Campylobacter* bloodstream infections in humans are with *Campylobacter fetus* (former subspecies, *C. fetus* subsp. *intestinalis*),⁷² *C. jejuni* commonly causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ *Campylobacter* enteritis results from oral transmission, and a variable but relatively small dose ($5\text{--}8 \times 10^2$ cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of *Campylobacter* enteritis have been associated with ingestion of contaminated water, raw milk, and uncooked meat and poultry. A review of common source outbreaks of *Campylobacter* infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by *Campylobacter* spp., and the specific balance between these genes and host susceptibility to *Campylobacter* enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; secretion of cytokines; chemokines and arachidonic acid metabolites; and production of substance P and reactive oxygen intermediates. Some *C. difficile* strains, including the epidemic ribotype 027, produce binary toxin that may have a role in increased virulence through host attachment and colonization, which type 2 immunity in the host may protect against.⁸³ Clinical presentation ranges from asymptomatic colonization to severe disease with pseudomembranous colitis (Fig. 101.1) and septic shock. Increased fecal lactoferrin, IL-1 β , and IL-8 were observed in patients with CDI.⁸⁴ Leukocytosis suggests severe infection, and leukemoid reactions have been described in colitis caused by *C. difficile*.⁸⁵ Further details of CDI are provided in Chapter 249.

Vibriosis

In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90}

FIG. 101.1 Proctoscopic view of pseudomembranous colitis in a patient who received clindamycin. Note the 4- to 8-mm raised white plaques overlying an erythematous mucosa. From Tedesco FJ, Barton RW, Alpers DH. Clindamycin-associated colitis. *Ann Intern Med.* 1974;81:429–433.

Follow for extended description

Salmonellosis and Enteric Fever

Salmonella enterocolitis is characterized by fever, cramping, abdominal pain, and diarrhea that begin 8 to 48 hours after ingestion of an infective dose, usually with food, and usually lasts 3 to 5 days (see Chapter 228). The diarrheal stools of patients with salmonellosis often contain a moderate number of polymorphonuclear leukocytes, usually fewer than is typical of shigellosis.

Although *Salmonella enteritis* predominantly involves the lamina propria in the small bowel, several reports have noted *Salmonella enterica* serovar Typhimurium as a cause of colitis, with crypt abscesses and erosion and ulcerations of the colonic mucosa, which result in blood and pus in the stool.⁹¹ Certain other strains of *Salmonella*—including *S. enterica* serovar Choleraesuis and *S. enterica* serovar Paratyphi, and especially *S. enterica* serovar Typhi—elicit a mononuclear response and cause the bacteremia that is characteristic of enteric fever. Animal models for *S. Typhimurium* continue to advance our understanding of the pathogenesis of intestinal and systemic *Salmonella* infections.⁹² *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5 (TLR5), nucleotide oligomerization domain–like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein. TLR5 is located in the plasma and phagosomal membranes, and the other two are located in the cytoplasm of the host's Peyer patch cells. Flagellin activates TLR5 and leads to the production of IL-12, whereas the activation of IPAF leads to the production of IL-1 β and IL-18. Both pathways act synergistically to activate interferon- γ -producing Th1 cells. These proinflammatory cytokines play important roles in the induction of inflammation with *Salmonella* infections.

Typhoid fever may lead to an erosion of the blood vessels in Peyer patches that, without appropriate treatment, could result in gross blood in the feces in 10% to 20% of patients. Severe intestinal hemorrhage may complicate approximately 2% of cases late in the course of untreated typhoid fever. Such intestinal bleeding may precede perforation, another complication of typhoid fever.⁹³ However, typhoid fever is primarily a systemic illness, not a diarrheal disease (see Chapter 102).

Yersiniosis

Y. enterocolitica (see Chapter 235) is an enteric pathogen that may be responsible for an enteric fever–like illness, mesenteric adenitis (which may mimic acute appendicitis), or an inflammatory ileitis or ulcerative colitis syndrome with fecal neutrophils and mononuclear cells.⁹⁴ *Y. enterocolitica* and *Yersinia pseudotuberculosis* may cause similar symptoms and signs. The virulence of *Yersinia* is associated with the direct injection of proteins, such as *Yersinia* protein kinase A Page 1296 (YpkA) and *Yersinia* outer protein E (YopE), into eukaryotic cells.^{95,96} *Yersinia* infection may also be associated with migratory polyarthritis, reactive arthritis, or erythema nodosum. A syndrome of acute diarrhea and vomiting is especially common in young children. The organism may cause disseminated abscesses in the liver and spleen⁹⁷ or an inflammatory colitis.⁹⁸ The causative agent, a gram-negative member of the family Enterobacteriaceae, is in the same genus as the plague bacillus *Yersinia pestis* and is sometimes mistaken for *Proteus* on initial culture plates.

FIG. 101.2 Methylene blue stain of fecal leukocytes from a patient with colitis. This exudative response may be seen in any active colitis syndrome, including those caused by the inflammatory or invasive pathogens discussed in this chapter.

Follow for extended description

Amebiasis

Entamoeba histolytica is one of the classic causes of dysenteric syndromes. Although the course and severity of the infection are variable, amebiasis is one infection that can cause sustained febrile diarrhea with stools becoming grossly bloody. For patients living in endemic areas or travelers who have been to such endemic areas and who have dysenteric symptoms, amebiasis should be considered in the differential diagnosis. For further details about amebiasis, please see Chapter 278.

Diagnosis

Diarrhea with fever, blood in the stool, severe abdominal pain, or signs of sepsis necessitates evaluation for enteric pathogens to guide disease management.⁹⁹ Clinical and epidemiologic information direct the specific tests to be requested. Examination for fecal leukocytes often reveals sheets of polymorphonuclear leukocytes in clumps of mucus, even in the absence of gross blood in the stool specimen (Fig. 101.2).^{100,101} Fewer pyknotic leukocytes are reported in amebic dysentery¹⁰²⁻¹⁰⁴; this may be attributable to the deeper undermining ulcers characteristic of intestinal amebiasis or to a contact-dependent cytolytic effect of the amoeba on leukocytes. The diagnosis of shigellosis is made using bacterial cultures or polymerase chain reaction (PCR) assay for the invasiveness plasmid in stool, rectal swab, or endoscopic biopsy specimens.¹⁰⁵ The use of fresh specimens promptly plated onto appropriate enteric culture media is very important to ensure the isolation of shigellae.¹⁰⁶

Specialized techniques are required to isolate *Vibrio* (thiosulfate citrate bile salts [TCBS] agar),¹⁰⁷ *Yersinia* (cold enrichment),¹⁰⁸ or *C. jejuni*.⁷³ The identification of toxigenic *C. difficile* is done by immunoassay for either *C. difficile* toxin B or PCR assay for *C. difficile* toxin B.¹⁰⁹

The diagnosis of infection with EHEC depends on culturing stool on MacConkey culture medium, on which the original O157:H7 EHEC strain yields sorbitol-negative colonies.¹¹⁰ The isolated *E. coli* is serotyped for somatic (O) and flagellar (H) antigens. Rapid methods for detection of Shiga toxins in the stool samples are now in use.^{111,112} The pathogenic Kanagawa-positive strains from patients produce β -hemolysis on special (Wagatsuma) medium—in contrast to environmental isolates—and are best isolated as blue-green colonies (alkaline) on TCBS agar.¹⁰⁷ A PCR method provides a species-specific probe for *V. parahaemolyticus*, *V. alginolyticus*, *Vibrio anguillarum*, and a cluster of related species.¹¹³ Cultivation may require cold enrichment techniques.¹⁰⁸ PCR methods for detection of pathogenic *Y. enterocolitica* are more sensitive than culture methods.¹¹⁴

Molecular approaches, including multiple pathogen nucleic acid amplification tests (NAAT) for simultaneous detection of multiple enteropathogens, have high sensitivity and specificity and are widely used for surveillance or clinical purposes.¹¹⁵ In outpatient settings, these have largely replaced culture methods in the United States, Canada, and western Europe. Not only do the NAAT approaches allow for higher sensitivity, but the turnaround time is also faster, facilitating hospital infection control measures and initiation of appropriate intervention. The broad, multiplex approach allows for a screening diagnostic approach for enteric pathogens, including viruses, bacteria, and parasites, without having to test for individual pathogens separately.¹¹⁶ There are several limitations inherent to this advanced molecular assay. In addition to relatively high cost, interpretation of the results of the assay is challenging. Detection of a pathogen does not necessarily mean identification of the cause of dysentery, as the test is not able to differentiate colonization from clinically relevant infection. Another reason is the high frequency of polymicrobial infections detected by this method, with up to 30% of positive samples detecting multiple pathogens, which can make determination of the true pathogen challenging.^{38,117,118} Additionally, specific pathogens such as EPEC or EAEC tend to be detected more commonly as observed in one study where both organisms were positive in 84% of the multipathogen samples.¹¹⁷ Another study showed a false positive rate around 20% with NAAT for organisms like *C. difficile*, EPEC, EAEC, and *Campylobacter*, which was largely caused by cross-reactivity with commensal organisms.³⁸ Quantitation of the burden of infection may assist in the interpretation of results from these molecular assays.¹¹⁹ Moreover, current commercially available GI enteric panels do not allow for detection of antibiotic resistance. The steady increase in the prevalence of antimicrobial resistant diarrhea-causing pathogens indicates the continuing need for traditional microbiologic methods for susceptibility testing to guide treatment.

Amebic dysentery is traditionally diagnosed by direct examination of wet mounts of fresh fecal or proctoscopic specimens, which reveal *E. histolytica* trophozoites or cysts characterized by four or fewer delicate nuclei with central karyosomes. Additional patients may be diagnosed by biopsy, which may reveal periodic acid–Schiff stain (PAS)-positive trophozoites or cysts in the undermining ulcer in the lamina propria, or by a fecal or serum *E. histolytica*-specific antigen test, which distinguishes virulent *E. histolytica* from avirulent *Entamoeba dispar* infections.¹²⁰ The use of multiplex NAAT panels including *E. histolytica* is largely replacing the historical diagnostic approaches to amoebiasis.¹²¹

Sigmoidoscopic examination, especially with biopsy, may be useful in the diagnosis of a pseudomembranous enterocolitis or in the identification of parasites such as *E. histolytica* (with a special PAS stain) or *B. coli*. Amebic colitis is associated with discrete small ulcerations with undermined edges amid relatively normal mucosa. Acute shigellosis causes more widespread, shallow, 3- to 7-mm ulcers with a more intense inflammatory exudate.

Other considerations in the differential diagnosis of inflammatory colitis include the potentially rapidly progressive necrotizing enterocolitis (NEC) syndromes (see Chapters 81 and 249) and noninfectious syndromes such as inflammatory bowel diseases, ulcerative colitis, and Crohn disease. These diagnoses are suggested by the setting, clinical course, history, and findings on radiologic and endoscopic examinations.

Finally, the presence of biomarkers of acute or prolonged intestinal inflammation, such as fecal myeloperoxidase and neopterin, may be associated with growth shortfalls in young children in developing areas.¹²²

Therapy

Therapy consists of supportive fluid and electrolyte management with antimicrobial therapy directed at a specific pathogen if suspected on Page 1297the basis of the epidemiologic setting or microbiologic test results. Fluid resuscitation is the primary treatment of all acute diarrheal disease syndromes. Oral rehydration salt solution (World Health Organization formula) is recommended for oral rehydration therapy.¹²³ Presumptive therapy for the inflammatory colitides varies greatly with the different organisms and is influenced by the increasing resistance of many enteric pathogens to antimicrobial therapy.¹²⁴ For example, an acute febrile dysenteric illness in a young child with daycare exposure or in an area in which shigellosis is common should be treated with an antimicrobial agent, such as one of the fluoroquinolones. Early administration of empirical antibiotic treatment is justified in children hospitalized for clinical dysentery.¹²⁵ Short-course (3 days) oral ciprofloxacin for *S. dysenteriae* type 1 dysentery in children has been effective.¹²⁶ If the *Shigella* organism is sensitive, prompt therapy can successfully stop the diarrhea, alleviate systemic symptoms, and reduce shedding of the organisms in the feces.^{127–129} Because shigellae are increasingly resistant to multiple antibiotics,^{130,131} the practitioner must be familiar with the local resistance pattern of shigellae to treat acute shigellosis appropriately when it is first suspected. Multidrug resistant, even extensively resistant, *Shigella* infections have been reported among MSM.^{132,133} Although ciprofloxacin may shorten the clinical course of *Salmonella* infections, it does not eradicate the organism.¹³⁴ Studies have shown increasing resistance of *Campylobacter* and even *Salmonella* to quinolones.^{135,136} Azithromycin may reduce the duration of symptoms of *Campylobacter* or *Shigella* enteritis. Zinc 20 mg daily (10 mg of elemental zinc in 5 mL) given in two equally divided doses for 2 weeks showed a significant reduction in the duration of acute shigellosis, promoted better weight gain, and reduced diarrheal morbidity in malnourished children at 6 months after infection.¹³⁷ Vitamin A has also been used to reduce severity and mortality from shigellosis.¹³⁸

Antibiotic therapy has not been shown to be effective for disease caused by EHEC or EIEC. Some EHEC strains may increase their phage-mediated production of Shiga-like toxins when exposed to antibiotics to which they are sensitive, such as ampicillin, tetracycline, sulfamethoxazole-trimethoprim, or quinolones.^{139–142}

Most patients with mild-to-moderate *C. jejuni* enterocolitis do not benefit from antibiotic treatment. Severely ill patients, especially those who are debilitated or immunocompromised, appear to benefit from antibiotic therapy, and erythromycin (or other macrolides) or ciprofloxacin are the drugs of choice. Increased fluoroquinolone resistance in *Campylobacter* spp. from

humans and animals, the result of mutations in topoisomerase genes, has been well documented.¹⁴³ Two studies now associate symptomatic and asymptomatic *Campylobacter* infections with reduced growth in Brazilian and Peruvian children.^{144,145}

Vancomycin and fidaxomicin are the drugs of choice for initial episode of CDI (see Chapter 249 for further details).^{146,147} Recurrent disease occurs in 20% to 65% of the patients, depending on the number of previous episodes of the disease.¹⁴⁸ Fidaxomicin is reported to be equivalent to vancomycin in treating acute CDI but may be superior in preventing relapses.¹⁴⁹ However, its advantage over vancomycin is lost in infections caused by the prevalent epidemic strain BI/NAP1/027. Alternative approaches to prevent recurrent disease include a tapering course of vancomycin; fecal microbiota transplantation (FMT)¹⁵⁰; or use of monoclonal antibodies.¹⁵¹ FMT is increasingly being used to prevent recurrent disease or to treat refractory CDI. More recently, the monoclonal antibody against TcdB has been shown to confer modest protection against recurrent CDI.⁹⁹ A vancomycin tapering course has been shown to be effective for treatment, and in some observational studies equal to FMT in preventing recurrence.¹⁵²

Antibiotic treatment with tetracycline or doxycycline can diminish the duration and severity of GI symptoms caused by *V. parahaemolyticus*. To prevent *V. parahaemolyticus* infections, consumption of raw or undercooked shellfish and exposure of wounds to seawater should be avoided.¹⁵³

Antibiotic therapy is not required for *Salmonella* enteritis, but patients who are severely ill or those with risk factors for developing extraintestinal spread of infection should be treated with fluoroquinolones or third-generation cephalosporins.

Yersinia infection is usually self-limiting; however, patients with severe illness or systemic infection are usually treated with tetracycline, chloramphenicol, or ciprofloxacin.

Prevention

Enteric pathogens are predominantly spread through contaminated food or water and sometimes via direct fecal-oral spread. Spread of infection occurs during ingestion of contaminated food or water or contact with a contaminated environment. Infection control measures include hand hygiene, proper food handling, and access to clean water. In some instances, isolation and treatment of infected individuals are necessary to prevent spread of infection, for example, in cases of CDI. Targeted health interventions for sexual networks of MSM are needed.

Immunization of susceptible individuals is an important component of prevention. However, for bacterial enteric pathogens, vaccine is only available for typhoid fever and cholera. Typhoid vaccination is recommended for people traveling to endemic areas. Ty21a is a live-attenuated oral vaccine containing *S. Typhi* in capsule formulation. Another typhoid vaccine is the parenteral Vi-polysaccharide. For individuals traveling to areas with recent or ongoing cholera outbreaks, CVD 103-HgR, a live-attenuated oral vaccine, is available. There is a need to develop effective vaccines for most infectious agents causing diarrhea worldwide.

Other Acute Infectious/Inflammatory Processes Involving the Intestinal Tract Gonococcal Proctitis

Neisseria gonorrhoeae (see Chapter 217) may cause ulcerative proctitis, usually acquired by anal intercourse.⁷ The resultant purulent proctitis is accompanied by an erythematous, friable mucous membrane in the rectal vault and occasional abscess or fistula formation. Although copious purulent discharge, tenesmus, and burning rectal pain may be noted, two-thirds of culture-positive patients with anorectal gonococcal infection are asymptomatic.¹⁵⁴ Additional diagnostic possibilities in cases of venereally acquired proctitis are syphilitic, herpetic, and chlamydial proctitis.¹⁵⁵

Unusual Causes of Dysentery

Spirillar or spirochetal dysentery has been reported to occur in southern France and has been attributed to *Spirillum* spp. Although severe mucoid diarrhea or dysentery has been associated with intestinal spirochetes, their frequency and role in causing enteric disease are unclear. A DNA probe for the 16S ribosomal RNA of the agent of swine dysentery, *Treponema hyodysenteriae*, has

been developed and may open new approaches to the recognition of similar infections in humans.¹⁵⁶

Other unusual causes of colitis include brucellosis¹⁵⁷ and adenovirus infections.^{158,159}

Necrotizing Enterocolitis in the Newborn

The syndrome of diffuse, fulminating, necrotizing colitis has been recognized in infants since the 1960s. This syndrome probably represents the same entity described as spontaneous intestinal perforation and peritonitis as early as 1838.¹⁶⁰ Although milder forms of the syndrome doubtless exist, the syndrome of NEC is defined by the presence of air in the wall of the intestine, portal venous system, or peritoneal cavity, or by necrosis of the bowel wall with mucosal sloughing. This fulminant syndrome often leads to intestinal perforation, peritonitis, and bacteremia. It is a major cause of mortality in low-birth-weight infants (<1500 g) after the first week of life.¹⁶¹ The diffuse necrotic changes that characterize this syndrome most often occur in the terminal ileum but may be seen in the colon or in the proximal portion of the GI tract.

Hypoxia-ischemia was previously thought to be the main mechanism for development of NEC, but this is uncertain,¹⁶² but current research suggests multifactorial pathogenesis. The three major risk factors associated with development of NEC are prematurity, bacterial colonization of the gut, and formula feeding¹⁶³; the pathogenesis involves a combination of immature intestinal barrier and abnormal intestinal microbiome.¹⁶⁴ Intestinal immaturity refers to Page 1298motility, absorption, barrier function, and circulatory regulation, but the most important aspect seems to be an excessive immune response to the gut microbiota which contributes to the development of NEC along with a dysbiotic intestinal microbiome. Studies in cell cultures and animal models suggest that the immature intestinal epithelium has an excessive inflammatory response to the microbiota.^{165,166} Studies comparing babies with NEC with control, or before and after development of NEC, suggests that a reduced diversity of the fecal microbiome and presence of opportunistic intestinal microbial species are associated with development of NEC.^{167,168}

Clinical features of this serious condition in newborn infants include apneic spells, vomiting, abdominal distention, and occasionally bloody diarrhea. Most infants are younger than 1 week, and there is an association with prematurity, maternal infections during delivery (e.g., amnionitis with prolonged postrupture of membranes), and exchange transfusion via the umbilical vein. There is no gender or seasonal predilection. The disease often progresses rapidly to intestinal perforation, shock, septicemia, and pneumatosis intestinalis. Air may also be evident in the portal venous system or biliary tract on plain radiographs. This syndrome is associated with mortality often greater than 70%.

The diagnosis of NEC should be considered in any premature infant with altered GI function, abdominal distention, or apneic spells. Further investigation should include examination of the stool for occult blood and for the presence of reducing substances. Plain abdominal radiographs may reveal air in the bowel wall, peritoneal cavity, or portal venous system, and there may be bloody diarrhea late in the course of the disease. Management must be initiated early and aggressively for any infant suspected of having NEC. Umbilical catheters should be removed, oral feeding should be stopped, and nasogastric aspiration should be initiated. Intravenous fluid therapy is of paramount importance. After obtaining specimens for blood culture, broad-spectrum intravenous antibiotics is recommended. Laparotomy and excision of the necrotic bowel are often necessary and should be done aggressively if there is any evidence of peritonitis or obstruction.¹⁶⁹

Prevention of NEC includes promotion of intestinal development, avoidance of risk factors and careful infection control measures in newborn ICUs. Antenatal corticosteroids, given for prevention of respiratory distress in the premature newborn, also promote intestinal development and appears protective against NEC development.¹⁷⁰ Hypertonic elemental formulas have been implicated and should be avoided in high-risk patients. Similarly, prolonged antibiotic use and gastric acid suppression increase NEC risk and should be avoided.¹⁷¹⁻¹⁷³ Conversely, use of human milk for preterm infants lowers the risk of NEC.^{174,175} Explanations for the advantage of human breast milk include the presence of lysozyme, antibodies, and cellular elements that may play a protective role against potential infectious agents, as well as promotion of growth for beneficial bacteria in the gut by human milk oligosaccharides (HMOs).^{176,177} Although oral

prophylactic nonabsorbable antibiotics have been suggested, serious questions remain about the use Page 1292

101

Acute Dysentery Syndromes (Diarrhea With Fever)

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Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.

- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.

- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis*, or *Treponema pallidum* as a cause of proctitis, or *Campylobacter* spp., *Shigella* spp., *C. trachomatis* (lymphogranuloma venereum serotypes), or *C. difficile* as a cause of colitis.

- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest *C. difficile* infection (CDI).

Microbiology

- Genomic studies of *Shigella* spp. have indicated that *Shigella* spp. and enteroinvasive *E. coli* are derived from multiple phyletic origins of *E. coli* despite their microbiological and clinical similarities.

- The cause of an outbreak of bloody diarrhea and severe hemolytic-uremic syndrome, unlike prior EHEC strains that had exhibited enteropathogenic *E. coli* traits of attachment and effacement, is a Shiga toxin-producing enteroaggregative *E. coli* strain.

- *Campylobacter* spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.

- The primary virulence factors that are known to cause disease in CDI are the two large toxins: *C. difficile* toxin (Tcd)A, or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.

- *Vibrio parahaemolyticus* has been recognized since 1950 in Japan and is a cause of seafood poisoning.

- *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.

- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.

- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

- Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although *Shigella* and perhaps *Campylobacter* durations of infection and symptoms may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic *Salmonella* infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

Definition

Diarrhea associated with fever typically suggests an acute inflammatory enteritis caused by one of several specific infectious agents. However, noninflammatory enteritides caused by viral or other agents may also present with diarrhea and fever. This chapter will address bacterial and amebic colitis as potential major causes of this syndrome. Viral and other parasitic etiologies will be addressed in other chapters. The acute inflammatory enteritides include several specific, distal small bowel, and colonic infections, such as those caused by *Campylobacter* spp., *Salmonella* spp., *Shigella* spp., and enteroinvasive *Escherichia coli* (EIEC), as well as the syndromes of necrotizing enteritis (NE) and antibiotic-associated pseudomembranous enterocolitis (*Clostridioides difficile* [formerly *Clostridium difficile*]). Several other infectious agents cause chronic enteric inflammatory processes that may result in syndromes of abdominal pain, weight loss, diarrhea, or malabsorption. These involve disease processes such as gastrointestinal (GI) mycoses; mycobacterioses; amoebiasis, or bacterial infections, such as those caused by enteropathogenic *E. coli* (EPEC), enterohemorrhagic (EHEC), and enteroaggregative *E. coli* (EAEC); and syphilis.

Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis* (including lymphogranuloma venereum serotypes), or *Treponema pallidum* as a cause of proctitis, or *Campylobacter*, *Shigella*, or *E. histolytica* as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care

facility would strongly suggest *C. difficile* infection (CDI), although community-acquired infection has been well recognized as well.

Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}

Potential causes of acute dysentery are listed in Table 101.1.

Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes *Shigella* spp. and Enteroinvasive *Escherichia coli*

Bacillary dysentery is caused by *Shigella* spp. and EIEC. It is estimated that *Shigella* spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four *Shigella* species are classified in subgroups A to D—*Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*, respectively. Genomic studies of *Shigella* spp. have indicated that *Shigella* and EIEC are derived from multiple phyletic origins within *E. coli*.¹¹ *S. dysenteriae* (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ *S. dysenteriae* is responsible for severe and epidemic disease, and *S. boydii* is mostly seen in the Indian subcontinent. *S. flexneri* and *S. sonnei* are the most common species around the world.¹² *Shigellae* are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. *Shigella* spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² *Shigella* spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with *Shigella*, and other enteric pathogens such as EAEC, *Campylobacter*, and *Giardia*, were associated with negative linear growth, which was sustained up to 5 years.¹⁴

Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

TABLE 101.1

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes

Bacillary dysentery (*Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Shigella boydii*; invasive *Escherichia coli*)

Campylobacteriosis (*Campylobacter jejuni*)

Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis

Gonococcal (*Neisseria gonorrhoeae*)

Herpetic (herpes simplex virus)

Chlamydial (*Chlamydia trachomatis*)

Syphilitic (*Treponema pallidum*)

Other Syndromes

Necrotizing enterocolitis of the newborn

Enteritis necroticans

Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)

Diverticulitis

Typhlitis

Chronic Inflammatory Processes

Enteropathogenic and enteroaggregative *E. coli*

Syphilis

Gastrointestinal tuberculosis

Gastrointestinal mycosis (including *Basidiobolus ranarum*)

Parasitic enteritis

Syndromes Without Known Infectious Cause

Idiopathic ulcerative colitis

Crohn disease

Radiation enteritis

Ischemic colitis

Allergic enteritis

Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC

strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin-producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:K60:H11, O103:H2, O91:H2, O145:H–, O111:K58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:K75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including aggR, aggA (encoding for the biofilm-producing aggregative adherence fimbriae I [AAF/I]), sigA, sepA, pic, aatA, aaiC, and aap, but it was negative for the eae (*E. coli* attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

TABLE 101.2

Enteroinvasive *Escherichia coli* Serotypes
 SEROTYPE DIFCO SEROGROUP TEST^a

O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

^a Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

Enteropathogenic and Enteroaggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.^{58–60} EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.^{61–63} EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such

as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other *Campylobacter* spp. have been published.^{69–71} Although there currently are more than 16 *Campylobacter* spp. and 6 subspecies, *C. jejuni* and *Campylobacter coli* are the most prevalent species associated with enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of *Campylobacter* bloodstream infections in humans are with *Campylobacter fetus* (former subspecies, *C. fetus* subsp. *intestinalis*),⁷² *C. jejuni* commonly causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ *Campylobacter* enteritis results from oral transmission, and a variable but relatively small dose ($5\text{--}8 \times 10^2$ cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of *Campylobacter* enteritis have been associated with ingestion of contaminated water, raw milk, and uncooked meat and poultry. A review of common source outbreaks of *Campylobacter* infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by *Campylobacter* spp., and the specific balance between these genes and host susceptibility to *Campylobacter* enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; secretion of cytokines; chemokines and arachidonic acid metabolites; and production of substance P and reactive oxygen intermediates. Some *C. difficile* strains, including the epidemic ribotype 027, produce binary toxin that may have a role in increased virulence through host attachment and colonization, which type 2 immunity in the host may protect against.⁸³ Clinical presentation ranges from asymptomatic colonization to severe disease with pseudomembranous colitis (Fig. 101.1) and septic shock. Increased fecal lactoferrin, IL-1 β , and IL-8 were observed in patients with CDI.⁸⁴ Leukocytosis suggests severe infection, and leukemoid reactions have been described in colitis caused by *C. difficile*.⁸⁵ Further details of CDI are provided in Chapter 249.

Vibriosis

In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and

occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90}

FIG. 101.1 Proctoscopic view of pseudomembranous colitis in a patient who received clindamycin. Note the 4- to 8-mm raised white plaques overlying an erythematous mucosa. From Tedesco FJ, Barton RW, Alpers DH. Clindamycin-associated colitis. *Ann Intern Med.* 1974;81:429–433.

Follow for extended description

Salmonellosis and Enteric Fever

Salmonella enterocolitis is characterized by fever, cramping, abdominal pain, and diarrhea that begin 8 to 48 hours after ingestion of an infective dose, usually with food, and usually lasts 3 to 5 days (see Chapter 228). The diarrheal stools of patients with salmonellosis often contain a moderate number of polymorphonuclear leukocytes, usually fewer than is typical of shigellosis.

Although *Salmonella enteritis* predominantly involves the lamina propria in the small bowel, several reports have noted *Salmonella enterica* serovar Typhimurium as a cause of colitis, with crypt abscesses and erosion and ulcerations of the colonic mucosa, which result in blood and pus in the stool.⁹¹ Certain other strains of *Salmonella*—including *S. enterica* serovar Choleraesuis and *S. enterica* serovar Paratyphi, and especially *S. enterica* serovar Typhi—elicit a mononuclear response and cause the bacteremia that is characteristic of enteric fever. Animal models for *S. Typhimurium* continue to advance our understanding of the pathogenesis of intestinal and systemic *Salmonella* infections.⁹² *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5 (TLR5), nucleotide oligomerization domain–like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein. TLR5 is located in the plasma and phagosomal membranes, and the other two are located in the cytoplasm of the host's Peyer patch cells. Flagellin activates TLR5 and leads to the production of IL-12, whereas the activation of IPAF leads to the production of IL-1 β and IL-18. Both pathways act synergistically to activate interferon- γ -producing Th1 cells. These proinflammatory cytokines play important roles in the induction of inflammation with *Salmonella* infections.

Typhoid fever may lead to an erosion of the blood vessels in Peyer patches that, without appropriate treatment, could result in gross blood in the feces in 10% to 20% of patients. Severe intestinal hemorrhage may complicate approximately 2% of cases late in the course of untreated typhoid fever. Such intestinal bleeding may precede perforation, another complication of typhoid fever.⁹³ However, typhoid fever is primarily a systemic illness, not a diarrheal disease (see Chapter 102).

Yersiniosis

Y. enterocolitica (see Chapter 235) is an enteric pathogen that may be responsible for an enteric fever–like illness, mesenteric adenitis (which may mimic acute appendicitis), or an inflammatory ileitis or ulcerative colitis syndrome with fecal neutrophils and mononuclear cells.⁹⁴ *Y. enterocolitica* and *Yersinia pseudotuberculosis* may cause similar symptoms and signs. The virulence of *Yersinia* is associated with the direct injection of proteins, such as *Yersinia* protein kinase A (YpkA) and *Yersinia* outer protein E (YopE), into eukaryotic cells.^{95,96} *Yersinia* infection may also be associated with migratory polyarthritis, reactive arthritis, or erythema

nodosum. A syndrome of acute diarrhea and vomiting is especially common in young children. The organism may cause disseminated abscesses in the liver and spleen⁹⁷ or an inflammatory colitis.⁹⁸ The causative agent, a gram-negative member of the family Enterobacteriaceae, is in the same genus as the plague bacillus *Yersinia pestis* and is sometimes mistaken for *Proteus* on initial culture plates.

FIG. 101.2 Methylene blue stain of fecal leukocytes from a patient with colitis. This exudative response may be seen in any active colitis syndrome, including those caused by the inflammatory or invasive pathogens discussed in this chapter.

Follow for extended description

Amebiasis

Entamoeba histolyticum is one of the classic causes of dysenteric syndromes. Although the course and severity of the infection are variable, amebiasis is one infection that can cause sustained febrile diarrhea with stools becoming grossly bloody. For patients living in endemic areas or travelers who have been to such endemic areas and who have dysenteric symptoms, amebiasis should be considered in the differential diagnosis. For further details about amebiasis, please see Chapter 278.

Diagnosis

Diarrhea with fever, blood in the stool, severe abdominal pain, or signs of sepsis necessitates evaluation for enteric pathogens to guide disease management.⁹⁹ Clinical and epidemiologic information direct the specific tests to be requested. Examination for fecal leukocytes often reveals sheets of polymorphonuclear leukocytes in clumps of mucus, even in the absence of gross blood in the stool specimen (Fig. 101.2).^{100,101} Fewer pyknotic leukocytes are reported in amebic dysentery¹⁰²⁻¹⁰⁴; this may be attributable to the deeper undermining ulcers characteristic of intestinal amebiasis or to a contact-dependent cytolytic effect of the amoeba on leukocytes. The diagnosis of shigellosis is made using bacterial cultures or polymerase chain reaction (PCR) assay for the invasiveness plasmid in stool, rectal swab, or endoscopic biopsy specimens.¹⁰⁵ The use of fresh specimens promptly plated onto appropriate enteric culture media is very important to ensure the isolation of shigellae.¹⁰⁶

Specialized techniques are required to isolate *Vibrio* (thiosulfate citrate bile salts [TCBS] agar),¹⁰⁷ *Yersinia* (cold enrichment),¹⁰⁸ or *C. jejuni*.⁷³ The identification of toxigenic *C. difficile* is done by immunoassay for either *C. difficile* toxin B or PCR assay for *C. difficile* toxin B.¹⁰⁹

The diagnosis of infection with EHEC depends on culturing stool on MacConkey culture medium, on which the original O157:H7 EHEC strain yields sorbitol-negative colonies.¹¹⁰ The isolated *E. coli* is serotyped for somatic (O) and flagellar (H) antigens. Rapid methods for detection of Shiga toxins in the stool samples are now in use.^{111,112} The pathogenic Kanagawa-positive strains from patients produce β -hemolysis on special (Wagatsuma) medium—in contrast to environmental isolates—and are best isolated as blue-green colonies (alkaline) on TCBS agar.¹⁰⁷ A PCR method provides a species-specific probe for *V. parahaemolyticus*, *V. alginolyticus*, *Vibrio anguillarum*, and a cluster of related species.¹¹³ Cultivation may require cold enrichment techniques.¹⁰⁸ PCR methods for detection of pathogenic *Y. enterocolitica* are more sensitive than culture methods.¹¹⁴

Molecular approaches, including multiple pathogen nucleic acid amplification tests (NAAT) for simultaneous detection of multiple enteropathogens, have high sensitivity and specificity and are widely used for surveillance or clinical purposes.¹¹⁵ In outpatient settings, these have largely replaced culture methods in the United States, Canada, and western Europe. Not only do the NAAT approaches allow for higher sensitivity, but the turnaround time is also faster, facilitating hospital infection control measures and initiation of appropriate intervention. The broad, multiplex approach allows for a screening diagnostic approach for enteric pathogens, including viruses, bacteria, and parasites, without having to test for individual pathogens separately.¹¹⁶ There are several limitations inherent to this advanced molecular assay. In addition to relatively high cost, interpretation of the results of the assay is challenging. Detection of a pathogen does not necessarily mean identification of the cause of dysentery, as the test is not able to differentiate colonization from clinically relevant infection. Another reason is the high frequency of polymicrobial infections detected by this method, with up to 30% of positive samples detecting

multiple pathogens, which can make determination of the true pathogen challenging.^{38,117,118} Additionally, specific pathogens such as EPEC or EAEC tend to be detected more commonly as observed in one study where both organisms were positive in 84% of the multipathogen samples.¹¹⁷ Another study showed a false positive rate around 20% with NAAT for organisms like *C. difficile*, EPEC, EAEC, and *Campylobacter*, which was largely caused by cross-reactivity with commensal organisms.³⁸ Quantitation of the burden of infection may assist in the interpretation of results from these molecular assays.¹¹⁹ Moreover, current commercially available GI enteric panels do not allow for detection of antibiotic resistance. The steady increase in the prevalence of antimicrobial resistant diarrhea-causing pathogens indicates the continuing need for traditional microbiologic methods for susceptibility testing to guide treatment.

Amebic dysentery is traditionally diagnosed by direct examination of wet mounts of fresh fecal or proctoscopic specimens, which reveal *E. histolytica* trophozoites or cysts characterized by four or fewer delicate nuclei with central karyosomes. Additional patients may be diagnosed by biopsy, which may reveal periodic acid–Schiff stain (PAS)-positive trophozoites or cysts in the undermining ulcer in the lamina propria, or by a fecal or serum *E. histolytica*-specific antigen test, which distinguishes virulent *E. histolytica* from avirulent *Entamoeba dispar* infections.¹²⁰ The use of multiplex NAAT panels including *E. histolytica* is largely replacing the historical diagnostic approaches to amoebiasis.¹²¹

Sigmoidoscopic examination, especially with biopsy, may be useful in the diagnosis of a pseudomembranous enterocolitis or in the identification of parasites such as *E. histolytica* (with a special PAS stain) or *B. coli*. Amebic colitis is associated with discrete small ulcerations with undermined edges amid relatively normal mucosa. Acute shigellosis causes more widespread, shallow, 3- to 7-mm ulcers with a more intense inflammatory exudate.

Other considerations in the differential diagnosis of inflammatory colitis include the potentially rapidly progressive necrotizing enterocolitis (NEC) syndromes (see Chapters 81 and 249) and noninfectious syndromes such as inflammatory bowel diseases, ulcerative colitis, and Crohn disease. These diagnoses are suggested by the setting, clinical course, history, and findings on radiologic and endoscopic examinations.

Finally, the presence of biomarkers of acute or prolonged intestinal inflammation, such as fecal myeloperoxidase and neopterin, may be associated with growth shortfalls in young children in developing areas.¹²²

Therapy

Therapy consists of supportive fluid and electrolyte management with antimicrobial therapy directed at a specific pathogen if suspected on Page 1297the basis of the epidemiologic setting or microbiologic test results. Fluid resuscitation is the primary treatment of all acute diarrheal disease syndromes. Oral rehydration salt solution (World Health Organization formula) is recommended for oral rehydration therapy.¹²³ Presumptive therapy for the inflammatory colitides varies greatly with the different organisms and is influenced by the increasing resistance of many enteric pathogens to antimicrobial therapy.¹²⁴ For example, an acute febrile dysenteric illness in a young child with daycare exposure or in an area in which shigellosis is common should be treated with an antimicrobial agent, such as one of the fluoroquinolones. Early administration of empirical antibiotic treatment is justified in children hospitalized for clinical dysentery.¹²⁵ Short-course (3 days) oral ciprofloxacin for *S. dysenteriae* type 1 dysentery in children has been effective.¹²⁶ If the *Shigella* organism is sensitive, prompt therapy can successfully stop the diarrhea, alleviate systemic symptoms, and reduce shedding of the organisms in the feces.^{127–129} Because shigellae are increasingly resistant to multiple antibiotics,^{130,131} the practitioner must be familiar with the local resistance pattern of shigellae to treat acute shigellosis appropriately when it is first suspected. Multidrug resistant, even extensively resistant, *Shigella* infections have been reported among MSM.^{132,133} Although ciprofloxacin may shorten the clinical course of *Salmonella* infections, it does not eradicate the organism.¹³⁴ Studies have shown increasing resistance of *Campylobacter* and even *Salmonella* to quinolones.^{135,136} Azithromycin may reduce the duration of symptoms of *Campylobacter* or *Shigella* enteritis. Zinc 20 mg daily (10 mg of elemental zinc in 5 mL) given in two equally divided doses for 2 weeks showed a significant reduction in the duration of acute shigellosis, promoted better weight gain, and reduced diarrheal

morbidity in malnourished children at 6 months after infection.¹³⁷ Vitamin A has also been used to reduce severity and mortality from shigellosis.¹³⁸

Antibiotic therapy has not been shown to be effective for disease caused by EHEC or EIEC. Some EHEC strains may increase their phage-mediated production of Shiga-like toxins when exposed to antibiotics to which they are sensitive, such as ampicillin, tetracycline, sulfamethoxazole-trimethoprim, or quinolones.¹³⁹⁻¹⁴²

Most patients with mild-to-moderate *C. jejuni* enterocolitis do not benefit from antibiotic treatment. Severely ill patients, especially those who are debilitated or immunocompromised, appear to benefit from antibiotic therapy, and erythromycin (or other macrolides) or ciprofloxacin are the drugs of choice. Increased fluoroquinolone resistance in *Campylobacter* spp. from humans and animals, the result of mutations in topoisomerase genes, has been well documented.¹⁴³ Two studies now associate symptomatic and asymptomatic *Campylobacter* infections with reduced growth in Brazilian and Peruvian children.^{144,145}

Vancomycin and fidaxomicin are the drugs of choice for initial episode of CDI (see Chapter 249 for further details).^{146,147} Recurrent disease occurs in 20% to 65% of the patients, depending on the number of previous episodes of the disease.¹⁴⁸ Fidaxomicin is reported to be equivalent to vancomycin in treating acute CDI but may be superior in preventing relapses.¹⁴⁹ However, its advantage over vancomycin is lost in infections caused by the prevalent epidemic strain BI/NAP1/027. Alternative approaches to prevent recurrent disease include a tapering course of vancomycin; fecal microbiota transplantation (FMT)¹⁵⁰; or use of monoclonal antibodies.¹⁵¹ FMT is increasingly being used to prevent recurrent disease or to treat refractory CDI. More recently, the monoclonal antibody against TcdB has been shown to confer modest protection against recurrent CDI.⁹⁹ A vancomycin tapering course has been shown to be effective for treatment, and in some observational studies equal to FMT in preventing recurrence.¹⁵²

Antibiotic treatment with tetracycline or doxycycline can diminish the duration and severity of GI symptoms caused by *V. parahaemolyticus*. To prevent *V. parahaemolyticus* infections, consumption of raw or undercooked shellfish and exposure of wounds to seawater should be avoided.¹⁵³

Antibiotic therapy is not required for *Salmonella* enteritis, but patients who are severely ill or those with risk factors for developing extraintestinal spread of infection should be treated with fluoroquinolones or third-generation cephalosporins.

Yersinia infection is usually self-limiting; however, patients with severe illness or systemic infection are usually treated with tetracycline, chloramphenicol, or ciprofloxacin.

Prevention

Enteric pathogens are predominantly spread through contaminated food or water and sometimes via direct fecal-oral spread. Spread of infection occurs during ingestion of contaminated food or water or contact with a contaminated environment. Infection control measures include hand hygiene, proper food handling, and access to clean water. In some instances, isolation and treatment of infected individuals are necessary to prevent spread of infection, for example, in cases of CDI. Targeted health interventions for sexual networks of MSM are needed.

Immunization of susceptible individuals is an important component of prevention. However, for bacterial enteric pathogens, vaccine is only available for typhoid fever and cholera. Typhoid vaccination is recommended for people traveling to endemic areas. Ty21a is a live-attenuated oral vaccine containing *S. Typhi* in capsule formulation. Another typhoid vaccine is the parenteral Vi-polysaccharide. For individuals traveling to areas with recent or ongoing cholera outbreaks, CVD 103-HgR, a live-attenuated oral vaccine, is available. There is a need to develop effective vaccines for most infectious agents causing diarrhea worldwide.

Other Acute Infectious/Inflammatory Processes Involving the Intestinal Tract Gonococcal Proctitis

Neisseria gonorrhoeae (see Chapter 217) may cause ulcerative proctitis, usually acquired by anal intercourse.⁷ The resultant purulent proctitis is accompanied by an erythematous, friable mucous membrane in the rectal vault and occasional abscess or fistula formation. Although copious purulent discharge, tenesmus, and burning rectal pain may be noted, two-thirds of culture-positive patients with anorectal gonococcal infection are asymptomatic.¹⁵⁴ Additional diagnostic possibilities in cases of venereally acquired proctitis are syphilitic, herpetic, and chlamydial proctitis.¹⁵⁵

Unusual Causes of Dysentery

Spirillar or spirochetal dysentery has been reported to occur in southern France and has been attributed to *Spirillum* spp. Although severe mucoid diarrhea or dysentery has been associated with intestinal spirochetes, their frequency and role in causing enteric disease are unclear. A DNA probe for the 16S ribosomal RNA of the agent of swine dysentery, *Treponema hyodysenteriae*, has been developed and may open new approaches to the recognition of similar infections in humans.¹⁵⁶

Other unusual causes of colitis include brucellosis¹⁵⁷ and adenovirus infections.^{158,159}

Necrotizing Enterocolitis in the Newborn

The syndrome of diffuse, fulminating, necrotizing colitis has been recognized in infants since the 1960s. This syndrome probably represents the same entity described as spontaneous intestinal perforation and peritonitis as early as 1838.¹⁶⁰ Although milder forms of the syndrome doubtless exist, the syndrome of NEC is defined by the presence of air in the wall of the intestine, portal venous system, or peritoneal cavity, or by necrosis of the bowel wall with mucosal sloughing. This fulminant syndrome often leads to intestinal perforation, peritonitis, and bacteremia. It is a major cause of mortality in low-birth-weight infants (<1500 g) after the first week of life.¹⁶¹ The diffuse necrotic changes that characterize this syndrome most often occur in the terminal ileum but may be seen in the colon or in the proximal portion of the GI tract.

Hypoxia-ischemia was previously thought to be the main mechanism for development of NEC, but this is uncertain,¹⁶² but current research suggests multifactorial pathogenesis. The three major risk factors associated with development of NEC are prematurity, bacterial colonization of the gut, and formula feeding¹⁶³; the pathogenesis involves a combination of immature intestinal barrier and abnormal intestinal microbiome.¹⁶⁴ Intestinal immaturity refers to Page 1298motility, absorption, barrier function, and circulatory regulation, but the most important aspect seems to be an excessive immune response to the gut microbiota which contributes to the development of NEC along with a dysbiotic intestinal microbiome. Studies in cell cultures and animal models suggest that the immature intestinal epithelium has an excessive inflammatory response to the microbiota.^{165,166} Studies comparing babies with NEC with control, or before and after development of NEC, suggests that a reduced diversity of the fecal microbiome and presence of opportunistic intestinal microbial species are associated with development of NEC.^{167,168}

Clinical features of this serious condition in newborn infants include apneic spells, vomiting, abdominal distention, and occasionally bloody diarrhea. Most infants are younger than 1 week, and there is an association with prematurity, maternal infections during delivery (e.g., amnionitis with prolonged postrupture of membranes), and exchange transfusion via the umbilical vein. There is no gender or seasonal predilection. The disease often progresses rapidly to intestinal perforation, shock, septicemia, and pneumatosis intestinalis. Air may also be evident in the portal venous system or biliary tract on plain radiographs. This syndrome is associated with mortality often greater than 70%.

The diagnosis of NEC should be considered in any premature infant with altered GI function, abdominal distention, or apneic spells. Further investigation should include examination of the stool for occult blood and for the presence of reducing substances. Plain abdominal radiographs may reveal air in the bowel wall, peritoneal cavity, or portal venous system, and there may be bloody diarrhea late in the course of the disease. Management must be initiated early and aggressively for any infant suspected of having NEC. Umbilical catheters should be removed, oral feeding should be stopped, and nasogastric aspiration should be initiated. Intravenous fluid therapy is of paramount importance. After obtaining specimens for blood culture, broad-spectrum intravenous antibiotics is recommended. Laparotomy and excision of the necrotic bowel are often

necessary and should be done aggressively if there is any evidence of peritonitis or obstruction.¹⁶⁹

Prevention of NEC includes promotion of intestinal development, avoidance of risk factors and careful infection control measures in newborn ICUs. Antenatal corticosteroids, given for prevention of respiratory distress in the premature newborn, also promote intestinal development and appears protective against NEC development.¹⁷⁰ Hypertonic elemental formulas have been implicated and should be avoided in high-risk patients. Similarly, prolonged antibiotic use and gastric acid suppression increase NEC risk and should be avoided.¹⁷¹⁻¹⁷³ Conversely, use of human milk for preterm infants lowers the risk of NEC.^{174,175} Explanations for the advantage of human breast milk include the presence of lysozyme, antibodies, and cellular elements that may play a protective role against potential infectious agents, as well as promotion of growth for beneficial bacteria in the gut by human milk oligosaccharides (HMOs).^{176,177} Although oral prophylactic nonabsorbable antibiotics have been suggested, serious questions remain about the use of prophylactic antibiotics, even in high-risk newborn infants weighing less than 1500 g. A few reports have shown that epidermal growth factor, which is found in high concentrations in breast milk, may have a cytoprotective effect, and probiotics may decrease or prevent neonatal NEC.^{178,179}

Darmbrand, Pig-Bel, Necrotizing Enteritis in Adults (Enteritis Necroticans)
First described as Darmbrand (meaning fire bowels) in epidemics of enteritis necroticans (EN) in northern Germany in the immediate postwar period in the mid-1940s, a severe necrotizing jejunitis has also been recognized in epidemic and sporadic forms after pork feasting in the highlands of Papua New Guinea.¹⁸⁰ Pig-bel was the name given to the syndrome of abdominal discomfort that followed a large pork meal, commonly eaten after a large pig kill, which takes place every 3 to 10 years among the highland Melanesians of Papua New Guinea. Sporadic cases have been reported from other parts of the world, including the United States.¹⁸¹

The pathologic findings are those of acute, patchy, necrotizing disease of the small bowel in previously healthy people, which may proceed rapidly to segmental gangrene, with small amounts of gas in the mucosa, mesentery, or nodes.

Several theories of pathogenesis have been suggested, most of which involve the toxic products of *Clostridium perfringens* type C, including α - and β -toxins. Sporadic cases of EN have been noted in association with nutritional disorders, alcoholism, and malabsorption, and after pancreatic or gastric resection.¹⁸² After gastric surgery, increased numbers of *C. perfringens* and increased levels of α -toxin have been noted in the upper small bowel and stomach. Whether α - or β -toxins are capable of causing EN alone, or whether they initiate the invasion of the mucosa by other organisms, such as gram-negative rods, is currently unclear. An attractive hypothesis was suggested by Lawrence and Walker,¹⁸² which could explain the association of EN with poor nutrition and episodic dietetic overindulgence. The low-protein diet of Papua New Guinea highlanders is associated with low levels of digestive proteases in the intestinal lumen, which inactivate the β -toxin. The proteases can be further blocked by the oral intake of trypsin inhibitors, which are found in this population in such dietary staples as sweet potatoes. Proteases return with improved diet, as occurred in postwar Germany. This hypothesis has been confirmed in an animal model that required protease inhibitors for symptomatic infection.¹⁸³

The clinical syndromes of EN range from anorexia, vomiting, severe abdominal pain, and bloody diarrhea to fulminant toxemia and shock. Acute complications that necessitate emergency surgery include paralytic ileus, bowel strangulation, and bowel perforation with peritonitis. These complications are common in the first 2 weeks of illness. Later complications that may also necessitate surgery include scarring, leading to stenosis; obstruction; malabsorption; and fistulas. EN occurs with greater frequency and greater severity in children younger than 10 years. In contrast to European control subjects who rarely have antibodies, 70% of the healthy adults in Papua New Guinea have detectable antibody to clostridial β -toxin.^{180,182}

The syndrome is defined by the pathologic findings but should be suspected in patients who develop severe abdominal pain, bloody diarrhea, ileus, and toxemia. The course is often too fulminant for radiographic detection of air in the bowel wall to be of any diagnostic value.

Agents held responsible for causing EN include *C. perfringens* type C, once designated as type F in the older classification of *Clostridium welchii*. Most surgically resected bowel specimens from patients with EN contain *C. perfringens*, more than half of which are type C. Furthermore, 12 of 21 cases described had a significant change in serum β -antitoxin titer after illness with pig-bel in Papua New Guinea.¹⁸⁰ Although polyvalent gas gangrene antiserum was ineffective, administration of type C antiserum resulted in a 30% decrease in the need for surgery, and mortality was reduced from 43% to 19%.¹⁸⁰ Furthermore, active immunization against the β -toxin has also proved effective in preventing pig-bel.¹⁸⁴

Some have suggested that type A *C. perfringens*, staphylococci, or even hepatitis virus may be responsible for EN. The syndrome of enteritis gravis has been described in association with infectious hepatitis, although no viral cause has been documented.

Considerations in the differential diagnosis of EN include acute shigellosis, acute food poisoning syndromes, antibiotic-associated pseudomembranous colitis, and acute ulcerative colitis. The absence of colonic involvement; the epidemiologic setting, especially in poorly nourished patients; and the rapid progression to toxemia and shock are strongly suggestive of EN.

Therapy for EN includes careful supportive care and bowel decompression. Fluid requirements may be substantially greater than what is indicated by fecal output. Resection of the involved bowel must be considered if there is a persistence of paralytic ileus, a rapid increase in signs of toxemia, localized or diffuse signs of peritonitis, persistent pain, or a palpable mass lesion. If subacute obstruction or malabsorption is suspected on the basis of weight loss, elective surgery may be required up to 6 months after the acute illness. Raw peanut or soybean diets should be avoided because they contain trypsin inhibitors. *C. perfringens* type C antiserum containing β -antitoxin or the active β -toxin vaccine should be available for use in areas where EN can be expected to occur. Page 1299

Chronic Infectious/Inflammatory Processes Involving the Intestinal Tract

Chronic inflammatory enteritides are often indolent, slowly progressive infections. Often, there is a history of weeks or months of fever, abdominal pain, weight loss, or other systemic manifestations. Recurring or relapsing symptoms may be seen with *C. jejuni* or *Salmonella* gastroenteritis. In addition, 16% of cases of shigellosis may become prolonged, lasting for 3 weeks or longer.¹⁸⁵ Any diarrheal illness that extends beyond 2 weeks is considered severe and is associated with a high risk of nutritional morbidity in children who live in tropical developing areas.¹⁸⁶⁻¹⁸⁸

Syphilis

Syphilis (see Chapter 243) can also involve the GI tract, usually in the upper part of the small bowel or stomach. An acute erosive and infiltrative gastritis with motile spirochetes and a positive specific response on treponemal immunofluorescence testing has been reported in late secondary syphilis.¹⁸⁹ The initial complaints are upper abdominal pain, vomiting, and weight loss. More classic are the late GI manifestations of lues: pyloric obstruction, hourglass constriction, and linitis plastica of the stomach. Less commonly, gumma may be seen in the small bowel or colon. More commonly, syphilis can cause proctitis in MSM, as the primary chancre. Anoscopy may reveal mucosal inflammation with ulcerations. Serology and if available, darkfield microscopy of the chancre would aid in the diagnosis.

Gastrointestinal Tuberculosis

Intestinal tuberculosis (TB), once considered common, had become a relatively rare disease but is now reemerging in association with HIV infections and with multidrug-resistant *Mycobacterium tuberculosis* (see Chapter 255).¹⁹⁰ Intestinal involvement with TB may be either primary, from ingestion of the organisms, or secondary (reactivation), usually from a pulmonary source.

Primary intestinal TB without pulmonary disease often results in hypertrophic mucosal changes. Patients with acute miliary TB may also have GI tract involvement.¹⁹¹ Patients with primary intestinal TB may present with abdominal pain, fever, and a tender, fixed, palpable mass in the ileocecal area. Primary hypertrophic intestinal TB continues to occur in the Middle East¹⁹² and in India.¹⁹³ Intestinal involvement secondary to pulmonary TB may result from swallowing infected sputum or from biliary excretion of the organism from an infected liver. The frequency of

secondary intestinal TB increases with far-advanced pulmonary disease. Hippocrates stated that “diarrhea attacking a person with phthisis is a mortal symptom.”

TB may involve any part of the GI tract, but most ulcerative and hypertrophic types occur in the ileocecal region, where there is a predominance of submucosal lymphatic tissue. The most common features are fever and abdominal pain that is often relieved by defecation or vomiting. Weight loss is more common in secondary intestinal TB. Only one-third of the patients with GI TB have diarrhea. Diarrhea may be related to exacerbations of abdominal pain and occasionally occurs with extensive involvement of the small intestine, which may cause steatorrhea and a malabsorption syndrome. Although ulceration and mucous diarrhea are relatively common with secondary intestinal TB, hemorrhage and the presence of gross blood in the stool are distinctly uncommon, perhaps because of the associated obliterative endarteritis.

The diagnosis of GI TB may be very difficult radiologically and even histologically. It must be distinguished from regional enteritis, sarcoidosis, actinomycosis, ameboma, carcinoma, and peri appendiceal abscess. In contrast to Crohn disease, GI TB rarely causes anal lesions, fistulas, or perforation. It is often associated with miliary nodules on the serosa, it rarely causes strictures longer than 3 cm, and it may cause circumferential transverse ulcers. TB may also cause fibrosis of the muscularis mucosa, pyloric metaplasia, and epithelial regeneration.¹⁹⁴ There may be minimal or no radiologic changes in the bowel mucosa. Small mucosal ulcerations may result in tiny, calcified nodules in the mucosa in association with calcified mesenteric lymph nodes analogous to those seen in the pulmonary Ghon complex. The ileocecal region often reveals radiologic evidence of irritability and hypermotility, with hypersegmentation of the mucosal folds or poor filling of the ileocecal region detected by barium enema. On occasion, frank ulcerations can be noted on contrast studies, and late in the course there is scarring. The diagnosis requires a careful examination of involved tissue for acid-fast bacilli by using stain and culture. Caseous necrosis is more frequently found in the mesenteric nodes than in intestinal tissue itself. Complications of intestinal TB include perforation, peritonitis, and obstruction from hypertrophy, scarring, or tuberculoma.

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Acute Dysentery Syndromes (Diarrhea With Fever)

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Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.
- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.
- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis*, or *Treponema pallidum* as a cause of proctitis, or *Campylobacter* spp., *Shigella* spp., *C. trachomatis* (lymphogranuloma venereum serotypes), or *C. difficile* as a cause of colitis.
- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest *C. difficile* infection (CDI).

Microbiology

- Genomic studies of *Shigella* spp. have indicated that *Shigella* spp. and enteroinvasive *E. coli* are derived from multiple phyletic origins of *E. coli* despite their microbiological and clinical similarities.
- The cause of an outbreak of bloody diarrhea and severe hemolytic-uremic syndrome, unlike prior EHEC strains that had exhibited enteropathogenic *E. coli* traits of attachment and effacement, is a Shiga toxin–producing enteroaggregative *E. coli* strain.
- *Campylobacter* spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.
- The primary virulence factors that are known to cause disease in CDI are the two large toxins: *C. difficile* toxin (Tcd)A, or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.
- *Vibrio parahaemolyticus* has been recognized since 1950 in Japan and is a cause of seafood poisoning.
- *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β –converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.
- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.
- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

• Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although *Shigella* and perhaps *Campylobacter* durations of infection and symptoms may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic *Salmonella* infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

Definition

Diarrhea associated with fever typically suggests an acute inflammatory enteritis caused by one of several specific infectious agents. However, noninflammatory enteritides caused by viral or other agents may also present with diarrhea and fever. This chapter will address bacterial and amebic colitis as potential major causes of this syndrome. Viral and other parasitic etiologies will be addressed in other chapters. The acute inflammatory enteritides include several specific, distal small bowel, and colonic infections, such as those caused by *Campylobacter* spp., *Salmonella* spp., *Shigella* spp., and enteroinvasive *Escherichia coli* (EIEC), as well as the syndromes of necrotizing enteritis (NE) and antibiotic-associated pseudomembranous enterocolitis (*Clostridioides difficile* [formerly *Clostridium difficile*]). Several other infectious agents cause chronic enteric inflammatory processes that may result in syndromes of abdominal pain, weight loss, diarrhea, or malabsorption. These involve disease processes such as gastrointestinal (GI) mycoses; mycobacterioses; amoebiasis, or bacterial infections, such as those caused by enteropathogenic *E. coli* (EPEC), enterohemorrhagic (EHEC), and enteroaggregative *E. coli* (EAEC); and syphilis.

Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever, implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis* (including lymphogranuloma venereum serotypes), or *Treponema pallidum* as a cause of proctitis, or *Campylobacter*, *Shigella*, or *E. histolytica* as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care

facility would strongly suggest *C. difficile* infection (CDI), although community-acquired infection has been well recognized as well.

Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}

Potential causes of acute dysentery are listed in Table 101.1.

Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes *Shigella* spp. and Enteroinvasive *Escherichia coli*

Bacillary dysentery is caused by *Shigella* spp. and EIEC. It is estimated that *Shigella* spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four *Shigella* species are classified in subgroups A to D—*Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*, respectively. Genomic studies of *Shigella* spp. have indicated that *Shigella* and EIEC are derived from multiple phyletic origins within *E. coli*.¹¹ *S. dysenteriae* (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ *S. dysenteriae* is responsible for severe and epidemic disease, and *S. boydii* is mostly seen in the Indian subcontinent. *S. flexneri* and *S. sonnei* are the most common species around the world.¹² *Shigellae* are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. *Shigella* spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² *Shigella* spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with *Shigella*, and other enteric pathogens such as EAEC, *Campylobacter*, and *Giardia*, were associated with negative linear growth, which was sustained up to 5 years.¹⁴

Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

TABLE 101.1

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes

Bacillary dysentery (*Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Shigella boydii*; invasive *Escherichia coli*)

Campylobacteriosis (*Campylobacter jejuni*)

Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis

Gonococcal (*Neisseria gonorrhoeae*)

Herpetic (herpes simplex virus)

Chlamydial (*Chlamydia trachomatis*)

Syphilitic (*Treponema pallidum*)

Other Syndromes

Necrotizing enterocolitis of the newborn

Enteritis necroticans

Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)

Diverticulitis

Typhlitis

Chronic Inflammatory Processes

Enteropathogenic and enteroaggregative *E. coli*

Syphilis

Gastrointestinal tuberculosis

Gastrointestinal mycosis (including *Basidiobolus ranarum*)

Parasitic enteritis

Syndromes Without Known Infectious Cause

Idiopathic ulcerative colitis

Crohn disease

Radiation enteritis

Ischemic colitis

Allergic enteritis

Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC

strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin-producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:K60:H11, O103:H2, O91:H2, O145:H–, O111:K58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:K75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including *aggR*, *aggA* (encoding for the biofilm-producing aggregative adherence fimbriae I [AAF/I]), *sigA*, *sepA*, *pic*, *aatA*, *aaiC*, and *aap*, but it was negative for the *eae* (*E. coli* attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

TABLE 101.2

Enteroinvasive *Escherichia coli* Serotypes
 SEROTYPE DIFCO SEROGROUP TEST^a

O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

^a Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

Enteropathogenic and Enteroaggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.^{58–60} EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.^{61–63} EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such

as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other Campylobacter spp. have been published.^{69–71} Although there currently are more than 16 Campylobacter spp. and 6 subspecies, *C. jejuni* and Campylobacter coli are the most prevalent species associated with enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of Campylobacter bloodstream infections in humans are with Campylobacter fetus (former subspecies, *C. fetus* subsp. *intestinalis*),⁷² *C. jejuni* commonly causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ Campylobacter enteritis results from oral transmission, and a variable but relatively small dose ($5\text{--}8 \times 10^2$ cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of Campylobacter enteritis have been associated with ingestion of contaminated water, raw milk, and uncooked meat and poultry. A review of common source outbreaks of Campylobacter infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by Campylobacter spp., and the specific balance between these genes and host susceptibility to Campylobacter enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; secretion of cytokines; chemokines and arachidonic acid metabolites; and production of substance P and reactive oxygen intermediates. Some *C. difficile* strains, including the epidemic ribotype 027, produce binary toxin that may have a role in increased virulence through host attachment and colonization, which type 2 immunity in the host may protect against.⁸³ Clinical presentation ranges from asymptomatic colonization to severe disease with pseudomembranous colitis (Fig. 101.1) and septic shock. Increased fecal lactoferrin, IL-1 β , and IL-8 were observed in patients with CDI.⁸⁴ Leukocytosis suggests severe infection, and leukemoid reactions have been described in colitis caused by *C. difficile*.⁸⁵ Further details of CDI are provided in Chapter 249.

Vibriosis

In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and

occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90}

FIG. 101.1 Proctoscopic view of pseudomembranous colitis in a patient who received clindamycin. Note the 4- to 8-mm raised white plaques overlying an erythematous mucosa. From Tedesco FJ, Barton RW, Alpers DH. Clindamycin-associated colitis. *Ann Intern Med.* 1974;81:429–433.

Follow for extended description

Salmonellosis and Enteric Fever

Salmonella enterocolitis is characterized by fever, cramping, abdominal pain, and diarrhea that begin 8 to 48 hours after ingestion of an infective dose, usually with food, and usually lasts 3 to 5 days (see Chapter 228). The diarrheal stools of patients with salmonellosis often contain a moderate number of polymorphonuclear leukocytes, usually fewer than is typical of shigellosis.

Although *Salmonella enteritis* predominantly involves the lamina propria in the small bowel, several reports have noted *Salmonella enterica* serovar Typhimurium as a cause of colitis, with crypt abscesses and erosion and ulcerations of the colonic mucosa, which result in blood and pus in the stool.⁹¹ Certain other strains of *Salmonella*—including *S. enterica* serovar Choleraesuis and *S. enterica* serovar Paratyphi, and especially *S. enterica* serovar Typhi—elicit a mononuclear response and cause the bacteremia that is characteristic of enteric fever. Animal models for *S. Typhimurium* continue to advance our understanding of the pathogenesis of intestinal and systemic *Salmonella* infections.⁹² *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5 (TLR5), nucleotide oligomerization domain–like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein. TLR5 is located in the plasma and phagosomal membranes, and the other two are located in the cytoplasm of the host's Peyer patch cells. Flagellin activates TLR5 and leads to the production of IL-12, whereas the activation of IPAF leads to the production of IL-1 β and IL-18. Both pathways act synergistically to activate interferon- γ -producing Th1 cells. These proinflammatory cytokines play important roles in the induction of inflammation with *Salmonella* infections.

Typhoid fever may lead to an erosion of the blood vessels in Peyer patches that, without appropriate treatment, could result in gross blood in the feces in 10% to 20% of patients. Severe intestinal hemorrhage may complicate approximately 2% of cases late in the course of untreated typhoid fever. Such intestinal bleeding may precede perforation, another complication of typhoid fever.⁹³ However, typhoid fever is primarily a systemic illness, not a diarrheal disease (see Chapter 102).

Yersiniosis

Y. enterocolitica (see Chapter 235) is an enteric pathogen that may be responsible for an enteric fever–like illness, mesenteric adenitis (which may mimic acute appendicitis), or an inflammatory ileitis or ulcerative colitis syndrome with fecal neutrophils and mononuclear cells.⁹⁴ *Y. enterocolitica* and *Yersinia pseudotuberculosis* may cause similar symptoms and signs. The virulence of *Yersinia* is associated with the direct injection of proteins, such as *Yersinia* protein kinase A (YpkA) and *Yersinia* outer protein E (YopE), into eukaryotic cells.^{95,96} *Yersinia* infection may also be associated with migratory polyarthritides, reactive arthritis, or erythema

nodosum. A syndrome of acute diarrhea and vomiting is especially common in young children. The organism may cause disseminated abscesses in the liver and spleen⁹⁷ or an inflammatory colitis.⁹⁸ The causative agent, a gram-negative member of the family Enterobacteriaceae, is in the same genus as the plague bacillus *Yersinia pestis* and is sometimes mistaken for *Proteus* on initial culture plates.

FIG. 101.2 Methylene blue stain of fecal leukocytes from a patient with colitis. This exudative response may be seen in any active colitis syndrome, including those caused by the inflammatory or invasive pathogens discussed in this chapter.

Follow for extended description

Amebiasis

Entamoeba histolyticum is one of the classic causes of dysenteric syndromes. Although the course and severity of the infection are variable, amebiasis is one infection that can cause sustained febrile diarrhea with stools becoming grossly bloody. For patients living in endemic areas or travelers who have been to such endemic areas and who have dysenteric symptoms, amebiasis should be considered in the differential diagnosis. For further details about amebiasis, please see Chapter 278.

Diagnosis

Diarrhea with fever, blood in the stool, severe abdominal pain, or signs of sepsis necessitates evaluation for enteric pathogens to guide disease management.⁹⁹ Clinical and epidemiologic information direct the specific tests to be requested. Examination for fecal leukocytes often reveals sheets of polymorphonuclear leukocytes in clumps of mucus, even in the absence of gross blood in the stool specimen (Fig. 101.2).^{100,101} Fewer pyknotic leukocytes are reported in amebic dysentery¹⁰²⁻¹⁰⁴; this may be attributable to the deeper undermining ulcers characteristic of intestinal amebiasis or to a contact-dependent cytolytic effect of the amoeba on leukocytes. The diagnosis of shigellosis is made using bacterial cultures or polymerase chain reaction (PCR) assay for the invasiveness plasmid in stool, rectal swab, or endoscopic biopsy specimens.¹⁰⁵ The use of fresh specimens promptly plated onto appropriate enteric culture media is very important to ensure the isolation of shigellae.¹⁰⁶

Specialized techniques are required to isolate *Vibrio* (thiosulfate citrate bile salts [TCBS] agar),¹⁰⁷ *Yersinia* (cold enrichment),¹⁰⁸ or *C. jejuni*.⁷³ The identification of toxigenic *C. difficile* is done by immunoassay for either *C. difficile* toxin B or PCR assay for *C. difficile* toxin B.¹⁰⁹

The diagnosis of infection with EHEC depends on culturing stool on MacConkey culture medium, on which the original O157:H7 EHEC strain yields sorbitol-negative colonies.¹¹⁰ The isolated *E. coli* is serotyped for somatic (O) and flagellar (H) antigens. Rapid methods for detection of Shiga toxins in the stool samples are now in use.^{111,112} The pathogenic Kanagawa-positive strains from patients produce β -hemolysis on special (Wagatsuma) medium—in contrast to environmental isolates—and are best isolated as blue-green colonies (alkaline) on TCBS agar.¹⁰⁷ A PCR method provides a species-specific probe for *V. parahaemolyticus*, *V. alginolyticus*, *Vibrio anguillarum*, and a cluster of related species.¹¹³ Cultivation may require cold enrichment techniques.¹⁰⁸ PCR methods for detection of pathogenic *Y. enterocolitica* are more sensitive than culture methods.¹¹⁴

Molecular approaches, including multiple pathogen nucleic acid amplification tests (NAAT) for simultaneous detection of multiple enteropathogens, have high sensitivity and specificity and are widely used for surveillance or clinical purposes.¹¹⁵ In outpatient settings, these have largely replaced culture methods in the United States, Canada, and western Europe. Not only do the NAAT approaches allow for higher sensitivity, but the turnaround time is also faster, facilitating hospital infection control measures and initiation of appropriate intervention. The broad, multiplex approach allows for a screening diagnostic approach for enteric pathogens, including viruses, bacteria, and parasites, without having to test for individual pathogens separately.¹¹⁶ There are several limitations inherent to this advanced molecular assay. In addition to relatively high cost, interpretation of the results of the assay is challenging. Detection of a pathogen does not necessarily mean identification of the cause of dysentery, as the test is not able to differentiate colonization from clinically relevant infection. Another reason is the high frequency of polymicrobial infections detected by this method, with up to 30% of positive samples detecting

multiple pathogens, which can make determination of the true pathogen challenging.^{38,117,118} Additionally, specific pathogens such as EPEC or EAEC tend to be detected more commonly as observed in one study where both organisms were positive in 84% of the multipathogen samples.¹¹⁷ Another study showed a false positive rate around 20% with NAAT for organisms like *C. difficile*, EPEC, EAEC, and *Campylobacter*, which was largely caused by cross-reactivity with commensal organisms.³⁸ Quantitation of the burden of infection may assist in the interpretation of results from these molecular assays.¹¹⁹ Moreover, current commercially available GI enteric panels do not allow for detection of antibiotic resistance. The steady increase in the prevalence of antimicrobial resistant diarrhea-causing pathogens indicates the continuing need for traditional microbiologic methods for susceptibility testing to guide treatment.

Amebic dysentery is traditionally diagnosed by direct examination of wet mounts of fresh fecal or proctoscopic specimens, which reveal *E. histolytica* trophozoites or cysts characterized by four or fewer delicate nuclei with central karyosomes. Additional patients may be diagnosed by biopsy, which may reveal periodic acid–Schiff stain (PAS)-positive trophozoites or cysts in the undermining ulcer in the lamina propria, or by a fecal or serum *E. histolytica*-specific antigen test, which distinguishes virulent *E. histolytica* from avirulent *Entamoeba dispar* infections.¹²⁰ The use of multiplex NAAT panels including *E. histolytica* is largely replacing the historical diagnostic approaches to amoebiasis.¹²¹

Sigmoidoscopic examination, especially with biopsy, may be useful in the diagnosis of a pseudomembranous enterocolitis or in the identification of parasites such as *E. histolytica* (with a special PAS stain) or *B. coli*. Amebic colitis is associated with discrete small ulcerations with undermined edges amid relatively normal mucosa. Acute shigellosis causes more widespread, shallow, 3- to 7-mm ulcers with a more intense inflammatory exudate.

Other considerations in the differential diagnosis of inflammatory colitis include the potentially rapidly progressive necrotizing enterocolitis (NEC) syndromes (see Chapters 81 and 249) and noninfectious syndromes such as inflammatory bowel diseases, ulcerative colitis, and Crohn disease. These diagnoses are suggested by the setting, clinical course, history, and findings on radiologic and endoscopic examinations.

Finally, the presence of biomarkers of acute or prolonged intestinal inflammation, such as fecal myeloperoxidase and neopterin, may be associated with growth shortfalls in young children in developing areas.¹²²

Therapy

Therapy consists of supportive fluid and electrolyte management with antimicrobial therapy directed at a specific pathogen if suspected on Page 1297the basis of the epidemiologic setting or microbiologic test results. Fluid resuscitation is the primary treatment of all acute diarrheal disease syndromes. Oral rehydration salt solution (World Health Organization formula) is recommended for oral rehydration therapy.¹²³ Presumptive therapy for the inflammatory colitides varies greatly with the different organisms and is influenced by the increasing resistance of many enteric pathogens to antimicrobial therapy.¹²⁴ For example, an acute febrile dysenteric illness in a young child with daycare exposure or in an area in which shigellosis is common should be treated with an antimicrobial agent, such as one of the fluoroquinolones. Early administration of empirical antibiotic treatment is justified in children hospitalized for clinical dysentery.¹²⁵ Short-course (3 days) oral ciprofloxacin for *S. dysenteriae* type 1 dysentery in children has been effective.¹²⁶ If the *Shigella* organism is sensitive, prompt therapy can successfully stop the diarrhea, alleviate systemic symptoms, and reduce shedding of the organisms in the feces.^{127–129} Because shigellae are increasingly resistant to multiple antibiotics,^{130,131} the practitioner must be familiar with the local resistance pattern of shigellae to treat acute shigellosis appropriately when it is first suspected. Multidrug resistant, even extensively resistant, *Shigella* infections have been reported among MSM.^{132,133} Although ciprofloxacin may shorten the clinical course of *Salmonella* infections, it does not eradicate the organism.¹³⁴ Studies have shown increasing resistance of *Campylobacter* and even *Salmonella* to quinolones.^{135,136} Azithromycin may reduce the duration of symptoms of *Campylobacter* or *Shigella* enteritis. Zinc 20 mg daily (10 mg of elemental zinc in 5 mL) given in two equally divided doses for 2 weeks showed a significant reduction in the duration of acute shigellosis, promoted better weight gain, and reduced diarrheal

morbidity in malnourished children at 6 months after infection.¹³⁷ Vitamin A has also been used to reduce severity and mortality from shigellosis.¹³⁸

Antibiotic therapy has not been shown to be effective for disease caused by EHEC or EIEC. Some EHEC strains may increase their phage-mediated production of Shiga-like toxins when exposed to antibiotics to which they are sensitive, such as ampicillin, tetracycline, sulfamethoxazole-trimethoprim, or quinolones.¹³⁹⁻¹⁴²

Most patients with mild-to-moderate *C. jejuni* enterocolitis do not benefit from antibiotic treatment. Severely ill patients, especially those who are debilitated or immunocompromised, appear to benefit from antibiotic therapy, and erythromycin (or other macrolides) or ciprofloxacin are the drugs of choice. Increased fluoroquinolone resistance in *Campylobacter* spp. from humans and animals, the result of mutations in topoisomerase genes, has been well documented.¹⁴³ Two studies now associate symptomatic and asymptomatic *Campylobacter* infections with reduced growth in Brazilian and Peruvian children.^{144,145}

Vancomycin and fidaxomicin are the drugs of choice for initial episode of CDI (see Chapter 249 for further details).^{146,147} Recurrent disease occurs in 20% to 65% of the patients, depending on the number of previous episodes of the disease.¹⁴⁸ Fidaxomicin is reported to be equivalent to vancomycin in treating acute CDI but may be superior in preventing relapses.¹⁴⁹ However, its advantage over vancomycin is lost in infections caused by the prevalent epidemic strain BI/NAP1/027. Alternative approaches to prevent recurrent disease include a tapering course of vancomycin; fecal microbiota transplantation (FMT)¹⁵⁰; or use of monoclonal antibodies.¹⁵¹ FMT is increasingly being used to prevent recurrent disease or to treat refractory CDI. More recently, the monoclonal antibody against TcdB has been shown to confer modest protection against recurrent CDI.⁹⁹ A vancomycin tapering course has been shown to be effective for treatment, and in some observational studies equal to FMT in preventing recurrence.¹⁵²

Antibiotic treatment with tetracycline or doxycycline can diminish the duration and severity of GI symptoms caused by *V. parahaemolyticus*. To prevent *V. parahaemolyticus* infections, consumption of raw or undercooked shellfish and exposure of wounds to seawater should be avoided.¹⁵³

Antibiotic therapy is not required for *Salmonella* enteritis, but patients who are severely ill or those with risk factors for developing extraintestinal spread of infection should be treated with fluoroquinolones or third-generation cephalosporins.

Yersinia infection is usually self-limiting; however, patients with severe illness or systemic infection are usually treated with tetracycline, chloramphenicol, or ciprofloxacin.

Prevention

Enteric pathogens are predominantly spread through contaminated food or water and sometimes via direct fecal-oral spread. Spread of infection occurs during ingestion of contaminated food or water or contact with a contaminated environment. Infection control measures include hand hygiene, proper food handling, and access to clean water. In some instances, isolation and treatment of infected individuals are necessary to prevent spread of infection, for example, in cases of CDI. Targeted health interventions for sexual networks of MSM are needed.

Immunization of susceptible individuals is an important component of prevention. However, for bacterial enteric pathogens, vaccine is only available for typhoid fever and cholera. Typhoid vaccination is recommended for people traveling to endemic areas. Ty21a is a live-attenuated oral vaccine containing *S. Typhi* in capsule formulation. Another typhoid vaccine is the parenteral Vi-polysaccharide. For individuals traveling to areas with recent or ongoing cholera outbreaks, CVD 103-HgR, a live-attenuated oral vaccine, is available. There is a need to develop effective vaccines for most infectious agents causing diarrhea worldwide.

Other Acute Infectious/Inflammatory Processes Involving the Intestinal Tract Gonococcal Proctitis

Neisseria gonorrhoeae (see Chapter 217) may cause ulcerative proctitis, usually acquired by anal intercourse.⁷ The resultant purulent proctitis is accompanied by an erythematous, friable mucous membrane in the rectal vault and occasional abscess or fistula formation. Although copious purulent discharge, tenesmus, and burning rectal pain may be noted, two-thirds of culture-positive patients with anorectal gonococcal infection are asymptomatic.¹⁵⁴ Additional diagnostic possibilities in cases of venereally acquired proctitis are syphilitic, herpetic, and chlamydial proctitis.¹⁵⁵

Unusual Causes of Dysentery

Spirillar or spirochetal dysentery has been reported to occur in southern France and has been attributed to *Spirillum* spp. Although severe mucoid diarrhea or dysentery has been associated with intestinal spirochetes, their frequency and role in causing enteric disease are unclear. A DNA probe for the 16S ribosomal RNA of the agent of swine dysentery, *Treponema hyodysenteriae*, has been developed and may open new approaches to the recognition of similar infections in humans.¹⁵⁶

Other unusual causes of colitis include brucellosis¹⁵⁷ and adenovirus infections.^{158,159}

Necrotizing Enterocolitis in the Newborn

The syndrome of diffuse, fulminating, necrotizing colitis has been recognized in infants since the 1960s. This syndrome probably represents the same entity described as spontaneous intestinal perforation and peritonitis as early as 1838.¹⁶⁰ Although milder forms of the syndrome doubtless exist, the syndrome of NEC is defined by the presence of air in the wall of the intestine, portal venous system, or peritoneal cavity, or by necrosis of the bowel wall with mucosal sloughing. This fulminant syndrome often leads to intestinal perforation, peritonitis, and bacteremia. It is a major cause of mortality in low-birth-weight infants (<1500 g) after the first week of life.¹⁶¹ The diffuse necrotic changes that characterize this syndrome most often occur in the terminal ileum but may be seen in the colon or in the proximal portion of the GI tract.

Hypoxia-ischemia was previously thought to be the main mechanism for development of NEC, but this is uncertain,¹⁶² but current research suggests multifactorial pathogenesis. The three major risk factors associated with development of NEC are prematurity, bacterial colonization of the gut, and formula feeding¹⁶³; the pathogenesis involves a combination of immature intestinal barrier and abnormal intestinal microbiome.¹⁶⁴ Intestinal immaturity refers to Page 1298motility, absorption, barrier function, and circulatory regulation, but the most important aspect seems to be an excessive immune response to the gut microbiota which contributes to the development of NEC along with a dysbiotic intestinal microbiome. Studies in cell cultures and animal models suggest that the immature intestinal epithelium has an excessive inflammatory response to the microbiota.^{165,166} Studies comparing babies with NEC with control, or before and after development of NEC, suggests that a reduced diversity of the fecal microbiome and presence of opportunistic intestinal microbial species are associated with development of NEC.^{167,168}

Clinical features of this serious condition in newborn infants include apneic spells, vomiting, abdominal distention, and occasionally bloody diarrhea. Most infants are younger than 1 week, and there is an association with prematurity, maternal infections during delivery (e.g., amnionitis with prolonged postrupture of membranes), and exchange transfusion via the umbilical vein. There is no gender or seasonal predilection. The disease often progresses rapidly to intestinal perforation, shock, septicemia, and pneumatosis intestinalis. Air may also be evident in the portal venous system or biliary tract on plain radiographs. This syndrome is associated with mortality often greater than 70%.

The diagnosis of NEC should be considered in any premature infant with altered GI function, abdominal distention, or apneic spells. Further investigation should include examination of the stool for occult blood and for the presence of reducing substances. Plain abdominal radiographs may reveal air in the bowel wall, peritoneal cavity, or portal venous system, and there may be bloody diarrhea late in the course of the disease. Management must be initiated early and aggressively for any infant suspected of having NEC. Umbilical catheters should be removed, oral feeding should be stopped, and nasogastric aspiration should be initiated. Intravenous fluid therapy is of paramount importance. After obtaining specimens for blood culture, broad-spectrum intravenous antibiotics is recommended. Laparotomy and excision of the necrotic bowel are often

necessary and should be done aggressively if there is any evidence of peritonitis or obstruction.¹⁶⁹

Prevention of NEC includes promotion of intestinal development, avoidance of risk factors and careful infection control measures in newborn ICUs. Antenatal corticosteroids, given for prevention of respiratory distress in the premature newborn, also promote intestinal development and appears protective against NEC development.¹⁷⁰ Hypertonic elemental formulas have been implicated and should be avoided in high-risk patients. Similarly, prolonged antibiotic use and gastric acid suppression increase NEC risk and should be avoided.¹⁷¹⁻¹⁷³ Conversely, use of human milk for preterm infants lowers the risk of NEC.^{174,175} Explanations for the advantage of human breast milk include the presence of lysozyme, antibodies, and cellular elements that may play a protective role against potential infectious agents, as well as promotion of growth for beneficial bacteria in the gut by human milk oligosaccharides (HMOs).^{176,177} Although oral prophylactic nonabsorbable antibiotics have been suggested, serious questions remain about the use of prophylactic antibiotics, even in high-risk newborn infants weighing less than 1500 g. A few reports have shown that epidermal growth factor, which is found in high concentrations in breast milk, may have a cytoprotective effect, and probiotics may decrease or prevent neonatal NEC.^{178,179}

Darmbrand, Pig-Bel, Necrotizing Enteritis in Adults (Enteritis Necroticans)
First described as Darmbrand (meaning fire bowels) in epidemics of enteritis necroticans (EN) in northern Germany in the immediate postwar period in the mid-1940s, a severe necrotizing jejunitis has also been recognized in epidemic and sporadic forms after pork feasting in the highlands of Papua New Guinea.¹⁸⁰ Pig-bel was the name given to the syndrome of abdominal discomfort that followed a large pork meal, commonly eaten after a large pig kill, which takes place every 3 to 10 years among the highland Melanesians of Papua New Guinea. Sporadic cases have been reported from other parts of the world, including the United States.¹⁸¹

The pathologic findings are those of acute, patchy, necrotizing disease of the small bowel in previously healthy people, which may proceed rapidly to segmental gangrene, with small amounts of gas in the mucosa, mesentery, or nodes.

Several theories of pathogenesis have been suggested, most of which involve the toxic products of *Clostridium perfringens* type C, including α - and β -toxins. Sporadic cases of EN have been noted in association with nutritional disorders, alcoholism, and malabsorption, and after pancreatic or gastric resection.¹⁸² After gastric surgery, increased numbers of *C. perfringens* and increased levels of α -toxin have been noted in the upper small bowel and stomach. Whether α - or β -toxins are capable of causing EN alone, or whether they initiate the invasion of the mucosa by other organisms, such as gram-negative rods, is currently unclear. An attractive hypothesis was suggested by Lawrence and Walker,¹⁸² which could explain the association of EN with poor nutrition and episodic dietetic overindulgence. The low-protein diet of Papua New Guinea highlanders is associated with low levels of digestive proteases in the intestinal lumen, which inactivate the β -toxin. The proteases can be further blocked by the oral intake of trypsin inhibitors, which are found in this population in such dietary staples as sweet potatoes. Proteases return with improved diet, as occurred in postwar Germany. This hypothesis has been confirmed in an animal model that required protease inhibitors for symptomatic infection.¹⁸³

The clinical syndromes of EN range from anorexia, vomiting, severe abdominal pain, and bloody diarrhea to fulminant toxemia and shock. Acute complications that necessitate emergency surgery include paralytic ileus, bowel strangulation, and bowel perforation with peritonitis. These complications are common in the first 2 weeks of illness. Later complications that may also necessitate surgery include scarring, leading to stenosis; obstruction; malabsorption; and fistulas. EN occurs with greater frequency and greater severity in children younger than 10 years. In contrast to European control subjects who rarely have antibodies, 70% of the healthy adults in Papua New Guinea have detectable antibody to clostridial β -toxin.^{180,182}

The syndrome is defined by the pathologic findings but should be suspected in patients who develop severe abdominal pain, bloody diarrhea, ileus, and toxemia. The course is often too fulminant for radiographic detection of air in the bowel wall to be of any diagnostic value.

Agents held responsible for causing EN include *C. perfringens* type C, once designated as type F in the older classification of *Clostridium welchii*. Most surgically resected bowel specimens from patients with EN contain *C. perfringens*, more than half of which are type C. Furthermore, 12 of 21 cases described had a significant change in serum β -antitoxin titer after illness with pig-bel in Papua New Guinea.¹⁸⁰ Although polyvalent gas gangrene antiserum was ineffective, administration of type C antiserum resulted in a 30% decrease in the need for surgery, and mortality was reduced from 43% to 19%.¹⁸⁰ Furthermore, active immunization against the β -toxin has also proved effective in preventing pig-bel.¹⁸⁴

Some have suggested that type A *C. perfringens*, staphylococci, or even hepatitis virus may be responsible for EN. The syndrome of enteritis gravis has been described in association with infectious hepatitis, although no viral cause has been documented.

Considerations in the differential diagnosis of EN include acute shigellosis, acute food poisoning syndromes, antibiotic-associated pseudomembranous colitis, and acute ulcerative colitis. The absence of colonic involvement; the epidemiologic setting, especially in poorly nourished patients; and the rapid progression to toxemia and shock are strongly suggestive of EN.

Therapy for EN includes careful supportive care and bowel decompression. Fluid requirements may be substantially greater than what is indicated by fecal output. Resection of the involved bowel must be considered if there is a persistence of paralytic ileus, a rapid increase in signs of toxemia, localized or diffuse signs of peritonitis, persistent pain, or a palpable mass lesion. If subacute obstruction or malabsorption is suspected on the basis of weight loss, elective surgery may be required up to 6 months after the acute illness. Raw peanut or soybean diets should be avoided because they contain trypsin inhibitors. *C. perfringens* type C antiserum containing β -antitoxin or the active β -toxin vaccine should be available for use in areas where EN can be expected to occur. Page 1299

Chronic Infectious/Inflammatory Processes Involving the Intestinal Tract

Chronic inflammatory enteritides are often indolent, slowly progressive infections. Often, there is a history of weeks or months of fever, abdominal pain, weight loss, or other systemic manifestations. Recurring or relapsing symptoms may be seen with *C. jejuni* or *Salmonella* gastroenteritis. In addition, 16% of cases of shigellosis may become prolonged, lasting for 3 weeks or longer.¹⁸⁵ Any diarrheal illness that extends beyond 2 weeks is considered severe and is associated with a high risk of nutritional morbidity in children who live in tropical developing areas.¹⁸⁶⁻¹⁸⁸

Syphilis

Syphilis (see Chapter 243) can also involve the GI tract, usually in the upper part of the small bowel or stomach. An acute erosive and infiltrative gastritis with motile spirochetes and a positive specific response on treponemal immunofluorescence testing has been reported in late secondary syphilis.¹⁸⁹ The initial complaints are upper abdominal pain, vomiting, and weight loss. More classic are the late GI manifestations of lues: pyloric obstruction, hourglass constriction, and linitis plastica of the stomach. Less commonly, gumma may be seen in the small bowel or colon. More commonly, syphilis can cause proctitis in MSM, as the primary chancre. Anoscopy may reveal mucosal inflammation with ulcerations. Serology and if available, darkfield microscopy of the chancre would aid in the diagnosis.

Gastrointestinal Tuberculosis

Intestinal tuberculosis (TB), once considered common, had become a relatively rare disease but is now reemerging in association with HIV infections and with multidrug-resistant *Mycobacterium tuberculosis* (see Chapter 255).¹⁹⁰ Intestinal involvement with TB may be either primary, from ingestion of the organisms, or secondary (reactivation), usually from a pulmonary source.

Primary intestinal TB without pulmonary disease often results in hypertrophic mucosal changes. Patients with acute miliary TB may also have GI tract involvement.¹⁹¹ Patients with primary intestinal TB may present with abdominal pain, fever, and a tender, fixed, palpable mass in the ileocecal area. Primary hypertrophic intestinal TB continues to occur in the Middle East¹⁹² and in India.¹⁹³ Intestinal involvement secondary to pulmonary TB may result from swallowing infected sputum or from biliary excretion of the organism from an infected liver. The frequency of

secondary intestinal TB increases with far-advanced pulmonary disease. Hippocrates stated that “diarrhea attacking a person with phthisis is a mortal symptom.”

TB may involve any part of the GI tract, but most ulcerative and hypertrophic types occur in the ileocecal region, where there is a predominance of submucosal lymphatic tissue. The most common features are fever and abdominal pain that is often relieved by defecation or vomiting. Weight loss is more common in secondary intestinal TB. Only one-third of the patients with GI TB have diarrhea. Diarrhea may be related to exacerbations of abdominal pain and occasionally occurs with extensive involvement of the small intestine, which may cause steatorrhea and a malabsorption syndrome. Although ulceration and mucous diarrhea are relatively common with secondary intestinal TB, hemorrhage and the presence of gross blood in the stool are distinctly uncommon, perhaps because of the associated obliterative endarteritis.

The diagnosis of GI TB may be very difficult radiologically and even histologically. It must be distinguished from regional enteritis, sarcoidosis, actinomycosis, ameboma, carcinoma, and peri appendiceal abscess. In contrast to Crohn disease, GI TB rarely causes anal lesions, fistulas, or perforation. It is often associated with miliary nodules on the serosa, it rarely causes strictures longer than 3 cm, and it may cause circumferential transverse ulcers. TB may also cause fibrosis of the muscularis mucosa, pyloric metaplasia, and epithelial regeneration.¹⁹⁴ There may be minimal or no radiologic changes in the bowel mucosa. Small mucosal ulcerations may result in tiny, calcified nodules in the mucosa in association with calcified mesenteric lymph nodes analogous to those seen in the pulmonary Ghon complex. The ileocecal region often reveals radiologic evidence of irritability and hypermotility, with hypersegmentation of the mucosal folds or poor filling of the ileocecal region detected by barium enema. On occasion, frank ulcerations can be noted on contrast studies, and late in the course there is scarring. The diagnosis requires a careful examination of involved tissue for acid-fast bacilli by using stain and culture. Caseous necrosis is more frequently found in the mesenteric nodes than in intestinal tissue itself. Complications of intestinal TB include perforation, peritonitis, and obstruction from hypertrophy, scarring, or tuberculoma.

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101

Acute Dysentery Syndromes (Diarrhea With Fever)

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Short View Summary

Definition

- Diarrhea associated with fever is typically seen in acute and chronic inflammatory enteritides that are caused by several specific infectious agents.

Epidemiology and Clinical Signals and Symptoms

- Acute dysenteric syndromes can be caused by any of several diverse pathogens, depending on exposure. *Campylobacter* spp., *Shigella* spp., and *Salmonella* spp. often cause febrile illnesses (with or without bloody stools) acquired in settings of suboptimal sanitation or food handling, as can be enterohemorrhagic *Escherichia coli* (EHEC), but with less likelihood of fever. *Shigella* spp. can be spread person to person by low inocula, and recent antibiotic or hospital exposure makes *Clostridioides difficile* (formerly *Clostridium difficile*) more likely.
- The bacterial enteric pathogens most associated with illnesses in children younger than 5 years in the United States are nontyphoidal *Salmonella* spp., followed by *Campylobacter* spp., *Yersinia enterocolitica*, and *E. coli* O157. In developing countries, norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. have the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, *Shigella* spp., and *Campylobacter* spp., and *Shigella* spp. were primarily associated with bloody diarrhea, and fever and vomiting were associated with rotavirus and vomiting with norovirus GII.

- Venereal exposure, particularly among men who have sex with men, may implicate gonococci, herpes simplex virus, Chlamydia trachomatis, or Treponema pallidum as a cause of proctitis, or Campylobacter spp., Shigella spp., C. trachomatis (lymphogranuloma venereum serotypes), or C. difficile as a cause of colitis.
- A history of antibiotic intake, colonic instrumentation, and/or recent admission to a health care facility strongly suggest C. difficile infection (CDI).

Microbiology

- Genomic studies of Shigella spp. have indicated that Shigella spp. and enteroinvasive E. coli are derived from multiple phyletic origins of E. coli despite their microbiological and clinical similarities.
- The cause of an outbreak of bloody diarrhea and severe hemolytic-uremic syndrome, unlike prior EHEC strains that had exhibited enteropathogenic E. coli traits of attachment and effacement, is a Shiga toxin-producing enteroaggregative E. coli strain.
- Campylobacter spp. have a small genome (1.6–2.0 megabases) and multiple species can cause both intestinal and systemic infections.
- The primary virulence factors that are known to cause disease in CDI are the two large toxins: C. difficile toxin (Tcd)A, or toxin A (308 kilodaltons [kDa]), and TcdB, or toxin B (270 kDa), although TcdB may be more clinically relevant than TcdA.
- Vibrio parahaemolyticus has been recognized since 1950 in Japan and is a cause of seafood poisoning.
- Salmonella flagellin is regulated by the fliC gene, which is the major ligand for Toll-like receptor 5, nucleotide oligomerization domain-like receptors, and interleukin-1 β -converting enzyme (ICE) protease-activating factor (IPAF) protein.

Diagnosis

- Any of the above-mentioned microorganisms may cause an acute dysentery syndrome with blood and/or pus in the stool.
- Examination for leukocytes or for fecal lactoferrin, myeloperoxidase, or calprotectin may suggest intestinal inflammation, even if blood is not present in the stool on gross examination.
- Multiple pathogen nucleic acid amplification tests for simultaneous detection of several enteropathogens with high accuracy, sensitivity, and specificity are increasingly being used for surveillance or clinical diagnosis.

Therapy and Prevention

- Fluid and electrolyte replacement are the cornerstones for treatment of all diarrheal diseases. Because there are many etiologic agents, the treatment and prevention rely on each specific cause of acute and chronic inflammatory enteritides, which are viewed in more detail in this chapter. Although Shigella and perhaps Campylobacter durations of infection and symptoms may be reduced by antimicrobials to which they are susceptible, antibiotics may also be indicated for systemic Salmonella infections for which foreign intravascular material or prostheses or severe atherosclerosis may pose increased risk. Some antimicrobials may worsen EHEC toxin production and illness severity.

Definition

Diarrhea associated with fever typically suggests an acute inflammatory enteritis caused by one of several specific infectious agents. However, noninflammatory enteritides caused by viral or other agents may also present with diarrhea and fever. This chapter will address bacterial and amebic colitis as potential major causes of this syndrome. Viral and other parasitic etiologies will be addressed in other chapters. The acute inflammatory enteritides include several specific, distal small bowel, and colonic infections, such as those caused by Campylobacter spp., Salmonella spp., Shigella spp., and enteroinvasive Escherichia coli (EIEC), as well as the syndromes of necrotizing enteritis (NE) and antibiotic-associated pseudomembranous enterocolitis (Clostridioides difficile [formerly Clostridium difficile]). Several other infectious agents cause chronic enteric inflammatory processes that may result in syndromes of abdominal pain, weight loss, diarrhea, or malabsorption. These involve disease processes such as gastrointestinal (GI) mycoses; mycobacterioses; amoebiasis, or bacterial infections, such as those caused by enteropathogenic E. coli (EPEC), enterohemorrhagic (EHEC), and enteroaggregative E. coli (EAEC); and syphilis.

Epidemiology

Syndromes of acute dysentery with fecal blood and pus have been well recognized since the days of Hippocrates. Dysentery is defined as frequent small bowel movements accompanied by blood and mucus, with tenesmus, defined as pain on defecation. This syndrome, with or without fever,

implies an inflammatory invasion of the colonic mucosa resulting from bacterial, cytotoxic, or parasitic tissue injury.

The epidemiologic patterns of acute dysenteric syndromes are influenced by the unusually low inoculum for infection required by organisms such as shigellae or amoebae. As few as 100 shigellae or 10 cysts of enteric protozoa, such as *Entamoeba coli*, have been found to result in infection in adult volunteers.^{1,2} In consequence, there is a substantial risk of person-to-person spread in daycare centers,³ institutions, or other areas where nonhygienic conditions may allow direct fecal-oral spread. The cysts of parasites such as *Entamoeba histolytica* or *Balantidium coli* often resist chlorination and therefore also may cause waterborne outbreaks of dysenteric illnesses. Saltwater or seafood exposure should lead to consideration of *Vibrio parahaemolyticus* as a potential cause of inflammatory colitis or of watery diarrhea, and farm or domestic animal exposure might lead to consideration of nontyphoid *Salmonella* spp., *Campylobacter jejuni* or *Campylobacter coli*, or *Yersinia enterocolitica*. A study in children younger than 5 years in the United States found nontyphoid *Salmonella*, followed by *Campylobacter*, *Y. enterocolitica*, and *E. coli* O157, were the most common bacterial enteric pathogens associated with illnesses.⁴ When typhoid fever is present with diarrhea in an endemic area, the diarrhea is often inflammatory, with many fecal polymorphonuclear or mononuclear leukocytes seen on microscopic examination.⁵ A multisite birth cohort study (Malnutrition and Enteric Disease Study [MAL-ED]) showed that norovirus GII, rotavirus, *Campylobacter* spp., astrovirus, and *Cryptosporidium* spp. exhibited the highest attributable burdens of diarrhea in the first year of life. In the second year of life, the major pathogens were *Campylobacter* spp., norovirus GII, rotavirus, astrovirus, and *Shigella* spp.⁶ *Campylobacter* spp. and EIEC were associated with dysentery. Rotavirus and *Shigella* spp. were associated with fever, whereas rotavirus and norovirus GII were most associated with vomiting.

A history of travel to areas of poor sanitation may implicate any of the listed pathogens. Finally, venereal exposure, particularly among men who have sex with men (MSM), may implicate gonococci, herpes simplex virus, *Chlamydia trachomatis* (including lymphogranuloma venereum serotypes), or *Treponema pallidum* as a cause of proctitis, or *Campylobacter*, *Shigella*, or *E. histolytica* as a cause of colitis.⁷ A history of antibiotic intake or recent admission to a health care facility would strongly suggest *C. difficile* infection (CDI), although community-acquired infection has been well recognized as well.

Although not grossly inflammatory, the still enigmatic Brainerd diarrhea syndrome, which has been associated with raw milk or unchlorinated water consumption in several outbreaks, may involve some proximal colonic lymphocytic infiltration.^{8,9}

Potential causes of acute dysentery are listed in Table 101.1.

Microbiology and Pathogenesis of the Major Causes of Acute Dysenteric Syndromes *Shigella* spp. and Enteroinvasive *Escherichia coli*

Bacillary dysentery is caused by *Shigella* spp. and EIEC. It is estimated that *Shigella* spp. (see Chapter 229) infected more than 200 million people and caused 650,000 deaths each year worldwide,¹⁰ but with proper control measures and improved treatments, the number of deaths has fallen by two-thirds. The four *Shigella* species are classified in subgroups A to D—*Shigella dysenteriae*, *Shigella flexneri*, *Shigella boydii*, and *Shigella sonnei*, respectively. Genomic studies of *Shigella* spp. have indicated that *Shigella* and EIEC are derived from multiple phyletic origins within *E. coli*.¹¹ *S. dysenteriae* (type 1) is capable of elaborating a potent toxin with enterotoxic, cytotoxic, and neurotoxic properties (Shiga toxin).¹¹ *S. dysenteriae* is responsible for severe and epidemic disease, and *S. boydii* is mostly seen in the Indian subcontinent. *S. flexneri* and *S. sonnei* are the most common species around the world.¹² *Shigellae* are facultative intracellular pathogens and may cause acute bloody dysentery with high fever and systemic manifestations of malaise, headache, and abdominal pain. The incubation period ranges from 6 hours to 9 days but is usually less than 72 hours. *Shigella* spp. is the most common cause of bloody diarrhea in children, and the syndrome may be particularly severe in poorly nourished children.¹² *Shigella* spp. is an important enteric pathogen with highest incidence in the second year of life.¹³ Subclinical infection with *Shigella*, and other enteric pathogens such as EAEC, *Campylobacter*, and *Giardia*, were associated with negative linear growth, which was sustained up to 5 years.¹⁴

Despite the intense superficial destructive process in the colonic epithelium that typifies acute shigellosis, bacteremia and disseminated infection are rare.¹⁵ All *Shigella* species invade and cause inflammatory destruction of the human colonic epithelium. The cell and tissue invasion results from a type III secretion system that delivers effector proteins into target eukaryotic cells.¹⁶ The acute infection by *S. dysenteriae* type 1 induces apoptotic cell death in colonic tissues associated with increased production of Fas/Fas ligand, perforin, caspase-1, and caspase-3, but reduced production of B-cell lymphoma 2 (Bcl-2) and interleukin-2 (IL-2).¹⁷ *Shigella* infection, especially with *S. dysenteriae* type 1, is associated with enteric protein loss that ceases after appropriate antimicrobial therapy.¹⁸ This protein loss may contribute to increased susceptibility to secondary infections or growth stunting.^{19,20} A complication of severe shigellosis in childhood is the hemolytic-uremic syndrome (HUS), which may be associated with a leukemoid reaction, pseudomembranous colitis, circulating immune complexes, and circulating endotoxin, usually in the absence of demonstrable bacteremia.²¹

Intestinal obstruction, which occurs uncommonly, is a poor prognostic sign, not infrequently associated with death or the development of HUS.²² Other, more common extraintestinal manifestations of shigellosis are headache, meningismus, and occasionally seizures, especially in young children.²³ These findings may relate to a neurotoxin that has been demonstrated with *S. dysenteriae* type 1,^{24,25} but may be seen with any *Shigella* infection. A serious arthritis similar to that seen in reactive arthritis has been described in up to 10% of patients 2 to 5 weeks after the dysenteric illness that characteristically occurs in patients with human leukocyte antigen (HLA)-B27.²⁶ Culture-positive conjunctivitis during acute shigellosis has also been described and may represent autoinoculation of the conjunctiva analogous to that induced in guinea pigs in the Sereny test.²⁷ Arthritis syndromes have also been described after inflammatory colitis with *Campylobacter jejuni*, *Y. enterocolitica*, *Salmonella enteritidis*, or *C. difficile*, again in association with HLA-B27.²⁸

TABLE 101.1

Differential Diagnosis of Acute Bacterial Dysentery and Inflammatory Enterocolitis

Specific Infectious Processes

Bacillary dysentery (*Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Shigella boydii*; invasive *Escherichia coli*)

Campylobacteriosis (*Campylobacter jejuni*)

Amebic dysentery (*Entamoeba histolytica*)

Ciliary dysentery (*Balantidium coli*)

Vibriosis (*Vibrio parahaemolyticus*)

Salmonellosis (*Salmonella enterica* serovar Typhimurium)

Typhoid fever (*S. enterica* serovar Typhi)

Enteric fever (*S. enterica* serovar Choleraesuis, *S. enterica* serovar Paratyphi)

Yersiniosis (*Yersinia enterocolitica*)

Proctitis

Gonococcal (*Neisseria gonorrhoeae*)

Herpetic (herpes simplex virus)

Chlamydial (*Chlamydia trachomatis*)

Syphilitic (*Treponema pallidum*)

Other Syndromes

Necrotizing enterocolitis of the newborn

Enteritis necroticans

Pseudomembranous colitis or *Clostridioides difficile* (formerly *Clostridium difficile*) colitis without overt pseudomembranes (*C. difficile*)

Diverticulitis

Typhlitis

Chronic Inflammatory Processes

Enteropathogenic and enteroaggregative *E. coli*

Syphilis

Gastrointestinal tuberculosis

Gastrointestinal mycosis (including *Basidiobolus ranarum*)

Parasitic enteritis
Syndromes Without Known Infectious Cause
Idiopathic ulcerative colitis
Crohn disease
Radiation enteritis
Ischemic colitis
Allergic enteritis
Brainerd diarrhea

Certain *E. coli* strains may produce a syndrome identical to that seen with acute shigellosis. The incubation period is usually 2 to 3 days after ingestion. Although invasive *E. coli* organisms appear to be limited to certain serotypes (Table 101.2),^{27,29,30} to confirm the presence of such organisms, their invasive potential should be demonstrated in research studies with the guinea pig conjunctivitis (Sereny) test³¹ or in HeLa cells,³² or the 120- to 140-megadalton plasmid associated with invasiveness in *Shigella* and invasive *E. coli* (EIEC) should be identified.³³ Under the appropriate conditions, EIEC has epidemic potential; these organisms were responsible for a widespread outbreak of dysentery associated with imported French Camembert cheese.^{30,34} Although they have been identified as occasional causes of diarrhea in Brazil,²⁷ EIEC do not appear to be frequent causes of sporadic diarrhea in the United States. Because they are often slow to ferment lactose in the laboratory, EIEC may be initially mistaken for shigellae,^{27,30,32} to which they are closely related. EIEC are also usually lysine negative, often nonmotile,³⁵ and antigenically related to *Shigella*³⁶; the commonly used NAAT tests do not discriminate between *Shigella* and EIEC species, so their true incidence is uncertain.^{37,38}

Shiga Toxin–Producing *Escherichia coli*

A significant cause of bloody diarrhea and potentially fatal HUS is now recognized to be because of EHEC organisms (see Chapter 98) that produce Shiga-like toxin, although the frequency with which they cause inflammatory diarrhea is not clear.^{39,40} These organisms have attracted particular attention because of widespread outbreaks in popular hamburger chain restaurants⁴¹ and with uncooked produce, such as spinach,^{42,43} and a widespread outbreak of bloody diarrhea and severe HUS, starting in Germany and extending across Europe, associated with eating imported Egyptian fenugreek sprouts in the summer of 2011. The latter, unlike prior EHEC strains that had exhibited EPEC traits of attachment and effacement, was a Shiga toxin–producing EAEC strain.^{44,45} Although it accounts for only 0.8% to 3.0% of all cases of diarrhea in the United States and Canada, EHEC (serotype O157) is estimated to account for 15% to 36% of cases of bloody diarrhea.³⁹ Most recognized EHEC strains are of serotype O157; others include O26:K60:H11, O103:H2, O91:H2, O145:H–, O111:K58:H–, O38:H21, O6:H–, O5:H–, O128, O139, O113:K75, O121, and O172.^{39,29,46} EHEC organisms were the most commonly recognized cause of diarrhea (3%) among 5415 patients studied in Calgary in Alberta, Canada, where such cases showed a summer seasonal peak.⁴⁷ EHEC are associated with 75% to 90% of cases of HUS in North America, a complication that develops in approximately 8% of recognized EHEC infections.³⁹

The incubation period is usually 3 to 4 days after the ingestion of contaminated food or water. Cattle are the major reservoir of the organism. The consumption of inadequately cooked beef, raw milk, or other products contaminated by the intestinal contents of cattle, and occasionally of contaminated water, has been recognized as an important source of O157:H7 infection.⁴⁸ A waterborne outbreak of *E. coli* O157:H7 was documented in upstate New York at a county fair.⁴⁹ The infection causes abdominal cramps and watery diarrhea, followed by bloody diarrhea.⁵⁰ In contrast to more inflammatory enteropathies (such as shigellosis or campylobacteriosis), fever is typically absent or minimal on presentation. The HUS and thrombocytopenia may occur 2 to 4 days after the onset of diarrhea, especially in children younger than 5 years and in older adults, as in nursing homes. EHEC O157:H7 forms characteristic actin pedestals on infected mammalian cells.⁵¹ Previously, typical EHEC organisms shared EPEC traits of secreting their own translocated intimin receptor (Tir) molecule (via a type II secretion system) into the plasma membranes of host cells.⁵² The attachment of the organism via its Tir receptor is required to trigger the assembly of actin into focused pedestals beneath bound bacteria. EHEC produces Stx1 or Stx2; the latter is more associated with HUS.⁵³ Another EHEC virulence factor, H7 flagellin, is also associated with activation of proinflammatory signals in human colonic epithelial cells in response to this noninvasive pathogen.⁵² The German outbreak EHEC strain was an EAEC O104:H4 strain that produced Stx2a and was positive for several EAEC genes, including

aggR, aggA (encoding for the biofilm-producing aggregative adherence fimbriae I [AAF/I]), sigA, sepA, pic, aatA, aaiC, and aap, but it was negative for the eae (E. coli attaching and effacing) gene and produced an extended-spectrum β -lactamase.^{54,55}

TABLE 101.2

Enteroinvasive *Escherichia coli* Serotypes

SEROTYPE DIFCO SEROGROUP TEST^a

O28	C
O29	—
O112	C
O124	B
O136	C (Trabulsi 193-T-64)
O143	—
O144	—
O152	(Trabulsi 185-T-64)

^a Difco Laboratories, Detroit.

Modified from references 27, 29, 30, and 195.

Enteropathogenic and Enteroaggregative *Escherichia coli*

EPEC organisms that cause acute diarrhea in infants may rarely be associated with an insidious persistent or relapsing illness.⁵⁶ *E. coli* organisms in O groups 1, 2, 4, 7, and 75 that produce hemolysin and necrotoxin have been isolated from patients with ulcerative colitis. These toxic organisms are not present in healthy people or in patients with acute diarrheal syndromes.⁵⁷ In addition, EAEC organisms are recognized as a cause of persistent diarrhea in India, Brazil, and Mexico.⁵⁸⁻⁶⁰ EAEC infections are an emerging problem (at least in terms of their recognition), and EPEC infections remain important in many parts of the world.⁶¹⁻⁶³ EAEC organisms have been associated with intestinal inflammation and malnutrition, even in the absence of diarrhea.^{64,65} The benefits of antibiotic treatment for severe or persistent diarrhea from EPEC and EAEC remain unclear. Increasing rates of resistance to commonly prescribed antibiotics such as ampicillin, trimethoprim-sulfamethoxazole, and doxycycline, and decreasing susceptibility to ciprofloxacin, ceftriaxone, and azithromycin has been reported among international travelers.^{66,67} Multidrug resistant EAEC have been detected in both asymptomatic or symptomatic children with or without malnutrition.⁶⁸

Campylobacter Enteritis

Campylobacter spp. have a small genome (1.6–2.0 megabases) and may persist in the GI tract of their natural zoonotic hosts, but can cause acute intestinal and systemic infection in humans. The genome sequence from multiple *C. jejuni* and several other Campylobacter spp. have been published.⁶⁹⁻⁷¹ Although there currently are more than 16 Campylobacter spp. and 6 subspecies, *C. jejuni* and Campylobacter coli are the most prevalent species associated with enteric infections. *C. jejuni* is the most prevalent bacterial pathogen isolated from enteric infections in many industrialized countries. *C. jejuni* and *C. coli* are highly prevalent in developing countries, especially in children younger than 2 years with enteric infection or in asymptomatic carriers. *C. jejuni* systemic infections (see Chapter 221) have been recognized for many years. Although the majority of Campylobacter bloodstream infections in humans are with Campylobacter fetus (former subspecies, *C. fetus* subsp. intestinalis),⁷² *C. jejuni* commonly causes an enteric infection in all age groups. Commercially available techniques have enabled the culture of *C. jejuni* on highly selective media at 42°C (108°F) from fecal specimens of patients with diarrhea.⁷³ Campylobacter enteritis results from oral transmission, and a variable but relatively small dose (5–8 × 10² cells/mL) is sufficient to cause infection.⁷⁴ This organism causes a syndrome of severe abdominal pain, fever, and acute inflammatory enteritis that may range from watery diarrhea to severe dysentery with blood and pus in the stools.⁷⁵ Infection has also been associated with sequelae of reactive arthritis and the Guillain-Barré syndrome.⁷⁶ Cross-reacting antibodies recognizing both *C. jejuni* lipopolysaccharide and antigenic determinants of nerve gangliosides are speculated to contribute to the development of the axonal nerve damage after *C. jejuni* infection. Reports from many parts of the world have revealed that in 5% to 14% of unselected cases of diarrhea, *C. jejuni* is present,^{75,77} and outbreaks of Campylobacter enteritis have been associated with ingestion of contaminated water, raw milk,

and uncooked meat and poultry. A review of common source outbreaks of *Campylobacter* infection in the United States between 1997 and 2008 showed that unlike sporadic illnesses, which are primarily attributable to poultry, dairy products were the vehicle most often associated with these outbreaks.⁷⁸ Although several genes are involved in the adhesion, colonization, invasion, and toxin production by *Campylobacter* spp., and the specific balance between these genes and host susceptibility to *Campylobacter* enteritis remains uncertain.⁶⁹

Clostridioides difficile Infection

CDI is the most common cause of antibiotic-associated diarrhea. In the 2000s, the incidence and mortality rates of CDI increased because of the emergence of strains characterized as BI/NAP1/O27/III, which have heightened toxinogenesis.⁷⁹ Advanced age is a critical risk factor for severe disease, recurrence, and mortality,⁸⁰ although CDI is also now appearing in younger, lower-risk populations.⁸¹ Disruption of normal intestinal microbiota from prior antibiotic use allows *C. difficile* colonization, germination of spores, and production of toxins. The primary virulence factors in CDI are two large toxins: *C. difficile* toxin A (TcdA), or toxin A (308 kDa) and TcdB, or toxin B (270 kDa).⁸² Both toxins are glucosyltransferases that inactivate Rho, Rac, and Cdc42 proteins and result in actin condensation and subsequent cytoskeletal changes, apoptosis, and cell death of target cells. Tcds also induce an intense inflammatory response characterized by infiltration of inflammatory cells, especially neutrophils; activation of submucosal neurons; secretion of cytokines; chemokines and arachidonic acid metabolites; and production of substance P and reactive oxygen intermediates. Some *C. difficile* strains, including the epidemic ribotype 027, produce binary toxin that may have a role in increased virulence through host attachment and colonization, which type 2 immunity in the host may protect against.⁸³ Clinical presentation ranges from asymptomatic colonization to severe disease with pseudomembranous colitis (Fig. 101.1) and septic shock. Increased fecal lactoferrin, IL-1 β , and IL-8 were observed in patients with CDI.⁸⁴ Leukocytosis suggests severe infection, and leukemoid reactions have been described in colitis caused by *C. difficile*.⁸⁵ Further details of CDI are provided in Chapter 249.

Vibriosis

In addition to classic and El Tor *Vibrio cholerae* O1 and O139, non-O1 *V. cholerae* and several halophilic *Vibrio* spp. (see Chapters 219 and 220) are now recognized to cause diarrhea and occasional wound or bloodstream infection.⁸⁶ The most common and best characterized is *V. parahaemolyticus*, which has been recognized since 1950 in Japan and was identified as a *Vibrio* spp. in 1963. *V. parahaemolyticus* is a cause of seafood poisoning, with onset of signs and symptoms 9 to 25 hours after the ingestion of inadequately cooked fish or shellfish. This has been reported throughout the coastal areas of the United States, especially in warmer waters and on cruise ships, and is the most common form of food poisoning in Japan where raw seafood is commonly eaten as sushi, although recently the incidence has been decreasing because of careful food preparation practices.^{87,88} Diarrhea may be explosive and watery or may be characterized by full-blown dysentery with blood and pus and superficial ulceration on proctoscopic examination.⁸⁹ The latter syndrome may be associated with cramps, nausea, vomiting, headache, and fever. The illness is usually self-limited, resolving within 3 to 4 days.

Other halophilic vibrios include *Vibrio alginolyticus*, *Vibrio fluvialis*, *Vibrio hollisae*, *Vibrio damsela*, and *Vibrio vulnificus*, which have been associated with enteric, wound, or systemic infections in humans.⁸⁶ *V. vulnificus* has been associated with life-threatening septicemia, often in persons with preexisting liver disease, occurring within 24 hours of ingesting raw oysters.^{86,90}

FIG. 101.1 Proctoscopic view of pseudomembranous colitis in a patient who received clindamycin. Note the 4- to 8-mm raised white plaques overlying an erythematous mucosa. From Tedesco FJ, Barton RW, Alpers DH. Clindamycin-associated colitis. *Ann Intern Med.* 1974;81:429–433.

Follow for extended description

Salmonellosis and Enteric Fever

Salmonella enterocolitis is characterized by fever, cramping, abdominal pain, and diarrhea that begin 8 to 48 hours after ingestion of an infective dose, usually with food, and usually lasts 3 to 5 days (see Chapter 228). The diarrheal stools of patients with salmonellosis often contain a moderate number of polymorphonuclear leukocytes, usually fewer than is typical of shigellosis.

Although *Salmonella* enteritis predominantly involves the lamina propria in the small bowel, several reports have noted *Salmonella enterica* serovar Typhimurium as a cause of colitis, with crypt abscesses and erosion and ulcerations of the colonic mucosa, which result in blood and pus in the stool.⁹¹ Certain other strains of *Salmonella*—including *S. enterica* serovar Choleraesuis and *S. enterica* serovar Paratyphi, and especially *S. enterica* serovar Typhi—elicit a mononuclear response and cause the bacteremia that is characteristic of enteric fever. Animal models for *S. Typhimurium* continue to advance our understanding of the pathogenesis of intestinal and systemic *Salmonella* infections.⁹² *Salmonella* flagellin is regulated by the *fliC* gene, which is the major ligand for Toll-like receptor 5 (TLR5), nucleotide oligomerization domain–like receptors, and interleukin-1 β –converting enzyme (ICE) protease-activating factor (IPAF) protein. TLR5 is located in the plasma and phagosomal membranes, and the other two are located in the cytoplasm of the host's Peyer patch cells. Flagellin activates TLR5 and leads to the production of IL-12, whereas the activation of IPAF leads to the production of IL-1 β and IL-18. Both pathways act synergistically to activate interferon- γ –producing Th1 cells. These proinflammatory cytokines play important roles in the induction of inflammation with *Salmonella* infections.

Typhoid fever may lead to an erosion of the blood vessels in Peyer patches that, without appropriate treatment, could result in gross blood in the feces in 10% to 20% of patients. Severe intestinal hemorrhage may complicate approximately 2% of cases late in the course of untreated typhoid fever. Such intestinal bleeding may precede perforation, another complication of typhoid fever.⁹³ However, typhoid fever is primarily a systemic illness, not a diarrheal disease (see Chapter 102).

Yersiniosis

Y. enterocolitica (see Chapter 235) is an enteric pathogen that may be responsible for an enteric fever–like illness, mesenteric adenitis (which may mimic acute appendicitis), or an inflammatory ileitis or ulcerative colitis syndrome with fecal neutrophils and mononuclear cells.⁹⁴ *Y. enterocolitica* and *Yersinia pseudotuberculosis* may cause similar symptoms and signs. The virulence of *Yersinia* is associated with the direct injection of proteins, such as *Yersinia* protein kinase A Page 1296(YpkA) and *Yersinia* outer protein E (YopE), into eukaryotic cells.^{95,96} *Yersinia* infection may also be associated with migratory polyarthritis, reactive arthritis, or erythema nodosum. A syndrome of acute diarrhea and vomiting is especially common in young children. The organism may cause disseminated abscesses in the liver and spleen⁹⁷ or an inflammatory colitis.⁹⁸ The causative agent, a gram-negative member of the family Enterobacteriaceae, is in the same genus as the plague bacillus *Yersinia pestis* and is sometimes mistaken for *Proteus* on initial culture plates.

FIG. 101.2 Methylene blue stain of fecal leukocytes from a patient with colitis. This exudative response may be seen in any active colitis syndrome, including those caused by the inflammatory or invasive pathogens discussed in this chapter.

Follow for extended description

Amebiasis

Entamoeba histolyticum is one of the classic causes of dysenteric syndromes. Although the course and severity of the infection are variable, amebiasis is one infection that can cause sustained febrile diarrhea with stools becoming grossly bloody. For patients living in endemic areas or travelers who have been to such endemic areas and who have dysenteric symptoms, amebiasis should be considered in the differential diagnosis. For further details about amebiasis, please see Chapter 278.

Diagnosis

Diarrhea with fever, blood in the stool, severe abdominal pain, or signs of sepsis necessitates evaluation for enteric pathogens to guide disease management.⁹⁹ Clinical and epidemiologic information direct the specific tests to be requested. Examination for fecal leukocytes often reveals sheets of polymorphonuclear leukocytes in clumps of mucus, even in the absence of gross blood in the stool specimen (Fig. 101.2).^{100,101} Fewer pyknotic leukocytes are reported in amebic dysentery^{102–104}; this may be attributable to the deeper undermining ulcers characteristic of intestinal amebiasis or to a contact-dependent cytolytic effect of the amoeba on leukocytes. The diagnosis of shigellosis is made using bacterial cultures or polymerase chain reaction (PCR) assay for the invasiveness plasmid in stool, rectal swab, or endoscopic biopsy

specimens.¹⁰⁵ The use of fresh specimens promptly plated onto appropriate enteric culture media is very important to ensure the isolation of shigellae.¹⁰⁶

Specialized techniques are required to isolate *Vibrio* (thiosulfate citrate bile salts [TCBS] agar),¹⁰⁷ *Yersinia* (cold enrichment),¹⁰⁸ or *C. jejuni*.⁷³ The identification of toxigenic *C. difficile* is done by immunoassay for either *C. difficile* toxin B or PCR assay for *C. difficile* toxin B.¹⁰⁹

The diagnosis of infection with EHEC depends on culturing stool on MacConkey culture medium, on which the original O157:H7 EHEC strain yields sorbitol-negative colonies.¹¹⁰ The isolated *E. coli* is serotyped for somatic (O) and flagellar (H) antigens. Rapid methods for detection of Shiga toxins in the stool samples are now in use.^{111,112} The pathogenic Kanagawa-positive strains from patients produce β -hemolysis on special (Wagatsuma) medium—in contrast to environmental isolates—and are best isolated as blue-green colonies (alkaline) on TCBS agar.¹⁰⁷ A PCR method provides a species-specific probe for *V. parahaemolyticus*, *V. alginolyticus*, *Vibrio anguillarum*, and a cluster of related species.¹¹³ Cultivation may require cold enrichment techniques.¹⁰⁸ PCR methods for detection of pathogenic *Y. enterocolitica* are more sensitive than culture methods.¹¹⁴

Molecular approaches, including multiple pathogen nucleic acid amplification tests (NAAT) for simultaneous detection of multiple enteropathogens, have high sensitivity and specificity and are widely used for surveillance or clinical purposes.¹¹⁵ In outpatient settings, these have largely replaced culture methods in the United States, Canada, and western Europe. Not only do the NAAT approaches allow for higher sensitivity, but the turnaround time is also faster, facilitating hospital infection control measures and initiation of appropriate intervention. The broad, multiplex approach allows for a screening diagnostic approach for enteric pathogens, including viruses, bacteria, and parasites, without having to test for individual pathogens separately.¹¹⁶ There are several limitations inherent to this advanced molecular assay. In addition to relatively high cost, interpretation of the results of the assay is challenging. Detection of a pathogen does not necessarily mean identification of the cause of dysentery, as the test is not able to differentiate colonization from clinically relevant infection. Another reason is the high frequency of polymicrobial infections detected by this method, with up to 30% of positive samples detecting multiple pathogens, which can make determination of the true pathogen challenging.^{38,117,118} Additionally, specific pathogens such as EPEC or EAEC tend to be detected more commonly as observed in one study where both organisms were positive in 84% of the multipathogen samples.¹¹⁷ Another study showed a false positive rate around 20% with NAAT for organisms like *C. difficile*, EPEC, EAEC, and *Campylobacter*, which was largely caused by cross-reactivity with commensal organisms.³⁸ Quantitation of the burden of infection may assist in the interpretation of results from these molecular assays.¹¹⁹ Moreover, current commercially available GI enteric panels do not allow for detection of antibiotic resistance. The steady increase in the prevalence of antimicrobial resistant diarrhea-causing pathogens indicates the continuing need for traditional microbiologic methods for susceptibility testing to guide treatment.

Amebic dysentery is traditionally diagnosed by direct examination of wet mounts of fresh fecal or proctoscopic specimens, which reveal *E. histolytica* trophozoites or cysts characterized by four or fewer delicate nuclei with central karyosomes. Additional patients may be diagnosed by biopsy, which may reveal periodic acid–Schiff stain (PAS)-positive trophozoites or cysts in the undermining ulcer in the lamina propria, or by a fecal or serum *E. histolytica*-specific antigen test, which distinguishes virulent *E. histolytica* from avirulent *Entamoeba dispar* infections.¹²⁰ The use of multiplex NAAT panels including *E. histolytica* is largely replacing the historical diagnostic approaches to amoebiasis.¹²¹

Sigmoidoscopic examination, especially with biopsy, may be useful in the diagnosis of a pseudomembranous enterocolitis or in the identification of parasites such as *E. histolytica* (with a special PAS stain) or *B. coli*. Amebic colitis is associated with discrete small ulcerations with undermined edges amid relatively normal mucosa. Acute shigellosis causes more widespread, shallow, 3- to 7-mm ulcers with a more intense inflammatory exudate.

Other considerations in the differential diagnosis of inflammatory colitis include the potentially rapidly progressive necrotizing enterocolitis (NEC) syndromes (see Chapters 81 and 249) and noninfectious syndromes such as inflammatory bowel diseases, ulcerative colitis, and Crohn

disease. These diagnoses are suggested by the setting, clinical course, history, and findings on radiologic and endoscopic examinations.

Finally, the presence of biomarkers of acute or prolonged intestinal inflammation, such as fecal myeloperoxidase and neopterin, may be associated with growth shortfalls in young children in developing areas.¹²²

Therapy

Therapy consists of supportive fluid and electrolyte management with antimicrobial therapy directed at a specific pathogen if suspected on the basis of the epidemiologic setting or microbiologic test results. Fluid resuscitation is the primary treatment of all acute diarrheal disease syndromes. Oral rehydration salt solution (World Health Organization formula) is recommended for oral rehydration therapy.¹²³ Presumptive therapy for the inflammatory colitides varies greatly with the different organisms and is influenced by the increasing resistance of many enteric pathogens to antimicrobial therapy.¹²⁴ For example, an acute febrile dysenteric illness in a young child with daycare exposure or in an area in which shigellosis is common should be treated with an antimicrobial agent, such as one of the fluoroquinolones. Early administration of empirical antibiotic treatment is justified in children hospitalized for clinical dysentery.¹²⁵ Short-course (3 days) oral ciprofloxacin for *S. dysenteriae* type 1 dysentery in children has been effective.¹²⁶ If the *Shigella* organism is sensitive, prompt therapy can successfully stop the diarrhea, alleviate systemic symptoms, and reduce shedding of the organisms in the feces.¹²⁷⁻¹²⁹ Because shigellae are increasingly resistant to multiple antibiotics,^{130,131} the practitioner must be familiar with the local resistance pattern of shigellae to treat acute shigellosis appropriately when it is first suspected. Multidrug resistant, even extensively resistant, *Shigella* infections have been reported among MSM.^{132,133} Although ciprofloxacin may shorten the clinical course of *Salmonella* infections, it does not eradicate the organism.¹³⁴ Studies have shown increasing resistance of *Campylobacter* and even *Salmonella* to quinolones.^{135,136} Azithromycin may reduce the duration of symptoms of *Campylobacter* or *Shigella* enteritis. Zinc 20 mg daily (10 mg of elemental zinc in 5 mL) given in two equally divided doses for 2 weeks showed a significant reduction in the duration of acute shigellosis, promoted better weight gain, and reduced diarrheal morbidity in malnourished children at 6 months after infection.¹³⁷ Vitamin A has also been used to reduce severity and mortality from shigellosis.¹³⁸

Antibiotic therapy has not been shown to be effective for disease caused by EHEC or EIEC. Some EHEC strains may increase their phage-mediated production of Shiga-like toxins when exposed to antibiotics to which they are sensitive, such as ampicillin, tetracycline, sulfamethoxazole-trimethoprim, or quinolones.¹³⁹⁻¹⁴²

Most patients with mild-to-moderate *C. jejuni* enterocolitis do not benefit from antibiotic treatment. Severely ill patients, especially those who are debilitated or immunocompromised, appear to benefit from antibiotic therapy, and erythromycin (or other macrolides) or ciprofloxacin are the drugs of choice. Increased fluoroquinolone resistance in *Campylobacter* spp. from humans and animals, the result of mutations in topoisomerase genes, has been well documented.¹⁴³ Two studies now associate symptomatic and asymptomatic *Campylobacter* infections with reduced growth in Brazilian and Peruvian children.^{144,145}

Vancomycin and fidaxomicin are the drugs of choice for initial episode of CDI (see Chapter 249 for further details).^{146,147} Recurrent disease occurs in 20% to 65% of the patients, depending on the number of previous episodes of the disease.¹⁴⁸ Fidaxomicin is reported to be equivalent to vancomycin in treating acute CDI but may be superior in preventing relapses.¹⁴⁹ However, its advantage over vancomycin is lost in infections caused by the prevalent epidemic strain BI/NAP1/027. Alternative approaches to prevent recurrent disease include a tapering course of vancomycin; fecal microbiota transplantation (FMT)¹⁵⁰; or use of monoclonal antibodies.¹⁵¹ FMT is increasingly being used to prevent recurrent disease or to treat refractory CDI. More recently, the monoclonal antibody against TcdB has been shown to confer modest protection against recurrent CDI.⁹⁹ A vancomycin tapering course has been shown to be effective for treatment, and in some observational studies equal to FMT in preventing recurrence.¹⁵²

Antibiotic treatment with tetracycline or doxycycline can diminish the duration and severity of GI symptoms caused by *V. parahaemolyticus*. To prevent *V. parahaemolyticus* infections,

consumption of raw or undercooked shellfish and exposure of wounds to seawater should be avoided.¹⁵³

Antibiotic therapy is not required for *Salmonella enteritis*, but patients who are severely ill or those with risk factors for developing extraintestinal spread of infection should be treated with fluoroquinolones or third-generation cephalosporins.

Yersinia infection is usually self-limiting; however, patients with severe illness or systemic infection are usually treated with tetracycline, chloramphenicol, or ciprofloxacin.

Prevention

Enteric pathogens are predominantly spread through contaminated food or water and sometimes via direct fecal-oral spread. Spread of infection occurs during ingestion of contaminated food or water or contact with a contaminated environment. Infection control measures include hand hygiene, proper food handling, and access to clean water. In some instances, isolation and treatment of infected individuals are necessary to prevent spread of infection, for example, in cases of CDI. Targeted health interventions for sexual networks of MSM are needed.

Immunization of susceptible individuals is an important component of prevention. However, for bacterial enteric pathogens, vaccine is only available for typhoid fever and cholera. Typhoid vaccination is recommended for people traveling to endemic areas. Ty21a is a live-attenuated oral vaccine containing *S. Typhi* in capsule formulation. Another typhoid vaccine is the parenteral Vi-polysaccharide. For individuals traveling to areas with recent or ongoing cholera outbreaks, CVD 103-HgR, a live-attenuated oral vaccine, is available. There is a need to develop effective vaccines for most infectious agents causing diarrhea worldwide.

Other Acute Infectious/Inflammatory Processes Involving the Intestinal Tract Gonococcal Proctitis

Neisseria gonorrhoeae (see Chapter 217) may cause ulcerative proctitis, usually acquired by anal intercourse.⁷ The resultant purulent proctitis is accompanied by an erythematous, friable mucous membrane in the rectal vault and occasional abscess or fistula formation. Although copious purulent discharge, tenesmus, and burning rectal pain may be noted, two-thirds of culture-positive patients with anorectal gonococcal infection are asymptomatic.¹⁵⁴ Additional diagnostic possibilities in cases of venereally acquired proctitis are syphilitic, herpetic, and chlamydial proctitis.¹⁵⁵

Unusual Causes of Dysentery

Spirillar or spirochetal dysentery has been reported to occur in southern France and has been attributed to *Spirillum* spp. Although severe mucoid diarrhea or dysentery has been associated with intestinal spirochetes, their frequency and role in causing enteric disease are unclear. A DNA probe for the 16S ribosomal RNA of the agent of swine dysentery, *Treponema hyodysenteriae*, has been developed and may open new approaches to the recognition of similar infections in humans.¹⁵⁶

Other unusual causes of colitis include brucellosis¹⁵⁷ and adenovirus infections.^{158,159}

Necrotizing Enterocolitis in the Newborn

The syndrome of diffuse, fulminating, necrotizing colitis has been recognized in infants since the 1960s. This syndrome probably represents the same entity described as spontaneous intestinal perforation and peritonitis as early as 1838.¹⁶⁰ Although milder forms of the syndrome doubtless exist, the syndrome of NEC is defined by the presence of air in the wall of the intestine, portal venous system, or peritoneal cavity, or by necrosis of the bowel wall with mucosal sloughing. This fulminant syndrome often leads to intestinal perforation, peritonitis, and bacteremia. It is a major cause of mortality in low-birth-weight infants (<1500 g) after the first week of life.¹⁶¹ The diffuse necrotic changes that characterize this syndrome most often occur in the terminal ileum but may be seen in the colon or in the proximal portion of the GI tract.

Hypoxia-ischemia was previously thought to be the main mechanism for development of NEC, but this is uncertain,¹⁶² but current research suggests multifactorial pathogenesis. The three

major risk factors associated with development of NEC are prematurity, bacterial colonization of the gut, and formula feeding¹⁶³; the pathogenesis involves a combination of immature intestinal barrier and abnormal intestinal microbiome.¹⁶⁴ Intestinal immaturity refers to Page 1298 motility, absorption, barrier function, and circulatory regulation, but the most important aspect seems to be an excessive immune response to the gut microbiota which contributes to the development of NEC along with a dysbiotic intestinal microbiome. Studies in cell cultures and animal models suggest that the immature intestinal epithelium has an excessive inflammatory response to the microbiota.^{165,166} Studies comparing babies with NEC with control, or before and after development of NEC, suggests that a reduced diversity of the fecal microbiome and presence of opportunistic intestinal microbial species are associated with development of NEC.^{167,168}

Clinical features of this serious condition in newborn infants include apneic spells, vomiting, abdominal distention, and occasionally bloody diarrhea. Most infants are younger than 1 week, and there is an association with prematurity, maternal infections during delivery (e.g., amnionitis with prolonged postrupture of membranes), and exchange transfusion via the umbilical vein. There is no gender or seasonal predilection. The disease often progresses rapidly to intestinal perforation, shock, septicemia, and pneumatosis intestinalis. Air may also be evident in the portal venous system or biliary tract on plain radiographs. This syndrome is associated with mortality often greater than 70%.

The diagnosis of NEC should be considered in any premature infant with altered GI function, abdominal distention, or apneic spells. Further investigation should include examination of the stool for occult blood and for the presence of reducing substances. Plain abdominal radiographs may reveal air in the bowel wall, peritoneal cavity, or portal venous system, and there may be bloody diarrhea late in the course of the disease. Management must be initiated early and aggressively for any infant suspected of having NEC. Umbilical catheters should be removed, oral feeding should be stopped, and nasogastric aspiration should be initiated. Intravenous fluid therapy is of paramount importance. After obtaining specimens for blood culture, broad-spectrum intravenous antibiotics is recommended. Laparotomy and excision of the necrotic bowel are often necessary and should be done aggressively if there is any evidence of peritonitis or obstruction.¹⁶⁹

Prevention of NEC includes promotion of intestinal development, avoidance of risk factors and careful infection control measures in newborn ICUs. Antenatal corticosteroids, given for prevention of respiratory distress in the premature newborn, also promote intestinal development and appears protective against NEC development.¹⁷⁰ Hypertonic elemental formulas have been implicated and should be avoided in high-risk patients. Similarly, prolonged antibiotic use and gastric acid suppression increase NEC risk and should be avoided.¹⁷¹⁻¹⁷³ Conversely, use of human milk for preterm infants lowers the risk of NEC.^{174,175} Explanations for the advantage of human breast milk include the presence of lysozyme, antibodies, and cellular elements that may play a protective role against potential infectious agents, as well as promotion of growth for beneficial bacteria in the gut by human milk oligosaccharides (HMOs).^{176,177} Although oral prophylactic nonabsorbable antibiotics have been suggested, serious questions remain about the use of prophylactic antibiotics, even in high-risk newborn infants weighing less than 1500 g. A few reports have shown that epidermal growth factor, which is found in high concentrations in breast milk, may have a cytoprotective effect, and probiotics may decrease or prevent neonatal NEC.^{178,179}

Darmbrand, Pig-Bel, Necrotizing Enteritis in Adults (Enteritis Necroticans)
First described as Darmbrand (meaning fire bowels) in epidemics of enteritis necroticans (EN) in northern Germany in the immediate postwar period in the mid-1940s, a severe necrotizing jejunitis has also been recognized in epidemic and sporadic forms after pork feasting in the highlands of Papua New Guinea.¹⁸⁰ Pig-bel was the name given to the syndrome of abdominal discomfort that followed a large pork meal, commonly eaten after a large pig kill, which takes place every 3 to 10 years among the highland Melanesians of Papua New Guinea. Sporadic cases have been reported from other parts of the world, including the United States.¹⁸¹

The pathologic findings are those of acute, patchy, necrotizing disease of the small bowel in previously healthy people, which may proceed rapidly to segmental gangrene, with small amounts of gas in the mucosa, mesentery, or nodes.

Several theories of pathogenesis have been suggested, most of which involve the toxic products of *Clostridium perfringens* type C, including α - and β -toxins. Sporadic cases of EN have been noted in association with nutritional disorders, alcoholism, and malabsorption, and after pancreatic or gastric resection.¹⁸² After gastric surgery, increased numbers of *C. perfringens* and increased levels of α -toxin have been noted in the upper small bowel and stomach. Whether α - or β -toxins are capable of causing EN alone, or whether they initiate the invasion of the mucosa by other organisms, such as gram-negative rods, is currently unclear. An attractive hypothesis was suggested by Lawrence and Walker,¹⁸² which could explain the association of EN with poor nutrition and episodic dietetic overindulgence. The low-protein diet of Papua New Guinea highlanders is associated with low levels of digestive proteases in the intestinal lumen, which inactivate the β -toxin. The proteases can be further blocked by the oral intake of trypsin inhibitors, which are found in this population in such dietary staples as sweet potatoes. Proteases return with improved diet, as occurred in postwar Germany. This hypothesis has been confirmed in an animal model that required protease inhibitors for symptomatic infection.¹⁸³

The clinical syndromes of EN range from anorexia, vomiting, severe abdominal pain, and bloody diarrhea to fulminant toxemia and shock. Acute complications that necessitate emergency surgery include paralytic ileus, bowel strangulation, and bowel perforation with peritonitis. These complications are common in the first 2 weeks of illness. Later complications that may also necessitate surgery include scarring, leading to stenosis; obstruction; malabsorption; and fistulas. EN occurs with greater frequency and greater severity in children younger than 10 years. In contrast to European control subjects who rarely have antibodies, 70% of the healthy adults in Papua New Guinea have detectable antibody to clostridial β -toxin.^{180,182}

The syndrome is defined by the pathologic findings but should be suspected in patients who develop severe abdominal pain, bloody diarrhea, ileus, and toxemia. The course is often too fulminant for radiographic detection of air in the bowel wall to be of any diagnostic value.

Agents held responsible for causing EN include *C. perfringens* type C, once designated as type F in the older classification of *Clostridium welchii*. Most surgically resected bowel specimens from patients with EN contain *C. perfringens*, more than half of which are type C. Furthermore, 12 of 21 cases described had a significant change in serum β -antitoxin titer after illness with pig-bel in Papua New Guinea.¹⁸⁰ Although polyvalent gas gangrene antiserum was ineffective, administration of type C antiserum resulted in a 30% decrease in the need for surgery, and mortality was reduced from 43% to 19%.¹⁸⁰ Furthermore, active immunization against the β -toxin has also proved effective in preventing pig-bel.¹⁸⁴

Some have suggested that type A *C. perfringens*, staphylococci, or even hepatitis virus may be responsible for EN. The syndrome of enteritis gravis has been described in association with infectious hepatitis, although no viral cause has been documented.

Considerations in the differential diagnosis of EN include acute shigellosis, acute food poisoning syndromes, antibiotic-associated pseudomembranous colitis, and acute ulcerative colitis. The absence of colonic involvement; the epidemiologic setting, especially in poorly nourished patients; and the rapid progression to toxemia and shock are strongly suggestive of EN.

Therapy for EN includes careful supportive care and bowel decompression. Fluid requirements may be substantially greater than what is indicated by fecal output. Resection of the involved bowel must be considered if there is a persistence of paralytic ileus, a rapid increase in signs of toxemia, localized or diffuse signs of peritonitis, persistent pain, or a palpable mass lesion. If subacute obstruction or malabsorption is suspected on the basis of weight loss, elective surgery may be required up to 6 months after the acute illness. Raw peanut or soybean diets should be avoided because they contain trypsin inhibitors. *C. perfringens* type C antiserum containing β -antitoxin or the active β -toxin vaccine should be available for use in areas where EN can be expected to occur. Page 1299

Chronic Infectious/Inflammatory Processes Involving the Intestinal Tract

Chronic inflammatory enteritides are often indolent, slowly progressive infections. Often, there is a history of weeks or months of fever, abdominal pain, weight loss, or other systemic

manifestations. Recurring or relapsing symptoms may be seen with *C. jejuni* or *Salmonella* gastroenteritis. In addition, 16% of cases of shigellosis may become prolonged, lasting for 3 weeks or longer.¹⁸⁵ Any diarrheal illness that extends beyond 2 weeks is considered severe and is associated with a high risk of nutritional morbidity in children who live in tropical developing areas.¹⁸⁶⁻¹⁸⁸

Syphilis

Syphilis (see Chapter 243) can also involve the GI tract, usually in the upper part of the small bowel or stomach. An acute erosive and infiltrative gastritis with motile spirochetes and a positive specific response on treponemal immunofluorescence testing has been reported in late secondary syphilis.¹⁸⁹ The initial complaints are upper abdominal pain, vomiting, and weight loss. More classic are the late GI manifestations of lues: pyloric obstruction, hourglass constriction, and linitis plastica of the stomach. Less commonly, gumma may be seen in the small bowel or colon. More commonly, syphilis can cause proctitis in MSM, as the primary chancre. Anoscopy may reveal mucosal inflammation with ulcerations. Serology and if available, darkfield microscopy of the chancre would aid in the diagnosis.

Gastrointestinal Tuberculosis

Intestinal tuberculosis (TB), once considered common, had become a relatively rare disease but is now reemerging in association with HIV infections and with multidrug-resistant *Mycobacterium tuberculosis* (see Chapter 255).¹⁹⁰ Intestinal involvement with TB may be either primary, from ingestion of the organisms, or secondary (reactivation), usually from a pulmonary source.

Primary intestinal TB without pulmonary disease often results in hypertrophic mucosal changes. Patients with acute miliary TB may also have GI tract involvement.¹⁹¹ Patients with primary intestinal TB may present with abdominal pain, fever, and a tender, fixed, palpable mass in the ileocecal area. Primary hypertrophic intestinal TB continues to occur in the Middle East¹⁹² and in India.¹⁹³ Intestinal involvement secondary to pulmonary TB may result from swallowing infected sputum or from biliary excretion of the organism from an infected liver. The frequency of secondary intestinal TB increases with far-advanced pulmonary disease. Hippocrates stated that “diarrhea attacking a person with phthisis is a mortal symptom.”

TB may involve any part of the GI tract, but most ulcerative and hypertrophic types occur in the ileocecal region, where there is a predominance of submucosal lymphatic tissue. The most common features are fever and abdominal pain that is often relieved by defecation or vomiting. Weight loss is more common in secondary intestinal TB. Only one-third of the patients with GI TB have diarrhea. Diarrhea may be related to exacerbations of abdominal pain and occasionally occurs with extensive involvement of the small intestine, which may cause steatorrhea and a malabsorption syndrome. Although ulceration and mucous diarrhea are relatively common with secondary intestinal TB, hemorrhage and the presence of gross blood in the stool are distinctly uncommon, perhaps because of the associated obliterative endarteritis.

The diagnosis of GI TB may be very difficult radiologically and even histologically. It must be distinguished from regional enteritis, sarcoidosis, actinomycosis, ameboma, carcinoma, and peri appendiceal abscess. In contrast to Crohn disease, GI TB rarely causes anal lesions, fistulas, or perforation. It is often associated with miliary nodules on the serosa, it rarely causes strictures longer than 3 cm, and it may cause circumferential transverse ulcers. TB may also cause fibrosis of the muscularis mucosa, pyloric metaplasia, and epithelial regeneration.¹⁹⁴ There may be minimal or no radiologic changes in the bowel mucosa. Small mucosal ulcerations may result in tiny, calcified nodules in the mucosa in association with calcified mesenteric lymph nodes analogous to those seen in the pulmonary Ghon complex. The ileocecal region often reveals radiologic evidence of irritability and hypermotility, with hypersegmentation of the mucosal folds or poor filling of the ileocecal region detected by barium enema. On occasion, frank ulcerations can be noted on contrast studies, and late in the course there is scarring. The diagnosis requires a careful examination of involved tissue for acid-fast bacilli by using stain and culture. Caseous necrosis is more frequently found in the mesenteric nodes than in intestinal tissue itself. Complications of intestinal TB include perforation, peritonitis, and obstruction from hypertrophy, scarring, or tuberculoma.

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Typhoid Fever, Paratyphoid Fever, and Typhoidal Fevers

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Short View Summary

Definition

- Enteric fever, or typhoid fever, is a nonspecific febrile illness caused by typhoidal *Salmonella* (*Salmonella enterica* serotypes Typhi and Paratyphi); the diagnosis should be considered in any patient with otherwise unexplained prolonged fever who has resided in or recently traveled in a typhoid-endemic setting.
- The term typhoidal fever is sometimes used more broadly to refer to a syndrome of persistent high-grade fevers, often with no localizing features. This chapter also covers the broader differential diagnosis and approach to patients with typhoidal fever, relapsing or prolonged fever, and fever accompanied by abdominal pain.

Epidemiology

- An estimated 12 to 24 million cases of enteric fever occur each year, largely in impoverished areas of Asia and Africa.
- Enteric fever is transmitted via the fecal-oral route, with contaminated municipal water supplies being commonly involved in transmission.
- Antibiotic-resistant strains of the major causative agents of enteric fever (*S. enterica* serotypes Typhi and Paratyphi A) are now common worldwide.

Clinical Manifestations

- Common life-threatening complications of enteric fever include intestinal hemorrhage, perforation, encephalopathy, and shock.
- Untreated, patients with enteric fever may be febrile for 3 to 4 weeks or longer, with mortality rates exceeding 10%, and prolonged asthenia and fatigue are common among survivors.
- Chronic biliary carriage of typhoidal *Salmonella* may occur after resolution of the acute illness.

Diagnosis

- Current diagnostic tests for enteric fever are imperfect. Blood cultures are 30% to 70% sensitive, depending on when in the natural history of the illness they are obtained. Bone marrow cultures are more sensitive but are impractical. Serologic assays lack both sensitivity and specificity, especially in enteric fever endemic areas; and nucleic acid amplification assays with sufficient sensitivity are not available.

Therapy

- Given the morbidity of typhoid fever, the risk of complications, and the lack of optimal diagnostic tests, the initiation of antibiotics for treating individuals with suspected enteric fever may be based on a presumptive diagnosis, particularly in resource-limited settings.
- The most commonly used agents for treating individuals with enteric fever are fluoroquinolones, azithromycin, and cefixime or ceftriaxone. Chloramphenicol, trimethoprim-sulfamethoxazole, and amoxicillin may be used to treat patients with susceptible strains.

Prevention

- An oral-attenuated typhoid vaccine and injectable polysaccharide vaccine are commercially available globally and provide 50% to 75% protection for 5 and 2 years, respectively. Two typhoid conjugate vaccines are now prequalified and recommended by the World Health Organization, with a protective efficacy of >80%. No vaccine effective against paratyphoid A is currently commercially available.

Enteric Fever (Typhoid and Paratyphoid Fever)

History

Historically termed putrid fever or dothienteritis, the name typhoid was coined in 1829 by Pierre Charles Alexander Louis. The name typhoid means typhus-like and reflects the difficulty in differentiating the illness from epidemic typhus, another common cause of prolonged fever in Europe during the 19th century. Louis used the term typhoid in a landmark report comparing the intestinal pathology in 50 patients who died of typhoid fever with 83 patients who died of noninfectious causes, and he described the characteristic inflammation of Peyer patches, intestinal ulceration, and mesenteric adenitis associated with typhoid fever.¹

William Budd is credited with the recognition of the fecal-oral transmission of typhoid fever in 1839, based on his meticulous observations during a large epidemic in the Taw Valley of England, although these findings were not widely recognized until many years later.²

The etiologic agent of typhoid fever was identified in pathologic specimens by Karl Eberth, who referred to it as *Bacillus typhosus*; the organism was subsequently first cultured in 1884 by Georg Gaffky at the Berlin Institute for Infectious Diseases. The term paratyphoid fever was first used in 1896 by Raoul Bensaude and Emile Achard, who isolated what would subsequently be recognized as *Salmonella Paratyphi B*. The role of asymptomatic carriers in the transmission of infectious disease was first recognized by Robert Koch, a theory that he derived from his longitudinal observations of patients recovered from typhoid but with continued shedding of *B. typhosus*.³ An understanding of the role of asymptomatic carriers in disease transmission was soon extended to other organisms and was an established mode of disease transmission at the time of George Soper's association of Mary Mallon (also known as Typhoid Mary) with repeated outbreaks of typhoid fever in households in New York, ultimately leading to her lifelong quarantine on North Brother Island.⁴

Widal developed the first serologic test for typhoid fever in 1896, an agglutination assay that detects the presence of antibodies against the O and H antigens of *Salmonella Typhi*.⁵ Despite inherent and significant limitations (see "Diagnosis"), the Widal test remains widely used in many resource-limited settings. The first vaccines for typhoid fever were killed whole-cell vaccines developed in 1896, credit for which is shared by Almroth Wright and Richard Pfeiffer.⁶ Despite advances in the development of a vaccine, typhoid fever remained a leading cause of morbidity and mortality in the early 20th century. A prominent example was a typhoid fever epidemic in the Spanish-American War, in which 20,000 American Army recruits contracted the disease and 1600 died. This defining event was linked to the first compulsory US military-wide vaccination program, in which a killed whole-cell, injectable typhoid vaccine developed by Frederick Russell was used.⁷

In the early 20th century, typhomalaria or typhomalarial fever remained an extremely common diagnosis, indicating that differentiating typhoid from malaria (and other causes of persistent fever) on clinical grounds was difficult. The difficulty distinguishing typhoid fever from typhus and other causes of fever proved clinically fortuitous. In 1948, Theodore Woodward and coworkers reported the effect of chloramphenicol in two patients who were referred for a study of the efficacy of the drug in scrub typhus.⁸ Both were subsequently found to have *Salmonella Typhi* bacteremia, and both patients improved rapidly after antibiotic therapy. Typhoid fever mortality has been reduced from 10% or greater to less than 1% with the use of antibiotics. However, successive pandemics of antibiotic-resistant organisms have occurred, including, most recently,

the emergence of fluoroquinolone- and ceftriaxone-resistant strains of *Salmonella* Typhi. In 2001, the complete sequence of a *Salmonella* Typhi strain (CT18, isolated from a patient in Vietnam in 1993) was published by Parkhill and coworkers at the Wellcome Trust Sanger Institute, ushering in a genomic era in our understanding of typhoid fever.⁹ Genomic analysis has opened up new insights into the history and epidemiology of typhoidal *Salmonella*, from evolutionary history^{10,11} to previously unrecognized historical outbreaks¹² and recent global dissemination.¹³

Etiologic Agents of Enteric Fever

Nomenclature and Classification of Typhoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella* Paratyphi A, B, and C are gram-negative bacilli that belong to the species *S. enterica* subspecies *enterica*. All *S. enterica* are categorized serologically by the Centers for Disease Control and Prevention (CDC) according to a modified Kauffman and White classification scheme, assigned based both on the O (the O polysaccharide) and H (flagellar) antigens.^{14,15} Although assigning a serogroup based on O antigen agglutination tests is a common procedure in many clinical microbiology laboratories, the approach has limited clinical utility. Complete serotyping based on both the O and H antigen is most commonly performed in reference laboratories. In addition to serogrouping and serotyping, standard microbiologic growth and biochemical parameters are also routinely used in the clinical laboratory for the presumptive identification of *S. enterica*. Serologic and selected biochemical characteristics that are used in the identification of *Salmonella* Typhi and *Salmonella* Paratyphi A, B, and C in microbiology laboratories are summarized in Table 102.1.

Clinical Distinction Between Typhoidal and Nontyphoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella* Paratyphi A and B are human-restricted pathogens, and *Salmonella* Paratyphi C is pathogenic for animals and humans. *Salmonella* Typhi and *Salmonella* Paratyphi are classified clinically as typhoidal *Salmonella* strains. This distinction separates these strains from all other pathogenic *S. enterica* serotypes, which are referred to collectively as nontyphoidal *Salmonella* (often abbreviated as NTS; see Chapter 228). Typhoidal strains cause enteric fever in all human hosts, whereas NTS strains are classically associated with inflammatory diarrhea in human hosts or disproportionately cause invasive disease in young children and immunocompromised individuals, with particular associations with malnutrition, HIV, and malaria. However, the clinical distinction between typhoidal and NTS is not absolute. NTS may cause invasive infection that includes prolonged bacteremia (with or without pyogenic foci) that may mimic the systemic illness caused by typhoidal strains.¹⁶ Although host factors influence susceptibility to invasive NTS infection, the likelihood of invasive infection is also dependent on characteristics of the bacterial strain. For example, the emergence of a dominant genotype associated with invasive NTS, *Salmonella* Typhimurium ST313, in sub-Saharan Africa illustrates the clinical overlap between typhoidal and nontyphoidal *S. enterica*.¹⁷

Genomic Features

Escherichia coli and *S. enterica* diverged over 100 million years ago, and *S. enterica* is now a genetically heterogeneous species.¹⁸ *Salmonella* Typhi is genetically homogeneous, meaning there is very limited genetic diversity across *Salmonella* Typhi isolates throughout the world. Based on standard rates of genetic drift, *Salmonella* Typhi likely emerged from a single-point origin within the last 50,000 years, coinciding with the time of human migrations out of Africa and into Asia.^{10,19} The genetic material of *Salmonella* Typhi consists of a single chromosome that contains approximately 5×10^6 base pairs encoding approximately 4000 genes. *Salmonella* Typhi often harbors additional plasmids that may be cryptic (confer an unknown function) or involved in pathogenesis. The latter include IncHI1 plasmids that comprise the majority of resistance plasmids found in *Salmonella* Typhi and that remain stable in the bacteria without ongoing pressure from antibiotic selection.^{20,21}

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Classification and Selected Microbiologic Characteristics of Typhoidal *Salmonella*

SEROTYPE NAME	<i>Salmonella</i> Typhi	<i>Salmonella</i> Paratyphi A	<i>Salmonella</i> Paratyphi B (schottmuelleri)	<i>Salmonella</i> Paratyphi C (hirschfeldii)	Comments
Serologic Classification ^a	D	A	B	C	
Serogroup	D	A	B	C	
O antigen	9, 12	1, 2, 12	1, 4, (5), 12	6, 7	
H antigen Phase 1:2	d: –	a: (1, 5)	b: (1, 2)	c: (1, 5)	Uncommon <i>Salmonella</i> Typhi H antigen variants include HJ:z66 found in Indonesia
K antigen	Vi	–	–	Vi	
Selected Biochemical Characteristics ^b					
Lactose fermentation	–	–	–	+	The majority of <i>Salmonella</i> Paratyphi C ferment lactose
Sulfide (H ₂ S) production	+ (weak)	–	+	+	3% of <i>Salmonella</i> Typhi do not produce hydrogen sulfide
Gas produced during glucose fermentation	–	+	+	+	
Lysine decarboxylase	+	–	+	+	
Ornithine decarboxylase	–	+	+	+	

^a Strong antigenic determinants are listed in bold; weaker or sometimes absent ones are in parentheses.
^b Isolates can be identified as *Salmonella* using traditional methods. A presumptive diagnosis of *Salmonella* Typhi can often be made for *Salmonella* isolates based on selected biochemical characteristics, in addition to agglutination with D and Vi antisera.

Source of Infection

Salmonella Typhi and *Salmonella* Paratyphi A and B are human restricted. There are no known animal reservoirs of these pathogens that cause enteric fever, and the source of infection is organisms shed Page 1303in the stool of infected humans. *Salmonella* Paratyphi C can be shed by humans and animals. In the preantibiotic era, patients recovering from enteric fever shed organisms in the stool during acute illness and typically for weeks during convalescence. In addition, a small portion of individuals convalescing from typhoid fever develop chronic asymptomatic carriage (defined as shedding of the bacteria in stool or urine for more than a year), which may persist for life.³⁵ In the preantibiotic era, up to 5% of typhoid fever survivors developed chronic asymptomatic bacterial shedding in the stool,^{36,37} with gallbladder disease being the major risk factor for carriage.^{38,39} Risk of carriage appears to be far greater in adults (3%–10%) than in young children (<1%).³⁶ Studies from Nepal, where enteric fever is highly endemic, found that 3.5% of individuals undergoing elective cholecystectomy grew *Salmonella* Typhi or *Salmonella* Paratyphi A from biliary cultures.⁴⁰ Even higher rates were observed in Santiago, Chile, another hyperendemic focus at the time, in the early 1980s.⁴¹ Urinary carriage may also occur, frequently in conjunction with urinary (*Salmonella* haematobium) schistosomiasis.⁴² Although chronic carriers are a source of transmission where enteric fever

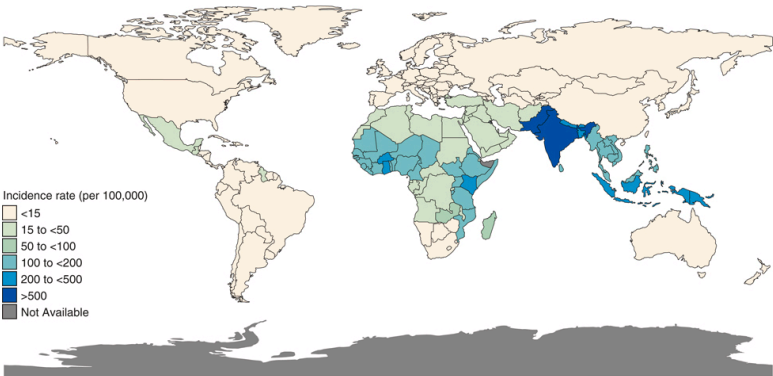


FIG. 102.1 Global distribution of typhoid and paratyphoid fever. Estimated incidence of typhoid and paratyphoid fever by country from the Global Burden of Disease Study 2019. Source: Institute for Health Metrics and Evaluation. Used with permission. All rights reserved.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

occurs sporadically, their importance in the transmission of enteric fever in highly endemic areas is uncertain.^{43,44}

Mode of Transmission

Infections with typhoidal *S. enterica* are most often acquired via ingestion of fecally contaminated water or food. *Salmonella* Typhi may persist for weeks after passage in water and may persist in a variety of contaminated food items—for example, dehydrated formula and iced beverages.⁴⁵⁻⁴⁷ Volunteer studies may not optimally mimic natural exposure; however, the inoculum required to produce disease in 50% of adult volunteers (ID₅₀) is 10⁵–7 organisms, although fewer than 10³ organisms may produce disease,^{48,49} which might be more reflective of actual exposures in nonepidemic settings.

Molecular epidemiologic studies based on high-resolution genotyping implicate waterborne transmission as the most likely route of transmission in the majority of cases of enteric fever.⁵⁰ A study in the Kathmandu valley of Nepal revealed that >80% of water collected from public taps contained typhoidal *Salmonella* DNA.⁵¹ Both waterborne and foodborne transmission may contribute to large epidemics, and in some cases, massive epidemics have been associated with water contamination from a single source.^{52,53} In contrast, direct person-to-person transmission of the etiologic agents of enteric fever appears less common, and more than 80% of cases of enteric fever occur in individuals with no known contact with an individual with symptomatic infection.⁵⁴ Even more remarkably, in households where multiple cases of enteric fever occur, only 20% of those infections shared a common bacterial genotype, suggesting that multiple infections within a household occur through a community-based exposure to multiple circulating genotypes, rather than through direct person-to-person transmission.⁵⁰ In highly endemic areas, consistent personal risk factors for infection with typhoidal *S. enterica* include drinking nonboiled water and eating food prepared outside the home, including street-vended foods and beverages.^{55,56} In South Asia, peak transmission occurs during and immediately after the monsoon season (peaks in June through October), although disease is present year-round.⁵⁷

Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically

In areas where enteric fever occurs sporadically, most cases are imported through travel. In the United States, >85% of cases occur in travelers to endemic areas.⁵⁸ The rate of typhoid fever is estimated at 10 per 1 million travelers arriving from Asia but increases to 89 per 1 million travelers arriving from India. More than 80% of all travel-associated cases in the United States relate to travel to South Asia.⁵⁸ The risk of travel-related enteric fever is higher among travelers visiting friends and relatives. Many of the remaining non-travel-associated cases in the United States are traceable to small, local foodborne outbreaks or a chronic carrier.^{59,60} In 2010, an outbreak of *Salmonella* Typhi occurred that involved at least 12 cases across three western US states and was attributed to imported fruit pulp from Guatemala.⁶¹ Also, in areas where enteric fever occurs sporadically, small clusters of cases of sexual transmission of *Salmonella* Typhi have been documented in men who have sex with men.⁶² Infections and clinical cases have occurred among microbiology laboratory staff, especially when *Salmonella* Typhi was used as an unknown for laboratory proficiency testing, and in at least once instance, secondary cases occurred among family members of laboratory personnel.⁶³

Antibiotic Resistance and Emergence of Pandemic *Salmonella* Typhi

Chloramphenicol-resistant *Salmonella* Typhi was first reported in 1950, 2 years after the antibiotic was first used to treat patients with typhoid fever.⁶⁴ However, widespread dissemination of antibiotic-resistant strains did not occur until the 1970s. Multidrug-resistant (MDR) *Salmonella* Typhi carrying IncHI1 plasmid-mediated resistance to chloramphenicol, ampicillin, trimethoprim, and sulfonamides became common in the 1980s. Nalidixic acid-resistant strains of *Salmonella* Typhi and *Salmonella* Paratyphi A, with Page 1300

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Typhoid Fever, Paratyphoid Fever, and Typhoidal Fevers

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Short View Summary

Definition

- Enteric fever, or typhoid fever, is a nonspecific febrile illness caused by typhoidal *Salmonella* (*Salmonella enterica* serotypes Typhi and Paratyphi); the diagnosis should be considered in any patient with otherwise unexplained prolonged fever who has resided in or recently traveled in a typhoid-endemic setting.
- The term typhoidal fever is sometimes used more broadly to refer to a syndrome of persistent high-grade fevers, often with no localizing features. This chapter also covers the broader differential diagnosis and approach to patients with typhoidal fever, relapsing or prolonged fever, and fever accompanied by abdominal pain.

Epidemiology

- An estimated 12 to 24 million cases of enteric fever occur each year, largely in impoverished areas of Asia and Africa.
- Enteric fever is transmitted via the fecal-oral route, with contaminated municipal water supplies being commonly involved in transmission.
- Antibiotic-resistant strains of the major causative agents of enteric fever (*S. enterica* serotypes Typhi and Paratyphi A) are now common worldwide.

Clinical Manifestations

- Common life-threatening complications of enteric fever include intestinal hemorrhage, perforation, encephalopathy, and shock.
- Untreated, patients with enteric fever may be febrile for 3 to 4 weeks or longer, with mortality rates exceeding 10%, and prolonged asthenia and fatigue are common among survivors.
- Chronic biliary carriage of typhoidal *Salmonella* may occur after resolution of the acute illness.

Diagnosis

- Current diagnostic tests for enteric fever are imperfect. Blood cultures are 30% to 70% sensitive, depending on when in the natural history of the illness they are obtained. Bone marrow cultures are more sensitive but are impractical. Serologic assays lack both sensitivity and specificity, especially in enteric fever endemic areas; and nucleic acid amplification assays with sufficient sensitivity are not available.

Therapy

- Given the morbidity of typhoid fever, the risk of complications, and the lack of optimal diagnostic tests, the initiation of antibiotics for treating individuals with suspected enteric fever may be based on a presumptive diagnosis, particularly in resource-limited settings.
- The most commonly used agents for treating individuals with enteric fever are fluoroquinolones, azithromycin, and cefixime or ceftriaxone. Chloramphenicol, trimethoprim-sulfamethoxazole, and amoxicillin may be used to treat patients with susceptible strains.

Prevention

- An oral-attenuated typhoid vaccine and injectable polysaccharide vaccine are commercially available globally and provide 50% to 75% protection for 5 and 2 years, respectively. Two typhoid conjugate vaccines are now prequalified and recommended by the World Health Organization, with a protective efficacy of >80%. No vaccine effective against paratyphoid A is currently commercially available.

Enteric Fever (Typhoid and Paratyphoid Fever)

History

Historically termed putrid fever or dothienteritis, the name typhoid was coined in 1829 by Pierre Charles Alexander Louis. The name typhoid means typhus-like and reflects the difficulty in differentiating the illness from epidemic typhus, another common cause of prolonged fever in Europe during the 19th century. Louis used the term typhoid in a landmark report comparing the intestinal pathology in 50 patients who died of typhoid fever with 83 patients who died of noninfectious causes, and he described the characteristic inflammation of Peyer patches, intestinal ulceration, and mesenteric adenitis associated with typhoid fever.¹

William Budd is credited with the recognition of the fecal-oral transmission of typhoid fever in 1839, based on his meticulous observations during a large epidemic in the Taw Valley of England, although these findings were not widely recognized until many years later.²

The etiologic agent of typhoid fever was identified in pathologic specimens by Karl Eberth, who referred to it as *Bacillus typhosus*; the organism was subsequently first cultured in 1884 by Georg Gaffky at the Berlin Institute for Infectious Diseases. The term paratyphoid fever was first used in 1896 by Raoul Bensaude and Emile Achard, who isolated what would subsequently be

recognized as *Salmonella Paratyphi B*. The role of asymptomatic carriers in the transmission of infectious disease was first recognized by Robert Koch, a theory that he derived from his longitudinal observations of patients recovered from typhoid but with continued shedding of *B. typhosus*.³ An understanding of the role of asymptomatic carriers in disease transmission was soon extended to other organisms and was an established mode of disease transmission at the time of George Soper's association of Mary Mallon (also known as Typhoid Mary) with repeated outbreaks of typhoid fever in households in New York, ultimately leading to her lifelong quarantine on North Brother Island.⁴

Widal developed the first serologic test for typhoid fever in 1896, an agglutination assay that detects the presence of antibodies against the O and H antigens of *Salmonella Typhi*.⁵ Despite inherent and significant limitations (see "Diagnosis"), the Widal test remains widely used in many resource-limited settings. The first vaccines for typhoid fever were killed whole-cell vaccines developed in 1896, credit for which is shared by Almroth Wright and Richard Pfeiffer.⁶ Despite advances in the development of a vaccine, typhoid fever remained a leading cause of morbidity and mortality in the early 20th century. A prominent example was a typhoid fever epidemic in the Spanish-American War, in which 20,000 American Army recruits contracted the disease and 1600 died. This defining event was linked to the first compulsory US military-wide vaccination program, in which a killed whole-cell, injectable typhoid vaccine developed by Frederick Russell was used.⁷

In the early 20th century, typhomalaria or typhomalarial fever remained an extremely common diagnosis, indicating that differentiating typhoid from malaria (and other causes of persistent fever) on clinical grounds was difficult. The difficulty distinguishing Page 1301 typhoid fever from typhus and other causes of fever proved clinically fortuitous. In 1948, Theodore Woodward and coworkers reported the effect of chloramphenicol in two patients who were referred for a study of the efficacy of the drug in scrub typhus.⁸ Both were subsequently found to have *Salmonella Typhi* bacteremia, and both patients improved rapidly after antibiotic therapy. Typhoid fever mortality has been reduced from 10% or greater to less than 1% with the use of antibiotics. However, successive pandemics of antibiotic-resistant organisms have occurred, including, most recently, the emergence of fluoroquinolone- and ceftriaxone-resistant strains of *Salmonella Typhi*. In 2001, the complete sequence of a *Salmonella Typhi* strain (CT18, isolated from a patient in Vietnam in 1993) was published by Parkhill and coworkers at the Wellcome Trust Sanger Institute, ushering in a genomic era in our understanding of typhoid fever.⁹ Genomic analysis has opened up new insights into the history and epidemiology of typhoidal *Salmonella*, from evolutionary history^{10,11} to previously unrecognized historical outbreaks¹² and recent global dissemination.¹³

Etiologic Agents of Enteric Fever

Nomenclature and Classification of Typhoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella Paratyphi A*, *B*, and *C* are gram-negative bacilli that belong to the species *S. enterica* subspecies *enterica*. All *S. enterica* are categorized serologically by the Centers for Disease Control and Prevention (CDC) according to a modified Kauffman and White classification scheme, assigned based both on the O (the O polysaccharide) and H (flagellar) antigens.^{14,15} Although assigning a serogroup based on O antigen agglutination tests is a common procedure in many clinical microbiology laboratories, the approach has limited clinical utility. Complete serotyping based on both the O and H antigen is most commonly performed in reference laboratories. In addition to serogrouping and serotyping, standard microbiologic growth and biochemical parameters are also routinely used in the clinical laboratory for the presumptive identification of *S. enterica*. Serologic and selected biochemical characteristics that are used in the identification of *Salmonella Typhi* and *Salmonella Paratyphi A*, *B*, and *C* in microbiology laboratories are summarized in Table 102.1.

Clinical Distinction Between Typhoidal and Nontyphoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella Paratyphi A* and *B* are human-restricted pathogens, and *Salmonella Paratyphi C* is pathogenic for animals and humans. *Salmonella Typhi* and *Salmonella Paratyphi* are classified clinically as typhoidal *Salmonella* strains. This distinction separates these strains from all other pathogenic *S. enterica* serotypes, which are referred to collectively as nontyphoidal *Salmonella* (often abbreviated as NTS; see Chapter 228). Typhoidal strains cause enteric fever in all human hosts, whereas NTS strains are classically associated with inflammatory diarrhea in human hosts or disproportionately cause invasive disease in young children and

immunocompromised individuals, with particular associations with malnutrition, HIV, and malaria. However, the clinical distinction between typhoidal and NTS is not absolute. NTS may cause invasive infection that includes prolonged bacteremia (with or without pyogenic foci) that may mimic the systemic illness caused by typhoidal strains.¹⁶ Although host factors influence susceptibility to invasive NTS infection, the likelihood of invasive infection is also dependent on characteristics of the bacterial strain. For example, the emergence of a dominant genotype associated with invasive NTS, *Salmonella* Typhimurium ST313, in sub-Saharan Africa illustrates the clinical overlap between typhoidal and nontyphoidal *S. enterica*.¹⁷

Genomic Features

Escherichia coli and *S. enterica* diverged over 100 million years ago, and *S. enterica* is now a genetically heterogeneous species.¹⁸ *Salmonella* Typhi is genetically homogeneous, meaning there is very limited genetic diversity across *Salmonella* Typhi isolates throughout the world. Based on standard rates of genetic drift, *Salmonella* Typhi likely emerged from a single-point origin within the last 50,000 years, coinciding with the time of human migrations out of Africa and into Asia.^{10,19} The genetic material of *Salmonella* Typhi consists of a single chromosome that contains approximately 5×10^6 base pairs encoding approximately 4000 genes. *Salmonella* Typhi often harbors additional plasmids that may be cryptic (confer an unknown function) or involved in pathogenesis. The latter include IncHI1 plasmids that comprise the Page 1302 majority of resistance plasmids found in *Salmonella* Typhi and that remain stable in the bacteria without ongoing pressure from antibiotic selection.^{20,21}

TABLE 102.1

Classification and Selected Microbiologic Characteristics of Typhoidal *Salmonella*

SEROTYPE NAME

Salmonella Typhi *Salmonella* Paratyphi A *Salmonella* Paratyphi B (*schottmuelleri*)

Salmonella Paratyphi C (*hirschfeldii*) Comments

Serologic Classification^a

Serogroup	D	A	B	C
O antigen	9, 12	1, 2, 12		1, 4, (5), 12
H antigen				
Phase 1:2				

d: –	a: (1, 5)	b: (1, 2)	c: (1, 5)	Uncommon <i>Salmonella</i> Typhi H antigen
variants include Hj:z66 found in Indonesia				

K antigen Vi – – Vi

Selected Biochemical Characteristics^b

Lactose fermentation	–	–	–	+	The majority of <i>Salmonella</i> Paratyphi C
ferment lactose					
Sulfide (H ₂ S) production		+	(weak)	–	+
produce hydrogen sulfide					+
Gas produced during glucose fermentation	–				+
Lysine decarboxylase	+	–	+	+	+
Ornithine decarboxylase		–	+	+	+

a Strong antigenic determinants are listed in bold; weaker or sometimes absent ones are in parentheses.

b Isolates can be identified as *Salmonella* using traditional methods. A presumptive diagnosis of *Salmonella* Typhi can often be made for *Salmonella* isolates based on selected biochemical characteristics, in addition to agglutination with D and Vi antisera.

From World Health Organization (WHO), Department of Vaccines and Biologicals. Background Document: The Diagnosis, Treatment and Prevention of Typhoid Fever. Geneva: WHO; 2003; and WHO, Department of Communicable Disease Surveillance and Response. *Salmonella* serotype Typhi. In: WHO Manual for the Laboratory Identification and Antimicrobial Susceptibility Testing of Bacterial Pathogens of Public Health Importance in the Developing World. Geneva: WHO; 2003:103–118.

The genetic core of *S. enterica* consists of genes that are conserved across Enterobacteriaceae and is approximately 90% similar to *E. coli*. In contrast, the genetic core varies approximately 1%

between *S. enterica* serovars. In addition to the genetic core, *Salmonella* Typhi contains pathogenicity islands—horizontally acquired genetic elements that are often associated with unique properties of each strain and are essential for specific phenotypes associated with pathogenesis of each serotype.¹⁸ In *S. enterica*, these are specifically identified as *Salmonella* pathogenicity islands, or SPIs, and are designated with a number. *Salmonella* Typhi possesses more than 20 SPIs, many of which are found in other *S. enterica* serotypes. For example, SPI-7 is a 134-kb genetic island that possesses elements encoding the *viaB* locus, which is involved in production of the Vi capsule and evasion of host immune responses.²² SPI-7 is widely conserved across *Salmonella* Typhi and is also found in *Salmonella* Paratyphi C.

One of the most remarkable genetic features of *Salmonella* Typhi is the accumulation of more than 200 presumed inactive genes called pseudogenes.⁹ This adaptation may play a major role in the host restriction of *Salmonella* Typhi. The accumulation of a large number of pseudogenes is a common feature shared with *Salmonella* Paratyphi A, B, and C, which emerged independently of *Salmonella* Typhi (although subsequent recombination events between *Salmonella* Typhi and *Salmonella* Paratyphi A appear to have occurred),²³ yet are also adapted specifically to cause disseminated infection in humans. The same phenomenon has been observed in *Salmonella* Typhimurium ST313 in sub-Saharan Africa; ST313 has accumulated extensive gene deletions and purported pseudogene formation (more than half of which are common to *Salmonella* Typhi and *Salmonella* Paratyphi A) compared with less invasive *Salmonella* Typhimurium strains.²⁴

Epidemiology

Burden and Distribution

Measures of the burden of enteric fever are limited by the absence of surveillance in many regions and also by the limited sensitivity of current diagnostic tests. Recognizing these limitations, estimates of the number of cases of typhoid and paratyphoid fever range from 12 to 16 million annually.²⁵ The typhoid and paratyphoid fevers were also responsible for 10 million disability-adjusted life years in 2017.²⁵ The economic toll of enteric fever is also devastating for affected families. A study from Malawi from 2019 to 2020 found that catastrophic illness, or an illness which cost over 40% of monthly household nonfood expenses, occurred in 44% of households affected by a case of typhoid fever.²⁶

The incidence of enteric fever and the mortality of the disease vary dramatically by region (Fig. 102.1). In South Asia and parts of sub-Saharan Africa, the incidence approaches 1000 cases per 100,000 person-years.^{27,28} In parts of northern Africa, South America, and Southeast Asia, the incidence is much lower and ranges between 10 and 100 cases per 100,000 person-years.²⁵ Disease is highly heterogeneous even within endemic countries, and the incidence tends to be considerably higher in urban than in rural areas.²⁷ In the United States and many other high-resource settings, the disease occurs sporadically, with less than 1 case per 100,000 person-years, and occurs most often in travelers returning from endemic zones (see “Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically”).

Salmonella Paratyphi A is the most common cause of paratyphoid fever and responsible for an estimated 2.6 to 4.2 million cases of enteric fever annually.²⁵ The proportional burden of enteric fever caused by *Salmonella* Paratyphi A has increased; *Salmonella* Paratyphi A is responsible for more than 20% to 50% of cases of enteric fever. This trend is expected to continue with the introduction of vaccines which protect against typhoid fever, but not enteric fever caused by *S. Paratyphi*.²⁹ In contrast, *Salmonella* Paratyphi B and C remain less-common causes of enteric fever globally.

Severity

In the preantibiotic era, 10% to 15% of clinically recognized cases of typhoid fever were fatal, whereas other cases resolved over a 3- to 4-week course of an often severe and debilitating illness.^{30,31} Prospective surveillance in the postantibiotic era has revealed that the majority of cases of enteric fever are uncomplicated and do not warrant hospitalization. Although the current mortality of typhoid fever is estimated at 0.2% to 2%, these estimates are based on limited data and there is significant variation by region.^{32,33} Reported mortality rates in hospitalized patients with enteric fever in resource-limited settings vary widely by location, ranging from 0% to 15%, with a median of 2%.³² In addition to fatal cases, in sub-Saharan Africa and other regions where

enteric fever is highly endemic, intestinal perforation caused by enteric fever is a leading cause of peritonitis and acute abdominal conditions requiring surgery in children.³⁴

Source of Infection

Salmonella Typhi and Salmonella Paratyphi A and B are human restricted. There are no known animal reservoirs of these pathogens that cause enteric fever, and the source of infection is organisms shed in the stool of infected humans. Salmonella Paratyphi C can be shed by humans and animals. In the preantibiotic era, patients recovering from enteric fever shed organisms in the stool during acute illness and typically for weeks during convalescence. In addition, a small portion of individuals convalescing from typhoid fever develop chronic asymptomatic carriage (defined as shedding of the bacteria in stool or urine for more than a year), which may persist for life.³⁵ In the preantibiotic era, up to 5% of typhoid fever survivors developed chronic asymptomatic bacterial shedding in the stool,^{36,37} with gallbladder disease being the major risk factor for carriage.^{38,39} Risk of carriage appears to be far greater in adults (3%–10%) than in young children (<1%).³⁶ Studies from Nepal, where enteric fever is highly endemic, found that 3.5% of individuals undergoing elective cholecystectomy grew Salmonella Typhi or Salmonella Paratyphi A from biliary cultures.⁴⁰ Even higher rates were observed in Santiago, Chile, another hyperendemic focus at the time, in the early 1980s.⁴¹ Urinary carriage may also occur, frequently in conjunction with urinary (Salmonella haematobium) schistosomiasis.⁴² Although chronic carriers are a source of transmission where enteric fever occurs sporadically, their importance in the transmission of enteric fever in highly endemic areas is uncertain.^{43,44}

FIG. 102.1 Global distribution of typhoid and paratyphoid fever. Estimated incidence of typhoid and paratyphoid fever by country from the Global Burden of Disease Study 2019. Source: Institute for Health Metrics and Evaluation. Used with permission. All rights reserved.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Follow for extended description

Mode of Transmission

Infections with typhoidal *S. enterica* are most often acquired via ingestion of fecally contaminated water or food. Salmonella Typhi may persist for weeks after passage in water and may persist in a variety of contaminated food items—for example, dehydrated formula and iced beverages.^{45–47} Volunteer studies may not optimally mimic natural exposure; however, the inoculum required to produce disease in 50% of adult volunteers (ID₅₀) is 10⁵–7 organisms, although fewer than 10³ organisms may produce disease,^{48,49} which might be more reflective of actual exposures in nonepidemic settings.

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In areas where enteric fever occurs sporadically, most cases are imported through travel. In the United States, >85% of cases occur in travelers to endemic areas.⁵⁸ The rate of typhoid fever is estimated at 10 per 1 million travelers arriving from Asia but increases to 89 per 1 million travelers arriving from India. More than 80% of all travel-associated cases in the United States relate to

travel to South Asia.⁵⁸ The risk of travel-related enteric fever is higher among travelers visiting friends and relatives. Many of the remaining non-travel-associated cases in the United States are traceable to small, local foodborne outbreaks or a chronic carrier.^{59,60} In 2010, an outbreak of *Salmonella* Typhi occurred that involved at least 12 cases across three western US states and was attributed to imported fruit pulp from Guatemala.⁶¹ Also, in areas where enteric fever occurs sporadically, small clusters of cases of sexual transmission of *Salmonella* Typhi have been documented in men who have sex with men.⁶² Infections and clinical cases have occurred among microbiology laboratory staff, especially when *Salmonella* Typhi was used as an unknown for laboratory proficiency testing, and in at least once instance, secondary cases occurred among family members of laboratory personnel.⁶³

Antibiotic Resistance and Emergence of Pandemic *Salmonella* Typhi

Chloramphenicol-resistant *Salmonella* Typhi was first reported in 1950, 2 years after the antibiotic was first used to treat patients with typhoid fever.⁶⁴ However, widespread dissemination of antibiotic-resistant strains did not occur until the 1970s. Multidrug-resistant (MDR) *Salmonella* Typhi carrying IncHI1 plasmid-mediated resistance to chloramphenicol, ampicillin, trimethoprim, and sulfonamides became common in the 1980s. Nalidixic acid-resistant strains of *Salmonella* Typhi and *Salmonella* Paratyphi A, with decreased susceptibility to fluoroquinolone antibiotics, emerged in the 1990s,⁶⁵ and now the vast majority of typhoidal *Salmonella* isolates from South Asia have reduced susceptibility or high-level resistance to fluoroquinolones.^{13,66} At present, reduced susceptibility to ciprofloxacin is less common and more sporadic in sub-Saharan Africa, though it appears to be on the rise.¹³

Although the lack of genetic diversity of *Salmonella* Typhi previously hindered efforts to characterize the global epidemiology of disease, the increasing use of high-density genotyping methods, including whole-genome sequencing, has circumvented these challenges and resulted in a better understanding of the forces driving the evolution and transmission of *Salmonella* Typhi. Although a general lack of diversity among *Salmonella* Typhi suggests a lack of strong selection pressure over millennia, the recent expansion of a single fluoroquinolone-resistant haplotype known as H58 into a pandemic strain found throughout Asia and Africa, and the emergence of several other lineages with diverse *gyrA* mutations, suggest that fluoroquinolones may be exerting a uniquely strong selection pressure that is shaping the global evolution of *Salmonella* Typhi.^{13,67} Evidence suggests that fluoroquinolone resistance-conferring mutations have arisen repeatedly in multiple *Salmonella* Typhi lineages, spread internationally and intercontinentally and replaced less resistant strains.¹³ More recently, *Salmonella* strains with resistance to third-generation cephalosporins (e.g., ceftriaxone) have been reported.^{68,69} In Pakistan, strains containing multidrug resistance and resistance to third-generation cephalosporins and fluoroquinolones have been termed “extensively drug-resistant” (XDR) typhoid, and within years of first appearance now account for the majority of isolates in the country.^{66,68,70} Most recently, azithromycin resistance has been identified in multiple lineages, arising from point mutations in an efflux pump (*acrB*).^{71,72} Azithromycin resistance in typhoidal *Salmonella* might become an increasing problem given the WHO recommendation for its use in mass drug administration in settings with high rates of child mortality.^{73,74}

Host Factors and Susceptibility to Infection

Historically, enteric fever was considered a disease that predominantly affected school-aged children and young adults. However, prospective community-based surveys in highly endemic areas have demonstrated that the incidence of *Salmonella* Typhi bacteremia is often highest in young children (<5 years old) and that this organism may be responsible for more than 75% of cases of occult bacteremia in these settings.⁷⁵ The higher incidence of enteric fever in older children among inpatients may reflect a predisposition to more severe disease in this older age group. In highly endemic areas, *Salmonella* Typhi bactericidal antibodies and anticapsular (Vi) antibodies increase significantly over the first decade of life.⁷⁶ In less endemic areas, the median age of patients with typhoid fever is higher, suggesting that the age distribution of patients may be influenced by acquired immunity. However, protection from a single episode of infection is limited, as demonstrated by frequent relapse and recurrent infections among patients who have recovered from typhoid fever.⁷⁷

Although invasive NTS infections classically occur in persons who are compromised by extremes of age, malnutrition, immunodeficiency, or genetic risk factors, there is not an overt association

between host Page 1304 risk factors and susceptibility to enteric fever. Various candidate gene analyses have drawn associations between single-nucleotide variations in Toll-like receptor 4 (TLR4), cystic fibrosis transmembrane conductance regulator (CFTR), and specific human leukocyte antigen (HLA) types with susceptibility to typhoid fever in Vietnam, as well as in human volunteers challenged with virulent organisms⁷⁸⁻⁸⁰; however, no detailed genome-wide study of typhoid fever associations has yet been reported.

Pathogenesis

Typhoidal *Salmonella* infection is acquired through ingestion of bacilli, typically in contaminated water, drinks, or food. The ID₅₀ is often cited to be 10⁵–7 organisms, but low inocula (<10³ organisms) may also produce infections.⁴⁸ After passing through the intestinal epithelium, typhoidal *Salmonella* are phagocytosed by macrophages and proliferate in the submucosa before dissemination via the lymphatic system and bloodstream. This period coincides with the onset of systemic symptoms and, in some cases, end organ manifestations. It is during this phase that bacilli enter the hepatobiliary system and may establish carriage in the gallbladder. Disseminated infection is terminated by immune clearance or administration of antibiotics, but bacterial persistence in the bone marrow is often longer.

It is important to note that enteric fever and NTS infections differ in fundamental aspects. With NTS, infections are typically self-limited in immunocompetent hosts and manifest largely as inflammatory gastroenteritis. In comparison, typhoidal serotypes are able to evade the normal host inflammatory response and cause prolonged bacteremia even in immunocompetent individuals, typically without overwhelming sepsis or pyogenic foci of infection. Specific bacterial adaptations used by *Salmonella* Typhi to produce persistent bacteremia in the absence of an immediately overwhelming inflammatory response include strategies to breach the intestinal epithelium in the absence of inflammatory diarrhea, to evade detection by host pattern-recognition receptors,⁸¹ to produce an AB₅ bacterial toxin that acts through cell-to-cell paracrine signaling pathways,⁸² and to persist and disseminate throughout the human host within phagocytic cells.⁸³ Much of our understanding of host-pathogen interactions in enteric fever is extrapolated from mouse models of enteric fever, which use Nramp1-deficient mice that develop disseminated infection from *Salmonella* Typhimurium or humanized mice altered to be susceptible to *Salmonella* Typhi.⁸⁴ However, controlled human infection models for *Salmonella* Typhi and Paratyphi A infection are now available and have added valuable insights into the host immune response to these organisms and disease pathogenesis.^{85,86}

Invasion

In contrast to NTS and other enteroinvasive pathogens, soon after ingestion, typhoidal *Salmonella* pass through the intestinal epithelium through specialized microfold cells (M cells) that transport the bacteria across the basolateral membrane, where they are phagocytosed by macrophages in the intestinal lymphoid tissue.⁸⁷ Such a stealth entry mechanism may explain why the invasion phase of *Salmonella* Typhi infection is usually asymptomatic and is accompanied by transient or mild diarrhea in only 10% to 20% of patients. Indeed, as *Salmonella* Typhi leaves the GI tract at this stage, stool cultures are often negative until days to weeks later when shedding through biliary system commences.

Latency and Dissemination

Typhoidal *Salmonella* deploy an array of virulence factors that enable them to persist and replicate in an intracellular compartment of phagocytic cells, primarily macrophages.^{88,89} Important cloaking mechanisms of *Salmonella* Typhi include suppression of flagellar protein synthesis, thus masking a potent inducer of TLR5 responses.⁹⁰ and synthesis of the Vi capsule that presumably masks the detection of lipopolysaccharide (LPS) and other outer membrane components, preventing early complement-mediated clearance.^{91,92} The Vi capsule, thus, helps *Salmonella* Typhi to limit inflammation and evade phagocytosis by neutrophils. The genes that repress flagellin biosynthesis and are responsible for Vi biosynthesis are encoded on the *viaB* locus located on SPI7, which is absent in most NTS strains.²² Induction of other bacterial defenses that are required for survival in the intracellular compartment is regulated by a two-component bacterial sensory-regulatory system, PhoP/PhoQ.^{93,94} The sensor PhoQ detects signals present inside the host phagosome and activates the transcriptional regulator PhoP, which in turn controls the expression of a large number of downstream genes involved in modification of lipid A, resistance to antimicrobial peptides, and acidification of the phagosome.^{95,96} Consistent with a

role in pathogenesis, PhoP/PhoQ mutants are nonvirulent in humans and have been evaluated as candidate live-attenuated vaccines.^{97,98}

Quantitative blood cultures demonstrate that more than 60% of culturable circulating *Salmonella* Typhi organisms reside in the intracellular compartment.⁹⁹ Presumably, this allows for dissemination throughout reticuloendothelial tissues via the blood and lymphatic systems. The heaviest burden of infection is established in the reticuloendothelial system (intestinal lymphoid tissue, liver, spleen, and bone marrow) and gallbladder. Patients with typhoid fever most often have very low-grade bacteremia, a feature that presents a formidable diagnostic challenge. The kinetics of quantitative bacterial cultures has been studied in Vietnam.⁹⁹ The median number of culturable bacteria is less than 1 colony-forming unit (CFU) per milliliter of whole blood in adults, and slightly higher (1.5 CFUs/mL) in children. Slightly more culturable organisms are present in the bone marrow, with a median of 10 per milliliter of bone marrow.¹⁰⁰ However, over the course of illness, the number of culturable *Salmonella* Typhi organisms in the blood decreases (median of 2 CFUs/mL in the first week to 0.1 CFUs/mL in the second and subsequent weeks), and the number of culturable bacilli increases in the bone marrow (4.8 CFUs/mL in the first week to 158 CFUs/mL in the third and subsequent weeks). As a result, the proportion of bacteria in the bone marrow increases from 5:1 (marrow to blood) in the first week of illness to more than 150:1 in the third week of illness, reflecting relative clearance of bacteria from peripheral blood but persistence in the marrow compartment.¹⁰⁰ Pathologic examination of bone marrow may demonstrate monocytic infiltrates typical of typhoid fever (see “Intestinal and Other Local Pathology”).

The initial period of replication and dissemination represents the incubation period of typhoidal *Salmonella* infection. This period typically lasts 1 to 2 weeks but can range widely (3–60 days), depending on the number of organisms ingested.¹⁰¹ Ultimately, infection results in the sufficient secretion of inflammatory cytokines, including pyrogenic cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), which result in fever. However, unlike what occurs during prototypical gram-negative bacteremia, septicemia with hypotension, neutrophilia, and disseminated intravascular coagulopathy are extremely uncommon in patients with typhoid or enteric fever.

Intestinal and Other Local Pathology

Intestinal lymphoid tissue is a predominant site of localized inflammation and persistent bacterial replication in cases of severe enteric fever (Fig. 102.2).^{102–105} Early in the course of illness, Peyer patches in the terminal ileum and draining mesenteric nodes are enlarged and contain infiltrates consisting of mononuclear cells, including macrophages and lymphocytes. As disease progresses, necrosis of intestinal lymphoid tissue occurs, with mixed inflammatory infiltrates, including neutrophils and ulceration and sloughing of the overlying intestinal mucosa.¹⁰² Immunohistochemical staining for the *Salmonella* Typhi O antigen reveals abundant organisms in necrotic areas.¹⁰³ The clinical manifestations of these changes include hemorrhage or intestinal perforation, two of the major

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Typhoid Fever, Paratyphoid Fever, and Typhoidal Fevers

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Short View Summary

Definition

- Enteric fever, or typhoid fever, is a nonspecific febrile illness caused by typhoidal *Salmonella* (*Salmonella enterica* serotypes Typhi and Paratyphi); the diagnosis should be considered in any patient with otherwise unexplained prolonged fever who has resided in or recently traveled in a typhoid-endemic setting.
- The term typhoidal fever is sometimes used more broadly to refer to a syndrome of persistent high-grade fevers, often with no localizing features. This chapter also covers the broader differential diagnosis and approach to patients with typhoidal fever, relapsing or prolonged fever, and fever accompanied by abdominal pain.

Epidemiology

- An estimated 12 to 24 million cases of enteric fever occur each year, largely in impoverished areas of Asia and Africa.
- Enteric fever is transmitted via the fecal-oral route, with contaminated municipal water supplies being commonly involved in transmission.
- Antibiotic-resistant strains of the major causative agents of enteric fever (*S. enterica* serotypes Typhi and Paratyphi A) are now common worldwide.

Clinical Manifestations

- Common life-threatening complications of enteric fever include intestinal hemorrhage, perforation, encephalopathy, and shock.
- Untreated, patients with enteric fever may be febrile for 3 to 4 weeks or longer, with mortality rates exceeding 10%, and prolonged asthenia and fatigue are common among survivors.
- Chronic biliary carriage of typhoidal *Salmonella* may occur after resolution of the acute illness.

Diagnosis

- Current diagnostic tests for enteric fever are imperfect. Blood cultures are 30% to 70% sensitive, depending on when in the natural history of the illness they are obtained. Bone marrow cultures are more sensitive but are impractical. Serologic assays lack both sensitivity and specificity, especially in enteric fever endemic areas; and nucleic acid amplification assays with sufficient sensitivity are not available.

Therapy

- Given the morbidity of typhoid fever, the risk of complications, and the lack of optimal diagnostic tests, the initiation of antibiotics for treating individuals with suspected enteric fever may be based on a presumptive diagnosis, particularly in resource-limited settings.
- The most commonly used agents for treating individuals with enteric fever are fluoroquinolones, azithromycin, and cefixime or ceftriaxone. Chloramphenicol, trimethoprim-sulfamethoxazole, and amoxicillin may be used to treat patients with susceptible strains.

Prevention

- An oral-attenuated typhoid vaccine and injectable polysaccharide vaccine are commercially available globally and provide 50% to 75% protection for 5 and 2 years, respectively. Two typhoid conjugate vaccines are now prequalified and recommended by the World Health Organization, with a protective efficacy of >80%. No vaccine effective against paratyphoid A is currently commercially available.

Enteric Fever (Typhoid and Paratyphoid Fever)

History

Historically termed putrid fever or dothienteritis, the name typhoid was coined in 1829 by Pierre Charles Alexander Louis. The name typhoid means typhus-like and reflects the difficulty in differentiating the illness from epidemic typhus, another common cause of prolonged fever in Europe during the 19th century. Louis used the term typhoid in a landmark report comparing the intestinal pathology in 50 patients who died of typhoid fever with 83 patients who died of noninfectious causes, and he described the characteristic inflammation of Peyer patches, intestinal ulceration, and mesenteric adenitis associated with typhoid fever.¹

William Budd is credited with the recognition of the fecal-oral transmission of typhoid fever in 1839, based on his meticulous observations during a large epidemic in the Taw Valley of England, although these findings were not widely recognized until many years later.²

The etiologic agent of typhoid fever was identified in pathologic specimens by Karl Eberth, who referred to it as *Bacillus typhosus*; the organism was subsequently first cultured in 1884 by Georg Gaffky at the Berlin Institute for Infectious Diseases. The term paratyphoid fever was first used in 1896 by Raoul Bensaude and Emile Achard, who isolated what would subsequently be recognized as *Salmonella* Paratyphi B. The role of asymptomatic carriers in the transmission of infectious disease was first recognized by Robert Koch, a theory that he derived from his longitudinal observations of patients recovered from typhoid but with continued shedding of *B. typhosus*.³ An understanding of the role of asymptomatic carriers in disease transmission was soon extended to other organisms and was an established mode of disease transmission at the time of George Soper's association of Mary Mallon (also known as Typhoid Mary) with repeated outbreaks of typhoid fever in households in New York, ultimately leading to her lifelong quarantine on North Brother Island.⁴

Widal developed the first serologic test for typhoid fever in 1896, an agglutination assay that detects the presence of antibodies against the O and H antigens of *Salmonella Typhi*.⁵ Despite inherent and significant limitations (see “Diagnosis”), the Widal test remains widely used in many resource-limited settings. The first vaccines for typhoid fever were killed whole-cell vaccines developed in 1896, credit for which is shared by Almroth Wright and Richard Pfeiffer.⁶ Despite advances in the development of a vaccine, typhoid fever remained a leading cause of morbidity and mortality in the early 20th century. A prominent example was a typhoid fever epidemic in the Spanish-American War, in which 20,000 American Army recruits contracted the disease and 1600 died. This defining event was linked to the first compulsory US military-wide vaccination program, in which a killed whole-cell, injectable typhoid vaccine developed by Frederick Russell was used.⁷

In the early 20th century, typhomalaria or typhomalarial fever remained an extremely common diagnosis, indicating that differentiating typhoid from malaria (and other causes of persistent fever) on clinical grounds was difficult. The difficulty distinguishing Page 1301 typhoid fever from typhus and other causes of fever proved clinically fortuitous. In 1948, Theodore Woodward and coworkers reported the effect of chloramphenicol in two patients who were referred for a study of the efficacy of the drug in scrub typhus.⁸ Both were subsequently found to have *Salmonella Typhi* bacteremia, and both patients improved rapidly after antibiotic therapy. Typhoid fever mortality has been reduced from 10% or greater to less than 1% with the use of antibiotics. However, successive pandemics of antibiotic-resistant organisms have occurred, including, most recently, the emergence of fluoroquinolone- and ceftriaxone-resistant strains of *Salmonella Typhi*. In 2001, the complete sequence of a *Salmonella Typhi* strain (CT18, isolated from a patient in Vietnam in 1993) was published by Parkhill and coworkers at the Wellcome Trust Sanger Institute, ushering in a genomic era in our understanding of typhoid fever.⁹ Genomic analysis has opened up new insights into the history and epidemiology of typhoidal *Salmonella*, from evolutionary history^{10,11} to previously unrecognized historical outbreaks¹² and recent global dissemination.¹³

Etiologic Agents of Enteric Fever

Nomenclature and Classification of Typhoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella Paratyphi* A, B, and C are gram-negative bacilli that belong to the species *S. enterica* subspecies *enterica*. All *S. enterica* are categorized serologically by the Centers for Disease Control and Prevention (CDC) according to a modified Kauffman and White classification scheme, assigned based both on the O (the O polysaccharide) and H (flagellar) antigens.^{14,15} Although assigning a serogroup based on O antigen agglutination tests is a common procedure in many clinical microbiology laboratories, the approach has limited clinical utility. Complete serotyping based on both the O and H antigen is most commonly performed in reference laboratories. In addition to serogrouping and serotyping, standard microbiologic growth and biochemical parameters are also routinely used in the clinical laboratory for the presumptive identification of *S. enterica*. Serologic and selected biochemical characteristics that are used in the identification of *Salmonella Typhi* and *Salmonella Paratyphi* A, B, and C in microbiology laboratories are summarized in Table 102.1.

Clinical Distinction Between Typhoidal and Nontyphoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella Paratyphi* A and B are human-restricted pathogens, and *Salmonella Paratyphi* C is pathogenic for animals and humans. *Salmonella Typhi* and *Salmonella Paratyphi* are classified clinically as typhoidal *Salmonella* strains. This distinction separates these strains from all other pathogenic *S. enterica* serotypes, which are referred to collectively as nontyphoidal *Salmonella* (often abbreviated as NTS; see Chapter 228). Typhoidal strains cause enteric fever in all human hosts, whereas NTS strains are classically associated with inflammatory diarrhea in human hosts or disproportionately cause invasive disease in young children and immunocompromised individuals, with particular associations with malnutrition, HIV, and malaria. However, the clinical distinction between typhoidal and NTS is not absolute. NTS may cause invasive infection that includes prolonged bacteremia (with or without pyogenic foci) that may mimic the systemic illness caused by typhoidal strains.¹⁶ Although host factors influence susceptibility to invasive NTS infection, the likelihood of invasive infection is also dependent on characteristics of the bacterial strain. For example, the emergence of a dominant genotype associated with invasive NTS, *Salmonella Typhimurium* ST313, in sub-Saharan Africa illustrates the clinical overlap between typhoidal and nontyphoidal *S. enterica*.¹⁷

Genomic Features

Escherichia coli and *S. enterica* diverged over 100 million years ago, and *S. enterica* is now a genetically heterogeneous species.¹⁸ *Salmonella Typhi* is genetically homogeneous, meaning there is very limited genetic diversity across *Salmonella Typhi* isolates throughout the world. Based on standard rates of genetic drift, *Salmonella Typhi* likely emerged from a single-point origin within the last 50,000 years, coinciding with the time of human migrations out of Africa and into Asia.^{10,19} The genetic material of *Salmonella Typhi* consists of a single chromosome that contains approximately 5×10^6 base pairs encoding approximately 4000 genes. *Salmonella Typhi* often harbors additional plasmids that may be cryptic (confer an unknown function) or involved in pathogenesis. The latter include IncHI1 plasmids that comprise the Page 1302 majority of resistance plasmids found in *Salmonella Typhi* and that remain stable in the bacteria without ongoing pressure from antibiotic selection.^{20,21}

TABLE 102.1

Classification and Selected Microbiologic Characteristics of Typhoidal *Salmonella*

SEROTYPE NAME

Salmonella Typhi *Salmonella Paratyphi A* *Salmonella Paratyphi B (schottmuelleri)*

Salmonella Paratyphi C (hirschfeldii) Comments

Serologic Classification^a

Serogroup	D	A	B	C
O antigen	9, 12	1, 2, 12		1, 4, (5), 12
H antigen				
Phase 1:2				
d: –	a: (1, 5)	b: (1, 2)		c: (1, 5)
Uncommon <i>Salmonella Typhi</i> H antigen				
variants include Hj:z66 found in Indonesia				
K antigen	Vi	–	–	Vi

^b Isolates can be identified as *Salmonella* using traditional methods. A presumptive diagnosis of *Salmonella Typhi* can often be made for *Salmonella* isolates based on selected biochemical characteristics, in addition to agglutination with D and Vi antisera.

From World Health Organization (WHO), Department of Vaccines and Biologicals. Background Document: The Diagnosis, Treatment and Prevention of Typhoid Fever. Geneva: WHO; 2003; and WHO, Department of Communicable Disease Surveillance and Response. *Salmonella* serotype Typhi. In: WHO Manual for the Laboratory Identification and Antimicrobial Susceptibility Testing of Bacterial Pathogens of Public Health Importance in the Developing World. Geneva: WHO; 2003:103–118.

The genetic core of *S. enterica* consists of genes that are conserved across Enterobacteriaceae and is approximately 90% similar to *E. coli*. In contrast, the genetic core varies approximately 1% between *S. enterica* serovars. In addition to the genetic core, *Salmonella Typhi* contains pathogenicity islands—horizontally acquired genetic elements that are often associated with unique properties of each strain and are essential for specific phenotypes associated with pathogenesis of each serotype.¹⁸ In *S. enterica*, these are specifically identified as *Salmonella* pathogenicity islands, or SPIs, and are designated with a number. *Salmonella Typhi* possesses more than 20 SPIs, many of which are found in other *S. enterica* serotypes. For example, SPI-7 is a 134-kb genetic island that possesses elements encoding the *viaB* locus, which is involved in production of the Vi capsule and evasion of host immune responses.²² SPI-7 is widely conserved across *Salmonella Typhi* and is also found in *Salmonella Paratyphi C*.

One of the most remarkable genetic features of *Salmonella* Typhi is the accumulation of more than 200 presumed inactive genes called pseudogenes.⁹ This adaptation may play a major role in the host restriction of *Salmonella* Typhi. The accumulation of a large number of pseudogenes is a common feature shared with *Salmonella* Paratyphi A, B, and C, which emerged independently of *Salmonella* Typhi (although subsequent recombination events between *Salmonella* Typhi and *Salmonella* Paratyphi A appear to have occurred),²³ yet are also adapted specifically to cause disseminated infection in humans. The same phenomenon has been observed in *Salmonella* Typhimurium ST313 in sub-Saharan Africa; ST313 has accumulated extensive gene deletions and purported pseudogene formation (more than half of which are common to *Salmonella* Typhi and *Salmonella* Paratyphi A) compared with less invasive *Salmonella* Typhimurium strains.²⁴

Epidemiology Burden and Distribution

Measures of the burden of enteric fever are limited by the absence of surveillance in many regions and also by the limited sensitivity of current diagnostic tests. Recognizing these limitations, estimates of the number of cases of typhoid and paratyphoid fever range from 12 to 16 million annually.²⁵ The typhoid and paratyphoid fevers were also responsible for 10 million disability-adjusted life years in 2017.²⁵ The economic toll of enteric fever is also devastating for affected families. A study from Malawi from 2019 to 2020 found that catastrophic illness, or an illness which cost over 40% of monthly household nonfood expenses, occurred in 44% of households affected by a case of typhoid fever.²⁶

The incidence of enteric fever and the mortality of the disease vary dramatically by region (Fig. 102.1). In South Asia and parts of sub-Saharan Africa, the incidence approaches 1000 cases per 100,000 person-years.^{27,28} In parts of northern Africa, South America, and Southeast Asia, the incidence is much lower and ranges between 10 and 100 cases per 100,000 person-years.²⁵ Disease is highly heterogeneous even within endemic countries, and the incidence tends to be considerably higher in urban than in rural areas.²⁷ In the United States and many other high-resource settings, the disease occurs sporadically, with less than 1 case per 100,000 person-years, and occurs most often in travelers returning from endemic zones (see “Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically”).

Salmonella Paratyphi A is the most common cause of paratyphoid fever and responsible for an estimated 2.6 to 4.2 million cases of enteric fever annually.²⁵ The proportional burden of enteric fever caused by *Salmonella* Paratyphi A has increased; *Salmonella* Paratyphi A is responsible for more than 20% to 50% of cases of enteric fever. This trend is expected to continue with the introduction of vaccines which protect against typhoid fever, but not enteric fever caused by *S. Paratyphi*.²⁹ In contrast, *Salmonella* Paratyphi B and C remain less-common causes of enteric fever globally.

Severity

In the preantibiotic era, 10% to 15% of clinically recognized cases of typhoid fever were fatal, whereas other cases resolved over a 3- to 4-week course of an often severe and debilitating illness.^{30,31} Prospective surveillance in the postantibiotic era has revealed that the majority of cases of enteric fever are uncomplicated and do not warrant hospitalization. Although the current mortality of typhoid fever is estimated at 0.2% to 2%, these estimates are based on limited data and there is significant variation by region.^{32,33} Reported mortality rates in hospitalized patients with enteric fever in resource-limited settings vary widely by location, ranging from 0% to 15%, with a median of 2%.³² In addition to fatal cases, in sub-Saharan Africa and other regions where enteric fever is highly endemic, intestinal perforation caused by enteric fever is a leading cause of peritonitis and acute abdominal conditions requiring surgery in children.³⁴

Source of Infection

Salmonella Typhi and *Salmonella* Paratyphi A and B are human restricted. There are no known animal reservoirs of these pathogens that cause enteric fever, and the source of infection is organisms shed in the stool of infected humans. *Salmonella* Paratyphi C can be shed by humans and animals. In the preantibiotic era, patients recovering from enteric fever shed organisms in the stool during acute illness and typically for weeks during convalescence. In

addition, a small portion of individuals convalescing from typhoid fever develop chronic asymptomatic carriage (defined as shedding of the bacteria in stool or urine for more than a year), which may persist for life.³⁵ In the preantibiotic era, up to 5% of typhoid fever survivors developed chronic asymptomatic bacterial shedding in the stool,^{36,37} with gallbladder disease being the major risk factor for carriage.^{38,39} Risk of carriage appears to be far greater in adults (3%–10%) than in young children (<1%).³⁶ Studies from Nepal, where enteric fever is highly endemic, found that 3.5% of individuals undergoing elective cholecystectomy grew *Salmonella* Typhi or *Salmonella* Paratyphi A from biliary cultures.⁴⁰ Even higher rates were observed in Santiago, Chile, another hyperendemic focus at the time, in the early 1980s.⁴¹ Urinary carriage may also occur, frequently in conjunction with urinary (*Salmonella* haematobium) schistosomiasis.⁴² Although chronic carriers are a source of transmission where enteric fever occurs sporadically, their importance in the transmission of enteric fever in highly endemic areas is uncertain.^{43,44}

FIG. 102.1 Global distribution of typhoid and paratyphoid fever. Estimated incidence of typhoid and paratyphoid fever by country from the Global Burden of Disease Study 2019. Source: Institute for Health Metrics and Evaluation. Used with permission. All rights reserved.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Follow for extended description

Mode of Transmission

Infections with typhoidal *S. enterica* are most often acquired via ingestion of fecally contaminated water or food. *Salmonella* Typhi may persist for weeks after passage in water and may persist in a variety of contaminated food items—for example, dehydrated formula and iced beverages.^{45–47} Volunteer studies may not optimally mimic natural exposure; however, the inoculum required to produce disease in 50% of adult volunteers (ID₅₀) is 10⁵–7 organisms, although fewer than 10³ organisms may produce disease,^{48,49} which might be more reflective of actual exposures in nonepidemic settings.

Molecular epidemiologic studies based on high-resolution genotyping implicate waterborne transmission as the most likely route of transmission in the majority of cases of enteric fever.⁵⁰ A study in the Kathmandu valley of Nepal revealed that >80% of water collected from public taps contained typhoidal *Salmonella* DNA.⁵¹ Both waterborne and foodborne transmission may contribute to large epidemics, and in some cases, massive epidemics have been associated with water contamination from a single source.^{52,53} In contrast, direct person-to-person transmission of the etiologic agents of enteric fever appears less common, and more than 80% of cases of enteric fever occur in individuals with no known contact with an individual with symptomatic infection.⁵⁴ Even more remarkably, in households where multiple cases of enteric fever occur, only 20% of those infections shared a common bacterial genotype, suggesting that multiple infections within a household occur through a community-based exposure to multiple circulating genotypes, rather than through direct person-to-person transmission.⁵⁰ In highly endemic areas, consistent personal risk factors for infection with typhoidal *S. enterica* include drinking nonboiled water and eating food prepared outside the home, including street-vended foods and beverages.^{55,56} In South Asia, peak transmission occurs during and immediately after the monsoon season (peaks in June through October), although disease is present year-round.⁵⁷

Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically

In areas where enteric fever occurs sporadically, most cases are imported through travel. In the United States, >85% of cases occur in travelers to endemic areas.⁵⁸ The rate of typhoid fever is estimated at 10 per 1 million travelers arriving from Asia but increases to 89 per 1 million travelers arriving from India. More than 80% of all travel-associated cases in the United States relate to travel to South Asia.⁵⁸ The risk of travel-related enteric fever is higher among travelers visiting friends and relatives. Many of the remaining non-travel-associated cases in the United States are traceable to small, local foodborne outbreaks or a chronic carrier.^{59,60} In 2010, an outbreak of *Salmonella* Typhi occurred that involved at least 12 cases across three western US states and was attributed to imported fruit pulp from Guatemala.⁶¹ Also, in areas where enteric fever occurs sporadically, small clusters of cases of sexual transmission of *Salmonella* Typhi have been documented in men who have sex with men.⁶² Infections and clinical cases have occurred among microbiology laboratory staff, especially when *Salmonella* Typhi was used as an unknown

for laboratory proficiency testing, and in at least once instance, secondary cases occurred among family members of laboratory personnel.⁶³

Antibiotic Resistance and Emergence of Pandemic Salmonella Typhi

Chloramphenicol-resistant Salmonella Typhi was first reported in 1950, 2 years after the antibiotic was first used to treat patients with typhoid fever.⁶⁴ However, widespread dissemination of antibiotic-resistant strains did not occur until the 1970s. Multidrug-resistant (MDR) Salmonella Typhi carrying IncHI1 plasmid-mediated resistance to chloramphenicol, ampicillin, trimethoprim, and sulfonamides became common in the 1980s. Nalidixic acid-resistant strains of Salmonella Typhi and Salmonella Paratyphi A, with decreased susceptibility to fluoroquinolone antibiotics, emerged in the 1990s,⁶⁵ and now the vast majority of typhoidal Salmonella isolates from South Asia have reduced susceptibility or high-level resistance to fluoroquinolones.^{13,66} At present, reduced susceptibility to ciprofloxacin is less common and more sporadic in sub-Saharan Africa, though it appears to be on the rise.¹³

Although the lack of genetic diversity of Salmonella Typhi previously hindered efforts to characterize the global epidemiology of disease, the increasing use of high-density genotyping methods, including whole-genome sequencing, has circumvented these challenges and resulted in a better understanding of the forces driving the evolution and transmission of Salmonella Typhi. Although a general lack of diversity among Salmonella Typhi suggests a lack of strong selection pressure over millennia, the recent expansion of a single fluoroquinolone-resistant haplotype known as H58 into a pandemic strain found throughout Asia and Africa, and the emergence of several other lineages with diverse gyrA mutations, suggest that fluoroquinolones may be exerting a uniquely strong selection pressure that is shaping the global evolution of Salmonella Typhi.^{13,67} Evidence suggests that fluoroquinolone resistance-conferring mutations have arisen repeatedly in multiple Salmonella Typhi lineages, spread internationally and intercontinentally and replaced less resistant strains.¹³ More recently, Salmonella strains with resistance to third-generation cephalosporins (e.g., ceftriaxone) have been reported.^{68,69} In Pakistan, strains containing multidrug resistance and resistance to third-generation cephalosporins and fluoroquinolones have been termed “extensively drug-resistant” (XDR) typhoid, and within years of first appearance now account for the majority of isolates in the country.^{66,68,70} Most recently, azithromycin resistance has been identified in multiple lineages, arising from point mutations in an efflux pump (acrB).^{71,72} Azithromycin resistance in typhoidal Salmonella might become an increasing problem given the WHO recommendation for its use in mass drug administration in settings with high rates of child mortality.^{73,74}

Host Factors and Susceptibility to Infection

Historically, enteric fever was considered a disease that predominantly affected school-aged children and young adults. However, prospective community-based surveys in highly endemic areas have demonstrated that the incidence of Salmonella Typhi bacteremia is often highest in young children (<5 years old) and that this organism may be responsible for more than 75% of cases of occult bacteremia in these settings.⁷⁵ The higher incidence of enteric fever in older children among inpatients may reflect a predisposition to more severe disease in this older age group. In highly endemic areas, Salmonella Typhi bactericidal antibodies and anticapsular (Vi) antibodies increase significantly over the first decade of life.⁷⁶ In less endemic areas, the median age of patients with typhoid fever is higher, suggesting that the age distribution of patients may be influenced by acquired immunity. However, protection from a single episode of infection is limited, as demonstrated by frequent relapse and recurrent infections among patients who have recovered from typhoid fever.⁷⁷

Although invasive NTS infections classically occur in persons who are compromised by extremes of age, malnutrition, immunodeficiency, or genetic risk factors, there is not an overt association between host Page 1304 risk factors and susceptibility to enteric fever. Various candidate gene analyses have drawn associations between single-nucleotide variations in Toll-like receptor 4 (TLR4), cystic fibrosis transmembrane conductance regulator (CFTR), and specific human leukocyte antigen (HLA) types with susceptibility to typhoid fever in Vietnam, as well as in human volunteers challenged with virulent organisms⁷⁸⁻⁸⁰; however, no detailed genome-wide study of typhoid fever associations has yet been reported.

Pathogenesis

Typhoidal *Salmonella* infection is acquired through ingestion of bacilli, typically in contaminated water, drinks, or food. The ID₅₀ is often cited to be 10⁵–7 organisms, but low inocula (<10³ organisms) may also produce infections.⁴⁸ After passing through the intestinal epithelium, typhoidal *Salmonella* are phagocytosed by macrophages and proliferate in the submucosa before dissemination via the lymphatic system and bloodstream. This period coincides with the onset of systemic symptoms and, in some cases, end organ manifestations. It is during this phase that bacilli enter the hepatobiliary system and may establish carriage in the gallbladder. Disseminated infection is terminated by immune clearance or administration of antibiotics, but bacterial persistence in the bone marrow is often longer.

It is important to note that enteric fever and NTS infections differ in fundamental aspects. With NTS, infections are typically self-limited in immunocompetent hosts and manifest largely as inflammatory gastroenteritis. In comparison, typhoidal serotypes are able to evade the normal host inflammatory response and cause prolonged bacteremia even in immunocompetent individuals, typically without overwhelming sepsis or pyogenic foci of infection. Specific bacterial adaptations used by *Salmonella* Typhi to produce persistent bacteremia in the absence of an immediately overwhelming inflammatory response include strategies to breach the intestinal epithelium in the absence of inflammatory diarrhea, to evade detection by host pattern-recognition receptors,⁸¹ to produce an AB₅ bacterial toxin that acts through cell-to-cell paracrine signaling pathways,⁸² and to persist and disseminate throughout the human host within phagocytic cells.⁸³ Much of our understanding of host-pathogen interactions in enteric fever is extrapolated from mouse models of enteric fever, which use Nramp1-deficient mice that develop disseminated infection from *Salmonella* Typhimurium or humanized mice altered to be susceptible to *Salmonella* Typhi.⁸⁴ However, controlled human infection models for *Salmonella* Typhi and Paratyphi A infection are now available and have added valuable insights into the host immune response to these organisms and disease pathogenesis.^{85,86}

Invasion

In contrast to NTS and other enteroinvasive pathogens, soon after ingestion, typhoidal *Salmonella* pass through the intestinal epithelium through specialized microfold cells (M cells) that transport the bacteria across the basolateral membrane, where they are phagocytosed by macrophages in the intestinal lymphoid tissue.⁸⁷ Such a stealth entry mechanism may explain why the invasion phase of *Salmonella* Typhi infection is usually asymptomatic and is accompanied by transient or mild diarrhea in only 10% to 20% of patients. Indeed, as *Salmonella* Typhi leaves the GI tract at this stage, stool cultures are often negative until days to weeks later when shedding through biliary system commences.

Latency and Dissemination

Typhoidal *Salmonella* deploy an array of virulence factors that enable them to persist and replicate in an intracellular compartment of phagocytic cells, primarily macrophages.^{88,89} Important cloaking mechanisms of *Salmonella* Typhi include suppression of flagellar protein synthesis, thus masking a potent inducer of TLR5 responses.⁹⁰ and synthesis of the Vi capsule that presumably masks the detection of lipopolysaccharide (LPS) and other outer membrane components, preventing early complement-mediated clearance.^{91,92} The Vi capsule, thus, helps *Salmonella* Typhi to limit inflammation and evade phagocytosis by neutrophils. The genes that repress flagellin biosynthesis and are responsible for Vi biosynthesis are encoded on the *viaB* locus located on SPI7, which is absent in most NTS strains.²² Induction of other bacterial defenses that are required for survival in the intracellular compartment is regulated by a two-component bacterial sensory-regulatory system, PhoP/PhoQ.^{93,94} The sensor PhoQ detects signals present inside the host phagosome and activates the transcriptional regulator PhoP, which in turn controls the expression of a large number of downstream genes involved in modification of lipid A, resistance to antimicrobial peptides, and acidification of the phagosome.^{95,96} Consistent with a role in pathogenesis, PhoP/PhoQ mutants are nonvirulent in humans and have been evaluated as candidate live-attenuated vaccines.^{97,98}

Quantitative blood cultures demonstrate that more than 60% of culturable circulating *Salmonella* Typhi organisms reside in the intracellular compartment.⁹⁹ Presumably, this allows for dissemination throughout reticuloendothelial tissues via the blood and lymphatic systems. The heaviest burden of infection is established in the reticuloendothelial system (intestinal lymphoid tissue, liver, spleen, and bone marrow) and gallbladder. Patients with typhoid fever most often

have very low-grade bacteremia, a feature that presents a formidable diagnostic challenge. The kinetics of quantitative bacterial cultures has been studied in Vietnam.⁹⁹ The median number of culturable bacteria is less than 1 colony-forming unit (CFU) per milliliter of whole blood in adults, and slightly higher (1.5 CFUs/mL) in children. Slightly more culturable organisms are present in the bone marrow, with a median of 10 per milliliter of bone marrow.¹⁰⁰ However, over the course of illness, the number of culturable *Salmonella* Typhi organisms in the blood decreases (median of 2 CFUs/mL in the first week to 0.1 CFUs/mL in the second and subsequent weeks), and the number of culturable bacilli increases in the bone marrow (4.8 CFUs/mL in the first week to 158 CFUs/mL in the third and subsequent weeks). As a result, the proportion of bacteria in the bone marrow increases from 5:1 (marrow to blood) in the first week of illness to more than 150:1 in the third week of illness, reflecting relative clearance of bacteria from peripheral blood but persistence in the marrow compartment.¹⁰⁰ Pathologic examination of bone marrow may demonstrate monocytic infiltrates typical of typhoid fever (see “Intestinal and Other Local Pathology”).

The initial period of replication and dissemination represents the incubation period of typhoidal *Salmonella* infection. This period typically lasts 1 to 2 weeks but can range widely (3–60 days), depending on the number of organisms ingested.¹⁰¹ Ultimately, infection results in the sufficient secretion of inflammatory cytokines, including pyrogenic cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), which result in fever. However, unlike what occurs during prototypical gram-negative bacteremia, septicemia with hypotension, neutrophilia, and disseminated intravascular coagulopathy are extremely uncommon in patients with typhoid or enteric fever.

Intestinal and Other Local Pathology

Intestinal lymphoid tissue is a predominant site of localized inflammation and persistent bacterial replication in cases of severe enteric fever (Fig. 102.2).^{102–105} Early in the course of illness, Peyer patches in the terminal ileum and draining mesenteric nodes are enlarged and contain infiltrates consisting of mononuclear cells, including macrophages and lymphocytes. As disease progresses, necrosis of intestinal lymphoid tissue occurs, with mixed inflammatory infiltrates, including neutrophils and ulceration and sloughing of the overlying intestinal mucosa.¹⁰² Immunohistochemical staining for the *Salmonella* Typhi O antigen reveals abundant organisms in necrotic areas.¹⁰³ The clinical manifestations of these changes include hemorrhage or intestinal perforation, two of the major life-threatening complications of enteric fever (Fig. 102.3). Other affected organs include the liver, with monocytic infiltrates and foci of parenchymal necrosis, and the spleen, with nodular monocytic infiltrates in the red pulp.¹⁰⁶ Similar monocytic infiltrates may occur in the gallbladder. Gallbladder perforation, which may mimic intestinal perforation, is a rare complication of typhoid fever.¹⁰⁷

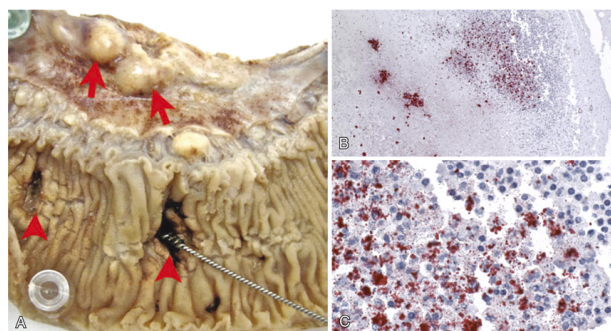


FIG. 102.2 Intestinal pathologic appearance of enteric fever. (A) The visualized mucosal surface of a surgically resected area demonstrates two perforations (arrowheads); the visualized serosal surface demonstrates significant enlargement of lymph nodes (arrows). (B and C) Lower- and higher-magnification views demonstrate immunohistochemical staining for the *Salmonella* O:9 antigen in a necrotic lymph node. From Neil KP, Sodha SV, Lukwago L, et al. A large outbreak of typhoid fever associated with a high rate of intestinal perforation in Kasese District, Uganda, 2008–2009. *Clin Infect Dis.* 2012;54:1096.

[flow for extended description](#)



FIG. 102.3 Intraoperative photograph of intestinal perforation caused by *Salmonella* Typhi. Intestinal perforation is visible on the antimesenteric border of the small bowel, which is inflamed with patchy exudates on the serosal surface. Courtesy Dr. Pukar Maskey, Patan Hospital, Kathmandu, Nepal. From Harris JB, Brooks WA. Typhoid and paratyphoid (enteric) fever. In: Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

Relapse
and
Chronic
Carriage

Untreated, typhoid fever may persist for up to 4 weeks, and relapse occurs in up to 10% of untreated cases, usually within 2 weeks after initial resolution of fever.¹⁰⁸ Relapse is usually caused by the same strain as that associated with the original infection.¹⁰⁹ The gallbladder is the primary site of chronic carriage, although some asymptomatic carriers continue to shed the organism in stool even after cholecystectomy. The in vitro and clinical observations that *Salmonella* Typhi forms biofilms on cholesterol gallstones may explain the strong epidemiologic association between gallstones and carriage.^{110,111} An association has been reproducibly observed between chronic carriage of *Salmonella* Typhi and risk of gallbladder carcinoma. It is unknown if carriage causes cancer or reflects underlying biliary epithelial surface abnormalities^{112,113}; chronic low-grade biliary inflammation and emerging data on the genotoxic activity of *Salmonella* CdtB support an etiological role.¹¹⁴

Clinical Manifestations

The clinical manifestations of enteric fever are nonspecific. Fever is reported in the vast majority of clinically apparent cases, and many other symptoms are variably reported, including headache, cough, nausea, vomiting, constipation, and diarrhea.¹¹⁵ Although paratyphoid fever was considered less severe than typhoid fever, more recent comparisons suggest that typhoid fever and paratyphoid fever caused by *Salmonella* Paratyphi A are generally indistinguishable on clinical grounds.¹¹⁶ Most patients with enteric fever are diagnosed in the ambulatory setting, and up to 90% are treated as outpatients. In contrast, classic descriptions of the features of enteric fever are derived from a series of hospitalized patients with more severe disease.³⁰ Complications of enteric fever were generally thought to occur late in the course of disease, usually after 2 weeks of fever. However, experience has demonstrated that major complications of enteric fever, including intestinal perforation and encephalopathy, may occur within days of onset of fever.^{103,104,117}

Uncomplicated Typhoid Fever

Clinical features of enteric fever diagnosed in the ambulatory setting are listed in Table 102.2. Fever without localizing signs or symptoms may be the sole manifestation of enteric fever. The onset of fever may be insidious, and fevers typically increase over the first week of illness.¹¹⁸ A variety of other nonspecific flulike symptoms are common early in the course of enteric fever, and headache, anorexia, myalgias, and malaise may precede the onset of fever. Mild confusion may be seen even in ambulatory patients, and a nonproductive cough is also a common feature of uncomplicated enteric fever.¹¹⁵ Abdominal complaints may include diarrhea, constipation, and abdominal pain. Invasive diarrhea, such as seen in NTS gastroenteritis, does not typically occur in enteric fever.

The physical findings of uncomplicated enteric fever are also nonspecific. Although relative bradycardia, or pulse-temperature dissociation, is a classic sign of enteric fever, it may not be a clinically useful predictor of enteric fever for individual patients, and is absent in the majority of patients.¹¹⁹ Rose spots, the classic cutaneous manifestation of enteric fever, are 1- to 4-mm blanching pink macules, and are most often seen on the chest, back, and abdomen during the second week of fever. However, rose spots are uncommon in uncomplicated typhoid fever. Mild, nonlocalizing, abdominal tenderness may be present. Hepatomegaly and splenomegaly, if present, are usually modest. A white or yellowish-brown coating of the tongue that spares the tongue's edges is a common physical finding.¹¹⁶

Laboratory Findings in Uncomplicated Infection

Although abnormal white blood cell counts in peripheral blood are common in patients with bacteremia, the majority of patients with uncomplicated enteric fever do not have leukocytosis, neutrophilia, or increased immature neutrophils.¹¹⁵ However, either leukocytosis or leukopenia may be present. Both the hematocrit and platelet counts are typically normal or slightly low. Elevated serum aspartate transaminase and alanine transaminase are very common in enteric fever; values two to three times above the upper limit of the normal range are typical.¹²⁰ On occasion, more severe hepatitis is observed, but very high levels of blood transaminases (>500 IU/L) should prompt concern for other causes, including viral hepatitis or drug toxicity.¹²⁰

Severe Illness

Classic descriptions of the features of enteric fever are drawn from observations of a series of hospitalized patients in the preantibiotic era, in which mortality rates were consistently in the 10%

to 15% range. The natural history of untreated disease included progressively increasing fevers over the first week of illness, followed by increasing abdominal complaints and rash over the second week of illness, followed by complications, including intestinal hemorrhage and perforation, or gradual resolution in the third and fourth weeks of illness.³⁰

In the present antibiotic era, it remains true that because enteric fever progresses gradually, patients who are severely ill with enteric fever may have been febrile for longer than a week when they first come to clinical attention. Patients with severe enteric fever may appear toxic, and characteristically would have moderate abdominal pain or tenderness, and constipation or diarrhea. Physical findings in severe enteric fever include hepatomegaly and splenomegaly. Patients with severe enteric fever are more likely to have major complications listed in Table 102.3. Complications associated with increased mortality in severe typhoid fever include intestinal hemorrhage and perforation, severe encephalopathy, seizures, and pneumonia.¹²¹

Gastrointestinal Complications

Clinical Features of Typhoid and Paratyphoid Fever		
	CLINICAL FEATURE	APPROXIMATE FREQUENCY *
Flulike symptoms	Fever	>95%
	Headache	80%
	Chills	40%
	Cough	30%
	Myalgia	20%
	Arthralgia	<5%
Abdominal symptoms	Anorexia	50%
	Abdominal pain	30%
	Diarrhea	20%
	Constipation	20%
	Coated tongue	50%
Physical findings	Hepatomegaly	10%
	Splenomegaly	10%
	Abdominal tenderness	5%
	Rash	<5%
	Generalized adenopathy	<5%

* The proportion of patients demonstrating these clinical features of enteric fever varies, depending on the time, the region, and the type of clinical population (hospitalized or ambulatory) assessed. Estimates are drawn from recent case series in an endemic area, with patients presenting for ambulatory or inpatient care. ¹⁰⁶⁻²⁰⁶

Intestinal hemorrhage occurs in up to 10% of hospitalized patients with severe enteric fever and is usually self-limited.^{30,32} Intestinal perforation, a major life-threatening complication of enteric fever, occurs in up to 3% of hospitalized patients.^{30-32,121} These complications arise directly from invasion of Salmonella into Peyer patches in the small intestines. A series of patients from Vietnam with intestinal perforation resulting from enteric fever demonstrated that the median length of illness preceding perforation was 9 days from the onset of fever, although some cases occurred within the first week.¹⁰⁴ Perforation is suggested by clinical signs of peritonitis, including tachycardia, leukocytosis, neutrophilia, and abdominal pain with guarding and rebound tenderness. The clinical diagnosis of perforation requires a high index of suspicion because patients with severe enteric fever may have a toxic appearance and significant abdominal tenderness even before perforation. Radiographic evidence of pneumoperitoneum may be present in only 50% of cases.

Complications of Typhoid and Paratyphoid Fever		
SYSTEM	COMPLICATION	NOTES
Gastrointestinal	Hemorrhage	10%–15% in hospitalized patients
	Perforation	3% in hospitalized patients
Hepatobiliary	Jaundice	1%–3% in hospitalized patients
	Hepatitis	Usually subclinical (↑ ALT/AST)
	Acute cholecystitis	Rare; gallbladder may perforate
Neurologic	Mild encephalopathy	Confusion or apathy common
	Severe encephalopathy	Delirium, stupor, or coma
	Seizures	Common in children ≤5 years
	Meningitis	Rare, primarily infants
	Guillain-Barré syndrome	Reported
Respiratory	Bronchitis	Dry cough is common
	Pneumonia	May be other concomitant bacterial infection (e.g., <i>Streptococcus pneumoniae</i>)
	Empyema	Rare reports
Cardiovascular	Myocarditis	Usually subclinical (ECG changes)
	Endocarditis	Rare reports
Hematologic	Anemia	Usually subclinical
	Disseminated intravascular coagulation	Usually subclinical (↑ PT/PTT)
Other	Musculoskeletal pyogenic infections	Osteomyelitis (particularly vertebral), psoas abscess and others reported
	Hemolytic-uremic syndrome	Reported
	Miscarriage	Reported

ALT, Alanine aminotransferase; AST, aspartate aminotransferase; ECG, electrocardiogram; PT, prothrombin time; PTT, partial thromboplastin time.

Modified from Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

Neurologic Complications

Various case series have documented a widely varying prevalence of diverse neuropsychiatric manifestations during enteric fever, ranging from encephalopathy, encephalomyelitis, transverse myelitis, meningitis, ataxia, and Guillain-Barré syndrome.^{117,122} Although most of these manifestations are uncommon, patients with severe enteric fever often present with some component of encephalopathy, and a history of confusion or an apathetic affect are common.³² Of note, typhoid was named for its clinical similarity to typhus, and the name typhus was from the Greek term for smoky or cloudy, reflecting the neurologic features that often accompanied this febrile illness. More severe encephalopathy, manifesting with delirium, stupor, and coma, occurs in a smaller number of hospitalized patients and is associated with a high risk of mortality.^{121,123,124} Osler described this as a “pseudo-wakeful state” of “muttering” delirium, known as “coma vigil,” and noted that it portends poor outcomes. Even in cases of severe encephalopathy, microbiologic evaluation of cerebral spinal fluid is usually unrevealing, and pleocytosis, if present, usually reveals fewer than 35 cells/ μ L.^{123,124} Encephalopathy during enteric fever may be most common among older children and young adults.¹²⁵ Seizures during enteric fever are most common in young children and are associated with increased mortality.¹²¹ Meningitis caused by *Salmonella Typhi* is seen primarily in infants.¹²⁶

Metastatic Pyogenic Complication

Enteric fever is a disseminated bacterial infection, but pyogenic abscess formation during enteric fever is notably infrequent, although *Salmonella Paratyphi C* is more likely to cause abscesses and focal infections than the other typhoidal strains. This contrasts with what occurs during invasive nontyphoidal salmonellosis, in which osteomyelitis, joint infection, abscess formation, and endovascular infection more frequently occur. Although uncommon, pyogenic complications during typhoid have been described and include empyema, osteomyelitis, hepatosplenic, and other organ system and soft tissue abscesses, as well as endovascular infections and endocarditis.¹²⁷

Diagnosis

The laboratory diagnosis of enteric fever is often challenging because current culture-based, serologic, and molecular diagnostic tests for enteric fever fail to achieve an optimal combination of sensitivity and specificity.¹²⁸⁻¹³¹ For this reason, it is often appropriate to treat patients with suspected enteric fever with empirical antibiotic therapy. The diagnosis of enteric fever should be considered in any person with fever, especially in those with fever lasting longer than 3 days and who have had exposure in the last 1 to 6 weeks to an area where enteric fever is endemic. In endemic areas, other clinical factors that are associated with a higher likelihood of enteric fever include a temperature greater than 39°C (102.2°F), ill appearance, young age (<5 years), and any abdominal complaints, including abdominal pain, diarrhea, or constipation.⁷⁵ As the duration of fever increases, the likelihood of enteric fever increases.¹³² Despite this, it is important to recognize that the majority of individuals suspected of having enteric fever, even in endemic settings, do not have enteric fever but rather an alternative infection^{133,134}; therefore, clinical suspicion of and empirical treatment for enteric fever should not halt the search for alternative causes of fever.

Culture-Based Diagnostics

Although a presumptive diagnosis of enteric fever may be sufficient grounds for initiating and continuing antimicrobial therapy, a definitive diagnosis of enteric fever is made only through the isolation of a typhoidal serotype of *S. enterica* (*Salmonella Typhi* or *Salmonella Paratyphi A, B, and C*) from the blood, bone marrow, stool, urine, or other clinical specimen of a febrile patient. Isolation of the causative organism also allows antimicrobial resistance testing, which facilitates optimal management.

Blood culture is the most common method of diagnosis, where adequate microbiologic facilities exist. The sensitivity of blood culture varies from approximately 40% to 80%,^{135,136} and a recent meta-analysis estimated an average sensitivity of 59%.¹³⁷ The low-grade bacteremia in enteric fever is an inherent limiting factor to the sensitivity of blood cultures, as is prior receipt of antibiotics.⁹⁹ To maximize the sensitivity of blood cultures, the World Health Organization (WHO) recommends that cultures be performed using 10 to 15 mL of blood from schoolchildren and adults, and 2 to 4 mL of blood from toddlers and preschool-aged children.¹³⁸ The use of a nonselective broth media, as is standard for most blood cultures, is encouraged because it

maximizes detection of a wide range of pathogens. The sole use of *Salmonella* selective or enriching media, such as ox bile medium, is discouraged.¹³⁸

Although less sensitive than blood cultures, stool cultures are positive in more than 50% of children and 30% of adults with enteric fever, and they increase the diagnostic yield over blood cultures alone by approximately 5%.¹³⁶ To maximize detection, WHO recommends the use of greater than 1 g of stool and selenite enrichment broth.¹³⁸ Culturing of bone marrow has the highest sensitivity (80%–95%) of all the culture-based approaches, with culturing of a 1-mL aspirate of bone marrow having the sensitivity of a 15-mL peripheral blood culture in an adult.^{99,100} Culturing of bone marrow is, however, not clinically practical for most patients with suspected enteric fever, although microbiologic and hematologic analysis of a bone marrow aspirate may be part of an evaluation of a patient with a prolonged fever of unknown origin when less invasive testing is unrevealing. Other specimens that may yield growth of *Salmonella* Typhi or *Salmonella* Paratyphi A, B, and C include urine, duodenal aspirates, and specimens from skin biopsy of rose spots. However, these approaches are not routinely recommended. Cultures made from intestinal biopsy specimens and peritoneal fluid of patients with perforation are rarely positive.^{103,104}

Serologic Tests

The Widal test detects agglutinating antibodies against the somatic (O) and flagellar (H) antigens of *Salmonella* Typhi. The Widal test was developed over a century ago and remains one of the world's most widely used diagnostic tests, but suffers from significant limitations in its sensitivity, specificity, and reliability. As with most serologic tests, a false-negative Widal test result may occur early in the course of illness, and a false-positive Widal test result may be caused by past infection or previous exposure to cross-reactive antigens or vaccination. There are no universal standards that define the cutoff dilution of agglutinating antibodies to indicate a positive Widal test result. The very low sensitivity and specificity of the assay (50%–80%) and the inability to discern active from previous infection or vaccination means that the assay should rarely, if ever, be used.¹³¹

Other rapid serologic tests for enteric fever have been developed. Major commercial tests include the IDL Tubex and Typhidot assays. The IDL Tubex test (IDL Biotech AB, Bromma, Sweden) is a rapid immunochromatographic test that detects immunoglobulin M (IgM) antibodies to the O:9 antigen (the major antigenic determinant of *Salmonella* Typhi LPS). An advantage of this approach is that T-cell-independent IgM responses that target *S. enterica* O polysaccharide develop early in the clinical course of illness. Studies have suggested that the sensitivity of the IDL Tubex is approximately 70% to 80%, with a specificity of 80% to 90% when evaluated against standards involving patients with blood culture-proven typhoid fever versus other known causes of bacteremia.^{139–142} The Typhidot assay (Malaysian Biodiagnostic Research, Bangi, Malaysia) that detects both IgM and IgG to a 50-kDa outer membrane protein antigen of *Salmonella* Typhi has similar performance characteristics.^{130,143}

Several newer serologic approaches have been developed and evaluated in clinical studies. These include detection of anti-*Salmonella* IgA in secretions of peripheral blood lymphocytes¹⁴⁴ and new antigen targets, common to *Salmonella* Typhi and *Salmonella* Paratyphi A, for enzyme-linked immunosorbent assays (ELISAs) and immunochromatographic assays.^{145–147} These diagnostics have demonstrated greater accuracy in distinguishing typhoid/paratyphoid from other febrile illnesses than currently available serologic tests, but these newer assays have not yet been commercialized.

Molecular Approaches

Molecular diagnostic approaches based on nucleic acid detection have met with only mixed and limited success, presumably reflecting the low-level bacteremia that has hindered culture-based methods, and the action of inhibitors in human blood; such assays are not currently commercially available. Although some promising results with new approaches have been seen in small studies,^{148,149} further clinical validation is needed to characterize their accuracy.

Screening for Chronic Carriage

Identifying chronic carriers of typhoidal *Salmonella* may be important for prevention of transmission, particularly in the context of enteric fever outbreaks in nonendemic settings in which

the source of infection is unknown. The conventional approach to identify carriers involves collection of at least three stool specimens (separated by days to weeks) for microbiologic culture to isolate *Salmonella*. Although specificity is excellent, the sensitivity of this approach is estimated at 70% to 80%.¹⁵⁰ Screening by duodenal string capsules is an alternative which may improve this yield, although the string test is not currently commercially available.¹⁵¹ There is an extensive body of literature on the use of Vi antibody testing to identify carriers, but the results have been mixed,¹⁵²⁻¹⁵⁵ and using Vi as a carrier detection target will be further compromised as Vi-based typhoid conjugate vaccines become key components of global typhoid-control strategies.

Management

Appropriate antibiotic management (Table 102.4) reduces the mortality rate of enteric fever from 10% to 15% to less than 1%, and shortens the duration of fever from 3 to 4 weeks to 3 to 5 days after Page 1308initiation of therapy.¹⁰⁸ Uncomplicated enteric fever is treated with a single antibacterial drug. However, the widespread emergence of antibiotic resistance in *Salmonella* Typhi and *Salmonella* Paratyphi A limits the arsenal of effective antibiotics in many areas. The selection of an appropriate antibiotic is dependent not only on antibiotic resistance patterns but also on the severity of illness, the age of the patient, and the availability and cost of antibiotics in resource-limited settings. In addition to antibiotic therapy, supportive care, monitoring for complications, and adjunctive therapy with pulse-dose steroids for severe disease (see later) are important aspects of management of patients with enteric fever.

Chloramphenicol, Ampicillin, and Trimethoprim-Sulfamethoxazole

Chloramphenicol, ampicillin, and trimethoprim-sulfamethoxazole were the original first-line antibiotics for treating patients with enteric fever. Widespread resistance emerged in the 1980s, through dissemination of strains containing IncH1 plasmids conferring simultaneous resistance to all three agents (Fig. 102.4).¹⁵⁶ Such strains remain widely disseminated and are termed MDR *Salmonella* Typhi. However, over the past several years, there has been a global decline in MDR *Salmonella* Typhi concomitant with the rise in fluoroquinolone resistance.¹³

Because of the risk of irreversible bone marrow aplasia in up to 1 in 10,000 recipients,¹⁵⁷ chloramphenicol is no longer routinely used in many areas, and oral chloramphenicol is not available in the United States. However, in many resource-limited areas, oral chloramphenicol is still available, has excellent bioavailability and central nervous system penetration, and has broad-spectrum activity against many common causes of serious childhood bacterial infections, including enteric fever. Chloramphenicol reduces the mortality rate of enteric fever substantially and decreases the time to defervescence to 3 to 5 days, and was shown to be equivalent to gatifloxacin in a randomized trial,¹⁵⁸ but its use is associated with a high relapse rate and risk of chronic carriage.^{159,160} Oral trimethoprim-sulfamethoxazole may also be used to treat individuals with susceptible strains, although delayed clearance and relapses are common.¹⁶¹ Oral amoxicillin may also be used to treat individuals with susceptible strains and may result in a

Antibiotics Commonly Used in Treating Patients with Enteric Fever *

	ANTIBIOTIC	ROUTE	TYPICAL PEDIATRIC DOSAGE *	ADULT DOSAGE	TYPICAL DURATION	COMMENTS
Original first-line agents	Chloramphenicol	PO/IV	12.5–25 mg/kg qid	2–3 g/day (divided qid)	14–21 days	These agents are effective for susceptible strains; however, multidrug resistance to all of these agents is common in many areas.
	Amoxicillin	PO	25–35 mg/kg tid	1 g tid	14 days	
	Ampicillin	IV	25–50 mg/kg q6h	2 g q6h	14 days	
	TMP/SMX	PO/IV	8–12 mg/kg/day (TMP) 40–60 mg/kg/day (SMX) (divided qid)	160 mg/800 mg qid	14 days	
Fluoroquinolones	Ofloxacin	PO/IV	15 mg/kg bid	400 mg PO bid	5–14 days	Ofloxacin and ciprofloxacin are effective for nalidixic acid–susceptible strains. Short courses of 5–7 days are acceptable in uncomplicated cases. Ten- to 14-day courses are recommended in patients requiring hospitalization or parenteral therapy. Gatifloxacin appears effective for uncomplicated cases caused by NaR strains.
	Ciprofloxacin	PO/IV	15 mg/kg bid	500 mg PO bid	5–14 days	
	Gatifloxacin	PO	10 mg/kg qd	10 mg/kg/day qd	7 days	
Third-generation cephalosporins	Ceftriaxone	IV	50–100 mg/kg qd	1–2 g IV qd	7–14 days ^c	These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.
	Cefixime	PO	10 mg/kg bid	200 mg PO bid	7–14 days	
	Aztreonam	IV	50 mg/kg q8h	2g q8h		
Macrolides	Azithromycin	PO	20 mg/kg qd	500–1000 mg PO qd	7 days	Alternative to ceftriaxone for cephalosporin-allergic patients. These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.
Carbapenems	Meropenem	IV	20–40 mg/kg q8h	1–2 g q8h	10 days	Case series data only (in XDR typhoid); no trials have been reported
Eradication of carriage	Ciprofloxacin	PO	—	500–750 mg bid	28 days	If eradication is indicated for public health reasons, a trial of medical therapy is justified, even in patients with evidence of cholelithiasis. Cholecystectomy may be considered.
	Amoxicillin	PO	75–100 mg/kg/day in 3 divided doses	2 g tid	4–6 weeks	Lower efficacy than fluoroquinolones but may be considered for treatment of carriage in individuals harboring strains with fluoroquinolone resistance.

NaR, Nalidixic acid resistant (see text); TMP-SMX, trimethoprim-sulfamethoxazole.

* See text for references.

^b Weight-based pediatric dosage should not exceed adult maximum.

roughly comparable likelihood of relapse.¹⁶²

Fluoroquinolones

Fluoroquinolone antibiotics are a highly potent therapy for enteric fever, although increasing antimicrobial resistance is occurring globally, limiting their effectiveness.^{13,163-165} For

Because of the availability and widespread use of fluoroquinolone antibiotics in many areas of the world, fluoroquinolone resistance has become increasingly common in *Salmonella* Typhi and *Salmonella* Paratyphi A in areas where enteric fever is highly endemic.^{13,66} In South and Southeast Asia, strains with reduced susceptibility to fluoroquinolones now predominate, and in sub-Saharan Africa, such strains are on the rise.¹³ Fever clearance times and treatment response have worsened with this rise in fluoroquinolone nonsusceptibility.¹⁶⁷

Before 2012, breakpoints for fluoroquinolone susceptibility were standard across all Enterobacteriaceae, although resistance to nalidixic acid (a nonfluorinated quinolone antibiotic) was associated with decreased clinical responsiveness to fluoroquinolone antibiotics, such as ofloxacin and ciprofloxacin.¹⁶⁸⁻¹⁷¹ Nalidixic acid resistance (NaR) thus became a surrogate marker for predicting clinical failure of fluoroquinolones, even for *Salmonella* Typhi and *Salmonella* Paratyphi strains that met Clinical Laboratory and Standards Institute (CLSI) criteria for fluoroquinolone susceptibility.^{65,172} The CLSI amended its breakpoints for *Salmonella* Typhi and *Salmonella* Paratyphi A, B, and C for ciprofloxacin in 2012 and for levofloxacin and ofloxacin in 2013. For ofloxacin, strains with a minimal inhibitory concentration (MIC) of less than or equal to 0.12 µg/mL are considered susceptible. Patients with strains with an MIC of 0.12 µg/mL defervesce within an average of less than 5 days after initiation of ofloxacin therapy, and patients with strains with an MIC of 0.064 µg/mL or less defervesce in less than 4 days.¹⁷¹ For ciprofloxacin, the amended breakpoint for defining susceptibility is now less than or equal to 0.06 µg/mL. A complicating factor is that many laboratories are not currently able to implement testing of these lower breakpoints for typhoidal *S. enterica* isolates. For such strains, NaR remains an appropriate marker of reduced susceptibility to fluoroquinolone antibiotics.

Although fluoroquinolones are generally well tolerated, concerns about fluoroquinolone-induced articular toxicity in immature animals have limited their use in children in some areas; however, several studies support the safety of their use in children.¹⁶⁶ Gatifloxacin is a dual mechanism-of-action fluoroquinolone that inhibits both DNA gyrase and topoisomerase IV. Gatifloxacin has been associated with increased frequency of hypoglycemia and hyperglycemia, compared with other fluoroquinolones, and for this reason it was withdrawn from the US market. However, gatifloxacin remains available in many countries where enteric fever is endemic and is reported to be clinically effective against some NaR *Salmonella* Typhi.¹⁷³ However, in Nepal, a trial comparing gatifloxacin with ceftriaxone was stopped early owing to higher treatment failure in the gatifloxacin arm as a result of fluoroquinolone resistance.¹⁵⁸ In documented cases of fluoroquinolone nonsusceptibility or for empirical treatment in cases acquired in regions where such strains are common, alternative agents should be used.

Cephalosporins

The emergence of *Salmonella* Typhi and *Salmonella* Paratyphi A strains with decreased susceptibility to fluoroquinolones has shifted the choice for empirical treatment of individuals with suspected severe enteric fever toward third-generation cephalosporins. However, although third-generation cephalosporins, such as ceftriaxone and cefixime, may be used, they appear inferior to fluoroquinolones for fully fluoroquinolone-susceptible strains.¹⁶⁴ Of note, both cefixime and short courses of parenteral ceftriaxone (≤ 7 days) are associated with high relapse rates during enteric fever.^{164,174} In addition, time to defervescence is longer in patients treated with cefixime and ceftriaxone, compared with responses after administration of fluoroquinolones.¹⁶⁴ Resistance of typhoidal *S. enterica* to extended-spectrum β -lactam antibiotics had been rare and sporadically reported^{175,176}; however, in 2017, a large outbreak of extended-spectrum β -lactam *Salmonella* Typhi was reported in Pakistan¹⁷⁷ and has spread throughout the country, displacing less resistant strains and now accounting for the majority of cases in the country.⁶⁶ These strains are additionally MDR and fluoroquinolone-resistant and are termed XDR. Treatment of XDR strains has been primarily with carbapenems or azithromycin. Travel-related spread of XDR strains raises the concern that highly drug-resistant strains may become increasingly common globally.¹³

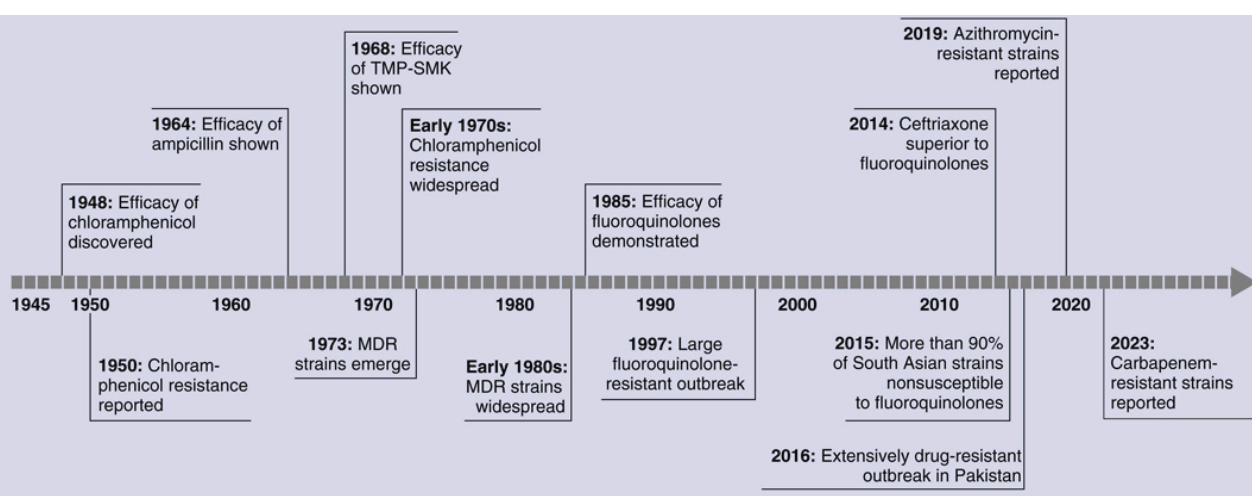
Azithromycin

Azithromycin is a highly effective drug for treating patients with enteric fever; favorable attributes include its efficacy, its established safety in children, and its excellent bioavailability and pharmacokinetics, with once-daily dosing. Azithromycin reaches high intracellular concentrations that may contribute to its high rates of cure. Duration of treatment is typically 5 to 7 days.^{163,174} Azithromycin may reduce risk of relapse compared with third-generation cephalosporins;

however, there is a trend towards higher treatment failure and slower time to fever clearance.¹⁶⁴ In a human challenge model, compared with ciprofloxacin, participants receiving azithromycin had significantly longer duration of bacteremia (20.1 vs. 90.8 hours) and fever.¹⁷⁸ In recent years, there have been increasing reports of azithromycin-resistant strains, associated with point mutations in an efflux pump.⁷¹ Such strains have been observed in several countries and have arisen independently in multiple *Salmonella* Typhi lineages.⁷² However, the global frequency of azithromycin resistance remains low, and it is an excellent option for treating patients with enteric fever caused by third-generation cephalosporin-resistant and fluoroquinolone-nonsusceptible *Salmonella* Typhi and *Salmonella* Paratyphi A. Page 1310

Other Antibiotics

Four small clinical trials have demonstrated the efficacy of aztreonam for treatment of enteric fever with comparable rates of efficacy and relapse to first-line antibiotics.^{179,180} In the setting of cephalosporin-resistant *Salmonella* Typhi, carbapenems have also been successfully used to treat patients with typhoid fever, although no detailed data on efficacy and relapse rates are available.¹⁸¹ Susceptible isolates, fluoroquinolone antibiotics, such as ciprofloxacin, and the less costly, more widely available ofloxacin, result in defervescence within 3 to 4 days, which is similar to the



response

associated with chloramphenicol and more rapid than that for third-generation cephalosporins.^{163,166} Administration of a fluoroquinolone also results in lower rates of relapse and carriage compared with those associated with use of chloramphenicol or cefixime.^{163,164,166}

Adjunctive Therapy

Adjunctive therapy with high-dose dexamethasone has been shown to decrease mortality in cases of severe typhoid fever.¹²⁴ A double-blind trial involving 38 adult and pediatric patients with severe typhoid fever in Indonesia, conducted in 1981 and 1982, demonstrated that treatment with chloramphenicol, along with 3 mg/kg of dexamethasone administered over 30 minutes, followed by eight doses at 1 mg/kg every 6 hours, significantly reduced mortality (2 of 20 patients died in the dexamethasone arm) compared with chloramphenicol and placebo (10 of 18 patients died in the placebo arm, $P = .003$).¹²⁴ In this study, severe typhoid fever was defined as typhoid fever with shock or profound encephalopathy manifesting with delirium or obtundation. More recent studies continue to support the concurrent use of dexamethasone and antibiotics in both children and adults with severe typhoid fever.^{123,182} For this reason, administering high-dose steroids to patients who present with shock or, more often, significant mental status changes, and in whom there is strong suspicion of enteric fever, is recommended.

Supportive Care and Management of Complications

Supportive care of patients with severe enteric fever includes rehydration, nutritional management, and mobilization. Contrary to a prevailing historical opinion, it appears that fevers and malaise can be managed safely and more comfortably with a standard regimen of antipyretics.¹⁸³ One small double-blind trial showed that ibuprofen was superior to paracetamol in reducing the morbidity of fevers in enteric fever.¹⁸³

Patients with intestinal hemorrhage should be monitored, ideally in an intensive care setting, but can generally be managed through supportive care. Blood products, including packed red cells and plasma, may be needed in severe cases. Intestinal perforation in enteric fever is associated with a high mortality rate, but prompt surgical intervention is associated with improved survival.^{31,104,184} Although the majority of perforations are single, careful inspection is required to determine if multiple perforations are present. Although simple closure may be performed in some cases, a wedge excision or segmental resection is more often necessary because the surrounding bowel is often necrotic.^{185,186} Drainage of collections and peritoneal lavage should be performed before wound closure. Antibiotic regimens after perforation should be expanded to cover other intestinal microbiota, including anaerobes.

Treatment of Relapse and Chronic Carriage

Relapse usually occurs within 2 weeks of discontinuation of antibiotics, and relapsing patients typically present with milder symptoms than during the primary episode. Isolates obtained after relapse typically have the same antibiotic susceptibility as during the initial episode. The approach to treating patients with relapsed enteric fever is usually the same as for the primary episode.

Chronic carriage is defined as asymptomatic shedding of typhoidal *S. enterica* for 1 year or more. Asymptomatic shedding does not indicate a recurrence of illness or a high risk of recurrent illness. Although chronic carriage is associated with an increased risk of gallbladder carcinoma,¹¹² it is unknown whether the eradication of carriage reduces this risk of carcinoma, so the decision to attempt to eradicate carriage is usually based on public health considerations and is essential for individuals whose occupation involves food handling. Several antibiotics used to treat individuals with acute enteric fever reach high concentrations in bile, including fluoroquinolones. A 28-day course of ciprofloxacin has demonstrated 80% to 90% efficacy in eradication of carriage.¹⁸⁷ However, cholecystectomy may be necessary to eradicate carriage in individuals with cholelithiasis who continue to shed typhoidal *S. enterica* after prolonged antimicrobial therapy.^{188,189} In an era of widespread nonsusceptibility to fluoroquinolones, there is a limited evidence base for successful treatment of carriage with antibiotics other than fluoroquinolones. Amoxicillin, administered at high doses (75–100 mg/day for 4–6 weeks), can be used, but tolerability is poor and overall eradication rates in many studies are modest.¹⁹⁰ There are even fewer data for treatment of carriers with trimethoprim-sulfamethoxazole with or without rifampin.¹⁹¹

Prevention

Because enteric fever is transmitted primarily through fecally contaminated water and food, the major preventive strategy to prevent enteric fever is the provision of safe water and adequate sanitation. At a municipal public infrastructure level, this requires appropriate facilities for waste disposal and sewage treatment in addition to provision of a safe water supply. In the United States, the introduction of chlorination and filtration of municipal water supplies in the early 1900s led to rapid and dramatic declines in the burden of typhoid.¹⁹² At a household level, interventions to prevent enteric fever include hand washing with soap before preparing or eating food, avoidance of raw or uncooked foods, and local disinfection of unsafe drinking water, followed by storage in narrow-mouthed containers. Unfortunately, WHO currently estimates that 2 billion people use a drinking water source that is contaminated with feces.¹⁹³

Historically, a heat-phenol-inactivated, whole-cell, killed typhoid vaccine was available for parenteral administration. The vaccine required multiple administrations, induced only short-term protection, and was associated with a high adverse event profile.¹⁹⁴ That vaccine is no longer commercially available in the United States.¹⁹⁵ Currently, two more effective and better-tolerated typhoid fever vaccines (Table 102.5) are commercially available in many areas: a capsular polysaccharide vaccine, licensed in the United States as Typhim Vi (Sanofi Pasteur, Swiftwater, PA), and an oral live-attenuated vaccine containing the *Salmonella* Typhi strain Ty21a, licensed in the United States as Vivotif (Valneva, Lyon, France). The polysaccharide vaccine is composed of the Vi capsular polysaccharide antigen from *Salmonella* Typhi strain Ty2, and the Vi vaccine confers 50% to 80% protection against *Salmonella* Typhi in children 2 years of age and older when administered as a single parenteral dose.^{194,196–198} The Vi vaccine is not boosted by additional doses, and protection wanes rapidly after 2 years, at which point revaccination is recommended if risk is ongoing. Ty21a is administered orally and also affords 50% to 80%

protection against *Salmonella* Typhi.^{194,199,200} Although Ty21a enteric-coated capsules are often used for travelers, a liquid formulation prepared from two sachets, one containing vaccine and a second containing a buffer, have been used for oral administration to children in resource-limited settings. The duration of immunity to the Ty21a vaccine is dependent on the number of doses received. In the United States, a four-dose every-other-day series of Ty21a is recommended and confers significant immunity for up to 7 years, although revaccination is recommended every 5 years if risk is ongoing.¹⁹⁵ Antibacterial drugs should not be given any time from 3 days before until 3 days after administration of a series of Ty21a. For most patients with significant primary immunodeficiencies, Ty21a should not be given; however, the Strategic Advisory Group of Experts for the WHO advises that Ty21a can be safely given to asymptomatic individuals living with human immunodeficiency virus (HIV) infection and a CD4+ T-cell count greater than 200/mm³.¹⁹⁴

TABLE 102.5

Typhoid Fever Vaccines Currently Commercially Available Internationally

VACCINE	TYPE	ROUTE	DOSE AND INTERVAL	MINIMUM AGE	PROTECTION AGAINST <i>SALMONELLA</i> TYPHI	BOOSTING INTERVAL IN TRAVELERS
Ty21a	Live attenuated	Oral	Four doses (in United States) Administer one dose every other day until complete	6 years ^a	50%–80% ^b	Every 5 years
Vi capsule antigen	Polysaccharide	Intramuscular	1	2 years	50%–80%	Every 2 years
Vi tetanus toxoid conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6 months	80%–85%	Not yet defined
Vi CRM-197 conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6 months	Not yet defined	Not yet defined

^a Per World Health Organization and US Advisory Committee on Immunization Practices. ^{194, 195}

^b Has some efficacy against *Salmonella* Paratyphi B (see text). Additional vaccines are under development, including conjugate vaccines.

Modified from Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

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Typhoid Fever, Paratyphoid Fever, and Typhoidal Fevers

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Short View Summary

Definition

- Enteric fever, or typhoid fever, is a nonspecific febrile illness caused by typhoidal *Salmonella* (*Salmonella enterica* serotypes Typhi and Paratyphi); the diagnosis should be considered in any patient with otherwise unexplained prolonged fever who has resided in or recently traveled in a typhoid-endemic setting.
- The term typhoidal fever is sometimes used more broadly to refer to a syndrome of persistent high-grade fevers, often with no localizing features. This chapter also covers the broader differential diagnosis and approach to patients with typhoidal fever, relapsing or prolonged fever, and fever accompanied by abdominal pain.

Epidemiology

- An estimated 12 to 24 million cases of enteric fever occur each year, largely in impoverished areas of Asia and Africa.
- Enteric fever is transmitted via the fecal-oral route, with contaminated municipal water supplies being commonly involved in transmission.
- Antibiotic-resistant strains of the major causative agents of enteric fever (*S. enterica* serotypes Typhi and Paratyphi A) are now common worldwide.

Clinical Manifestations

- Common life-threatening complications of enteric fever include intestinal hemorrhage, perforation, encephalopathy, and shock.
- Untreated, patients with enteric fever may be febrile for 3 to 4 weeks or longer, with mortality rates exceeding 10%, and prolonged asthenia and fatigue are common among survivors.
- Chronic biliary carriage of typhoidal *Salmonella* may occur after resolution of the acute illness.

Diagnosis

- Current diagnostic tests for enteric fever are imperfect. Blood cultures are 30% to 70% sensitive, depending on when in the natural history of the illness they are obtained. Bone marrow cultures are more sensitive but are impractical. Serologic assays lack both sensitivity and specificity, especially in enteric fever endemic areas; and nucleic acid amplification assays with sufficient sensitivity are not available.

Therapy

- Given the morbidity of typhoid fever, the risk of complications, and the lack of optimal diagnostic tests, the initiation of antibiotics for treating individuals with suspected enteric fever may be based on a presumptive diagnosis, particularly in resource-limited settings.
- The most commonly used agents for treating individuals with enteric fever are fluoroquinolones, azithromycin, and cefixime or ceftriaxone. Chloramphenicol, trimethoprim-sulfamethoxazole, and amoxicillin may be used to treat patients with susceptible strains.

Prevention

- An oral-attenuated typhoid vaccine and injectable polysaccharide vaccine are commercially available globally and provide 50% to 75% protection for 5 and 2 years, respectively. Two typhoid conjugate vaccines are now prequalified and recommended by the World Health Organization, with a protective efficacy of >80%. No vaccine effective against paratyphoid A is currently commercially available.

Enteric Fever (Typhoid and Paratyphoid Fever)

History

Historically termed putrid fever or dothienteritis, the name typhoid was coined in 1829 by Pierre Charles Alexander Louis. The name typhoid means typhus-like and reflects the difficulty in differentiating the illness from epidemic typhus, another common cause of prolonged fever in Europe during the 19th century. Louis used the term typhoid in a landmark report comparing the intestinal pathology in 50 patients who died of typhoid fever with 83 patients who died of noninfectious causes, and he described the characteristic inflammation of Peyer patches, intestinal ulceration, and mesenteric adenitis associated with typhoid fever.¹

William Budd is credited with the recognition of the fecal-oral transmission of typhoid fever in 1839, based on his meticulous observations during a large epidemic in the Taw Valley of England, although these findings were not widely recognized until many years later.²

The etiologic agent of typhoid fever was identified in pathologic specimens by Karl Eberth, who referred to it as *Bacillus typhosus*; the organism was subsequently first cultured in 1884 by Georg Gaffky at the Berlin Institute for Infectious Diseases. The term paratyphoid fever was first used in 1896 by Raoul Bensaude and Emile Achard, who isolated what would subsequently be recognized as *Salmonella Paratyphi B*. The role of asymptomatic carriers in the transmission of infectious disease was first recognized by Robert Koch, a theory that he derived from his longitudinal observations of patients recovered from typhoid but with continued shedding of *B. typhosus*.³ An understanding of the role of asymptomatic carriers in disease transmission was soon extended to other organisms and was an established mode of disease transmission at the time of George Soper's association of Mary Mallon (also known as Typhoid Mary) with repeated outbreaks of typhoid fever in households in New York, ultimately leading to her lifelong quarantine on North Brother Island.⁴

Widal developed the first serologic test for typhoid fever in 1896, an agglutination assay that detects the presence of antibodies against the O and H antigens of *Salmonella Typhi*.⁵ Despite inherent and significant limitations (see "Diagnosis"), the Widal test remains widely used in many resource-limited settings. The first vaccines for typhoid fever were killed whole-cell vaccines developed in 1896, credit for which is shared by Almroth Wright and Richard Pfeiffer.⁶ Despite advances in the development of a vaccine, typhoid fever remained a leading cause of morbidity and mortality in the early 20th century. A prominent example was a typhoid fever epidemic in the Spanish-American War, in which 20,000 American Army recruits contracted the disease and 1600 died. This defining event was linked to the first compulsory US military-wide vaccination program, in which a killed whole-cell, injectable typhoid vaccine developed by Frederick Russell was used.⁷

In the early 20th century, typhomalaria or typhomalarial fever remained an extremely common diagnosis, indicating that differentiating typhoid from malaria (and other causes of persistent fever) on clinical grounds was difficult. The difficulty distinguishing Page 1301 typhoid fever from typhus and other causes of fever proved clinically fortuitous. In 1948, Theodore Woodward and coworkers reported the effect of chloramphenicol in two patients who were referred for a study of the efficacy of the drug in scrub typhus.⁸ Both were subsequently found to have *Salmonella Typhi* bacteremia, and both patients improved rapidly after antibiotic therapy. Typhoid fever mortality has been reduced from 10% or greater to less than 1% with the use of antibiotics. However,

successive pandemics of antibiotic-resistant organisms have occurred, including, most recently, the emergence of fluoroquinolone- and ceftriaxone-resistant strains of *Salmonella* Typhi. In 2001, the complete sequence of a *Salmonella* Typhi strain (CT18, isolated from a patient in Vietnam in 1993) was published by Parkhill and coworkers at the Wellcome Trust Sanger Institute, ushering in a genomic era in our understanding of typhoid fever.⁹ Genomic analysis has opened up new insights into the history and epidemiology of typhoidal *Salmonella*, from evolutionary history^{10,11} to previously unrecognized historical outbreaks¹² and recent global dissemination.¹³

Etiologic Agents of Enteric Fever
Nomenclature and Classification of Typhoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella* Paratyphi A, B, and C are gram-negative bacilli that belong to the species *S. enterica* subspecies *enterica*. All *S. enterica* are categorized serologically by the Centers for Disease Control and Prevention (CDC) according to a modified Kauffman and White classification scheme, assigned based both on the O (the O polysaccharide) and H (flagellar) antigens.^{14,15} Although assigning a serogroup based on O antigen agglutination tests is a common procedure in many clinical microbiology laboratories, the approach has limited clinical utility. Complete serotyping based on both the O and H antigen is most commonly performed in reference laboratories. In addition to serogrouping and serotyping, standard microbiologic growth and biochemical parameters are also routinely used in the clinical laboratory for the presumptive identification of *S. enterica*. Serologic and selected biochemical characteristics that are used in the identification of *Salmonella* Typhi and *Salmonella* Paratyphi A, B, and C in microbiology laboratories are summarized in Table 102.1.

Clinical Distinction Between Typhoidal and Nontyphoidal *Salmonella enterica*
Salmonella Typhi and *Salmonella* Paratyphi A and B are human-restricted pathogens, and *Salmonella* Paratyphi C is pathogenic for animals and humans. *Salmonella* Typhi and *Salmonella* Paratyphi are classified clinically as typhoidal *Salmonella* strains. This distinction separates these strains from all other pathogenic *S. enterica* serotypes, which are referred to collectively as nontyphoidal *Salmonella* (often abbreviated as NTS; see Chapter 228). Typhoidal strains cause enteric fever in all human hosts, whereas NTS strains are classically associated with inflammatory diarrhea in human hosts or disproportionately cause invasive disease in young children and immunocompromised individuals, with particular associations with malnutrition, HIV, and malaria. However, the clinical distinction between typhoidal and NTS is not absolute. NTS may cause invasive infection that includes prolonged bacteremia (with or without pyogenic foci) that may mimic the systemic illness caused by typhoidal strains.¹⁶ Although host factors influence susceptibility to invasive NTS infection, the likelihood of invasive infection is also dependent on characteristics of the bacterial strain. For example, the emergence of a dominant genotype associated with invasive NTS, *Salmonella* Typhimurium ST313, in sub-Saharan Africa illustrates the clinical overlap between typhoidal and nontyphoidal *S. enterica*.¹⁷

Genomic Features
Escherichia coli and *S. enterica* diverged over 100 million years ago, and *S. enterica* is now a genetically heterogeneous species.¹⁸ *Salmonella* Typhi is genetically homogeneous, meaning there is very limited genetic diversity across *Salmonella* Typhi isolates throughout the world. Based on standard rates of genetic drift, *Salmonella* Typhi likely emerged from a single-point origin within the last 50,000 years, coinciding with the time of human migrations out of Africa and into Asia.^{10,19} The genetic material of *Salmonella* Typhi consists of a single chromosome that contains approximately 5×10^6 base pairs encoding approximately 4000 genes. *Salmonella* Typhi often harbors additional plasmids that may be cryptic (confer an unknown function) or involved in pathogenesis. The latter include IncHI1 plasmids that comprise the Page 1302majority of resistance plasmids found in *Salmonella* Typhi and that remain stable in the bacteria without ongoing pressure from antibiotic selection.^{20,21}

TABLE 102.1

Classification and Selected Microbiologic Characteristics of Typhoidal <i>Salmonella</i>			
SERO TYPE NAME			
<i>Salmonella</i> Typhi	<i>Salmonella</i> Paratyphi A	<i>Salmonella</i> Paratyphi B (<i>schottmuelleri</i>)	
<i>Salmonella</i> Paratyphi C (<i>hirschfeldii</i>)	Comments		

Serologic Classification^a

Serogroup	D	A	B	C		
O antigen	9, 12	1, 2, 12		1, 4, (5), 12	6, 7	
H antigen						
Phase 1:2						
d: –	a: (1, 5)	b: (1, 2)	c: (1, 5)		Uncommon Salmonella Typhi H antigen variants include Hj:z66 found in Indonesia	
K antigen	Vi	–	–	Vi		
Selected Biochemical Characteristics ^b						
Lactose fermentation	–	–	–	+	The majority of Salmonella Paratyphi C ferment lactose	
Sulfide (H ₂ S) production		+ (weak)	–	+	+	3% of Salmonella Typhi do not produce hydrogen sulfide
Gas produced during glucose fermentation	–			+	+	+
Lysine decarboxylase	+	–	+	+		
Ornithine decarboxylase	–	+	+	+		
a Strong antigenic determinants are listed in bold; weaker or sometimes absent ones are in parentheses.						

a Strong antigenic determinants are listed in bold; weaker or sometimes absent ones are in parentheses.

b Isolates can be identified as Salmonella using traditional methods. A presumptive diagnosis of Salmonella Typhi can often be made for Salmonella isolates based on selected biochemical characteristics, in addition to agglutination with D and Vi antisera.

From World Health Organization (WHO), Department of Vaccines and Biologicals. Background Document: The Diagnosis, Treatment and Prevention of Typhoid Fever. Geneva: WHO; 2003; and WHO, Department of Communicable Disease Surveillance and Response. Salmonella serotype Typhi. In: WHO Manual for the Laboratory Identification and Antimicrobial Susceptibility Testing of Bacterial Pathogens of Public Health Importance in the Developing World. Geneva: WHO; 2003:103–118.

The genetic core of *S. enterica* consists of genes that are conserved across Enterobacteriaceae and is approximately 90% similar to *E. coli*. In contrast, the genetic core varies approximately 1% between *S. enterica* serovars. In addition to the genetic core, Salmonella Typhi contains pathogenicity islands—horizontally acquired genetic elements that are often associated with unique properties of each strain and are essential for specific phenotypes associated with pathogenesis of each serotype.¹⁸ In *S. enterica*, these are specifically identified as Salmonella pathogenicity islands, or SPIs, and are designated with a number. Salmonella Typhi possesses more than 20 SPIs, many of which are found in other *S. enterica* serotypes. For example, SPI-7 is a 134-kb genetic island that possesses elements encoding the *viaB* locus, which is involved in production of the Vi capsule and evasion of host immune responses.²² SPI-7 is widely conserved across Salmonella Typhi and is also found in Salmonella Paratyphi C.

One of the most remarkable genetic features of Salmonella Typhi is the accumulation of more than 200 presumed inactive genes called pseudogenes.⁹ This adaptation may play a major role in the host restriction of Salmonella Typhi. The accumulation of a large number of pseudogenes is a common feature shared with Salmonella Paratyphi A, B, and C, which emerged independently of Salmonella Typhi (although subsequent recombination events between Salmonella Typhi and Salmonella Paratyphi A appear to have occurred),²³ yet are also adapted specifically to cause disseminated infection in humans. The same phenomenon has been observed in Salmonella Typhimurium ST313 in sub-Saharan Africa; ST313 has accumulated extensive gene deletions and purported pseudogene formation (more than half of which are common to Salmonella Typhi and Salmonella Paratyphi A) compared with less invasive Salmonella Typhimurium strains.²⁴

Epidemiology Burden and Distribution

Measures of the burden of enteric fever are limited by the absence of surveillance in many regions and also by the limited sensitivity of current diagnostic tests. Recognizing these limitations, estimates of the number of cases of typhoid and paratyphoid fever range from 12 to 16 million annually.²⁵ The typhoid and paratyphoid fevers were also responsible for 10 million disability-

adjusted life years in 2017.²⁵ The economic toll of enteric fever is also devastating for affected families. A study from Malawi from 2019 to 2020 found that catastrophic illness, or an illness which cost over 40% of monthly household nonfood expenses, occurred in 44% of households affected by a case of typhoid fever.²⁶

The incidence of enteric fever and the mortality of the disease vary dramatically by region (Fig. 102.1). In South Asia and parts of sub-Saharan Africa, the incidence approaches 1000 cases per 100,000 person-years.^{27,28} In parts of northern Africa, South America, and Southeast Asia, the incidence is much lower and ranges between 10 and 100 cases per 100,000 person-years.²⁵ Disease is highly heterogeneous even within endemic countries, and the incidence tends to be considerably higher in urban than in rural areas.²⁷ In the United States and many other high-resource settings, the disease occurs sporadically, with less than 1 case per 100,000 person-years, and occurs most often in travelers returning from endemic zones (see “Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically”).

Salmonella Paratyphi A is the most common cause of paratyphoid fever and responsible for an estimated 2.6 to 4.2 million cases of enteric fever annually.²⁵ The proportional burden of enteric fever caused by *Salmonella* Paratyphi A has increased; *Salmonella* Paratyphi A is responsible for more than 20% to 50% of cases of enteric fever. This trend is expected to continue with the introduction of vaccines which protect against typhoid fever, but not enteric fever caused by *S. Paratyphi*.²⁹ In contrast, *Salmonella* Paratyphi B and C remain less-common causes of enteric fever globally.

Severity

In the preantibiotic era, 10% to 15% of clinically recognized cases of typhoid fever were fatal, whereas other cases resolved over a 3- to 4-week course of an often severe and debilitating illness.^{30,31} Prospective surveillance in the postantibiotic era has revealed that the majority of cases of enteric fever are uncomplicated and do not warrant hospitalization. Although the current mortality of typhoid fever is estimated at 0.2% to 2%, these estimates are based on limited data and there is significant variation by region.^{32,33} Reported mortality rates in hospitalized patients with enteric fever in resource-limited settings vary widely by location, ranging from 0% to 15%, with a median of 2%.³² In addition to fatal cases, in sub-Saharan Africa and other regions where enteric fever is highly endemic, intestinal perforation caused by enteric fever is a leading cause of peritonitis and acute abdominal conditions requiring surgery in children.³⁴

Source of Infection

Salmonella Typhi and *Salmonella* Paratyphi A and B are human restricted. There are no known animal reservoirs of these pathogens that cause enteric fever, and the source of infection is organisms shed in the stool of infected humans. *Salmonella* Paratyphi C can be shed by humans and animals. In the preantibiotic era, patients recovering from enteric fever shed organisms in the stool during acute illness and typically for weeks during convalescence. In addition, a small portion of individuals convalescing from typhoid fever develop chronic asymptomatic carriage (defined as shedding of the bacteria in stool or urine for more than a year), which may persist for life.³⁵ In the preantibiotic era, up to 5% of typhoid fever survivors developed chronic asymptomatic bacterial shedding in the stool,^{36,37} with gallbladder disease being the major risk factor for carriage.^{38,39} Risk of carriage appears to be far greater in adults (3%–10%) than in young children (<1%).³⁶ Studies from Nepal, where enteric fever is highly endemic, found that 3.5% of individuals undergoing elective cholecystectomy grew *Salmonella* Typhi or *Salmonella* Paratyphi A from biliary cultures.⁴⁰ Even higher rates were observed in Santiago, Chile, another hyperendemic focus at the time, in the early 1980s.⁴¹ Urinary carriage may also occur, frequently in conjunction with urinary (*Salmonella* haematobium) schistosomiasis.⁴² Although chronic carriers are a source of transmission where enteric fever occurs sporadically, their importance in the transmission of enteric fever in highly endemic areas is uncertain.^{43,44}

FIG. 102.1 Global distribution of typhoid and paratyphoid fever. Estimated incidence of typhoid and paratyphoid fever by country from the Global Burden of Disease Study 2019. Source: Institute for Health Metrics and Evaluation. Used with permission. All rights reserved.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Follow for extended description

Mode of Transmission

Infections with typhoidal *S. enterica* are most often acquired via ingestion of fecally contaminated water or food. *Salmonella* Typhi may persist for weeks after passage in water and may persist in a variety of contaminated food items—for example, dehydrated formula and iced beverages.⁴⁵⁻⁴⁷ Volunteer studies may not optimally mimic natural exposure; however, the inoculum required to produce disease in 50% of adult volunteers (ID₅₀) is 10⁵–7 organisms, although fewer than 10³ organisms may produce disease,^{48,49} which might be more reflective of actual exposures in nonepidemic settings.

Molecular epidemiologic studies based on high-resolution genotyping implicate waterborne transmission as the most likely route of transmission in the majority of cases of enteric fever.⁵⁰ A study in the Kathmandu valley of Nepal revealed that >80% of water collected from public taps contained typhoidal *Salmonella* DNA.⁵¹ Both waterborne and foodborne transmission may contribute to large epidemics, and in some cases, massive epidemics have been associated with water contamination from a single source.^{52,53} In contrast, direct person-to-person transmission of the etiologic agents of enteric fever appears less common, and more than 80% of cases of enteric fever occur in individuals with no known contact with an individual with symptomatic infection.⁵⁴ Even more remarkably, in households where multiple cases of enteric fever occur, only 20% of those infections shared a common bacterial genotype, suggesting that multiple infections within a household occur through a community-based exposure to multiple circulating genotypes, rather than through direct person-to-person transmission.⁵⁰ In highly endemic areas, consistent personal risk factors for infection with typhoidal *S. enterica* include drinking nonboiled water and eating food prepared outside the home, including street-vended foods and beverages.^{55,56} In South Asia, peak transmission occurs during and immediately after the monsoon season (peaks in June through October), although disease is present year-round.⁵⁷

Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically

In areas where enteric fever occurs sporadically, most cases are imported through travel. In the United States, >85% of cases occur in travelers to endemic areas.⁵⁸ The rate of typhoid fever is estimated at 10 per 1 million travelers arriving from Asia but increases to 89 per 1 million travelers arriving from India. More than 80% of all travel-associated cases in the United States relate to travel to South Asia.⁵⁸ The risk of travel-related enteric fever is higher among travelers visiting friends and relatives. Many of the remaining non-travel-associated cases in the United States are traceable to small, local foodborne outbreaks or a chronic carrier.^{59,60} In 2010, an outbreak of *Salmonella* Typhi occurred that involved at least 12 cases across three western US states and was attributed to imported fruit pulp from Guatemala.⁶¹ Also, in areas where enteric fever occurs sporadically, small clusters of cases of sexual transmission of *Salmonella* Typhi have been documented in men who have sex with men.⁶² Infections and clinical cases have occurred among microbiology laboratory staff, especially when *Salmonella* Typhi was used as an unknown for laboratory proficiency testing, and in at least once instance, secondary cases occurred among family members of laboratory personnel.⁶³

Antibiotic Resistance and Emergence of Pandemic *Salmonella* Typhi

Chloramphenicol-resistant *Salmonella* Typhi was first reported in 1950, 2 years after the antibiotic was first used to treat patients with typhoid fever.⁶⁴ However, widespread dissemination of antibiotic-resistant strains did not occur until the 1970s. Multidrug-resistant (MDR) *Salmonella* Typhi carrying IncHI1 plasmid-mediated resistance to chloramphenicol, ampicillin, trimethoprim, and sulfonamides became common in the 1980s. Nalidixic acid-resistant strains of *Salmonella* Typhi and *Salmonella* Paratyphi A, with decreased susceptibility to fluoroquinolone antibiotics, emerged in the 1990s,⁶⁵ and now the vast majority of typhoidal *Salmonella* isolates from South Asia have reduced susceptibility or high-level resistance to fluoroquinolones.^{13,66} At present, reduced susceptibility to ciprofloxacin is less common and more sporadic in sub-Saharan Africa, though it appears to be on the rise.¹³

Although the lack of genetic diversity of *Salmonella* Typhi previously hindered efforts to characterize the global epidemiology of disease, the increasing use of high-density genotyping methods, including whole-genome sequencing, has circumvented these challenges and resulted in a better understanding of the forces driving the evolution and transmission of *Salmonella* Typhi. Although a general lack of diversity among *Salmonella* Typhi suggests a lack of strong selection

pressure over millennia, the recent expansion of a single fluoroquinolone-resistant haplotype known as H58 into a pandemic strain found throughout Asia and Africa, and the emergence of several other lineages with diverse *gyrA* mutations, suggest that fluoroquinolones may be exerting a uniquely strong selection pressure that is shaping the global evolution of *Salmonella* Typhi.^{13,67} Evidence suggests that fluoroquinolone resistance-conferring mutations have arisen repeatedly in multiple *Salmonella* Typhi lineages, spread internationally and intercontinentally and replaced less resistant strains.¹³ More recently, *Salmonella* strains with resistance to third-generation cephalosporins (e.g., ceftriaxone) have been reported.^{68,69} In Pakistan, strains containing multidrug resistance and resistance to third-generation cephalosporins and fluoroquinolones have been termed “extensively drug-resistant” (XDR) typhoid, and within years of first appearance now account for the majority of isolates in the country.^{66,68,70} Most recently, azithromycin resistance has been identified in multiple lineages, arising from point mutations in an efflux pump (*acrB*).^{71,72} Azithromycin resistance in typhoidal *Salmonella* might become an increasing problem given the WHO recommendation for its use in mass drug administration in settings with high rates of child mortality.^{73,74}

Host Factors and Susceptibility to Infection

Historically, enteric fever was considered a disease that predominantly affected school-aged children and young adults. However, prospective community-based surveys in highly endemic areas have demonstrated that the incidence of *Salmonella* Typhi bacteremia is often highest in young children (<5 years old) and that this organism may be responsible for more than 75% of cases of occult bacteremia in these settings.⁷⁵ The higher incidence of enteric fever in older children among inpatients may reflect a predisposition to more severe disease in this older age group. In highly endemic areas, *Salmonella* Typhi bactericidal antibodies and anticapsular (Vi) antibodies increase significantly over the first decade of life.⁷⁶ In less endemic areas, the median age of patients with typhoid fever is higher, suggesting that the age distribution of patients may be influenced by acquired immunity. However, protection from a single episode of infection is limited, as demonstrated by frequent relapse and recurrent infections among patients who have recovered from typhoid fever.⁷⁷

Although invasive NTS infections classically occur in persons who are compromised by extremes of age, malnutrition, immunodeficiency, or genetic risk factors, there is not an overt association between host Page 1304 risk factors and susceptibility to enteric fever. Various candidate gene analyses have drawn associations between single-nucleotide variations in Toll-like receptor 4 (TLR4), cystic fibrosis transmembrane conductance regulator (CFTR), and specific human leukocyte antigen (HLA) types with susceptibility to typhoid fever in Vietnam, as well as in human volunteers challenged with virulent organisms⁷⁸⁻⁸⁰; however, no detailed genome-wide study of typhoid fever associations has yet been reported.

Pathogenesis

Typhoidal *Salmonella* infection is acquired through ingestion of bacilli, typically in contaminated water, drinks, or food. The ID₅₀ is often cited to be 10⁵–7 organisms, but low inocula (<10³ organisms) may also produce infections.⁴⁸ After passing through the intestinal epithelium, typhoidal *Salmonella* are phagocytosed by macrophages and proliferate in the submucosa before dissemination via the lymphatic system and bloodstream. This period coincides with the onset of systemic symptoms and, in some cases, end organ manifestations. It is during this phase that bacilli enter the hepatobiliary system and may establish carriage in the gallbladder. Disseminated infection is terminated by immune clearance or administration of antibiotics, but bacterial persistence in the bone marrow is often longer.

It is important to note that enteric fever and NTS infections differ in fundamental aspects. With NTS, infections are typically self-limited in immunocompetent hosts and manifest largely as inflammatory gastroenteritis. In comparison, typhoidal serotypes are able to evade the normal host inflammatory response and cause prolonged bacteremia even in immunocompetent individuals, typically without overwhelming sepsis or pyogenic foci of infection. Specific bacterial adaptations used by *Salmonella* Typhi to produce persistent bacteremia in the absence of an immediately overwhelming inflammatory response include strategies to breach the intestinal epithelium in the absence of inflammatory diarrhea, to evade detection by host pattern-recognition receptors,⁸¹ to produce an AB₅ bacterial toxin that acts through cell-to-cell paracrine signaling pathways,⁸² and to persist and disseminate throughout the human host within

phagocytic cells.⁸³ Much of our understanding of host-pathogen interactions in enteric fever is extrapolated from mouse models of enteric fever, which use Nramp1-deficient mice that develop disseminated infection from *Salmonella* Typhimurium or humanized mice altered to be susceptible to *Salmonella* Typhi.⁸⁴ However, controlled human infection models for *Salmonella* Typhi and Paratyphi A infection are now available and have added valuable insights into the host immune response to these organisms and disease pathogenesis.^{85,86}

Invasion

In contrast to NTS and other enteroinvasive pathogens, soon after ingestion, typhoidal *Salmonella* pass through the intestinal epithelium through specialized microfold cells (M cells) that transport the bacteria across the basolateral membrane, where they are phagocytosed by macrophages in the intestinal lymphoid tissue.⁸⁷ Such a stealth entry mechanism may explain why the invasion phase of *Salmonella* Typhi infection is usually asymptomatic and is accompanied by transient or mild diarrhea in only 10% to 20% of patients. Indeed, as *Salmonella* Typhi leaves the GI tract at this stage, stool cultures are often negative until days to weeks later when shedding through biliary system commences.

Latency and Dissemination

Typhoidal *Salmonella* deploy an array of virulence factors that enable them to persist and replicate in an intracellular compartment of phagocytic cells, primarily macrophages.^{88,89} Important cloaking mechanisms of *Salmonella* Typhi include suppression of flagellar protein synthesis, thus masking a potent inducer of TLR5 responses.⁹⁰ and synthesis of the Vi capsule that presumably masks the detection of lipopolysaccharide (LPS) and other outer membrane components, preventing early complement-mediated clearance.^{91,92} The Vi capsule, thus, helps *Salmonella* Typhi to limit inflammation and evade phagocytosis by neutrophils. The genes that repress flagellin biosynthesis and are responsible for Vi biosynthesis are encoded on the *viaB* locus located on SPI7, which is absent in most NTS strains.²² Induction of other bacterial defenses that are required for survival in the intracellular compartment is regulated by a two-component bacterial sensory-regulatory system, PhoP/PhoQ.^{93,94} The sensor PhoQ detects signals present inside the host phagosome and activates the transcriptional regulator PhoP, which in turn controls the expression of a large number of downstream genes involved in modification of lipid A, resistance to antimicrobial peptides, and acidification of the phagosome.^{95,96} Consistent with a role in pathogenesis, PhoP/PhoQ mutants are nonvirulent in humans and have been evaluated as candidate live-attenuated vaccines.^{97,98}

Quantitative blood cultures demonstrate that more than 60% of culturable circulating *Salmonella* Typhi organisms reside in the intracellular compartment.⁹⁹ Presumably, this allows for dissemination throughout reticuloendothelial tissues via the blood and lymphatic systems. The heaviest burden of infection is established in the reticuloendothelial system (intestinal lymphoid tissue, liver, spleen, and bone marrow) and gallbladder. Patients with typhoid fever most often have very low-grade bacteremia, a feature that presents a formidable diagnostic challenge. The kinetics of quantitative bacterial cultures has been studied in Vietnam.⁹⁹ The median number of culturable bacteria is less than 1 colony-forming unit (CFU) per milliliter of whole blood in adults, and slightly higher (1.5 CFUs/mL) in children. Slightly more culturable organisms are present in the bone marrow, with a median of 10 per milliliter of bone marrow.¹⁰⁰ However, over the course of illness, the number of culturable *Salmonella* Typhi organisms in the blood decreases (median of 2 CFUs/mL in the first week to 0.1 CFUs/mL in the second and subsequent weeks), and the number of culturable bacilli increases in the bone marrow (4.8 CFUs/mL in the first week to 158 CFUs/mL in the third and subsequent weeks). As a result, the proportion of bacteria in the bone marrow increases from 5:1 (marrow to blood) in the first week of illness to more than 150:1 in the third week of illness, reflecting relative clearance of bacteria from peripheral blood but persistence in the marrow compartment.¹⁰⁰ Pathologic examination of bone marrow may demonstrate monocytic infiltrates typical of typhoid fever (see “Intestinal and Other Local Pathology”).

The initial period of replication and dissemination represents the incubation period of typhoidal *Salmonella* infection. This period typically lasts 1 to 2 weeks but can range widely (3–60 days), depending on the number of organisms ingested.¹⁰¹ Ultimately, infection results in the sufficient secretion of inflammatory cytokines, including pyrogenic cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), which result in fever. However, unlike what occurs during prototypical gram-negative bacteremia, septicemia with hypotension, neutrophilia, and

disseminated intravascular coagulopathy are extremely uncommon in patients with typhoid or enteric fever.

Intestinal and Other Local Pathology

Intestinal lymphoid tissue is a predominant site of localized inflammation and persistent bacterial replication in cases of severe enteric fever (Fig. 102.2).¹⁰²⁻¹⁰⁵ Early in the course of illness, Peyer patches in the terminal ileum and draining mesenteric nodes are enlarged and contain infiltrates consisting of mononuclear cells, including macrophages and lymphocytes. As disease progresses, necrosis of intestinal lymphoid tissue occurs, with mixed inflammatory infiltrates, including neutrophils and ulceration and sloughing of the overlying intestinal mucosa.¹⁰² Immunohistochemical staining for the *Salmonella* Typhi O antigen reveals abundant organisms in necrotic areas.¹⁰³ The clinical manifestations of these changes include hemorrhage or intestinal perforation, two of the major life-threatening complications of enteric fever (Fig. 102.3). Other affected organs include the liver, with monocytic infiltrates and foci of parenchymal necrosis, and the spleen, with nodular monocytic infiltrates in the red pulp.¹⁰⁶ Similar monocytic infiltrates may occur in the gallbladder. Gallbladder perforation, which may mimic intestinal perforation, is a rare complication of typhoid fever.¹⁰⁷

Relapse and Chronic Carriage

Untreated, typhoid fever may persist for up to 4 weeks, and relapse occurs in up to 10% of untreated cases, usually within 2 weeks after initial resolution of fever.¹⁰⁸ Relapse is usually caused by the same strain as that associated with the original infection.¹⁰⁹ The gallbladder is the primary site of chronic carriage, although some asymptomatic carriers continue to shed the organism in stool even after cholecystectomy. The *in vitro* and clinical observations that *Salmonella* Typhi forms biofilms on cholesterol gallstones may explain the strong epidemiologic association between gallstones and carriage.^{110,111} An association has been reproducibly observed between chronic carriage of *Salmonella* Typhi and risk of gallbladder carcinoma. It is unknown if carriage causes cancer or reflects underlying biliary epithelial surface abnormalities^{112,113}; chronic low-grade biliary inflammation and emerging data on the genotoxic activity of *Salmonella* CdtB support an etiological role.¹¹⁴

FIG. 102.2 Intestinal pathologic appearance of enteric fever. (A) The visualized mucosal surface of a surgically resected area demonstrates two perforations (arrowheads); the visualized serosal surface demonstrates significant enlargement of lymph nodes (arrows). (B and C) Lower- and higher-magnification views demonstrate immunohistochemical staining for the *Salmonella* O:9 antigen in a necrotic lymph node. From Neil KP, Sodha SV, Lukwago L, et al. A large outbreak of typhoid fever associated with a high rate of intestinal perforation in Kasese District, Uganda, 2008–2009. *Clin Infect Dis*. 2012;54:1096.

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FIG. 102.3 Intraoperative photograph of intestinal perforation caused by *Salmonella* Typhi. Intestinal perforation is visible on the antimesenteric border of the small bowel, which is inflamed with patchy exudates on the serosal surface. Courtesy Dr. Pukar Maskey, Patan Hospital, Kathmandu, Nepal. From Harris JB, Brooks WA. Typhoid and paratyphoid (enteric) fever. In: Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

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Clinical Manifestations

The clinical manifestations of enteric fever are nonspecific. Fever is reported in the vast majority of clinically apparent cases, and many other symptoms are variably reported, including headache, cough, nausea, vomiting, constipation, and diarrhea.¹¹⁵ Although paratyphoid fever was considered less severe than typhoid fever, more recent comparisons suggest that typhoid fever and paratyphoid fever caused by *Salmonella* Paratyphi A are generally indistinguishable on clinical grounds.¹¹⁶ Most patients with enteric fever are diagnosed in the ambulatory setting, and up to 90% are treated as outpatients. In contrast, classic descriptions of the features of enteric fever are derived from a series of hospitalized patients with more severe disease.³⁰ Complications of enteric fever were generally thought to occur late in the course of disease, usually after 2 weeks of fever. However, experience has demonstrated that major complications of enteric fever, including intestinal perforation and encephalopathy, may occur within days of onset of fever.^{103,104,117}

Uncomplicated Typhoid Fever

Clinical features of enteric fever diagnosed in the ambulatory setting are listed in Table 102.2. Fever without localizing signs or symptoms may be the sole manifestation of enteric fever. The onset of fever may be insidious, and fevers typically increase over the first week of illness.¹¹⁸ A variety of other nonspecific flulike symptoms are common early in the course of enteric fever, and headache, anorexia, myalgias, and malaise may precede the onset of fever. Mild confusion may be seen even in ambulatory patients, and a nonproductive cough is also a common feature of uncomplicated enteric fever.¹¹⁵ Abdominal complaints may include diarrhea, constipation, and abdominal pain. Invasive diarrhea, such as seen in NTS gastroenteritis, does not typically occur in enteric fever.

The physical findings of uncomplicated enteric fever are also nonspecific. Although relative bradycardia, or pulse-temperature dissociation, is a classic sign of enteric fever, it may not be a clinically useful predictor of enteric fever for individual patients, and is absent in the majority of patients.¹¹⁹ Rose spots, the classic cutaneous manifestation of enteric fever, are 1- to 4-mm blanching pink macules, and are most often seen on the chest, back, and abdomen during the second week of fever. However, rose spots are uncommon in uncomplicated typhoid fever. Mild, nonlocalizing, abdominal tenderness may be present. Hepatomegaly and splenomegaly, if present, are usually modest. A white or yellowish-brown coating of the tongue that spares the tongue's edges is a common physical finding.¹¹⁶

Laboratory Findings in Uncomplicated Infection

Although abnormal white blood cell counts in peripheral blood are common in patients with bacteremia, the majority of patients with uncomplicated enteric fever do not have leukocytosis, neutrophilia, or increased immature neutrophils.¹¹⁵ However, either leukocytosis or Page 1306leukopenia may be present. Both the hematocrit and platelet counts are typically normal or slightly low. Elevated serum aspartate transaminase and alanine transaminase are very common in enteric fever; values two to three times above the upper limit of the normal range are typical.¹²⁰ On occasion, more severe hepatitis is observed, but very high levels of blood transaminases (>500 IU/L) should prompt concern for other causes, including viral hepatitis or drug toxicity.¹²⁰

TABLE 102.2

Clinical Features of Typhoid and Paratyphoid Fever

CLINICAL FEATURE APPROXIMATE FREQUENCY^a

Flulike symptoms	Fever	>95%
Headache	80%	
Chills	40%	
Cough	30%	
Myalgia	20%	
Arthralgia	<5%	
Abdominal symptoms	Anorexia	50%
Abdominal pain	30%	
Diarrhea	20%	
Constipation	20%	
Physical findings	Coated tongue	50%
Hepatomegaly	10%	
Splenomegaly	10%	
Abdominal tenderness	5%	
Rash	<5%	
Generalized adenopathy	<5%	

^a The proportion of patients demonstrating these clinical features of enteric fever varies, depending on the time, the region, and the type of clinical population (hospitalized or ambulatory) assessed. Estimates are drawn from recent case series in an endemic area, with patients presenting for ambulatory or inpatient care.^{116,216}

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Severe Illness

Classic descriptions of the features of enteric fever are drawn from observations of a series of hospitalized patients in the preantibiotic era, in which mortality rates were consistently in the 10% to 15% range. The natural history of untreated disease included progressively increasing fevers over the first week of illness, followed by increasing abdominal complaints and rash over the second week of illness, followed by complications, including intestinal hemorrhage and perforation, or gradual resolution in the third and fourth weeks of illness.³⁰

In the present antibiotic era, it remains true that because enteric fever progresses gradually, patients who are severely ill with enteric fever may have been febrile for longer than a week when they first come to clinical attention. Patients with severe enteric fever may appear toxic, and characteristically would have moderate abdominal pain or tenderness, and constipation or diarrhea. Physical findings in severe enteric fever include hepatomegaly and splenomegaly. Patients with severe enteric fever are more likely to have major complications listed in Table 102.3. Complications associated with increased mortality in severe typhoid fever include intestinal hemorrhage and perforation, severe encephalopathy, seizures, and pneumonia.¹²¹

Gastrointestinal Complications

Intestinal hemorrhage occurs in up to 10% of hospitalized patients with severe enteric fever and is usually self-limited.^{30,32} Intestinal perforation, a major life-threatening complication of enteric fever, occurs in up to 3% of hospitalized patients.^{30-32,121} These complications arise directly from invasion of *Salmonella* into Peyer patches in the small intestines. A series of patients from Vietnam with intestinal perforation resulting from enteric fever demonstrated that the median length of illness preceding perforation was 9 days from the onset of fever, although some cases occurred within the first week.¹⁰⁴ Perforation is suggested by clinical signs of peritonitis, including tachycardia, leukocytosis, neutrophilia, and abdominal pain with guarding and rebound tenderness. The clinical diagnosis of perforation requires a high index of suspicion because patients with severe enteric fever may have a toxic appearance and significant abdominal tenderness even before perforation. Radiographic evidence of pneumoperitoneum may be present in only 50% of cases.

TABLE 102.3

Complications of Typhoid and Paratyphoid Fever

SYSTEM	COMPLICATION	NOTES
Gastrointestinal	Hemorrhage	10%–15% in hospitalized patients
Perforation		3% in hospitalized patients
Hepatobiliary	Jaundice	1%–3% in hospitalized patients
Hepatitis	Usually subclinical (↑ ALT/AST)	
Acute cholecystitis	Rare; gallbladder may perforate	
Neurologic	Mild encephalopathy	Confusion or apathy common
	Severe encephalopathy	Delirium, stupor, or coma
Seizures	Common in children ≤5years	
Meningitis	Rare, primarily infants	
Guillain-Barré syndrome	Reported	
Respiratory	Bronchitis	Dry cough is common
Pneumonia	May be other concomitant bacterial infection (e.g., <i>Streptococcus pneumoniae</i>)	
Empyema	Rare reports	
Cardiovascular	Myocarditis	Usually subclinical (ECG changes)
Endocarditis	Rare reports	
Hematologic	Anemia	Usually subclinical
Disseminated intravascular coagulation		Usually subclinical (↑ PT/PTT)
Other	Musculoskeletal pyogenic infections	Osteomyelitis (particularly vertebral), psoas abscess and others reported
Hemolytic-uremic syndrome	Reported	
Miscarriage	Reported	

ALT, Alanine aminotransferase; AST, aspartate aminotransferase; ECG, electrocardiogram; PT, prothrombin time; PTT, partial thromboplastin time.

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Neurologic Complications

Various case series have documented a widely varying prevalence of diverse neuropsychiatric manifestations during enteric fever, ranging from encephalopathy, encephalomyelitis, transverse myelitis, meningitis, ataxia, and Guillain-Barré syndrome.^{117,122} Although most of these manifestations are uncommon, patients with severe enteric fever often present with some component of encephalopathy, and a history of confusion or an apathetic affect are common.³² Of note, typhoid was named for its clinical similarity to typhus, and the name typhus was from the Greek term for smoky or cloudy, reflecting the neurologic features that often accompanied this febrile illness. More severe encephalopathy, manifesting with delirium, stupor, and coma, occurs in a smaller number of hospitalized patients and is associated with a high risk of mortality.^{121,123,124} Osler described this as a “pseudo-wakeful state” of “muttering” delirium, known as “coma vigil,” and noted that it portends poor outcomes. Even in cases of severe encephalopathy, microbiologic evaluation of cerebral spinal fluid is usually unrevealing, and pleocytosis, if present, usually reveals fewer than 35 cells/ μ L.^{123,124} Encephalopathy during enteric fever may be most common among older children and young adults.¹²⁵ Seizures during enteric fever are most common in young children and are associated with increased mortality.¹²¹ Meningitis caused by *Salmonella Typhi* is seen primarily in infants.¹²⁶

Metastatic Pyogenic Complication

Enteric fever is a disseminated bacterial infection, but pyogenic abscess formation during enteric fever is notably infrequent, although *Salmonella Paratyphi C* is more likely to cause abscesses and focal infections than the other typhoidal strains. This contrasts with what occurs during invasive nontyphoidal salmonellosis, in which osteomyelitis, joint infection, abscess formation, and endovascular infection more frequently occur. Although uncommon, pyogenic complications during typhoid have been described and include empyema, osteomyelitis, hepatosplenic, and other organ system and soft tissue abscesses, as well as endovascular infections and endocarditis.¹²⁷

Diagnosis

The laboratory diagnosis of enteric fever is often challenging because current culture-based, serologic, and molecular diagnostic tests for enteric fever fail to achieve an optimal combination of sensitivity and specificity.^{128–131} For this reason, it is often appropriate to treat patients with suspected enteric fever with empirical antibiotic therapy. The diagnosis of enteric fever should be considered in any person with fever, especially in those with fever lasting longer than 3 days and who have had exposure in the last 1 to 6 weeks to an area where enteric fever is endemic. In endemic areas, other clinical factors that are associated with a higher likelihood of enteric fever include a temperature greater than 39°C (102.2°F), ill appearance, young age (<5 years), and any abdominal complaints, including abdominal pain, diarrhea, or constipation.⁷⁵ As the duration of fever increases, the likelihood of enteric fever increases.¹³² Despite this, it is important to recognize that the majority of individuals suspected of having enteric fever, even in endemic settings, do not have enteric fever but rather an alternative infection^{133,134}; therefore, clinical suspicion of and empirical treatment for enteric fever should not halt the search for alternative causes of fever.

Culture-Based Diagnostics

Although a presumptive diagnosis of enteric fever may be sufficient grounds for initiating and continuing antimicrobial therapy, a definitive diagnosis of enteric fever is made only through the isolation of a typhoidal serotype of *S. enterica* (*Salmonella Typhi* or *Salmonella Paratyphi A, B, and C*) from the blood, bone marrow, stool, urine, or other clinical specimen of a febrile patient. Isolation of the causative organism also allows antimicrobial resistance testing, which facilitates optimal management.

Blood culture is the most common method of diagnosis, where adequate microbiologic facilities exist. The sensitivity of blood culture varies from approximately 40% to 80%,^{135,136} and a recent meta-analysis estimated an average sensitivity of 59%.¹³⁷ The low-grade bacteremia in enteric fever is an inherent limiting factor to the sensitivity of blood cultures, as is prior receipt of antibiotics.⁹⁹ To maximize the sensitivity of blood cultures, the World Health Organization (WHO)

recommends that cultures be performed using 10 to 15 mL of blood from schoolchildren and adults, and 2 to 4 mL of blood from toddlers and preschool-aged children.¹³⁸ The use of a nonselective broth media, as is standard for most blood cultures, is encouraged because it maximizes detection of a wide range of pathogens. The sole use of *Salmonella* selective or enriching media, such as ox bile medium, is discouraged.¹³⁸

Although less sensitive than blood cultures, stool cultures are positive in more than 50% of children and 30% of adults with enteric fever, and they increase the diagnostic yield over blood cultures alone by approximately 5%.¹³⁶ To maximize detection, WHO recommends the use of greater than 1 g of stool and selenite enrichment broth.¹³⁸ Culturing of bone marrow has the highest sensitivity (80%–95%) of all the culture-based approaches, with culturing of a 1-mL aspirate of bone marrow having the sensitivity of a 15-mL peripheral blood culture in an adult.^{99,100} Culturing of bone marrow is, however, not clinically practical for most patients with suspected enteric fever, although microbiologic and hematologic analysis of a bone marrow aspirate may be part of an evaluation of a patient with a prolonged fever of unknown origin when less invasive testing is unrevealing. Other specimens that may yield growth of *Salmonella* Typhi or *Salmonella* Paratyphi A, B, and C include urine, duodenal aspirates, and specimens from skin biopsy of rose spots. However, these approaches are not routinely recommended. Cultures made from intestinal biopsy specimens and peritoneal fluid of patients with perforation are rarely positive.^{103,104}

Serologic Tests

The Widal test detects agglutinating antibodies against the somatic (O) and flagellar (H) antigens of *Salmonella* Typhi. The Widal test was developed over a century ago and remains one of the world's most widely used diagnostic tests, but suffers from significant limitations in its sensitivity, specificity, and reliability. As with most serologic tests, a false-negative Widal test result may occur early in the course of illness, and a false-positive Widal test result may be caused by past infection or previous exposure to cross-reactive antigens or vaccination. There are no universal standards that define the cutoff dilution of agglutinating antibodies to indicate a positive Widal test result. The very low sensitivity and specificity of the assay (50%–80%) and the inability to discern active from previous infection or vaccination means that the assay should rarely, if ever, be used.¹³¹

Other rapid serologic tests for enteric fever have been developed. Major commercial tests include the IDL Tubex and Typhidot assays. The IDL Tubex test (IDL Biotech AB, Bromma, Sweden) is a rapid immunochromatographic test that detects immunoglobulin M (IgM) antibodies to the O:9 antigen (the major antigenic determinant of *Salmonella* Typhi LPS). An advantage of this approach is that T-cell-independent IgM responses that target *S. enterica* O polysaccharide develop early in the clinical course of illness. Studies have suggested that the sensitivity of the IDL Tubex is approximately 70% to 80%, with a specificity of 80% to 90% when evaluated against standards involving patients with blood culture-proven typhoid fever versus other known causes of bacteremia.^{139–142} The Typhidot assay (Malaysian Biodiagnostic Research, Bangi, Malaysia) that detects both IgM and IgG to a 50-kDa outer membrane protein antigen of *Salmonella* Typhi has similar performance characteristics.^{130,143}

Several newer serologic approaches have been developed and evaluated in clinical studies. These include detection of anti-*Salmonella* IgA in secretions of peripheral blood lymphocytes¹⁴⁴ and new antigen targets, common to *Salmonella* Typhi and *Salmonella* Paratyphi A, for enzyme-linked immunosorbent assays (ELISAs) and immunochromatographic assays.^{145–147} These diagnostics have demonstrated greater accuracy in distinguishing typhoid/paratyphoid from other febrile illnesses than currently available serologic tests, but these newer assays have not yet been commercialized.

Molecular Approaches

Molecular diagnostic approaches based on nucleic acid detection have met with only mixed and limited success, presumably reflecting the low-level bacteremia that has hindered culture-based methods, and the action of inhibitors in human blood; such assays are not currently commercially available. Although some promising results with new approaches have been seen in small studies,^{148,149} further clinical validation is needed to characterize their accuracy.

Screening for Chronic Carriage

Identifying chronic carriers of typhoidal *Salmonella* may be important for prevention of transmission, particularly in the context of enteric fever outbreaks in nonendemic settings in which the source of infection is unknown. The conventional approach to identify carriers involves collection of at least three stool specimens (separated by days to weeks) for microbiologic culture to isolate *Salmonella*. Although specificity is excellent, the sensitivity of this approach is estimated at 70% to 80%.¹⁵⁰ Screening by duodenal string capsules is an alternative which may improve this yield, although the string test is not currently commercially available.¹⁵¹ There is an extensive body of literature on the use of Vi antibody testing to identify carriers, but the results have been mixed,¹⁵²⁻¹⁵⁵ and using Vi as a carrier detection target will be further compromised as Vi-based typhoid conjugate vaccines become key components of global typhoid-control strategies.

Management

Appropriate antibiotic management (Table 102.4) reduces the mortality rate of enteric fever from 10% to 15% to less than 1%, and shortens the duration of fever from 3 to 4 weeks to 3 to 5 days after Page 1308initiation of therapy.¹⁰⁸ Uncomplicated enteric fever is treated with a single antibacterial drug. However, the widespread emergence of antibiotic resistance in *Salmonella* Typhi and *Salmonella* Paratyphi A limits the arsenal of effective antibiotics in many areas. The selection of an appropriate antibiotic is dependent not only on antibiotic resistance patterns but also on the severity of illness, the age of the patient, and the availability and cost of antibiotics in resource-limited settings. In addition to antibiotic therapy, supportive care, monitoring for complications, and adjunctive therapy with pulse-dose steroids for severe disease (see later) are important aspects of management of patients with enteric fever.

TABLE 102.4

Antibiotics Commonly Used in Treating Patients with Enteric Fevera

ANTIBIOTIC	ROUTE	TYPICAL PEDIATRIC DOSAGEb	ADULT DOSAGE	TYPICAL DURATION	COMMENTS
Original first-line agents		Chloramphenicol	PO/IV 12.5–25mg/kg qid	2–3g/day	
(divided qid)	14–21 days	These agents are effective for susceptible strains; however, multidrug resistance to all of these agents is common in many areas.			
Amoxicillin	PO	25–35mg/kg tid	1g tid	14 days	
Ampicillin	IV	25–50mg/kg q6h	2g q6h	14 days	
TMP/SMX	PO/IV	8–12mg/kg/day (TMP)	40–60mg/kg/day (SMX) (divided qid)	160mg/800mg qid	14 days
Fluoroquinolones	Ofloxacin	PO/IV 15mg/kg bid	400mg PO bid	5–14 days	
Ofloxacin and ciprofloxacin are effective for nalidixic acid–susceptible strains. Short courses of 5–7 days are acceptable in uncomplicated cases. Ten- to 14-day courses are recommended in patients requiring hospitalization or parenteral therapy.					
Gatifloxacin appears effective for uncomplicated cases caused by NaR strains.					
Ciprofloxacin	PO/IV	15mg/kg bid	500mg PO bid	5–14 days	
Gatifloxacin	PO	10mg/kg qd	10mg/kg/day qd	7 days	
Third-generation cephalosporins	Ceftriaxone	IV	50–100mg/kg qd	1–2g IV qd	7–14 daysc
These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.					
Cefixime	PO	10mg/kg bid	200mg PO bid	7–14 days	
Aztreonam	IV	50mg/kg q8h	2g q8h	Alternative to ceftriaxone for cephalosporin-allergic patients.	
Macrolides	Azithromycin	PO	20mg/kg qd	500–1000mg PO qd	7 days
These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.					
Carbapenems	Meropenem	IV	20–40 mg/kg q8h	1–2 g q8h	10 days
Case series data only (in XDR typhoid); no trials have been reported					
Eradication of carriage	Ciprofloxacin	PO	—	500–750mg bid	28 days
If eradication is indicated for public health reasons, a trial of medical therapy is justified, even in patients with evidence of cholelithiasis. Cholecystectomy may be considered.					

Amoxicillin PO 75–100mg/kg/day in 3 divided doses 2g tid 4–6 weeks Lower efficacy than fluoroquinolones but may be considered for treatment of carriage in individuals harboring strains with fluoroquinolone resistance.
NaR, Nalidixic acid resistant (see text); TMP-SMX, trimethoprim-sulfamethoxazole.

a See text for references.

b Weight-based pediatric dosage should not exceed adult maximum.

c Recommended that treatment continue for at least 7 days after defervescence to minimize relapse.

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Chloramphenicol, Ampicillin, and Trimethoprim-Sulfamethoxazole

Chloramphenicol, ampicillin, and trimethoprim-sulfamethoxazole were the original first-line antibiotics for treating patients with enteric fever. Widespread resistance emerged in the 1980s, through dissemination of strains containing IncH1 plasmids conferring simultaneous resistance to all three agents (Fig. 102.4).¹⁵⁶ Such strains remain widely disseminated and are termed MDR *Salmonella* Typhi. However, over the past several years, there has been a global decline in MDR *Salmonella* Typhi concomitant with the rise in fluoroquinolone resistance.¹³

Because of the risk of irreversible bone marrow aplasia in up to 1 in 10,000 recipients,¹⁵⁷ chloramphenicol is no longer routinely used in many areas, and oral chloramphenicol is not available in the United States. However, in many resource-limited areas, oral chloramphenicol is still available, has excellent bioavailability and central nervous system penetration, and has broad-spectrum activity against many common causes of serious childhood bacterial infections, including enteric fever. Chloramphenicol reduces the mortality rate of enteric fever substantially and decreases the time to defervescence to 3 to 5 days, and was shown to be equivalent to ciprofloxacin in a randomized trial,¹⁵⁸ but its use is associated with a high relapse rate and risk of chronic carriage.^{159,160} Oral trimethoprim-sulfamethoxazole may also be used to treat individuals with susceptible strains, although delayed clearance and relapses are common.¹⁶¹ Oral amoxicillin may also be used to treat individuals with susceptible strains and may result in a roughly comparable likelihood of relapse.¹⁶²

Fluoroquinolones

Fluoroquinolone antibiotics are a highly potent therapy for enteric fever, although increasing antimicrobial resistance is occurring globally, limiting their effectiveness.^{13,163–165} For susceptible isolates, fluoroquinolone antibiotics, such as ciprofloxacin, and the less costly, more widely available ofloxacin, result in defervescence within 3 to 4 days, which is similar to the response associated with chloramphenicol and more rapid than that for third-generation cephalosporins.^{163,166} Administration of a fluoroquinolone also results in lower rates of relapse and carriage compared with those associated with use of chloramphenicol or cefixime.^{163,164,166}

FIG. 102.4 History of antibiotic efficacy studies and the emergence of antimicrobial resistance in *Salmonella* Typhi. MDR, Multidrug resistant; TMP-SMX, trimethoprim-sulfamethoxazole. From Andrews JR, Qamar FN, Charles RC, et al. Extensively drug-resistant typhoid—are conjugate vaccines arriving just in time? *N Engl J Med*. 2018;379:1493–1495.

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Because of the availability and widespread use of fluoroquinolone antibiotics in many areas of the world, fluoroquinolone resistance has become increasingly common in *Salmonella* Typhi and *Salmonella* Paratyphi A in areas where enteric fever is highly endemic.^{13,66} In South and Southeast Asia, strains with reduced susceptibility to fluoroquinolones now predominate, and in sub-Saharan Africa, such strains are on the rise.¹³ Fever clearance times and treatment response have worsened with this rise in fluoroquinolone nonsusceptibility.¹⁶⁷

Before 2012, breakpoints for fluoroquinolone susceptibility were standard across all Enterobacteriaceae, although resistance to nalidixic acid (a nonfluorinated quinolone antibiotic) was associated with decreased clinical responsiveness to fluoroquinolone antibiotics, such as ofloxacin and ciprofloxacin.¹⁶⁸⁻¹⁷¹ Nalidixic acid resistance (NaR) thus became a surrogate marker for predicting clinical failure of fluoroquinolones, even for *Salmonella* Typhi and *Salmonella* Paratyphi strains that met Clinical Laboratory and Standards Institute (CLSI) criteria for fluoroquinolone susceptibility.^{65,172} The CLSI amended its breakpoints for *Salmonella* Typhi and *Salmonella* Paratyphi A, B, and C for ciprofloxacin in 2012 and for levofloxacin and ofloxacin in 2013. For ofloxacin, strains with a minimal inhibitory concentration (MIC) of less than or equal to 0.12 µg/mL are considered susceptible. Patients with strains with an MIC of 0.12 µg/mL defervesce within an average of less than 5 days after initiation of ofloxacin therapy, and patients with strains with an MIC of 0.064 µg/mL or less defervesce in less than 4 days.¹⁷¹ For ciprofloxacin, the amended breakpoint for defining susceptibility is now less than or equal to 0.06 µg/mL. A complicating factor is that many laboratories are not currently able to implement testing of these lower breakpoints for typhoidal *S. enterica* isolates. For such strains, NaR remains an appropriate marker of reduced susceptibility to fluoroquinolone antibiotics.

Although fluoroquinolones are generally well tolerated, concerns about fluoroquinolone-induced articular toxicity in immature animals have limited their use in children in some areas; however, several studies support the safety of their use in children.¹⁶⁶ Gatifloxacin is a dual mechanism-of-action fluoroquinolone that inhibits both DNA gyrase and topoisomerase IV. Gatifloxacin has been associated with increased frequency of hypoglycemia and hyperglycemia, compared with other fluoroquinolones, and for this reason it was withdrawn from the US market. However, gatifloxacin remains available in many countries where enteric fever is endemic and is reported to be clinically effective against some NaR *Salmonella* Typhi.¹⁷³ However, in Nepal, a trial comparing gatifloxacin with ceftriaxone was stopped early owing to higher treatment failure in the gatifloxacin arm as a result of fluoroquinolone resistance.¹⁵⁸ In documented cases of fluoroquinolone nonsusceptibility or for empirical treatment in cases acquired in regions where such strains are common, alternative agents should be used.

Cephalosporins

The emergence of *Salmonella* Typhi and *Salmonella* Paratyphi A strains with decreased susceptibility to fluoroquinolones has shifted the choice for empirical treatment of individuals with suspected severe enteric fever toward third-generation cephalosporins. However, although third-generation cephalosporins, such as ceftriaxone and cefixime, may be used, they appear inferior to fluoroquinolones for fully fluoroquinolone-susceptible strains.¹⁶⁴ Of note, both cefixime and short courses of parenteral ceftriaxone (≤ 7 days) are associated with high relapse rates during enteric fever.^{164,174} In addition, time to defervescence is longer in patients treated with cefixime and ceftriaxone, compared with responses after administration of fluoroquinolones.¹⁶⁴ Resistance of typhoidal *S. enterica* to extended-spectrum β -lactam antibiotics had been rare and sporadically reported^{175,176}; however, in 2017, a large outbreak of extended-spectrum β -lactam *Salmonella* Typhi was reported in Pakistan¹⁷⁷ and has spread throughout the country, displacing less resistant strains and now accounting for the majority of cases in the country.⁶⁶ These strains are additionally MDR and fluoroquinolone-resistant and are termed XDR. Treatment of XDR strains has been primarily with carbapenems or azithromycin. Travel-related spread of XDR strains raises the concern that highly drug-resistant strains may become increasingly common globally.¹³

Azithromycin

Azithromycin is a highly effective drug for treating patients with enteric fever; favorable attributes include its efficacy, its established safety in children, and its excellent bioavailability and pharmacokinetics, with once-daily dosing. Azithromycin reaches high intracellular concentrations that may contribute to its high rates of cure. Duration of treatment is typically 5 to 7 days.^{163,174} Azithromycin may reduce risk of relapse compared with third-generation cephalosporins; however, there is a trend towards higher treatment failure and slower time to fever clearance.¹⁶⁴ In a human challenge model, compared with ciprofloxacin, participants receiving azithromycin had significantly longer duration of bacteremia (20.1 vs. 90.8 hours) and fever.¹⁷⁸ In recent years, there have been increasing reports of azithromycin-resistant strains, associated with point mutations in an efflux pump.⁷¹ Such strains have been observed in several countries and have arisen independently in multiple *Salmonella* Typhi lineages.⁷² However, the global frequency of azithromycin resistance remains low, and it is an excellent option for treating patients with enteric

fever caused by third-generation cephalosporin-resistant and fluoroquinolone-nonsusceptible *Salmonella* Typhi and *Salmonella* Paratyphi A. Page 1310

Other Antibiotics

Four small clinical trials have demonstrated the efficacy of aztreonam for treatment of enteric fever with comparable rates of efficacy and relapse to first-line antibiotics.^{179,180} In the setting of cephalosporin-resistant *Salmonella* Typhi, carbapenems have also been successfully used to treat patients with typhoid fever, although no detailed data on efficacy and relapse rates are available.¹⁸¹

Adjunctive Therapy

Adjunctive therapy with high-dose dexamethasone has been shown to decrease mortality in cases of severe typhoid fever.¹²⁴ A double-blind trial involving 38 adult and pediatric patients with severe typhoid fever in Indonesia, conducted in 1981 and 1982, demonstrated that treatment with chloramphenicol, along with 3 mg/kg of dexamethasone administered over 30 minutes, followed by eight doses at 1 mg/kg every 6 hours, significantly reduced mortality (2 of 20 patients died in the dexamethasone arm) compared with chloramphenicol and placebo (10 of 18 patients died in the placebo arm, $P = .003$).¹²⁴ In this study, severe typhoid fever was defined as typhoid fever with shock or profound encephalopathy manifesting with delirium or obtundation. More recent studies continue to support the concurrent use of dexamethasone and antibiotics in both children and adults with severe typhoid fever.^{123,182} For this reason, administering high-dose steroids to patients who present with shock or, more often, significant mental status changes, and in whom there is strong suspicion of enteric fever, is recommended.

Supportive Care and Management of Complications

Supportive care of patients with severe enteric fever includes rehydration, nutritional management, and mobilization. Contrary to a prevailing historical opinion, it appears that fevers and malaise can be managed safely and more comfortably with a standard regimen of antipyretics.¹⁸³ One small double-blind trial showed that ibuprofen was superior to paracetamol in reducing the morbidity of fevers in enteric fever.¹⁸³

Patients with intestinal hemorrhage should be monitored, ideally in an intensive care setting, but can generally be managed through supportive care. Blood products, including packed red cells and plasma, may be needed in severe cases. Intestinal perforation in enteric fever is associated with a high mortality rate, but prompt surgical intervention is associated with improved survival.^{31,104,184} Although the majority of perforations are single, careful inspection is required to determine if multiple perforations are present. Although simple closure may be performed in some cases, a wedge excision or segmental resection is more often necessary because the surrounding bowel is often necrotic.^{185,186} Drainage of collections and peritoneal lavage should be performed before wound closure. Antibiotic regimens after perforation should be expanded to cover other intestinal microbiota, including anaerobes.

Treatment of Relapse and Chronic Carriage

Relapse usually occurs within 2 weeks of discontinuation of antibiotics, and relapsing patients typically present with milder symptoms than during the primary episode. Isolates obtained after relapse typically have the same antibiotic susceptibility as during the initial episode. The approach to treating patients with relapsed enteric fever is usually the same as for the primary episode.

Chronic carriage is defined as asymptomatic shedding of typhoidal *S. enterica* for 1 year or more. Asymptomatic shedding does not indicate a recurrence of illness or a high risk of recurrent illness. Although chronic carriage is associated with an increased risk of gallbladder carcinoma,¹¹² it is unknown whether the eradication of carriage reduces this risk of carcinoma, so the decision to attempt to eradicate carriage is usually based on public health considerations and is essential for individuals whose occupation involves food handling. Several antibiotics used to treat individuals with acute enteric fever reach high concentrations in bile, including fluoroquinolones. A 28-day course of ciprofloxacin has demonstrated 80% to 90% efficacy in eradication of carriage.¹⁸⁷ However, cholecystectomy may be necessary to eradicate carriage in individuals with cholelithiasis who continue to shed typhoidal *S. enterica* after prolonged antimicrobial therapy.^{188,189} In an era of widespread nonsusceptibility to fluoroquinolones, there is a limited evidence base for successful treatment of carriage with antibiotics other than fluoroquinolones.

Amoxicillin, administered at high doses (75–100 mg/day for 4–6 weeks), can be used, but tolerability is poor and overall eradication rates in many studies are modest.¹⁹⁰ There are even fewer data for treatment of carriers with trimethoprim-sulfamethoxazole with or without rifampin.¹⁹¹

Because enteric fever is transmitted primarily through fecally contaminated water and food, the major preventive strategy to prevent enteric fever is the provision of safe water and adequate sanitation. At a municipal public infrastructure level, this requires appropriate facilities for waste disposal and sewage treatment in addition to provision of a safe water supply. In the United States, the introduction of chlorination and filtration of municipal water supplies in the early 1900s led to rapid and dramatic declines in the burden of typhoid.¹⁹² At a household level, interventions to prevent enteric fever include hand washing with soap before preparing or eating food, avoidance of raw or uncooked foods, and local disinfection of unsafe drinking water, followed by storage in narrow-mouthed containers. Unfortunately, WHO currently estimates that 2 billion people use a drinking water source that is contaminated with feces.¹⁹³

TABLE 102.5

VACCINE	TYPE	ROUTE	DOSE AND INTERVAL	MINIMUM AGE	BOOSTING INTERVAL IN TRAVELERS
Ty21a	Live attenuated	Oral			
Four doses (in United States)					
Administer one dose every other day until complete					
6years ^a	50%–80% ^b	Every 5years			
Vi capsule antigen	Polysaccharide	Intramuscular	1	2years	50%–80%
Every 2years					
Vi tetanus toxoid conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6months	
80%–85%	Not yet defined				
Vi CRM-197 conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6 months	
Not yet defined	Not yet defined				

b Has some efficacy against *Salmonella Paratyphi B* (see text). Additional vaccines are under development, including conjugate vaccines.

Modified from Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

In 2018, WHO recommended that typhoid vaccination programs be undertaken in settings with high typhoid incidence or high burden of antimicrobial-resistant *Salmonella Typhi*.¹⁹⁴ However, typhoid vaccines are not currently widely used in most endemic regions and are more often used by travelers to endemic areas. For travelers receiving either the Vi or Ty21a vaccine, it is optimal that typhoid vaccination be completed at least 7 days before travel, if possible. Limitations of currently licensed vaccines include limited immunity in young children (<2 years of age) and lack of cross-protection against other causes of enteric fever, primarily *Salmonella Paratyphi A*, although Ty21a appears to provide protection against *Salmonella Paratyphi B*.²⁰¹ To improve both immunogenicity in children younger than 2 years and to induce longer-lasting protection, a number of conjugate Vi vaccines are in various stages of development. One version that conjugated Vi to *Pseudomonas aeruginosa* recombinant exoprotein A (Vi-rEPA) achieved 90% protective efficacy against typhoid fever in a large randomized, double-blind clinical trial involving 11,000 children aged 2 to 5 years in Vietnam,²⁰² and has been demonstrated to be safe and immunogenic in phase II trials in infants.²⁰³ However, this vaccine has not been licensed internationally. A Vi-tetanus toxoid conjugate vaccine (Typbar-TCV; Bharat Biotech, Hyderabad, India) was licensed in India in 2015 on the basis of immunogenicity data.²⁰⁴ This vaccine was found to have 79% to 85% effectiveness in three large randomized trials^{205–207} and was prequalified by the WHO in 2018. A second Vi conjugate vaccine (TyphiBEV; Biological E, Hyderabad, India), using a modified diphtheria toxin (CRM 197) was prequalified by the WHO in 2020 based on immunogenicity,²⁰⁸ though clinical effectiveness data are not yet available. Both vaccines can be given to children as young as 6 months of age and have been introduced into routine immunization programs in select high burden countries. Live-attenuated and O-specific polysaccharide-based conjugate vaccines of paratyphoid A have also been reported; however, efficacy data and valid correlates of protection are lacking.²⁰⁹

In non–highly endemic areas, other efforts to prevent enteric fever are aimed at limiting the role of carriers in transmission, particularly for carriers who present a significant public health risk, including food handlers and those working in childcare or health care settings. In the United States, local and state health departments provide screening and work exclusion policies for individuals with enteric fever. The precise policies vary by location but may require screening of the patient and selected contacts to document clearance of bacterial shedding before discharge from public health follow-up and clearance for the performance of certain occupations.

Differential Diagnosis of Enteric and Typhoidal Fevers

Given the nonspecific clinical features of enteric fever and the current limitations of diagnostic testing, it is important that clinicians consider the broader differential diagnosis when approaching patients with possible enteric fever. Such patients usually present with prolonged or relapsing fever, an acute febrile episode in a resource-limited area, or fever and abdominal symptoms and signs. Considerations in the differential diagnosis of enteric fever include epidemiologic risk factors, such as the relative local prevalence of invasive nontyphoidal salmonellosis, malaria, and other febrile diseases, and the type and timing of possible exposures, the duration and magnitude of fevers, the types of abdominal symptoms and signs, and laboratory features at the time of clinical presentation.

Bacteremia Associated With Nontyphoidal *Salmonella*

Although febrile gastroenteritis is the most common clinical manifestation of human nontyphoidal salmonellosis, nontyphoidal serotypes can also cause prolonged fever and bacteremia, particularly in immunocompromised patients with cell-mediated immunodeficiencies or with concomitant malaria. NTS bacteremia is a leading cause of death of young children and adults with AIDS in sub-Saharan Africa.^{17,210} NTS can also cause abscess formation (spleen, kidneys, liver), and *Salmonella* and staphylococci are the leading causes of aortic atheroma seeding, mycotic aneurysm formation, and prolonged bacteremia.²¹¹ Individuals with sickle cell disease and individuals with concomitant schistosomiasis are also at risk of prolonged and invasive nontyphoidal salmonellosis (see Chapter 228).^{212,213}

Typhoidal Fever

A number of non-*Salmonella* infections have features that may mimic typhoid or enteric fever (Table 102.6). These illnesses have sometimes been referred to as typhoidal fevers and are usually characterized as nonspecific systemic illnesses associated with fever that lasts longer than a week. Indeed, typhoid itself was named for its clinical similarity to typhus, caused by *Rickettsia* species. Typhus and typhoid are both characterized as febrile illnesses that may be prolonged and associated with encephalopathy or confusion, and if untreated, both diseases are associated with high mortality rates. In South and Southeast Asia, scrub typhus (*Orientia tsutsugamushi*) and murine typhus (*Rickettsia typhi*) are highly endemic, and are often mistaken for typhoid.²¹⁴⁻²¹⁶ In sub-Saharan Africa, various *Rickettsia* species, including *Rickettsia africae* (the causative agent of African tick-bite fever), are common causes of nonspecific febrile illness. Leptospirosis is also a common cause of nonspecific febrile illness in many typhoid-endemic areas, and is often underdiagnosed owing to lack or underuse of diagnostics.²¹⁴ A clinical diagnosis of typhomalarial fever, a term that appeared in the Page 1312mid-19th century, highlights the overlap of the clinical syndromes of typhoid with malaria.²¹⁷ Individuals with malaria can also present with a wide range of symptoms, including prolonged or recurrent fever, headache, abdominal pain, and cough. Similarly, individuals with brucellosis may present with nonspecific fever, malaise, and abdominal pain. Fever may be prolonged, and infection during brucellosis is often characterized by its chronicity. Manifestations include osteoarticular and genitourinary involvement, granulomatous hepatitis, uveitis, and, rarely, endocarditis. Tularemia is a zoonotic

Differential Diagnoses of Infectious Diseases to Consider in Patients with Possible Typhoid or Enteric Fever

Common Infectious Causes of Nonspecific Acute Febrile Illness in Resource-Limited Areas
<i>Salmonella</i> bacteremia (typhoidal and nontyphoidal serotypes) Malaria Dengue and other arboviral fevers Leptospirosis Influenza Rickettsioses Bartonellosis
Infectious Causes of Typhoidal Illnesses, Including Prolonged Nonlocalizing Febrile Illnesses
Persistent nontyphoidal bacteremia or abscess formation Typhus (rickettsiosis) Brucellosis Typhoidal tularemia Q fever Bartonellosis (including trench fever) Meloidosis Rat-bite fever, Haverhill fever (<i>Streptobacillus moniliformis</i>) Relapsing fever (borreliosis) Infective endocarditis Malaria Babesiosis Visceral leishmaniasis West African trypanosomiasis Tuberculosis Endemic mycoses (histoplasmosis)

infection caused by *Francisella tularensis*. Most human infections are because of exposure to small rodents or lagomorphs (rabbits), and humans usually become infected through direct contact with infected animals or animal parts, or through the bite of an arthropod as a transmission vehicle. Similar to *Salmonella*, *F. tularensis* is an intracellular pathogen that is able to survive and multiply within phagocytic cells, including macrophages, and infection spreads throughout the reticuloendothelial system. Without appropriate treatment, typhoidal tularemia is often fatal.

Persistent fever with nonspecific symptoms may also be seen with melioidosis, an infection caused by *Burkholderia pseudomallei*. Melioidosis is usually reported in South and East Asia and northern Australia. Melioidosis may manifest in a number of ways, ranging from a chronic febrile illness to an acute septicemia,²¹⁸ and it may include abscess formation (often in the spleen, liver, lungs, brain, or muscles) or pneumonia. Melioidosis is associated with water and soil exposure and can affect anyone, although individuals with diabetes, immunosuppression, chronic renal disease, alcoholism, and cirrhosis are at particular risk. Clinical manifestations may occur years after initial exposure.

A typhoidal-like illness can also be seen with Q fever, caused by the intracellular bacterium *Coxiella burnetii*. Q fever is a zoonotic infection usually associated with exposure to sheep or cattle, and humans become infected through exposure to pseudospores from animal products, especially placental tissues. Individuals with Q fever may present with prolonged fever and a wide range of symptoms, including headache, pneumonia, hepatitis, and endocarditis. Rat-bite fever, caused by *Streptobacillus moniliformis*, may also cause a nonspecific febrile illness that may mimic enteric fever, especially when the route of infection is foodborne (Haverhill fever); arthritis and rash can occur.²¹⁹

A number of *Bartonella* species can cause infections characterized by recurrent or prolonged fever. *Bartonella bacilliformis* is the cause of Carrion disease and is transmitted by sand flies in focal areas of Peru, Colombia, and Ecuador. During the acute illness (Oroya fever), patients present with fever and hemolytic anemia. In survivors, disease can progress to verruga peruana, a chronic illness characterized by vascular, proliferative, nodular skin lesions. In addition, *Bartonella quintana* is the cause of trench fever, a relapsing febrile illness transmitted by body lice. Trench fever usually affects individuals who are displaced or severely impoverished; onset is usually acute with high fever, headache, and retro-orbital pain, bone pain of the shins, and myalgia. Each episode of fever may last for 2 to 4 days, with relapses after 4 to 6 days. Prolonged bacteremia can be detected in 20% to 30% of patients.²²⁰ Relapsing fever can also be caused by a number of *Borrelia* species and can be either louse-borne or tick-borne. Louse-borne relapsing fever is a human-restricted infection caused by *Borrelia recurrentis* and is associated with human body lice. Most cases of louse-borne relapsing fever are reported in Africa. Tick-borne relapsing fever has been reported in Asia, Africa, and the Americas. Relapsing fevers are characterized by relapsing and recurrent episodes of fever associated with spirochetemia. Diagnosis usually rests on detection of *Borrelia* on a thick blood smear with microscopy, although polymerase chain reaction (PCR)-based diagnosis is available. Untreated, louse-borne relapsing fever is associated with a high mortality rate. Fever may be prolonged, with many recurrences over several months.²²¹

An enteric fever-like syndrome with abdominal pain may also be caused by a number of enteric pathogens, including *Yersinia enterocolitica*, *Yersinia pseudotuberculosis*, and *Campylobacter fetus* and *Campylobacter jejuni*. Individuals with these infections may present with a prolonged febrile illness, headache, and abdominal pain. Individuals with *Y. enterocolitica* or *Y. pseudotuberculosis* may have acute diarrhea as a major manifestation.^{222,223} Most important, the prolonged enteric fever-like syndromes caused by these agents tend to occur in individuals with significant underlying disease, especially cirrhosis and iron-overloaded states.²²² Diagnosis is usually based on microbiologic culturing of blood.

Additional Causes of Prolonged or Persistent Fever

Most acute febrile illnesses resolve within 3 to 7 days. Individuals with typhoid or enteric fever may have fever that lasts for 1 to 4 weeks before resolution if left untreated. Clinicians assessing individuals with fever that lasts longer than 7 to 10 days need to consider a wide differential, both for infectious and noninfectious processes (see Chapter 58). Noninfectious causes that also need to be considered include lymphoma, malignancy, drug fever, Addison disease, and inflammatory conditions and connective tissue disorders. As described earlier, prolonged fever may also be associated with malaria, brucellosis, yersiniosis, tularemia, Q fever, melioidosis, endocarditis, tuberculosis, and endemic mycoses such as histoplasmosis. Recurrent or relapsing fever may be associated with trench fever and relapsing fever. Clinicians caring for individuals with prolonged fever in resource-limited or tropical areas or caring for individuals who have traveled to such regions need to consider these and other diagnoses. For instance, individuals with babesiosis, caused by an intraerythrocytic protozoal parasite transmitted by ticks, may also present with a nonspecific and prolonged febrile syndrome. Microscopy can confirm the diagnosis. Chronic and relapsing fever are also part of visceral leishmaniasis, a protozoal parasitic infection of the reticuloendothelial system. Individuals with visceral leishmaniasis often present with prolonged and ongoing fever, hepatosplenomegaly, pancytopenia, hypergammaglobulinemia, and absence of eosinophilia. Serologic tests and direct parasitic detection are usually used to confirm the diagnosis. Individuals with human African trypanosomiasis (sleeping sickness) can present with chronic and recurrent fever, weight loss, and lymphadenopathy before neurologic manifestations occur. Serologic tests and direct parasitic detection are usually used to confirm the diagnosis. Prolonged fever can also be associated with a number of viral syndromes, including Epstein-Barr virus (mononucleosis), characterized by prolonged fever, headache, weakness, and fatigue and may include pharyngitis, hepatosplenomegaly, and lymphadenopathy. Serologic assays are available.

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Typhoid Fever, Paratyphoid Fever, and Typhoidal Fevers

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Short View Summary

Definition

- Enteric fever, or typhoid fever, is a nonspecific febrile illness caused by typhoidal *Salmonella* (*Salmonella enterica* serotypes Typhi and Paratyphi); the diagnosis should be considered in any patient with otherwise unexplained prolonged fever who has resided in or recently traveled in a typhoid-endemic setting.
- The term typhoidal fever is sometimes used more broadly to refer to a syndrome of persistent high-grade fevers, often with no localizing features. This chapter also covers the broader differential diagnosis and approach to patients with typhoidal fever, relapsing or prolonged fever, and fever accompanied by abdominal pain.

Epidemiology

- An estimated 12 to 24 million cases of enteric fever occur each year, largely in impoverished areas of Asia and Africa.
- Enteric fever is transmitted via the fecal-oral route, with contaminated municipal water supplies being commonly involved in transmission.
- Antibiotic-resistant strains of the major causative agents of enteric fever (*S. enterica* serotypes Typhi and Paratyphi A) are now common worldwide.

Clinical Manifestations

- Common life-threatening complications of enteric fever include intestinal hemorrhage, perforation, encephalopathy, and shock.

- Untreated, patients with enteric fever may be febrile for 3 to 4 weeks or longer, with mortality rates exceeding 10%, and prolonged asthenia and fatigue are common among survivors.
- Chronic biliary carriage of typhoidal *Salmonella* may occur after resolution of the acute illness.

Diagnosis

- Current diagnostic tests for enteric fever are imperfect. Blood cultures are 30% to 70% sensitive, depending on when in the natural history of the illness they are obtained. Bone marrow cultures are more sensitive but are impractical. Serologic assays lack both sensitivity and specificity, especially in enteric fever endemic areas; and nucleic acid amplification assays with sufficient sensitivity are not available.

Therapy

- Given the morbidity of typhoid fever, the risk of complications, and the lack of optimal diagnostic tests, the initiation of antibiotics for treating individuals with suspected enteric fever may be based on a presumptive diagnosis, particularly in resource-limited settings.
- The most commonly used agents for treating individuals with enteric fever are fluoroquinolones, azithromycin, and cefixime or ceftriaxone. Chloramphenicol, trimethoprim-sulfamethoxazole, and amoxicillin may be used to treat patients with susceptible strains.

Prevention

- An oral-attenuated typhoid vaccine and injectable polysaccharide vaccine are commercially available globally and provide 50% to 75% protection for 5 and 2 years, respectively. Two typhoid conjugate vaccines are now prequalified and recommended by the World Health Organization, with a protective efficacy of >80%. No vaccine effective against paratyphoid A is currently commercially available.

Enteric Fever (Typhoid and Paratyphoid Fever)

History

Historically termed putrid fever or dothienteritis, the name typhoid was coined in 1829 by Pierre Charles Alexander Louis. The name typhoid means typhus-like and reflects the difficulty in differentiating the illness from epidemic typhus, another common cause of prolonged fever in Europe during the 19th century. Louis used the term typhoid in a landmark report comparing the intestinal pathology in 50 patients who died of typhoid fever with 83 patients who died of noninfectious causes, and he described the characteristic inflammation of Peyer patches, intestinal ulceration, and mesenteric adenitis associated with typhoid fever.¹

William Budd is credited with the recognition of the fecal-oral transmission of typhoid fever in 1839, based on his meticulous observations during a large epidemic in the Taw Valley of England, although these findings were not widely recognized until many years later.²

The etiologic agent of typhoid fever was identified in pathologic specimens by Karl Eberth, who referred to it as *Bacillus typhosus*; the organism was subsequently first cultured in 1884 by Georg Gaffky at the Berlin Institute for Infectious Diseases. The term paratyphoid fever was first used in 1896 by Raoul Bensaude and Emile Achard, who isolated what would subsequently be recognized as *Salmonella Paratyphi B*. The role of asymptomatic carriers in the transmission of infectious disease was first recognized by Robert Koch, a theory that he derived from his longitudinal observations of patients recovered from typhoid but with continued shedding of *B. typhosus*.³ An understanding of the role of asymptomatic carriers in disease transmission was soon extended to other organisms and was an established mode of disease transmission at the time of George Soper's association of Mary Mallon (also known as Typhoid Mary) with repeated outbreaks of typhoid fever in households in New York, ultimately leading to her lifelong quarantine on North Brother Island.⁴

Widal developed the first serologic test for typhoid fever in 1896, an agglutination assay that detects the presence of antibodies against the O and H antigens of *Salmonella Typhi*.⁵ Despite inherent and significant limitations (see "Diagnosis"), the Widal test remains widely used in many resource-limited settings. The first vaccines for typhoid fever were killed whole-cell vaccines developed in 1896, credit for which is shared by Almroth Wright and Richard Pfeiffer.⁶ Despite advances in the development of a vaccine, typhoid fever remained a leading cause of morbidity and mortality in the early 20th century. A prominent example was a typhoid fever epidemic in the Spanish-American War, in which 20,000 American Army recruits contracted the disease and 1600 died. This defining event was linked to the first compulsory US military-wide vaccination program, in which a killed whole-cell, injectable typhoid vaccine developed by Frederick Russell was used.⁷

In the early 20th century, typhomalaria or typhomalarial fever remained an extremely common diagnosis, indicating that differentiating typhoid from malaria (and other causes of persistent fever) on clinical grounds was difficult. The difficulty distinguishing Page 1301 typhoid fever from typhus and other causes of fever proved clinically fortuitous. In 1948, Theodore Woodward and coworkers reported the effect of chloramphenicol in two patients who were referred for a study of the efficacy of the drug in scrub typhus.⁸ Both were subsequently found to have *Salmonella* Typhi bacteremia, and both patients improved rapidly after antibiotic therapy. Typhoid fever mortality has been reduced from 10% or greater to less than 1% with the use of antibiotics. However, successive pandemics of antibiotic-resistant organisms have occurred, including, most recently, the emergence of fluoroquinolone- and ceftriaxone-resistant strains of *Salmonella* Typhi. In 2001, the complete sequence of a *Salmonella* Typhi strain (CT18, isolated from a patient in Vietnam in 1993) was published by Parkhill and coworkers at the Wellcome Trust Sanger Institute, ushering in a genomic era in our understanding of typhoid fever.⁹ Genomic analysis has opened up new insights into the history and epidemiology of typhoidal *Salmonella*, from evolutionary history^{10,11} to previously unrecognized historical outbreaks¹² and recent global dissemination.¹³

Etiologic Agents of Enteric Fever

Nomenclature and Classification of Typhoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella* Paratyphi A, B, and C are gram-negative bacilli that belong to the species *S. enterica* subspecies *enterica*. All *S. enterica* are categorized serologically by the Centers for Disease Control and Prevention (CDC) according to a modified Kauffman and White classification scheme, assigned based both on the O (the O polysaccharide) and H (flagellar) antigens.^{14,15} Although assigning a serogroup based on O antigen agglutination tests is a common procedure in many clinical microbiology laboratories, the approach has limited clinical utility. Complete serotyping based on both the O and H antigen is most commonly performed in reference laboratories. In addition to serogrouping and serotyping, standard microbiologic growth and biochemical parameters are also routinely used in the clinical laboratory for the presumptive identification of *S. enterica*. Serologic and selected biochemical characteristics that are used in the identification of *Salmonella* Typhi and *Salmonella* Paratyphi A, B, and C in microbiology laboratories are summarized in Table 102.1.

Clinical Distinction Between Typhoidal and Nontyphoidal *Salmonella enterica*

Salmonella Typhi and *Salmonella* Paratyphi A and B are human-restricted pathogens, and *Salmonella* Paratyphi C is pathogenic for animals and humans. *Salmonella* Typhi and *Salmonella* Paratyphi are classified clinically as typhoidal *Salmonella* strains. This distinction separates these strains from all other pathogenic *S. enterica* serotypes, which are referred to collectively as nontyphoidal *Salmonella* (often abbreviated as NTS; see Chapter 228). Typhoidal strains cause enteric fever in all human hosts, whereas NTS strains are classically associated with inflammatory diarrhea in human hosts or disproportionately cause invasive disease in young children and immunocompromised individuals, with particular associations with malnutrition, HIV, and malaria. However, the clinical distinction between typhoidal and NTS is not absolute. NTS may cause invasive infection that includes prolonged bacteremia (with or without pyogenic foci) that may mimic the systemic illness caused by typhoidal strains.¹⁶ Although host factors influence susceptibility to invasive NTS infection, the likelihood of invasive infection is also dependent on characteristics of the bacterial strain. For example, the emergence of a dominant genotype associated with invasive NTS, *Salmonella* Typhimurium ST313, in sub-Saharan Africa illustrates the clinical overlap between typhoidal and nontyphoidal *S. enterica*.¹⁷

Genomic Features

Escherichia coli and *S. enterica* diverged over 100 million years ago, and *S. enterica* is now a genetically heterogeneous species.¹⁸ *Salmonella* Typhi is genetically homogeneous, meaning there is very limited genetic diversity across *Salmonella* Typhi isolates throughout the world. Based on standard rates of genetic drift, *Salmonella* Typhi likely emerged from a single-point origin within the last 50,000 years, coinciding with the time of human migrations out of Africa and into Asia.^{10,19} The genetic material of *Salmonella* Typhi consists of a single chromosome that contains approximately 5×10^6 base pairs encoding approximately 4000 genes. *Salmonella* Typhi often harbors additional plasmids that may be cryptic (confer an unknown function) or involved in pathogenesis. The latter include IncHI1 plasmids that comprise the Page 1302 majority of

resistance plasmids found in *Salmonella* Typhi and that remain stable in the bacteria without ongoing pressure from antibiotic selection.^{20,21}

TABLE 102.1

Classification and Selected Microbiologic Characteristics of Typhoidal *Salmonella*

SEROTYPE NAME

Salmonella Typhi *Salmonella* Paratyphi A *Salmonella* Paratyphi B (*schottmuelleri*)

Salmonella Paratyphi C (*hirschfeldii*) Comments

Serologic Classification^a

Serogroup	D	A	B	C	
O antigen	9, 12	1, 2, 12		1, 4, (5), 12	6, 7

H antigen

Phase 1:2

d: –	a: (1, 5)	b: (1, 2)	c: (1, 5)	Uncommon <i>Salmonella</i> Typhi H antigen
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variants include H_j:z66 found in Indonesia

K antigen	Vi	–	–	Vi
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Selected Biochemical Characteristics^b

Lactose fermentation	–	–	–	+	The majority of <i>Salmonella</i> Paratyphi C
ferment lactose					

Sulfide (H ₂ S) production		+	(weak)	–	+	+	3% of <i>Salmonella</i> Typhi do not
produce hydrogen sulfide							

Gas produced during glucose fermentation	–				+	+	+
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Lysine decarboxylase	+	–		+	+		
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Ornithine decarboxylase		–		+	+	+	
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a Strong antigenic determinants are listed in bold; weaker or sometimes absent ones are in parentheses.

b Isolates can be identified as *Salmonella* using traditional methods. A presumptive diagnosis of *Salmonella* Typhi can often be made for *Salmonella* isolates based on selected biochemical characteristics, in addition to agglutination with D and Vi antisera.

From World Health Organization (WHO), Department of Vaccines and Biologicals. Background Document: The Diagnosis, Treatment and Prevention of Typhoid Fever. Geneva: WHO; 2003; and WHO, Department of Communicable Disease Surveillance and Response. *Salmonella* serotype Typhi. In: WHO Manual for the Laboratory Identification and Antimicrobial Susceptibility Testing of Bacterial Pathogens of Public Health Importance in the Developing World. Geneva: WHO; 2003:103–118.

The genetic core of *S. enterica* consists of genes that are conserved across Enterobacteriaceae and is approximately 90% similar to *E. coli*. In contrast, the genetic core varies approximately 1% between *S. enterica* serovars. In addition to the genetic core, *Salmonella* Typhi contains pathogenicity islands—horizontally acquired genetic elements that are often associated with unique properties of each strain and are essential for specific phenotypes associated with pathogenesis of each serotype.¹⁸ In *S. enterica*, these are specifically identified as *Salmonella* pathogenicity islands, or SPIs, and are designated with a number. *Salmonella* Typhi possesses more than 20 SPIs, many of which are found in other *S. enterica* serotypes. For example, SPI-7 is a 134-kb genetic island that possesses elements encoding the *viaB* locus, which is involved in production of the Vi capsule and evasion of host immune responses.²² SPI-7 is widely conserved across *Salmonella* Typhi and is also found in *Salmonella* Paratyphi C.

One of the most remarkable genetic features of *Salmonella* Typhi is the accumulation of more than 200 presumed inactive genes called pseudogenes.⁹ This adaptation may play a major role in the host restriction of *Salmonella* Typhi. The accumulation of a large number of pseudogenes is a common feature shared with *Salmonella* Paratyphi A, B, and C, which emerged independently of *Salmonella* Typhi (although subsequent recombination events between *Salmonella* Typhi and *Salmonella* Paratyphi A appear to have occurred),²³ yet are also adapted specifically to cause disseminated infection in humans. The same phenomenon has been observed in *Salmonella* Typhimurium ST313 in sub-Saharan Africa; ST313 has accumulated extensive gene deletions and

purported pseudogene formation (more than half of which are common to *Salmonella* Typhi and *Salmonella* Paratyphi A) compared with less invasive *Salmonella* Typhimurium strains.²⁴

Epidemiology Burden and Distribution

Measures of the burden of enteric fever are limited by the absence of surveillance in many regions and also by the limited sensitivity of current diagnostic tests. Recognizing these limitations, estimates of the number of cases of typhoid and paratyphoid fever range from 12 to 16 million annually.²⁵ The typhoid and paratyphoid fevers were also responsible for 10 million disability-adjusted life years in 2017.²⁵ The economic toll of enteric fever is also devastating for affected families. A study from Malawi from 2019 to 2020 found that catastrophic illness, or an illness which cost over 40% of monthly household nonfood expenses, occurred in 44% of households affected by a case of typhoid fever.²⁶

The incidence of enteric fever and the mortality of the disease vary dramatically by region (Fig. 102.1). In South Asia and parts of sub-Saharan Africa, the incidence approaches 1000 cases per 100,000 person-years.^{27,28} In parts of northern Africa, South America, and Southeast Asia, the incidence is much lower and ranges between 10 and 100 cases per 100,000 person-years.²⁵ Disease is highly heterogeneous even within endemic countries, and the incidence tends to be considerably higher in urban than in rural areas.²⁷ In the United States and many other high-resource settings, the disease occurs sporadically, with less than 1 case per 100,000 person-years, and occurs most often in travelers returning from endemic zones (see “Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically”).

Salmonella Paratyphi A is the most common cause of paratyphoid fever and responsible for an estimated 2.6 to 4.2 million cases of enteric fever annually.²⁵ The proportional burden of enteric fever caused by *Salmonella* Paratyphi A has increased; *Salmonella* Paratyphi A is responsible for more than 20% to 50% of cases of enteric fever. This trend is expected to continue with the introduction of vaccines which protect against typhoid fever, but not enteric fever caused by *S. Paratyphi*.²⁹ In contrast, *Salmonella* Paratyphi B and C remain less-common causes of enteric fever globally.

Severity

In the preantibiotic era, 10% to 15% of clinically recognized cases of typhoid fever were fatal, whereas other cases resolved over a 3- to 4-week course of an often severe and debilitating illness.^{30,31} Prospective surveillance in the postantibiotic era has revealed that the majority of cases of enteric fever are uncomplicated and do not warrant hospitalization. Although the current mortality of typhoid fever is estimated at 0.2% to 2%, these estimates are based on limited data and there is significant variation by region.^{32,33} Reported mortality rates in hospitalized patients with enteric fever in resource-limited settings vary widely by location, ranging from 0% to 15%, with a median of 2%.³² In addition to fatal cases, in sub-Saharan Africa and other regions where enteric fever is highly endemic, intestinal perforation caused by enteric fever is a leading cause of peritonitis and acute abdominal conditions requiring surgery in children.³⁴

Source of Infection

Salmonella Typhi and *Salmonella* Paratyphi A and B are human restricted. There are no known animal reservoirs of these pathogens that cause enteric fever, and the source of infection is organisms shed in the stool of infected humans. *Salmonella* Paratyphi C can be shed by humans and animals. In the preantibiotic era, patients recovering from enteric fever shed organisms in the stool during acute illness and typically for weeks during convalescence. In addition, a small portion of individuals convalescing from typhoid fever develop chronic asymptomatic carriage (defined as shedding of the bacteria in stool or urine for more than a year), which may persist for life.³⁵ In the preantibiotic era, up to 5% of typhoid fever survivors developed chronic asymptomatic bacterial shedding in the stool,^{36,37} with gallbladder disease being the major risk factor for carriage.^{38,39} Risk of carriage appears to be far greater in adults (3%–10%) than in young children (<1%).³⁶ Studies from Nepal, where enteric fever is highly endemic, found that 3.5% of individuals undergoing elective cholecystectomy grew *Salmonella* Typhi or *Salmonella* Paratyphi A from biliary cultures.⁴⁰ Even higher rates were observed in Santiago, Chile, another hyperendemic focus at the time, in the early 1980s.⁴¹ Urinary carriage

may also occur, frequently in conjunction with urinary (*Salmonella haematobium*) schistosomiasis.⁴² Although chronic carriers are a source of transmission where enteric fever occurs sporadically, their importance in the transmission of enteric fever in highly endemic areas is uncertain.^{43,44}

FIG. 102.1 Global distribution of typhoid and paratyphoid fever. Estimated incidence of typhoid and paratyphoid fever by country from the Global Burden of Disease Study 2019. Source: Institute for Health Metrics and Evaluation. Used with permission. All rights reserved.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

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Mode of Transmission

Infections with typhoidal *S. enterica* are most often acquired via ingestion of fecally contaminated water or food. *Salmonella* Typhi may persist for weeks after passage in water and may persist in a variety of contaminated food items—for example, dehydrated formula and iced beverages.^{45–47} Volunteer studies may not optimally mimic natural exposure; however, the inoculum required to produce disease in 50% of adult volunteers (ID₅₀) is 10⁵–7 organisms, although fewer than 10³ organisms may produce disease,^{48,49} which might be more reflective of actual exposures in nonepidemic settings.

Molecular epidemiologic studies based on high-resolution genotyping implicate waterborne transmission as the most likely route of transmission in the majority of cases of enteric fever.⁵⁰ A study in the Kathmandu valley of Nepal revealed that >80% of water collected from public taps contained typhoidal *Salmonella* DNA.⁵¹ Both waterborne and foodborne transmission may contribute to large epidemics, and in some cases, massive epidemics have been associated with water contamination from a single source.^{52,53} In contrast, direct person-to-person transmission of the etiologic agents of enteric fever appears less common, and more than 80% of cases of enteric fever occur in individuals with no known contact with an individual with symptomatic infection.⁵⁴ Even more remarkably, in households where multiple cases of enteric fever occur, only 20% of those infections shared a common bacterial genotype, suggesting that multiple infections within a household occur through a community-based exposure to multiple circulating genotypes, rather than through direct person-to-person transmission.⁵⁰ In highly endemic areas, consistent personal risk factors for infection with typhoidal *S. enterica* include drinking nonboiled water and eating food prepared outside the home, including street-vended foods and beverages.^{55,56} In South Asia, peak transmission occurs during and immediately after the monsoon season (peaks in June through October), although disease is present year-round.⁵⁷

Acquisition of Disease in Areas Where Enteric Fever Occurs Sporadically

In areas where enteric fever occurs sporadically, most cases are imported through travel. In the United States, >85% of cases occur in travelers to endemic areas.⁵⁸ The rate of typhoid fever is estimated at 10 per 1 million travelers arriving from Asia but increases to 89 per 1 million travelers arriving from India. More than 80% of all travel-associated cases in the United States relate to travel to South Asia.⁵⁸ The risk of travel-related enteric fever is higher among travelers visiting friends and relatives. Many of the remaining non-travel-associated cases in the United States are traceable to small, local foodborne outbreaks or a chronic carrier.^{59,60} In 2010, an outbreak of *Salmonella* Typhi occurred that involved at least 12 cases across three western US states and was attributed to imported fruit pulp from Guatemala.⁶¹ Also, in areas where enteric fever occurs sporadically, small clusters of cases of sexual transmission of *Salmonella* Typhi have been documented in men who have sex with men.⁶² Infections and clinical cases have occurred among microbiology laboratory staff, especially when *Salmonella* Typhi was used as an unknown for laboratory proficiency testing, and in at least once instance, secondary cases occurred among family members of laboratory personnel.⁶³

Antibiotic Resistance and Emergence of Pandemic *Salmonella* Typhi

Chloramphenicol-resistant *Salmonella* Typhi was first reported in 1950, 2 years after the antibiotic was first used to treat patients with typhoid fever.⁶⁴ However, widespread dissemination of antibiotic-resistant strains did not occur until the 1970s. Multidrug-resistant (MDR) *Salmonella* Typhi carrying IncHI1 plasmid-mediated resistance to chloramphenicol, ampicillin, trimethoprim, and sulfonamides became common in the 1980s. Nalidixic acid-resistant strains of *Salmonella* Typhi and *Salmonella* Paratyphi A, with decreased susceptibility to fluoroquinolone antibiotics,

emerged in the 1990s,⁶⁵ and now the vast majority of typhoidal *Salmonella* isolates from South Asia have reduced susceptibility or high-level resistance to fluoroquinolones.^{13,66} At present, reduced susceptibility to ciprofloxacin is less common and more sporadic in sub-Saharan Africa, though it appears to be on the rise.¹³

Although the lack of genetic diversity of *Salmonella* Typhi previously hindered efforts to characterize the global epidemiology of disease, the increasing use of high-density genotyping methods, including whole-genome sequencing, has circumvented these challenges and resulted in a better understanding of the forces driving the evolution and transmission of *Salmonella* Typhi. Although a general lack of diversity among *Salmonella* Typhi suggests a lack of strong selection pressure over millennia, the recent expansion of a single fluoroquinolone-resistant haplotype known as H58 into a pandemic strain found throughout Asia and Africa, and the emergence of several other lineages with diverse *gyrA* mutations, suggest that fluoroquinolones may be exerting a uniquely strong selection pressure that is shaping the global evolution of *Salmonella* Typhi.^{13,67} Evidence suggests that fluoroquinolone resistance-conferring mutations have arisen repeatedly in multiple *Salmonella* Typhi lineages, spread internationally and intercontinentally and replaced less resistant strains.¹³ More recently, *Salmonella* strains with resistance to third-generation cephalosporins (e.g., ceftriaxone) have been reported.^{68,69} In Pakistan, strains containing multidrug resistance and resistance to third-generation cephalosporins and fluoroquinolones have been termed “extensively drug-resistant” (XDR) typhoid, and within years of first appearance now account for the majority of isolates in the country.^{66,68,70} Most recently, azithromycin resistance has been identified in multiple lineages, arising from point mutations in an efflux pump (*acrB*).^{71,72} Azithromycin resistance in typhoidal *Salmonella* might become an increasing problem given the WHO recommendation for its use in mass drug administration in settings with high rates of child mortality.^{73,74}

Host Factors and Susceptibility to Infection

Historically, enteric fever was considered a disease that predominantly affected school-aged children and young adults. However, prospective community-based surveys in highly endemic areas have demonstrated that the incidence of *Salmonella* Typhi bacteremia is often highest in young children (<5 years old) and that this organism may be responsible for more than 75% of cases of occult bacteremia in these settings.⁷⁵ The higher incidence of enteric fever in older children among inpatients may reflect a predisposition to more severe disease in this older age group. In highly endemic areas, *Salmonella* Typhi bactericidal antibodies and anticapsular (Vi) antibodies increase significantly over the first decade of life.⁷⁶ In less endemic areas, the median age of patients with typhoid fever is higher, suggesting that the age distribution of patients may be influenced by acquired immunity. However, protection from a single episode of infection is limited, as demonstrated by frequent relapse and recurrent infections among patients who have recovered from typhoid fever.⁷⁷

Although invasive NTS infections classically occur in persons who are compromised by extremes of age, malnutrition, immunodeficiency, or genetic risk factors, there is not an overt association between host Page 1304 risk factors and susceptibility to enteric fever. Various candidate gene analyses have drawn associations between single-nucleotide variations in Toll-like receptor 4 (TLR4), cystic fibrosis transmembrane conductance regulator (CFTR), and specific human leukocyte antigen (HLA) types with susceptibility to typhoid fever in Vietnam, as well as in human volunteers challenged with virulent organisms⁷⁸⁻⁸⁰; however, no detailed genome-wide study of typhoid fever associations has yet been reported.

Pathogenesis

Typhoidal *Salmonella* infection is acquired through ingestion of bacilli, typically in contaminated water, drinks, or food. The ID₅₀ is often cited to be 10⁵–7 organisms, but low inocula (<10³ organisms) may also produce infections.⁴⁸ After passing through the intestinal epithelium, typhoidal *Salmonella* are phagocytosed by macrophages and proliferate in the submucosa before dissemination via the lymphatic system and bloodstream. This period coincides with the onset of systemic symptoms and, in some cases, end organ manifestations. It is during this phase that bacilli enter the hepatobiliary system and may establish carriage in the gallbladder. Disseminated infection is terminated by immune clearance or administration of antibiotics, but bacterial persistence in the bone marrow is often longer.

It is important to note that enteric fever and NTS infections differ in fundamental aspects. With NTS, infections are typically self-limited in immunocompetent hosts and manifest largely as inflammatory gastroenteritis. In comparison, typhoidal serotypes are able to evade the normal host inflammatory response and cause prolonged bacteremia even in immunocompetent individuals, typically without overwhelming sepsis or pyogenic foci of infection. Specific bacterial adaptations used by *Salmonella* Typhi to produce persistent bacteremia in the absence of an immediately overwhelming inflammatory response include strategies to breach the intestinal epithelium in the absence of inflammatory diarrhea, to evade detection by host pattern-recognition receptors,⁸¹ to produce an AB5 bacterial toxin that acts through cell-to-cell paracrine signaling pathways,⁸² and to persist and disseminate throughout the human host within phagocytic cells.⁸³ Much of our understanding of host-pathogen interactions in enteric fever is extrapolated from mouse models of enteric fever, which use Nramp1-deficient mice that develop disseminated infection from *Salmonella* Typhimurium or humanized mice altered to be susceptible to *Salmonella* Typhi.⁸⁴ However, controlled human infection models for *Salmonella* Typhi and Paratyphi A infection are now available and have added valuable insights into the host immune response to these organisms and disease pathogenesis.^{85,86}

Invasion

In contrast to NTS and other enteroinvasive pathogens, soon after ingestion, typhoidal *Salmonella* pass through the intestinal epithelium through specialized microfold cells (M cells) that transport the bacteria across the basolateral membrane, where they are phagocytosed by macrophages in the intestinal lymphoid tissue.⁸⁷ Such a stealth entry mechanism may explain why the invasion phase of *Salmonella* Typhi infection is usually asymptomatic and is accompanied by transient or mild diarrhea in only 10% to 20% of patients. Indeed, as *Salmonella* Typhi leaves the GI tract at this stage, stool cultures are often negative until days to weeks later when shedding through biliary system commences.

Latency and Dissemination

Typhoidal *Salmonella* deploy an array of virulence factors that enable them to persist and replicate in an intracellular compartment of phagocytic cells, primarily macrophages.^{88,89} Important cloaking mechanisms of *Salmonella* Typhi include suppression of flagellar protein synthesis, thus masking a potent inducer of TLR5 responses.⁹⁰ and synthesis of the Vi capsule that presumably masks the detection of lipopolysaccharide (LPS) and other outer membrane components, preventing early complement-mediated clearance.^{91,92} The Vi capsule, thus, helps *Salmonella* Typhi to limit inflammation and evade phagocytosis by neutrophils. The genes that repress flagellin biosynthesis and are responsible for Vi biosynthesis are encoded on the *viaB* locus located on SPI7, which is absent in most NTS strains.²² Induction of other bacterial defenses that are required for survival in the intracellular compartment is regulated by a two-component bacterial sensory-regulatory system, PhoP/PhoQ.^{93,94} The sensor PhoQ detects signals present inside the host phagosome and activates the transcriptional regulator PhoP, which in turn controls the expression of a large number of downstream genes involved in modification of lipid A, resistance to antimicrobial peptides, and acidification of the phagosome.^{95,96} Consistent with a role in pathogenesis, PhoP/PhoQ mutants are nonvirulent in humans and have been evaluated as candidate live-attenuated vaccines.^{97,98}

Quantitative blood cultures demonstrate that more than 60% of culturable circulating *Salmonella* Typhi organisms reside in the intracellular compartment.⁹⁹ Presumably, this allows for dissemination throughout reticuloendothelial tissues via the blood and lymphatic systems. The heaviest burden of infection is established in the reticuloendothelial system (intestinal lymphoid tissue, liver, spleen, and bone marrow) and gallbladder. Patients with typhoid fever most often have very low-grade bacteremia, a feature that presents a formidable diagnostic challenge. The kinetics of quantitative bacterial cultures has been studied in Vietnam.⁹⁹ The median number of culturable bacteria is less than 1 colony-forming unit (CFU) per milliliter of whole blood in adults, and slightly higher (1.5 CFUs/mL) in children. Slightly more culturable organisms are present in the bone marrow, with a median of 10 per milliliter of bone marrow.¹⁰⁰ However, over the course of illness, the number of culturable *Salmonella* Typhi organisms in the blood decreases (median of 2 CFUs/mL in the first week to 0.1 CFUs/mL in the second and subsequent weeks), and the number of culturable bacilli increases in the bone marrow (4.8 CFUs/mL in the first week to 158 CFUs/mL in the third and subsequent weeks). As a result, the proportion of bacteria in the bone marrow increases from 5:1 (marrow to blood) in the first week of illness to more than 150:1 in the third

week of illness, reflecting relative clearance of bacteria from peripheral blood but persistence in the marrow compartment.100 Pathologic examination of bone marrow may demonstrate monocytic infiltrates typical of typhoid fever (see “Intestinal and Other Local Pathology”).

The initial period of replication and dissemination represents the incubation period of typhoidal *Salmonella* infection. This period typically lasts 1 to 2 weeks but can range widely (3–60 days), depending on the number of organisms ingested.101 Ultimately, infection results in the sufficient secretion of inflammatory cytokines, including pyrogenic cytokines such as tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6), which result in fever. However, unlike what occurs during prototypical gram-negative bacteremia, septicemia with hypotension, neutrophilia, and disseminated intravascular coagulopathy are extremely uncommon in patients with typhoid or enteric fever.

Intestinal and Other Local Pathology

Intestinal lymphoid tissue is a predominant site of localized inflammation and persistent bacterial replication in cases of severe enteric fever (Fig. 102.2).102–105 Early in the course of illness, Peyer patches in the terminal ileum and draining mesenteric nodes are enlarged and contain infiltrates consisting of mononuclear cells, including macrophages and lymphocytes. As disease progresses, necrosis of intestinal lymphoid tissue occurs, with mixed inflammatory infiltrates, including neutrophils and ulceration and sloughing of the overlying intestinal mucosa.102 Immunohistochemical staining for the *Salmonella* Typhi O antigen reveals abundant organisms in necrotic areas.103 The clinical manifestations of these changes include hemorrhage or intestinal perforation, two of the major life-threatening complications of enteric fever (Fig. 102.3). Other affected organs include the liver, with monocytic infiltrates and foci of parenchymal necrosis, and the spleen, with nodular monocytic infiltrates in the red pulp.106 Similar monocytic infiltrates may occur in the gallbladder. Gallbladder perforation, which may mimic intestinal perforation, is a rare complication of typhoid fever.107

Relapse and Chronic Carriage

Untreated, typhoid fever may persist for up to 4 weeks, and relapse occurs in up to 10% of untreated cases, usually within 2 weeks after initial resolution of fever.108 Relapse is usually caused by the same strain as that associated with the original infection.109 The gallbladder is the primary site of chronic carriage, although some asymptomatic carriers continue to shed the organism in stool even after cholecystectomy. The *in vitro* and clinical observations that *Salmonella* Typhi forms biofilms on cholesterol gallstones may explain the strong epidemiologic association between gallstones and carriage.110,111 An association has been reproducibly observed between chronic carriage of *Salmonella* Typhi and risk of gallbladder carcinoma. It is unknown if carriage causes cancer or reflects underlying biliary epithelial surface abnormalities112,113; chronic low-grade biliary inflammation and emerging data on the genotoxic activity of *Salmonella* CdtB support an etiological role.114

FIG. 102.2 Intestinal pathologic appearance of enteric fever.(A) The visualized mucosal surface of a surgically resected area demonstrates two perforations (arrowheads); the visualized serosal surface demonstrates significant enlargement of lymph nodes (arrows). (B and C) Lower- and higher-magnification views demonstrate immunohistochemical staining for the *Salmonella* O:9 antigen in a necrotic lymph node. From Neil KP, Sodha SV, Lukwago L, et al. A large outbreak of typhoid fever associated with a high rate of intestinal perforation in Kasese District, Uganda, 2008–2009. *Clin Infect Dis*. 2012;54:1096.

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FIG. 102.3 Intraoperative photograph of intestinal perforation caused by *Salmonella* Typhi. Intestinal perforation is visible on the antimesenteric border of the small bowel, which is inflamed with patchy exudates on the serosal surface. Courtesy Dr. Pukar Maskey, Patan Hospital, Kathmandu, Nepal. From Harris JB, Brooks WA. Typhoid and paratyphoid (enteric) fever. In: Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

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Clinical Manifestations

The clinical manifestations of enteric fever are nonspecific. Fever is reported in the vast majority of clinically apparent cases, and many other symptoms are variably reported, including headache, cough, nausea, vomiting, constipation, and diarrhea.¹¹⁵ Although paratyphoid fever was considered less severe than typhoid fever, more recent comparisons suggest that typhoid fever and paratyphoid fever caused by *Salmonella Paratyphi A* are generally indistinguishable on clinical grounds.¹¹⁶ Most patients with enteric fever are diagnosed in the ambulatory setting, and up to 90% are treated as outpatients. In contrast, classic descriptions of the features of enteric fever are derived from a series of hospitalized patients with more severe disease.³⁰ Complications of enteric fever were generally thought to occur late in the course of disease, usually after 2 weeks of fever. However, experience has demonstrated that major complications of enteric fever, including intestinal perforation and encephalopathy, may occur within days of onset of fever.^{103,104,117}

Uncomplicated Typhoid Fever

Clinical features of enteric fever diagnosed in the ambulatory setting are listed in Table 102.2. Fever without localizing signs or symptoms may be the sole manifestation of enteric fever. The onset of fever may be insidious, and fevers typically increase over the first week of illness.¹¹⁸ A variety of other nonspecific flulike symptoms are common early in the course of enteric fever, and headache, anorexia, myalgias, and malaise may precede the onset of fever. Mild confusion may be seen even in ambulatory patients, and a nonproductive cough is also a common feature of uncomplicated enteric fever.¹¹⁵ Abdominal complaints may include diarrhea, constipation, and abdominal pain. Invasive diarrhea, such as seen in NTS gastroenteritis, does not typically occur in enteric fever.

The physical findings of uncomplicated enteric fever are also nonspecific. Although relative bradycardia, or pulse-temperature dissociation, is a classic sign of enteric fever, it may not be a clinically useful predictor of enteric fever for individual patients, and is absent in the majority of patients.¹¹⁹ Rose spots, the classic cutaneous manifestation of enteric fever, are 1- to 4-mm blanching pink macules, and are most often seen on the chest, back, and abdomen during the second week of fever. However, rose spots are uncommon in uncomplicated typhoid fever. Mild, nonlocalizing, abdominal tenderness may be present. Hepatomegaly and splenomegaly, if present, are usually modest. A white or yellowish-brown coating of the tongue that spares the tongue's edges is a common physical finding.¹¹⁶

Laboratory Findings in Uncomplicated Infection

Although abnormal white blood cell counts in peripheral blood are common in patients with bacteremia, the majority of patients with uncomplicated enteric fever do not have leukocytosis, neutrophilia, or increased immature neutrophils.¹¹⁵ However, either leukocytosis or Page 1306leukopenia may be present. Both the hematocrit and platelet counts are typically normal or slightly low. Elevated serum aspartate transaminase and alanine transaminase are very common in enteric fever; values two to three times above the upper limit of the normal range are typical.¹²⁰ On occasion, more severe hepatitis is observed, but very high levels of blood transaminases (>500 IU/L) should prompt concern for other causes, including viral hepatitis or drug toxicity.¹²⁰

TABLE 102.2

Clinical Features of Typhoid and Paratyphoid Fever

CLINICAL FEATURE	APPROXIMATE FREQUENCY ^a	
Flulike symptoms	Fever	>95%
Headache	80%	
Chills	40%	
Cough	30%	
Myalgia	20%	
Arthralgia	<5%	
Abdominal symptoms	Anorexia	50%
Abdominal pain	30%	
Diarrhea	20%	
Constipation	20%	
Physical findings	Coated tongue	50%
Hepatomegaly	10%	

Splenomegaly 10%
 Abdominal tenderness 5%
 Rash <5%
 Generalized adenopathy <5%

a The proportion of patients demonstrating these clinical features of enteric fever varies, depending on the time, the region, and the type of clinical population (hospitalized or ambulatory) assessed. Estimates are drawn from recent case series in an endemic area, with patients presenting for ambulatory or inpatient care.^{116,216}

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Severe Illness

Classic descriptions of the features of enteric fever are drawn from observations of a series of hospitalized patients in the preantibiotic era, in which mortality rates were consistently in the 10% to 15% range. The natural history of untreated disease included progressively increasing fevers over the first week of illness, followed by increasing abdominal complaints and rash over the second week of illness, followed by complications, including intestinal hemorrhage and perforation, or gradual resolution in the third and fourth weeks of illness.³⁰

In the present antibiotic era, it remains true that because enteric fever progresses gradually, patients who are severely ill with enteric fever may have been febrile for longer than a week when they first come to clinical attention. Patients with severe enteric fever may appear toxic, and characteristically would have moderate abdominal pain or tenderness, and constipation or diarrhea. Physical findings in severe enteric fever include hepatomegaly and splenomegaly. Patients with severe enteric fever are more likely to have major complications listed in Table 102.3. Complications associated with increased mortality in severe typhoid fever include intestinal hemorrhage and perforation, severe encephalopathy, seizures, and pneumonia.¹²¹

Gastrointestinal Complications

Intestinal hemorrhage occurs in up to 10% of hospitalized patients with severe enteric fever and is usually self-limited.^{30,32} Intestinal perforation, a major life-threatening complication of enteric fever, occurs in up to 3% of hospitalized patients.^{30–32,121} These complications arise directly from invasion of *Salmonella* into Peyer patches in the small intestines. A series of patients from Vietnam with intestinal perforation resulting from enteric fever demonstrated that the median length of illness preceding perforation was 9 days from the onset of fever, although some cases occurred within the first week.¹⁰⁴ Perforation is suggested by clinical signs of peritonitis, including tachycardia, leukocytosis, neutrophilia, and abdominal pain with guarding and rebound tenderness. The clinical diagnosis of perforation requires a high index of suspicion because patients with severe enteric fever may have a toxic appearance and significant abdominal tenderness even before perforation. Radiographic evidence of pneumoperitoneum may be present in only 50% of cases.

TABLE 102.3

Complications of Typhoid and Paratyphoid Fever

SYSTEM	COMPLICATION	NOTES
Gastrointestinal	Hemorrhage	10%–15% in hospitalized patients
Perforation		3% in hospitalized patients
Hepatobiliary	Jaundice	1%–3% in hospitalized patients
Hepatitis	Usually subclinical (↑ ALT/AST)	
Acute cholecystitis	Rare; gallbladder may perforate	
Neurologic	Mild encephalopathy	Confusion or apathy common
Severe encephalopathy		Delirium, stupor, or coma
Seizures		Common in children ≤5years
Meningitis		Rare, primarily infants
Guillain-Barré syndrome		Reported
Respiratory	Bronchitis	Dry cough is common
Pneumonia		May be other concomitant bacterial infection (e.g., <i>Streptococcus pneumoniae</i>)
Empyema		Rare reports

Cardiovascular	Myocarditis	Usually subclinical (ECG changes)
Endocarditis	Rare reports	
Hematologic	Anemia	Usually subclinical
Disseminated intravascular coagulation	Usually subclinical (↑ PT/PTT)	
Other	Musculoskeletal pyogenic infections Osteomyelitis (particularly vertebral), psoas abscess and others reported	
Hemolytic-uremic syndrome	Reported	
Miscarriage	Reported	

ALT, Alanine aminotransferase; AST, aspartate aminotransferase; ECG, electrocardiogram; PT, prothrombin time; PTT, partial thromboplastin time.

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Neurologic Complications

Various case series have documented a widely varying prevalence of diverse neuropsychiatric manifestations during enteric fever, ranging from encephalopathy, encephalomyelitis, transverse myelitis, meningitis, ataxia, and Guillain-Barré syndrome.^{117,122} Although most of these manifestations are uncommon, patients with severe enteric fever often present with some component of encephalopathy, and a history of confusion or an apathetic affect are common.³² Of note, typhoid was named for its clinical similarity to typhus, and the name typhus was from the Greek term for smoky or cloudy, reflecting the neurologic features that often accompanied this febrile illness. More severe encephalopathy, manifesting with delirium, stupor, and coma, occurs in a smaller number of hospitalized patients and is associated with a high risk of mortality.^{121,123,124} Osler described this as a “pseudo-wakeful state” of “muttering” delirium, known as “coma vigil,” and noted that it portends poor outcomes. Even in cases of severe encephalopathy, microbiologic evaluation of cerebral spinal fluid is usually unrevealing, and pleocytosis, if present, usually reveals fewer than 35 cells/μL.^{123,124} Encephalopathy during enteric fever may be most common among older children and young adults.¹²⁵ Seizures during enteric fever are most common in young children and are associated with increased mortality.¹²¹ Meningitis caused by *Salmonella Typhi* is seen primarily in infants.¹²⁶

Metastatic Pyogenic Complication

Enteric fever is a disseminated bacterial infection, but pyogenic abscess formation during enteric fever is notably infrequent, although *Salmonella Paratyphi C* is more likely to cause abscesses and focal infections than the other typhoidal strains. This contrasts with what occurs during invasive nontyphoidal salmonellosis, in which osteomyelitis, joint infection, abscess formation, and endovascular infection more frequently occur. Although uncommon, pyogenic complications during typhoid have been described and include empyema, osteomyelitis, hepatosplenic, and other organ system and soft tissue abscesses, as well as endovascular infections and endocarditis.¹²⁷

Diagnosis

The laboratory diagnosis of enteric fever is often challenging because current culture-based, serologic, and molecular diagnostic tests for enteric fever fail to achieve an optimal combination of sensitivity and specificity.^{128–131} For this reason, it is often appropriate to treat patients with suspected enteric fever with empirical antibiotic therapy. The diagnosis of enteric fever should be considered in any person with fever, especially in those with fever lasting longer than 3 days and who have had exposure in the last 1 to 6 weeks to an area where enteric fever is endemic. In endemic areas, other clinical factors that are associated with a higher likelihood of enteric fever include a temperature greater than 39°C (102.2°F), ill appearance, young age (<5 years), and any abdominal complaints, including abdominal pain, diarrhea, or constipation.⁷⁵ As the duration of fever increases, the likelihood of enteric fever increases.¹³² Despite this, it is important to recognize that the majority of individuals suspected of having enteric fever, even in endemic settings, do not have enteric fever but rather an alternative infection^{133,134}; therefore, clinical suspicion of and empirical treatment for enteric fever should not halt the search for alternative causes of fever.

Culture-Based Diagnostics

Although a presumptive diagnosis of enteric fever may be sufficient grounds for initiating and continuing antimicrobial therapy, a definitive diagnosis of enteric fever is made only through the isolation of a typhoidal serotype of *S. enterica* (*Salmonella* Typhi or *Salmonella* Paratyphi A, B, and C) from the blood, bone marrow, stool, urine, or other clinical specimen of a febrile patient. Isolation of the causative organism also allows antimicrobial resistance testing, which facilitates optimal management.

Blood culture is the most common method of diagnosis, where adequate microbiologic facilities exist. The sensitivity of blood culture varies from approximately 40% to 80%,^{135,136} and a recent meta-analysis estimated an average sensitivity of 59%.¹³⁷ The low-grade bacteremia in enteric fever is an inherent limiting factor to the sensitivity of blood cultures, as is prior receipt of antibiotics.⁹⁹ To maximize the sensitivity of blood cultures, the World Health Organization (WHO) recommends that cultures be performed using 10 to 15 mL of blood from schoolchildren and adults, and 2 to 4 mL of blood from toddlers and preschool-aged children.¹³⁸ The use of a nonselective broth media, as is standard for most blood cultures, is encouraged because it maximizes detection of a wide range of pathogens. The sole use of *Salmonella* selective or enriching media, such as ox bile medium, is discouraged.¹³⁸

Although less sensitive than blood cultures, stool cultures are positive in more than 50% of children and 30% of adults with enteric fever, and they increase the diagnostic yield over blood cultures alone by approximately 5%.¹³⁶ To maximize detection, WHO recommends the use of greater than 1 g of stool and selenite enrichment broth.¹³⁸ Culturing of bone marrow has the highest sensitivity (80%–95%) of all the culture-based approaches, with culturing of a 1-mL aspirate of bone marrow having the sensitivity of a 15-mL peripheral blood culture in an adult.^{99,100} Culturing of bone marrow is, however, not clinically practical for most patients with suspected enteric fever, although microbiologic and hematologic analysis of a bone marrow aspirate may be part of an evaluation of a patient with a prolonged fever of unknown origin when less invasive testing is unrevealing. Other specimens that may yield growth of *Salmonella* Typhi or *Salmonella* Paratyphi A, B, and C include urine, duodenal aspirates, and specimens from skin biopsy of rose spots. However, these approaches are not routinely recommended. Cultures made from intestinal biopsy specimens and peritoneal fluid of patients with perforation are rarely positive.^{103,104}

Serologic Tests

The Widal test detects agglutinating antibodies against the somatic (O) and flagellar (H) antigens of *Salmonella* Typhi. The Widal test was developed over a century ago and remains one of the world's most widely used diagnostic tests, but suffers from significant limitations in its sensitivity, specificity, and reliability. As with most serologic tests, a false-negative Widal test result may occur early in the course of illness, and a false-positive Widal test result may be caused by past infection or previous exposure to cross-reactive antigens or vaccination. There are no universal standards that define the cutoff dilution of agglutinating antibodies to indicate a positive Widal test result. The very low sensitivity and specificity of the assay (50%–80%) and the inability to discern active from previous infection or vaccination means that the assay should rarely, if ever, be used.¹³¹

Other rapid serologic tests for enteric fever have been developed. Major commercial tests include the IDL Tubex and Typhidot assays. The IDL Tubex test (IDL Biotech AB, Bromma, Sweden) is a rapid immunochromatographic test that detects immunoglobulin M (IgM) antibodies to the O:9 antigen (the major antigenic determinant of *Salmonella* Typhi LPS). An advantage of this approach is that T-cell-independent IgM responses that target *S. enterica* O polysaccharide develop early in the clinical course of illness. Studies have suggested that the sensitivity of the IDL Tubex is approximately 70% to 80%, with a specificity of 80% to 90% when evaluated against standards involving patients with blood culture-proven typhoid fever versus other known causes of bacteremia.^{139–142} The Typhidot assay (Malaysian Biodiagnostic Research, Bangi, Malaysia) that detects both IgM and IgG to a 50-kDa outer membrane protein antigen of *Salmonella* Typhi has similar performance characteristics.^{130,143}

Several newer serologic approaches have been developed and evaluated in clinical studies. These include detection of anti-*Salmonella* IgA in secretions of peripheral blood lymphocytes¹⁴⁴ and new antigen targets, common to *Salmonella* Typhi and *Salmonella* Paratyphi A, for enzyme-linked

immunosorbent assays (ELISAs) and immunochromatographic assays.¹⁴⁵⁻¹⁴⁷ These diagnostics have demonstrated greater accuracy in distinguishing typhoid/paratyphoid from other febrile illnesses than currently available serologic tests, but these newer assays have not yet been commercialized.

Molecular Approaches

Molecular diagnostic approaches based on nucleic acid detection have met with only mixed and limited success, presumably reflecting the low-level bacteremia that has hindered culture-based methods, and the action of inhibitors in human blood; such assays are not currently commercially available. Although some promising results with new approaches have been seen in small studies,^{148,149} further clinical validation is needed to characterize their accuracy.

Screening for Chronic Carriage

Identifying chronic carriers of typhoidal *Salmonella* may be important for prevention of transmission, particularly in the context of enteric fever outbreaks in nonendemic settings in which the source of infection is unknown. The conventional approach to identify carriers involves collection of at least three stool specimens (separated by days to weeks) for microbiologic culture to isolate *Salmonella*. Although specificity is excellent, the sensitivity of this approach is estimated at 70% to 80%.¹⁵⁰ Screening by duodenal string capsules is an alternative which may improve this yield, although the string test is not currently commercially available.¹⁵¹ There is an extensive body of literature on the use of Vi antibody testing to identify carriers, but the results have been mixed,¹⁵²⁻¹⁵⁵ and using Vi as a carrier detection target will be further compromised as Vi-based typhoid conjugate vaccines become key components of global typhoid-control strategies.

Management

Appropriate antibiotic management (Table 102.4) reduces the mortality rate of enteric fever from 10% to 15% to less than 1%, and shortens the duration of fever from 3 to 4 weeks to 3 to 5 days after initiation of therapy.¹⁰⁸ Uncomplicated enteric fever is treated with a single antibacterial drug. However, the widespread emergence of antibiotic resistance in *Salmonella* Typhi and *Salmonella* Paratyphi A limits the arsenal of effective antibiotics in many areas. The selection of an appropriate antibiotic is dependent not only on antibiotic resistance patterns but also on the severity of illness, the age of the patient, and the availability and cost of antibiotics in resource-limited settings. In addition to antibiotic therapy, supportive care, monitoring for complications, and adjunctive therapy with pulse-dose steroids for severe disease (see later) are important aspects of management of patients with enteric fever.

TABLE 102.4

Antibiotics Commonly Used in Treating Patients with Enteric Fever^a

ANTIBIOTIC	ROUTE	TYPICAL PEDIATRIC DOSAGE ^b	ADULT DOSAGE	TYPICAL DURATION	COMMENTS
Original first-line agents		Chloramphenicol	PO/IV 12.5–25mg/kg qid	2–3g/day	
(divided qid)	14–21 days	These agents are effective for susceptible strains; however, multidrug resistance to all of these agents is common in many areas.			
Amoxicillin	PO	25–35mg/kg tid	1g tid	14 days	
Ampicillin	IV	25–50mg/kg q6h	2g q6h	14 days	
TMP/SMX	PO/IV	8–12mg/kg/day (TMP)	40–60mg/kg/day (SMX) (divided qid)	160mg/800mg qid	14 days
Fluoroquinolones	Ofloxacin	PO/IV 15mg/kg bid	400mg PO bid	5–14 days	
Ofloxacin and ciprofloxacin are effective for nalidixic acid–susceptible strains. Short courses of 5–7 days are acceptable in uncomplicated cases. Ten- to 14-day courses are recommended in patients requiring hospitalization or parenteral therapy.					
Gatifloxacin appears effective for uncomplicated cases caused by NaR strains.					
Ciprofloxacin	PO/IV	15mg/kg bid	500mg PO bid	5–14 days	
Gatifloxacin	PO	10mg/kg qd	10mg/kg/day qd	7 days	
Third-generation cephalosporins	Ceftriaxone	IV	50–100mg/kg qd	1–2g IV qd	7–14 days
These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.					
Cefixime	PO	10mg/kg bid	200mg PO bid	7–14 days	

Aztreonam	IV	50mg/kg q8h	2g q8h	Alternative to ceftriaxone for cephalosporin-allergic patients.
Macrolides	Azithromycin	PO	20mg/kg qd	500–1000mg PO qd 7 days These are an alternative to fluoroquinolones for strains with decreased fluoroquinolone susceptibility, and for empirical therapy in areas where decreased fluoroquinolone susceptibility is common.
Carbapenems	Meropenem	IV	20–40 mg/kg q8h	1–2 g q8h 10 days Case series data only (in XDR typhoid); no trials have been reported
Eradication of carriage	Ciprofloxacin	PO	—	500–750mg bid 28 days If eradication is indicated for public health reasons, a trial of medical therapy is justified, even in patients with evidence of cholelithiasis. Cholecystectomy may be considered.
Amoxicillin	PO	75–100mg/kg/day in 3 divided doses	2g tid	4–6 weeks Lower efficacy than fluoroquinolones but may be considered for treatment of carriage in individuals harboring strains with fluoroquinolone resistance.

NaR, Nalidixic acid resistant (see text); TMP-SMX, trimethoprim-sulfamethoxazole.

a See text for references.

b Weight-based pediatric dosage should not exceed adult maximum.

c Recommended that treatment continue for at least 7 days after defervescence to minimize relapse.

Modified from Magill AJ, et al., eds. Hunter's Tropical Medicine and Emerging Infectious Diseases. 9th ed. Philadelphia: Saunders; 2013:568–576.

Chloramphenicol, Ampicillin, and Trimethoprim-Sulfamethoxazole

Chloramphenicol, ampicillin, and trimethoprim-sulfamethoxazole were the original first-line antibiotics for treating patients with enteric fever. Widespread resistance emerged in the 1980s, through dissemination of strains containing IncH1 plasmids conferring simultaneous resistance to all three agents (Fig. 102.4).¹⁵⁶ Such strains remain widely disseminated and are termed MDR *Salmonella* Typhi. However, over the past several years, there has been a global decline in MDR *Salmonella* Typhi concomitant with the rise in fluoroquinolone resistance.¹³

Because of the risk of irreversible bone marrow aplasia in up to 1 in 10,000 recipients,¹⁵⁷ chloramphenicol is no longer routinely used in many areas, and oral chloramphenicol is not available in the United States. However, in many resource-limited areas, oral chloramphenicol is still available, has excellent bioavailability and central nervous system penetration, and has broad-spectrum activity against many common causes of serious childhood bacterial infections, including enteric fever. Chloramphenicol reduces the mortality rate of enteric fever substantially and decreases the time to defervescence to 3 to 5 days, and was shown to be equivalent to ciprofloxacin in a randomized trial,¹⁵⁸ but its use is associated with a high relapse rate and risk of chronic carriage.^{159,160} Oral trimethoprim-sulfamethoxazole may also be used to treat individuals with susceptible strains, although delayed clearance and relapses are common.¹⁶¹ Oral amoxicillin may also be used to treat individuals with susceptible strains and may result in a roughly comparable likelihood of relapse.¹⁶²

Fluoroquinolones

Fluoroquinolone antibiotics are a highly potent therapy for enteric fever, although increasing antimicrobial resistance is occurring globally, limiting their effectiveness.^{13,163–165} For susceptible isolates, fluoroquinolone antibiotics, such as ciprofloxacin, and the less costly, more widely available ofloxacin, result in defervescence within 3 to 4 days, which is similar to the response associated with chloramphenicol and more rapid than that for third-generation cephalosporins.^{163,166} Administration of a fluoroquinolone also results in lower rates of relapse and carriage compared with those associated with use of chloramphenicol or cefixime.^{163,164,166}

FIG. 102.4 History of antibiotic efficacy studies and the emergence of antimicrobial resistance in *Salmonella* Typhi. MDR, Multidrug resistant; TMP-SMX, trimethoprim-sulfamethoxazole. From

Andrews JR, Qamar FN, Charles RC, et al. Extensively drug-resistant typhoid—are conjugate vaccines arriving just in time? *N Engl J Med*. 2018;379:1493–1495.

Follow for extended description

Because of the availability and widespread use of fluoroquinolone antibiotics in many areas of the world, fluoroquinolone resistance has become increasingly common in *Salmonella Typhi* and *Salmonella Paratyphi A* in areas where enteric fever is highly endemic.^{13,66} In South and Southeast Asia, strains with reduced susceptibility to fluoroquinolones now predominate, and in sub-Saharan Africa, such strains are on the rise.¹³ Fever clearance times and treatment response have worsened with this rise in fluoroquinolone nonsusceptibility.¹⁶⁷

Before 2012, breakpoints for fluoroquinolone susceptibility were standard across all Enterobacteriaceae, although resistance to nalidixic acid (a nonfluorinated quinolone antibiotic) was associated with decreased clinical responsiveness to fluoroquinolone antibiotics, such as ofloxacin and ciprofloxacin.^{168–171} Nalidixic acid resistance (NaR) thus became a surrogate marker for predicting clinical failure of fluoroquinolones, even for *Salmonella Typhi* and *Salmonella Paratyphi* strains that met Clinical Laboratory and Standards Institute (CLSI) criteria for fluoroquinolone susceptibility.^{65,172} The CLSI amended its breakpoints for *Salmonella Typhi* and *Salmonella Paratyphi A*, B, and C for ciprofloxacin in 2012 and for levofloxacin and ofloxacin in 2013. For ofloxacin, strains with a minimal inhibitory concentration (MIC) of less than or equal to 0.12 µg/mL are considered susceptible. Patients with strains with an MIC of 0.12 µg/mL defervesce within an average of less than 5 days after initiation of ofloxacin therapy, and patients with strains with an MIC of 0.064 µg/mL or less defervesce in less than 4 days.¹⁷¹ For ciprofloxacin, the amended breakpoint for defining susceptibility is now less than or equal to 0.06 µg/mL. A complicating factor is that many laboratories are not currently able to implement testing of these lower breakpoints for typhoidal *S. enterica* isolates. For such strains, NaR remains an appropriate marker of reduced susceptibility to fluoroquinolone antibiotics.

Although fluoroquinolones are generally well tolerated, concerns about fluoroquinolone-induced articular toxicity in immature animals have limited their use in children in some areas; however, several studies support the safety of their use in children.¹⁶⁶ Gatifloxacin is a dual mechanism-of-action fluoroquinolone that inhibits both DNA gyrase and topoisomerase IV. Gatifloxacin has been associated with increased frequency of hypoglycemia and hyperglycemia, compared with other fluoroquinolones, and for this reason it was withdrawn from the US market. However, gatifloxacin remains available in many countries where enteric fever is endemic and is reported to be clinically effective against some NaR *Salmonella Typhi*.¹⁷³ However, in Nepal, a trial comparing gatifloxacin with ceftriaxone was stopped early owing to higher treatment failure in the gatifloxacin arm as a result of fluoroquinolone resistance.¹⁵⁸ In documented cases of fluoroquinolone nonsusceptibility or for empirical treatment in cases acquired in regions where such strains are common, alternative agents should be used.

Cephalosporins

The emergence of *Salmonella Typhi* and *Salmonella Paratyphi A* strains with decreased susceptibility to fluoroquinolones has shifted the choice for empirical treatment of individuals with suspected severe enteric fever toward third-generation cephalosporins. However, although third-generation cephalosporins, such as ceftriaxone and cefixime, may be used, they appear inferior to fluoroquinolones for fully fluoroquinolone-susceptible strains.¹⁶⁴ Of note, both cefixime and short courses of parenteral ceftriaxone (≤7 days) are associated with high relapse rates during enteric fever.^{164,174} In addition, time to defervescence is longer in patients treated with cefixime and ceftriaxone, compared with responses after administration of fluoroquinolones.¹⁶⁴ Resistance of typhoidal *S. enterica* to extended-spectrum β-lactam antibiotics had been rare and sporadically reported^{175,176}; however, in 2017, a large outbreak of extended-spectrum β-lactam *Salmonella Typhi* was reported in Pakistan¹⁷⁷ and has spread throughout the country, displacing less resistant strains and now accounting for the majority of cases in the country.⁶⁶ These strains are additionally MDR and fluoroquinolone-resistant and are termed XDR. Treatment of XDR strains has been primarily with carbapenems or azithromycin. Travel-related spread of XDR strains raises the concern that highly drug-resistant strains may become increasingly common globally.¹³

Azithromycin

Azithromycin is a highly effective drug for treating patients with enteric fever; favorable attributes include its efficacy, its established safety in children, and its excellent bioavailability and

pharmacokinetics, with once-daily dosing. Azithromycin reaches high intracellular concentrations that may contribute to its high rates of cure. Duration of treatment is typically 5 to 7 days.^{163,174} Azithromycin may reduce risk of relapse compared with third-generation cephalosporins; however, there is a trend towards higher treatment failure and slower time to fever clearance.¹⁶⁴ In a human challenge model, compared with ciprofloxacin, participants receiving azithromycin had significantly longer duration of bacteremia (20.1 vs. 90.8 hours) and fever.¹⁷⁸ In recent years, there have been increasing reports of azithromycin-resistant strains, associated with point mutations in an efflux pump.⁷¹ Such strains have been observed in several countries and have arisen independently in multiple *Salmonella* Typhi lineages.⁷² However, the global frequency of azithromycin resistance remains low, and it is an excellent option for treating patients with enteric fever caused by third-generation cephalosporin-resistant and fluoroquinolone-nonsusceptible *Salmonella* Typhi and *Salmonella* Paratyphi A. Page 1310

Other Antibiotics

Four small clinical trials have demonstrated the efficacy of aztreonam for treatment of enteric fever with comparable rates of efficacy and relapse to first-line antibiotics.^{179,180} In the setting of cephalosporin-resistant *Salmonella* Typhi, carbapenems have also been successfully used to treat patients with typhoid fever, although no detailed data on efficacy and relapse rates are available.¹⁸¹

Adjunctive Therapy

Adjunctive therapy with high-dose dexamethasone has been shown to decrease mortality in cases of severe typhoid fever.¹²⁴ A double-blind trial involving 38 adult and pediatric patients with severe typhoid fever in Indonesia, conducted in 1981 and 1982, demonstrated that treatment with chloramphenicol, along with 3 mg/kg of dexamethasone administered over 30 minutes, followed by eight doses at 1 mg/kg every 6 hours, significantly reduced mortality (2 of 20 patients died in the dexamethasone arm) compared with chloramphenicol and placebo (10 of 18 patients died in the placebo arm, $P = .003$).¹²⁴ In this study, severe typhoid fever was defined as typhoid fever with shock or profound encephalopathy manifesting with delirium or obtundation. More recent studies continue to support the concurrent use of dexamethasone and antibiotics in both children and adults with severe typhoid fever.^{123,182} For this reason, administering high-dose steroids to patients who present with shock or, more often, significant mental status changes, and in whom there is strong suspicion of enteric fever, is recommended.

Supportive Care and Management of Complications

Supportive care of patients with severe enteric fever includes rehydration, nutritional management, and mobilization. Contrary to a prevailing historical opinion, it appears that fevers and malaise can be managed safely and more comfortably with a standard regimen of antipyretics.¹⁸³ One small double-blind trial showed that ibuprofen was superior to paracetamol in reducing the morbidity of fevers in enteric fever.¹⁸³

Patients with intestinal hemorrhage should be monitored, ideally in an intensive care setting, but can generally be managed through supportive care. Blood products, including packed red cells and plasma, may be needed in severe cases. Intestinal perforation in enteric fever is associated with a high mortality rate, but prompt surgical intervention is associated with improved survival.^{31,104,184} Although the majority of perforations are single, careful inspection is required to determine if multiple perforations are present. Although simple closure may be performed in some cases, a wedge excision or segmental resection is more often necessary because the surrounding bowel is often necrotic.^{185,186} Drainage of collections and peritoneal lavage should be performed before wound closure. Antibiotic regimens after perforation should be expanded to cover other intestinal microbiota, including anaerobes.

Treatment of Relapse and Chronic Carriage

Relapse usually occurs within 2 weeks of discontinuation of antibiotics, and relapsing patients typically present with milder symptoms than during the primary episode. Isolates obtained after relapse typically have the same antibiotic susceptibility as during the initial episode. The approach to treating patients with relapsed enteric fever is usually the same as for the primary episode.

Chronic carriage is defined as asymptomatic shedding of typhoidal *S. enterica* for 1 year or more. Asymptomatic shedding does not indicate a recurrence of illness or a high risk of recurrent illness.

Although chronic carriage is associated with an increased risk of gallbladder carcinoma,¹¹² it is unknown whether the eradication of carriage reduces this risk of carcinoma, so the decision to attempt to eradicate carriage is usually based on public health considerations and is essential for individuals whose occupation involves food handling. Several antibiotics used to treat individuals with acute enteric fever reach high concentrations in bile, including fluoroquinolones. A 28-day course of ciprofloxacin has demonstrated 80% to 90% efficacy in eradication of carriage.¹⁸⁷ However, cholecystectomy may be necessary to eradicate carriage in individuals with cholelithiasis who continue to shed typhoidal *S. enterica* after prolonged antimicrobial therapy.^{188,189} In an era of widespread nonsusceptibility to fluoroquinolones, there is a limited evidence base for successful treatment of carriage with antibiotics other than fluoroquinolones. Amoxicillin, administered at high doses (75–100 mg/day for 4–6 weeks), can be used, but tolerability is poor and overall eradication rates in many studies are modest.¹⁹⁰ There are even fewer data for treatment of carriers with trimethoprim-sulfamethoxazole with or without rifampin.¹⁹¹

Prevention

Because enteric fever is transmitted primarily through fecally contaminated water and food, the major preventive strategy to prevent enteric fever is the provision of safe water and adequate sanitation. At a municipal public infrastructure level, this requires appropriate facilities for waste disposal and sewage treatment in addition to provision of a safe water supply. In the United States, the introduction of chlorination and filtration of municipal water supplies in the early 1900s led to rapid and dramatic declines in the burden of typhoid.¹⁹² At a household level, interventions to prevent enteric fever include hand washing with soap before preparing or eating food, avoidance of raw or uncooked foods, and local disinfection of unsafe drinking water, followed by storage in narrow-mouthed containers. Unfortunately, WHO currently estimates that 2 billion people use a drinking water source that is contaminated with feces.¹⁹³

Historically, a heat-phenol-inactivated, whole-cell, killed typhoid vaccine was available for parenteral administration. The vaccine required multiple administrations, induced only short-term protection, and was associated with a high adverse event profile.¹⁹⁴ That vaccine is no longer commercially available in the United States.¹⁹⁵ Currently, two more effective and better-tolerated typhoid fever vaccines (Table 102.5) are commercially available in many areas: a capsular polysaccharide vaccine, licensed in the United States as Typhim Vi (Sanofi Pasteur, Swiftwater, PA), and an oral live-attenuated vaccine containing the *Salmonella* Typhi strain Ty21a, licensed in the United States as Vivotif (Valneva, Lyon, France). The polysaccharide vaccine is composed of the Vi capsular polysaccharide antigen from *Salmonella* Typhi strain Ty2, and the Vi vaccine confers 50% to 80% protection against *Salmonella* Typhi in children 2 years of age and older when administered as a single parenteral dose.^{194,196–198} The Vi vaccine is not boosted by additional doses, and protection wanes rapidly after 2 years, at which point revaccination is recommended if risk is ongoing. Ty21a is administered orally and also affords 50% to 80% protection against *Salmonella* Typhi.^{194,199,200} Although Ty21a enteric-coated capsules are often used for travelers, a liquid formulation prepared from two sachets, one containing vaccine and a second containing a buffer, have been used for oral administration to children in resource-limited settings. The duration of immunity to the Ty21a vaccine is dependent on the number of doses received. In the United States, a four-dose every-other-day series of Ty21a is recommended and confers significant immunity for up to 7 years, although revaccination is recommended every 5 years if risk is ongoing.¹⁹⁵ Antibacterial drugs should not be given any time from 3 days before until 3 days after administration of a series of Ty21a. For most patients with significant primary immunodeficiencies, Ty21a should not be given; however, the Strategic Advisory Group of Experts for the WHO advises that Ty21a can be safely given to asymptomatic individuals living with human immunodeficiency virus (HIV) infection and a CD4+ T-cell count greater than 200/mm³.¹⁹⁴

TABLE 102.5

Typhoid Fever Vaccines Currently Commercially Available Internationally				
VACCINE	TYPE	ROUTE	DOSE AND INTERVAL	MINIMUM AGE
PROTECTION AGAINST <i>SALMONELLA</i> TYPHI			BOOSTING INTERVAL IN TRAVELERS	
Ty21a	Live attenuated	Oral		
Four doses (in United States)				

Administer one dose every other day until complete

6years^a 50%–80%^b Every 5years

Vi capsule antigen	Polysaccharide	Intramuscular	1	2years	50%–80%	Every
2years						

Vi tetanus toxoid conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6months
80%–85%	Not yet defined			

Vi CRM-197 conjugate vaccine	Polysaccharide conjugate	Intramuscular	1	6 months
Not yet defined	Not yet defined			

^a Per World Health Organization and US Advisory Committee on Immunization Practices.^{194,195}

^b Has some efficacy against *Salmonella Paratyphi B* (see text). Additional vaccines are under development, including conjugate vaccines.

Modified from Magill AJ, et al., eds. *Hunter's Tropical Medicine and Emerging Infectious Diseases*. 9th ed. Philadelphia: Saunders; 2013:568–576.

In 2018, WHO recommended that typhoid vaccination programs be undertaken in settings with high typhoid incidence or high burden of antimicrobial-resistant *Salmonella Typhi*.¹⁹⁴ However, typhoid vaccines are not currently widely used in most endemic regions and are more often used by travelers to endemic areas. For travelers receiving either the Vi or Ty21a vaccine, it is optimal that typhoid vaccination be completed at least 7 days before travel, if possible. Limitations of currently licensed vaccines include limited immunity in young children (<2 years of age) and lack of cross-protection against other causes of enteric fever, primarily *Salmonella Paratyphi A*, although Ty21a appears to provide protection against *Salmonella Paratyphi B*.²⁰¹ To improve both immunogenicity in children younger than 2 years and to induce longer-lasting protection, a number of conjugate Vi vaccines are in various stages of development. One version that conjugated Vi to *Pseudomonas aeruginosa* recombinant exoprotein A (Vi-rEPA) achieved 90% protective efficacy against typhoid fever in a large randomized, double-blind clinical trial involving 11,000 children aged 2 to 5 years in Vietnam,²⁰² and has been demonstrated to be safe and immunogenic in phase II trials in infants.²⁰³ However, this vaccine has not been licensed internationally. A Vi-tetanus toxoid conjugate vaccine (Typbar-TCV; Bharat Biotech, Hyderabad, India) was licensed in India in 2015 on the basis of immunogenicity data.²⁰⁴ This vaccine was found to have 79% to 85% effectiveness in three large randomized trials^{205–207} and was prequalified by the WHO in 2018. A second Vi conjugate vaccine (TyphiBEV; Biological E, Hyderabad, India), using a modified diphtheria toxin (CRM 197) was prequalified by the WHO in 2020 based on immunogenicity,²⁰⁸ though clinical effectiveness data are not yet available. Both vaccines can be given to children as young as 6 months of age and have been introduced into routine immunization programs in select high burden countries. Live-attenuated and O-specific polysaccharide-based conjugate vaccines of paratyphoid A have also been reported; however, efficacy data and valid correlates of protection are lacking.²⁰⁹

In non–highly endemic areas, other efforts to prevent enteric fever are aimed at limiting the role of carriers in transmission, particularly for carriers who present a significant public health risk, including food handlers and those working in childcare or health care settings. In the United States, local and state health departments provide screening and work exclusion policies for individuals with enteric fever. The precise policies vary by location but may require screening of the patient and selected contacts to document clearance of bacterial shedding before discharge from public health follow-up and clearance for the performance of certain occupations.

Differential Diagnosis of Enteric and Typhoidal Fevers

Given the nonspecific clinical features of enteric fever and the current limitations of diagnostic testing, it is important that clinicians consider the broader differential diagnosis when approaching patients with possible enteric fever. Such patients usually present with prolonged or relapsing fever, an acute febrile episode in a resource-limited area, or fever and abdominal symptoms and signs. Considerations in the differential diagnosis of enteric fever include epidemiologic risk factors, such as the relative local prevalence of invasive nontyphoidal salmonellosis, malaria, and other febrile diseases, and the type and timing of possible exposures, the duration and magnitude of fevers, the types of abdominal symptoms and signs, and laboratory features at the time of clinical presentation.

Bacteremia Associated With Nontyphoidal Salmonella

Although febrile gastroenteritis is the most common clinical manifestation of human nontyphoidal salmonellosis, nontyphoidal serotypes can also cause prolonged fever and bacteremia, particularly in immunocompromised patients with cell-mediated immunodeficiencies or with concomitant malaria. NTS bacteremia is a leading cause of death of young children and adults with AIDS in sub-Saharan Africa.^{17,210} NTS can also cause abscess formation (spleen, kidneys, liver), and *Salmonella* and staphylococci are the leading causes of aortic atheroma seeding, mycotic aneurysm formation, and prolonged bacteremia.²¹¹ Individuals with sickle cell disease and individuals with concomitant schistosomiasis are also at risk of prolonged and invasive nontyphoidal salmonellosis (see Chapter 228).^{212,213}

Typhoidal Fever

A number of non-*Salmonella* infections have features that may mimic typhoid or enteric fever (Table 102.6). These illnesses have sometimes been referred to as typhoidal fevers and are usually characterized as nonspecific systemic illnesses associated with fever that lasts longer than a week. Indeed, typhoid itself was named for its clinical similarity to typhus, caused by *Rickettsia* species. Typhus and typhoid are both characterized as febrile illnesses that may be prolonged and associated with encephalopathy or confusion, and if untreated, both diseases are associated with high mortality rates. In South and Southeast Asia, scrub typhus (*Orientia tsutsugamushi*) and murine typhus (*Rickettsia typhi*) are highly endemic, and are often mistaken for typhoid.²¹⁴⁻²¹⁶ In sub-Saharan Africa, various *Rickettsia* species, including *Rickettsia africae* (the causative agent of African tick-bite fever), are common causes of nonspecific febrile illness. Leptospirosis is also a common cause of nonspecific febrile illness in many typhoid-endemic areas, and is often underdiagnosed owing to lack or underuse of diagnostics.²¹⁴ A clinical diagnosis of typhomalarial fever, a term that appeared in the mid-19th century, highlights the overlap of the clinical syndromes of typhoid with malaria.²¹⁷ Individuals with malaria can also present with a wide range of symptoms, including prolonged or recurrent fever, headache, abdominal pain, and cough. Similarly, individuals with brucellosis may present with nonspecific fever, malaise, and abdominal pain. Fever may be prolonged, and infection during brucellosis is often characterized by its chronicity. Manifestations include osteoarticular and genitourinary involvement, granulomatous hepatitis, uveitis, and, rarely, endocarditis. Tularemia is a zoonotic infection caused by *Francisella tularensis*. Most human infections are because of exposure to small rodents or lagomorphs (rabbits), and humans usually become infected through direct contact with infected animals or animal parts, or through the bite of an arthropod as a transmission vehicle. Similar to *Salmonella*, *F. tularensis* is an intracellular pathogen that is able to survive and multiply within phagocytic cells, including macrophages, and infection spreads throughout the reticuloendothelial system. Without appropriate treatment, typhoidal tularemia is often fatal.

TABLE 102.6

Differential Diagnoses of Infectious Diseases to Consider in Patients with Possible Typhoid or Enteric Fever

Common Infectious Causes of Nonspecific Acute Febrile Illness in Resource-Limited Areas

Salmonella bacteremia (typhoidal and nontyphoidal serotypes)

Malaria

Dengue and other arboviral fevers

Leptospirosis

Influenza

Rickettsioses

Bartonellosis

Infectious Causes of Typhoidal Illnesses, Including Prolonged Nonlocalizing Febrile Illnesses

Persistent nontyphoidal bacteremia or abscess formation

Typhus (rickettsiosis)

Brucellosis

Typhoidal tularemia

Q fever

Bartonellosis (including trench fever)

Melioidosis

Rat-bite fever, Haverhill fever (*Streptobacillus moniliformis*)
Relapsing fever (borreliosis)
Infective endocarditis
Malaria
Babesiosis
Visceral leishmaniasis
West African trypanosomiasis
Tuberculosis
Endemic mycoses (histoplasmosis)

Persistent fever with nonspecific symptoms may also be seen with melioidosis, an infection caused by *Burkholderia pseudomallei*. Melioidosis is usually reported in South and East Asia and northern Australia. Melioidosis may manifest in a number of ways, ranging from a chronic febrile illness to an acute septicemia,²¹⁸ and it may include abscess formation (often in the spleen, liver, lungs, brain, or muscles) or pneumonia. Melioidosis is associated with water and soil exposure and can affect anyone, although individuals with diabetes, immunosuppression, chronic renal disease, alcoholism, and cirrhosis are at particular risk. Clinical manifestations may occur years after initial exposure.

A typhoidal-like illness can also be seen with Q fever, caused by the intracellular bacterium *Coxiella burnetii*. Q fever is a zoonotic infection usually associated with exposure to sheep or cattle, and humans become infected through exposure to pseudospores from animal products, especially placental tissues. Individuals with Q fever may present with prolonged fever and a wide range of symptoms, including headache, pneumonia, hepatitis, and endocarditis. Rat-bite fever, caused by *Streptobacillus moniliformis*, may also cause a nonspecific febrile illness that may mimic enteric fever, especially when the route of infection is foodborne (Haverhill fever); arthritis and rash can occur.²¹⁹

A number of *Bartonella* species can cause infections characterized by recurrent or prolonged fever. *Bartonella bacilliformis* is the cause of Carrion disease and is transmitted by sand flies in focal areas of Peru, Colombia, and Ecuador. During the acute illness (Oroya fever), patients present with fever and hemolytic anemia. In survivors, disease can progress to verruga peruana, a chronic illness characterized by vascular, proliferative, nodular skin lesions. In addition, *Bartonella quintana* is the cause of trench fever, a relapsing febrile illness transmitted by body lice. Trench fever usually affects individuals who are displaced or severely impoverished; onset is usually acute with high fever, headache, and retro-orbital pain, bone pain of the shins, and myalgia. Each episode of fever may last for 2 to 4 days, with relapses after 4 to 6 days. Prolonged bacteremia can be detected in 20% to 30% of patients.²²⁰ Relapsing fever can also be caused by a number of *Borrelia* species and can be either louse-borne or tick-borne. Louse-borne relapsing fever is a human-restricted infection caused by *Borrelia recurrentis* and is associated with human body lice. Most cases of louse-borne relapsing fever are reported in Africa. Tick-borne relapsing fever has been reported in Asia, Africa, and the Americas. Relapsing fevers are characterized by relapsing and recurrent episodes of fever associated with spirochetemia. Diagnosis usually rests on detection of *Borrelia* on a thick blood smear with microscopy, although polymerase chain reaction (PCR)-based diagnosis is available. Untreated, louse-borne relapsing fever is associated with a high mortality rate. Fever may be prolonged, with many recurrences over several months.²²¹

An enteric fever-like syndrome with abdominal pain may also be caused by a number of enteric pathogens, including *Yersinia enterocolitica*, *Yersinia pseudotuberculosis*, and *Campylobacter fetus* and *Campylobacter jejuni*. Individuals with these infections may present with a prolonged febrile illness, headache, and abdominal pain. Individuals with *Y. enterocolitica* or *Y. pseudotuberculosis* may have acute diarrhea as a major manifestation.^{222,223} Most important, the prolonged enteric fever-like syndromes caused by these agents tend to occur in individuals with significant underlying disease, especially cirrhosis and iron-overloaded states.²²² Diagnosis is usually based on microbiologic culturing of blood.

Additional Causes of Prolonged or Persistent Fever

Most acute febrile illnesses resolve within 3 to 7 days. Individuals with typhoid or enteric fever may have fever that lasts for 1 to 4 weeks before resolution if left untreated. Clinicians assessing individuals with fever that lasts longer than 7 to 10 days need to consider a wide differential, both for infectious and noninfectious processes (see Chapter 58). Noninfectious causes that also need

to be considered include lymphoma, malignancy, drug fever, Addison disease, and inflammatory conditions and connective tissue disorders. As described earlier, prolonged fever may also be associated with malaria, brucellosis, yersiniosis, tularemia, Q fever, melioidosis, endocarditis, tuberculosis, and endemic mycoses such as histoplasmosis. Recurrent or relapsing fever may be associated with trench fever and relapsing fever. Clinicians caring for individuals with prolonged fever in resource-limited or tropical areas or caring for individuals who have traveled to such regions need to consider these and other diagnoses. For instance, individuals with babesiosis, caused by an intraerythrocytic protozoal parasite transmitted by ticks, may also present with a nonspecific and prolonged febrile syndrome. Microscopy can confirm the diagnosis. Chronic and relapsing fever are also part of visceral leishmaniasis, a protozoal parasitic infection of the reticuloendothelial system. Individuals with visceral leishmaniasis often present with prolonged and ongoing fever, hepatosplenomegaly, pancytopenia, hypergammaglobulinemia, and absence of eosinophilia. Serologic tests and direct parasitic detection are usually used to confirm the diagnosis. Individuals with human African trypanosomiasis (sleeping sickness) can present with chronic and recurrent fever, weight loss, and lymphadenopathy before neurologic manifestations occur. Serologic tests and direct parasitic detection are usually used to confirm the diagnosis. Prolonged fever can also be associated with a number of viral syndromes, including Epstein-Barr virus (mononucleosis), characterized by prolonged fever, headache, weakness, and fatigue and may include pharyngitis, hepatosplenomegaly, and lymphadenopathy. Serologic assays are available.

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